

Human review packet: GolzHaghtalabYang2025Distortion

Generated from byte-pinned source excerpts, the expanded PaperInterface specifications, and hash-bound raw-source-to-Spec screening records.

Paper: GolzHaghtalabYang2025Distortion
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Source version: NeurIPS 2025 published conference PDF
Source record: audit/paper_statement_map.json
Rows: 16
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Source URL: [official source](#)

How to use this packet

For each source claim, compare the exact source excerpts with the expanded PaperInterface specification. Mark up this PDF or fill the blank reviewer box in the TeX source; return the annotated file for reconciliation into the paper-local human-review log.

Open the interactive dashboard

The interactive dashboard is an optional alternative to using this PDF; it presents the same review surface rather than an additional review requirement. After forking and cloning (or pulling) AppliedModelingLib, run the following from the repository root. The cache command is needed only the first time in a new clone.

```
cd <your-repository-checkout>
lake exe cache get
python3 scripts/review_dashboard.py --paper GolzHaghtalabYang2025Distortion --serve
```

Open <http://127.0.0.1:8765/> in a browser and keep that terminal running while reviewing. The dashboard records saved annotations in this paper's local reviewer-owned review record.

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Governing Lean declarations

These exact graph-bound declaration bodies are retained by the displayed specifications or reviewed prerequisites. They are supporting context and do not add source-result rows or semantic judgments.

AppliedModelingLib.Alignment.Welfare.contextFreeAlternativePolicy

Declaration location: reusable library

Lean declaration body

```
type:
{Alternative : Type u_1} → PMF Alternative → PUnit.{u_2 + 1} → PMF Alternative

value:
fun {Alternative : Type u_1} (policy : PMF Alternative) (a : PUnit.{u_2 + 1}) => policy
```

AppliedModelingLib.Alignment.Welfare.sigmoidChordSlope

Declaration location: reusable library

Lean declaration body

```
type:
ℝ → ℝ

value:
fun (beta : ℝ) => (beta.sigmoid - 1 / 2) / beta
```

AppliedModelingLib.Learning.HumanFeedback.BinaryPairwiseReport

Declaration location: reusable library

Lean declaration body

```
type:
Type u → Type u

value:
fun (Alternative : Type u) => (Alternative × Alternative) × Bool
```

AppliedModelingLib.Learning.HumanFeedback.CorrelatedPairSampling

Declaration location: reusable library

Lean declaration body

```
type:
Type u_1 → Type u_1

value:
fun (Response : Type u_1) => PMF (Response × Response)
```

AppliedModelingLib.Learning.HumanFeedback.CorrelatedPairSampling.HasFullOffDiagonalSupport

Declaration location: reusable library

Lean declaration body

```
type:
{Response : Type u_1} → AppliedModelingLib.Learning.HumanFeedback.CorrelatedPairSampling Response → Prop

value:
fun {Response : Type u_1} (sampling : AppliedModelingLib.Learning.HumanFeedback.CorrelatedPairSampling Response) =>
  ∀ (first second : Response), first ≠ second → 0 < (sampling (first, second)).toReal
```

Direct retained dependencies

- [AppliedModelingLib.Learning.HumanFeedback.CorrelatedPairSampling](#)

AppliedModelingLib.Learning.HumanFeedback.CorrelatedPairSampling.HasZeroDiagonal

Declaration location: reusable library

Lean declaration body

```
type:
{Response : Type u_1} → AppliedModelingLib.Learning.HumanFeedback.CorrelatedPairSampling Response → Prop

value:
fun {Response : Type u_1} (sampling : AppliedModelingLib.Learning.HumanFeedback.CorrelatedPairSampling Response) =>
  ∀ (response : Response), (sampling (response, response)).toReal = 0
```

Direct retained dependencies

- [AppliedModelingLib.Learning.HumanFeedback.CorrelatedPairSampling](#)

AppliedModelingLib.Learning.HumanFeedback.CorrelatedPairSampling.IsSymmetric

Declaration location: reusable library

Lean declaration body

```
type:
{Response : Type u_1} → AppliedModelingLib.Learning.HumanFeedback.CorrelatedPairSampling Response → Prop

value:
fun {Response : Type u_1} (sampling : AppliedModelingLib.Learning.HumanFeedback.CorrelatedPairSampling Response) =>
  ∀ (first second : Response), (sampling (first, second)).toReal = (sampling (second, first)).toReal
```

Direct retained dependencies

- [AppliedModelingLib.Learning.HumanFeedback.CorrelatedPairSampling](#)

AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset

Declaration location: reusable library

Lean declaration body

```
field count:
{Alternative : Type u_1} →
  AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset Alternative → Alternative → Alternative → ℕ

field diagonal_zero:
∀ {Alternative : Type u_1} (self : AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset Alternative)
  (alternative : Alternative), self.count alternative alternative = 0
```

AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset.CDFLikePairwiseLink

Declaration location: reusable library

Lean declaration body

```
field toFun:
AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset.CDFLikePairwiseLink → ℝ → ℝ

field nonneg:
∀ (self : AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset.CDFLikePairwiseLink) (gap : ℝ),
  0 ≤ self.toFun gap

field le_one:
∀ (self : AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset.CDFLikePairwiseLink) (gap : ℝ),
  self.toFun gap ≤ 1

field monotone:
∀ (self : AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset.CDFLikePairwiseLink), Monotone self.toFun

field complementary:
∀ (self : AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset.CDFLikePairwiseLink) (gap : ℝ),
  self.toFun (-gap) + self.toFun gap = 1

field tendsto_atBot_zero:
∀ (self : AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset.CDFLikePairwiseLink),
  Filter.Tendsto self.toFun Filter.atBot (nhds 0)

field tendsto_atTop_one:
∀ (self : AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset.CDFLikePairwiseLink),
  Filter.Tendsto self.toFun Filter.atTop (nhds 1)
```

AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset.ScoreVector

Declaration location: reusable library

Lean declaration body

```
type:
Type u_1 → Type u_1

value:
fun (Alternative : Type u_1) => Alternative → ℝ
```

AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset.isReferenceNormalized

Declaration location: reusable library

Lean declaration body

```
type:
{Alternative : Type u_1} →
Alternative → AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset.ScoreVector Alternative → Prop

value:
fun {Alternative : Type u_1} (reference : Alternative)
(score : AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset.ScoreVector Alternative) =>
score reference = 0
```

Direct retained dependencies

- [AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset.ScoreVector](#)

AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset.randomUtilityWinProbability

Declaration location: reusable library

Lean declaration body

```
type:
{Alternative : Type u_1} →
(ℝ → ℝ) →
AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset.ScoreVector Alternative →
Alternative → Alternative → ℝ

value:
fun {Alternative : Type u_1} (link : ℝ → ℝ)
(score : AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset.ScoreVector Alternative)
(first second : Alternative) =>
link (score first - score second)
```

Direct retained dependencies

- [AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset.ScoreVector](#)

AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset.pairwiseLogLikelihood

Declaration location: reusable library

Lean declaration body

```
type:
{Alternative : Type u_1} →
[Fintype Alternative] →
[DecidableEq Alternative] →
AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset Alternative →
(ℝ → ℝ) → AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset.ScoreVector Alternative → ℝ

value:
fun {Alternative : Type u_1} [Fintype Alternative] [DecidableEq Alternative]
(dataset : AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset Alternative) (link : ℝ → ℝ)
(score : AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset.ScoreVector Alternative) =>
∑ pair ∈ Finset.univ.offDiag,
↑(dataset.count pair.1 pair.2) *
Real.log
(AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset.randomUtilityWinProbability link score pair.1
pair.2)
```

Direct retained dependencies

- [AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset](#)
- [AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset.ScoreVector](#)
- [AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset.randomUtilityWinProbability](#)

AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset.isPairwiseMLE

Declaration location: reusable library

Lean declaration body

```
type:
{Alternative : Type u_1} →
[Fintype Alternative] →
[DecidableEq Alternative] →
  AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset Alternative →
  ( $\mathbb{R} \rightarrow \mathbb{R}$ ) →
    Alternative → AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset.ScoreVector Alternative → Prop

value:
fun {Alternative : Type u_1} [Fintype Alternative] [DecidableEq Alternative]
  (dataset : AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset Alternative) (link :  $\mathbb{R} \rightarrow \mathbb{R}$ )
  (reference : Alternative)
  (score : AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset.ScoreVector Alternative) =>
AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset.isReferenceNormalized reference score  $\wedge$ 
  IsMaxOn (dataset.pairwiseLogLikelihood link)
    {candidate : AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset.ScoreVector Alternative |
      AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset.isReferenceNormalized reference candidate}
  score
```

Direct retained dependencies

- [AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset](#)
- [AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset.ScoreVector](#)
- [AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset.isReferenceNormalized](#)
- [AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset.pairwiseLogLikelihood](#)

AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset.sigmoidCDFLikePairwiseLink

Declaration location: reusable library

Lean declaration body

```
type:
AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset.CDFLikePairwiseLink

value:
{ toFun := Real.sigmoid, nonneg := Real.sigmoid_nonneg, le_one := Real.sigmoid_le_one,
  monotone := Real.sigmoid_monotone,
  complementary := AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset.sigmoidCDFLikePairwiseLink._proof_1,
  tendsto_atBot_zero := Real.tendsto sigmoid_atBot, tendsto_atTop_one := Real.tendsto sigmoid_atTop }
```

Direct retained dependencies

- [AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset.CDFLikePairwiseLink](#)

AppliedModelingLib.Learning.HumanFeedback.RewardEquivalent

Declaration location: reusable library

Lean declaration body

```
type:
{Context : Type u_1} → {Response : Type u_2} → (Context → Response →  $\mathbb{R}$ ) → (Context → Response →  $\mathbb{R}$ ) → Prop

value:
fun {Context : Type u_1} {Response : Type u_2} (first second : Context → Response →  $\mathbb{R}$ ) =>
   $\exists$  (shift : Context →  $\mathbb{R}$ ),
     $\forall$  (context : Context) (response : Response), second context response = first context response + shift context
```

AppliedModelingLib.Learning.HumanFeedback.binaryReportWin

Declaration location: reusable library

Lean declaration body

```
type:
{Alternative : Type u_1} →
[DecidableEq Alternative] →
Alternative → Alternative → AppliedModelingLib.Learning.HumanFeedback.BinaryPairwiseReport Alternative → Prop

value:
fun {Alternative : Type u_1} [DecidableEq Alternative] (winner loser : Alternative)
  (report : AppliedModelingLib.Learning.HumanFeedback.BinaryPairwiseReport Alternative) =>
  winner ≠ loser ∧
  (report.1.1 = winner ∧ report.1.2 = loser ∧ report.2 = true ∨
   report.1.1 = loser ∧ report.1.2 = winner ∧ report.2 = false)
```

Direct retained dependencies

- [AppliedModelingLib.Learning.HumanFeedback.BinaryPairwiseReport](#)

AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset.ofBinaryReports

Declaration location: reusable library

Lean declaration body

```
type:
{Alternative : Type u_1} →
[DecidableEq Alternative] →
{horizon : ℕ} →
(Fin horizon → AppliedModelingLib.Learning.HumanFeedback.BinaryPairwiseReport Alternative) →
AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset Alternative

value:
fun {Alternative : Type u_1} [DecidableEq Alternative] {horizon : ℕ}
  (sample : Fin horizon → AppliedModelingLib.Learning.HumanFeedback.BinaryPairwiseReport Alternative) =>
  {
    count := fun (winner loser : Alternative) =>
      {index : Fin horizon |
        AppliedModelingLib.Learning.HumanFeedback.binaryReportWin winner loser (sample index)}.card,
    diagonal_zero := AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset.ofBinaryReports._proof_1 sample }
```

Direct retained dependencies

- [AppliedModelingLib.Learning.HumanFeedback.BinaryPairwiseReport](#)
- [AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset](#)
- [AppliedModelingLib.Learning.HumanFeedback.binaryReportWin](#)

AppliedModelingLib.Learning.HumanFeedback.correlatedBernoulliReport

Declaration location: reusable library

Lean declaration body

```
type:
(probability : ℝ) → 0 ≤ probability → probability ≤ 1 → PMF Bool

value:
fun (probability : ℝ) (hnonneg : 0 ≤ probability) (hle_one : probability ≤ 1) =>
  PMF.bernoulli (probability, hnonneg)
  (AppliedModelingLib.Learning.HumanFeedback.correlatedBernoulliReport._proof_1 probability hnonneg hle_one)
```

AppliedModelingLib.Learning.HumanFeedback.correlatedOneComparisonReportLaw

Declaration location: reusable library

Lean declaration body

```
type:
{Response : Type u_1} →
[Fintype Response] →
[DecidableEq Response] →
AppliedModelingLib.Learning.HumanFeedback.CorrelatedPairSampling Response →
AppliedModelingLib.Learning.HumanFeedback.PairwisePreference PUnit.{u_2 + 1} Response →
PMF (AppliedModelingLib.Learning.HumanFeedback.BinaryPairwiseReport Response)

value:
fun {Response : Type u_1} [Fintype Response] [DecidableEq Response]
  (sampling : AppliedModelingLib.Learning.HumanFeedback.CorrelatedPairSampling Response)
  (preference : AppliedModelingLib.Learning.HumanFeedback.PairwisePreference PUnit.{u_2 + 1} Response) =>
  PMF.bind sampling fun (pair : Response × Response) =>
  PMF.map (fun (outcome : Bool) => (pair, outcome))
  (AppliedModelingLib.Learning.HumanFeedback.correlatedBernoulliReport (preference.prob PUnit.unit pair.1 pair.2)
   (AppliedModelingLib.Learning.HumanFeedback.correlatedOneComparisonReportLaw._proof_1 preference pair)
   (AppliedModelingLib.Learning.HumanFeedback.correlatedOneComparisonReportLaw._proof_2 preference pair))
```

Direct retained dependencies

- [AppliedModelingLib.Learning.HumanFeedback.BinaryPairwiseReport](#)
- [AppliedModelingLib.Learning.HumanFeedback.CorrelatedPairSampling](#)
- [AppliedModelingLib.Learning.HumanFeedback.PairwisePreference](#)
- [AppliedModelingLib.Learning.HumanFeedback.correlatedBernoulliReport](#)

AppliedModelingLib.PMFKernel

Declaration location: reusable library

Lean declaration body

```
type:
Type u_1 → Type u_2 → Type (max u_1 u_2)

value:
fun (Input : Type u_1) (Output : Type u_2) => Input → PMF Output
```

AppliedModelingLib.Learning.HumanFeedback.FinitePolicy

Declaration location: reusable library

Lean declaration body

```
type:
Type u_1 → Type u_2 → Type (max u_1 u_2)

value:
fun (Context : Type u_1) (Response : Type u_2) => AppliedModelingLib.PMFKernel Context Response
```

Direct retained dependencies

- [AppliedModelingLib.PMFKernel](#)

AppliedModelingLib.Learning.HumanFeedback.policyLogRatio

Declaration location: reusable library

Lean declaration body

```
type:
{Context : Type u_1} →
{Response : Type u_2} →
AppliedModelingLib.Learning.HumanFeedback.FinitePolicy Context Response →
AppliedModelingLib.Learning.HumanFeedback.FinitePolicy Context Response → Context → Response → ℝ

value:
fun {Context : Type u_1} {Response : Type u_2}
(candidate reference : AppliedModelingLib.Learning.HumanFeedback.FinitePolicy Context Response) (context : Context)
(response : Response) =>
Real.log ((candidate context) response).toReal - Real.log ((reference context) response).toReal
```

Direct retained dependencies

- [AppliedModelingLib.Learning.HumanFeedback.FinitePolicy](#)

AppliedModelingLib.Learning.HumanFeedback.dpoImplicitReward

Declaration location: reusable library

Lean declaration body

```
type:
{Context : Type u_1} →
{Response : Type u_2} →
AppliedModelingLib.Learning.HumanFeedback.FinitePolicy Context Response →
AppliedModelingLib.Learning.HumanFeedback.FinitePolicy Context Response → ℝ → Context → Response → ℝ

value:
fun {Context : Type u_1} {Response : Type u_2}
(candidate reference : AppliedModelingLib.Learning.HumanFeedback.FinitePolicy Context Response) (klWeight : ℝ)
(a : Context) (a_1 : Response) =>
klWeight * AppliedModelingLib.Learning.HumanFeedback.policyLogRatio candidate reference a a_1
```

Direct retained dependencies

- [AppliedModelingLib.Learning.HumanFeedback.FinitePolicy](#)
- [AppliedModelingLib.Learning.HumanFeedback.policyLogRatio](#)

AppliedModelingLib.Probability.finiteIdScoreSum

Declaration location: reusable library

Lean declaration body

```
type:
{α : Type u_1} → {ι : Type u_2} → [Fintype ι] → (α → ℝ) → (ι → α) → ℝ

value:
fun {α : Type u_1} {ι : Type u_2} [Fintype ι] (score : α → ℝ) (sample : ι → α) => ∑ i : ι, score (sample i)
```

AppliedModelingLib.Probability.finiteMGF

Declaration location: reusable library

Lean declaration body

```
type:
{α : Type u_1} → [Fintype α] → PMF α → (α → ℝ) → ℝ → ℝ

value:
fun {α : Type u_1} [Fintype α] (μ : PMF α) (score : α → ℝ) (z : ℝ) => ∑ a : α, (μ a).toReal * Real.exp (z * score a)
```

AppliedModelingLib.binaryMixturePMF

Declaration location: reusable library

Lean declaration body

```
type:
{α : Type u_1} → (p : NNReal) → p ≤ 1 → PMF α → PMF α → PMF α

value:
fun {α : Type u_1} (p : NNReal) (hp : p ≤ 1) (selected unselected : PMF α) =>
(PMF.bernoulli p hp).bind fun (takeSelected : Bool) => if takeSelected = true then selected else unselected
```

AppliedModelingLib.exponentialTilt

Declaration location: reusable library

Lean declaration body

```
type:
{Outcome : Type u_1} → [Fintype Outcome] → [DecidableEq Outcome] → PMF Outcome → (Outcome → ℝ) → ℝ → PMF Outcome

value:
fun {Outcome : Type u_1} [Fintype Outcome] [DecidableEq Outcome] (reference : PMF Outcome) (utility : Outcome → ℝ)
(inverseTemperature : ℝ) =>
PMF.ofFintype
(fun (outcome : Outcome) =>
ENNReal.ofReal
((reference outcome).toReal * Real.exp (inverseTemperature * utility outcome) /
AppliedModelingLib.Probability.finiteMGF reference utility inverseTemperature))
(AppliedModelingLib.exponentialTilt._proof_1 reference utility inverseTemperature)
```

Direct retained dependencies

- [AppliedModelingLib.Probability.finiteMGF](#)

AppliedModelingLib.Learning.HumanFeedback.contextExponentialTiltPolicy

Declaration location: reusable library

Lean declaration body

```
type:
{Context : Type u_1} →
{Response : Type u_2} →
[Fintype Response] →
[DecidableEq Response] →
AppliedModelingLib.Learning.HumanFeedback.FinitePolicy Context Response →
(Context → Response → ℝ) → ℝ → Context → PMF Response

value:
fun {Context : Type u_1} {Response : Type u_2} [Fintype Response] [DecidableEq Response]
(reference : AppliedModelingLib.Learning.HumanFeedback.FinitePolicy Context Response)
(utility : Context → Response → ℝ) (inverseTemperature : ℝ) (a : Context) =>
AppliedModelingLib.exponentialTilt (reference a) (utility a) inverseTemperature
```

Direct retained dependencies

- [AppliedModelingLib.Learning.HumanFeedback.FinitePolicy](#)
- [AppliedModelingLib.exponentialTilt](#)

AppliedModelingLib.Learning.HumanFeedback.dpoOptimalPolicy

Declaration location: reusable library

Lean declaration body

```
type:
{Context : Type u_1} →
{Response : Type u_2} →
[Fintype Response] →
[DecidableEq Response] →
AppliedModelingLib.Learning.HumanFeedback.FinitePolicy Context Response →
(Context → Response → ℝ) → ℝ → Context → PMF Response

value:
fun {Context : Type u_1} {Response : Type u_2} [Fintype Response] [DecidableEq Response]
(reference : AppliedModelingLib.Learning.HumanFeedback.FinitePolicy Context Response)
(reward : Context → Response → ℝ) (klWeight : ℝ) (a : Context) =>
AppliedModelingLib.Learning.HumanFeedback.contextExponentialTiltPolicy reference reward klWeight-1 a
```

Direct retained dependencies

- [AppliedModelingLib.Learning.HumanFeedback.FinitePolicy](#)
- [AppliedModelingLib.Learning.HumanFeedback.contextExponentialTiltPolicy](#)

AppliedModelingLib.Learning.HumanFeedback.pointwisePolicyKLDivergence

Declaration location: reusable library

Lean declaration body

```
type:
{Context : Type u_1} →
{Response : Type u_2} →
[Fintype Response] →
[DecidableEq Response] →
AppliedModelingLib.Learning.HumanFeedback.FinitePolicy Context Response →
AppliedModelingLib.Learning.HumanFeedback.FinitePolicy Context Response → Context → ℝ

value:
fun {Context : Type u_1} {Response : Type u_2} [Fintype Response] [DecidableEq Response]
(first second : AppliedModelingLib.Learning.HumanFeedback.FinitePolicy Context Response) (context : Context) =>
AppliedModelingLib.finiteKLDivergence (first context) (second context)
```

Direct retained dependencies

- [AppliedModelingLib.Learning.HumanFeedback.FinitePolicy](#)
- [AppliedModelingLib.finiteKLDivergence](#)

AppliedModelingLib.finitePMFIdPathMeasure

Declaration location: reusable library

Lean declaration body

```
type:
{Signal : Type u_1} →
[inst : MeasurableSpace Signal] →
[Fintype Signal] → [DecidableEq Signal] → PMF Signal → MeasureTheory.Measure (ℕ → Signal)

value:
fun {Signal : Type u_1} [MeasurableSpace Signal] [Fintype Signal] [DecidableEq Signal] (law : PMF Signal) =>
MeasureTheory.Measure.infinitePi fun (x : ℕ) => law.toMeasure
```

AppliedModelingLib.finitePMFIdPathPrefix

Declaration location: reusable library

Lean declaration body

```
type:
{Signal : Type u_1} → (ℕ → Signal) → (N : ℕ) → Fin N → Signal

value:
fun {Signal : Type u_1} (path : ℕ → Signal) (N : ℕ) (a : Fin N) => path ↑ a
```

AppliedModelingLib.finiteWeightedPMF

Declaration location: reusable library

Lean declaration body

```
type:
{α : Type u_1} → [inst : Fintype α] → (weight : α → ℝ) → (∀ (a : α), 0 ≤ weight a) → 0 < ∑ a : α, weight a → PMF α

value:
fun {α : Type u_1} [Fintype α] (weight : α → ℝ) (hweight_nonneg : ∀ (a : α), 0 ≤ weight a)
(htotal_pos : 0 < ∑ a : α, weight a) =>
PMF.ofFintype (fun (a : α) => ENNReal.ofReal (weight a / ∑ b : α, weight b))
(AppliedModelingLib.finiteWeightedPMF_proof_1 weight hweight_nonneg htotal_pos)
```

AppliedModelingLib.pmfExp

Declaration location: reusable library

Lean declaration body

```
type:
{α : Type u_1} → [Fintype α] → PMF α → (α → ℝ) → ℝ

value:
fun {α : Type u_1} [Fintype α] (μ : PMF α) (f : α → ℝ) => ∑ a : α, (μ a).toReal * f a
```

AppliedModelingLib.Alignment.Welfare.pairwiseBordaScore

Declaration location: reusable library

Lean declaration body

```
type:
{Alternative : Type u_1} →
[Fintype Alternative] →
[DecidableEq Alternative] →
PMF Alternative →
AppliedModelingLib.Learning.HumanFeedback.PairwisePreference PUnit.{u_2 + 1} Alternative → Alternative → ℝ

value:
fun {Alternative : Type u_1} [Fintype Alternative] [DecidableEq Alternative] (sampling : PMF Alternative)
(preference : AppliedModelingLib.Learning.HumanFeedback.PairwisePreference PUnit.{u_2 + 1} Alternative)
(alternative : Alternative) =>
AppliedModelingLib.pmfExp sampling fun (opponent : Alternative) => preference.prob PUnit.unit alternative opponent
```

Direct retained dependencies

- [AppliedModelingLib.Learning.HumanFeedback.PairwisePreference](#)
- [AppliedModelingLib.pmfExp](#)

AppliedModelingLib.IsFiniteKLExtendedRegularizedUtilityMax

Declaration location: reusable library

Lean declaration body

```
type:
{Outcome : Type u_1} →
[Fintype Outcome] → [DecidableEq Outcome] → PMF Outcome → PMF Outcome → (Outcome → ℝ) → WithTop ℝ → Prop

value:
fun {Outcome : Type u_1} [Fintype Outcome] [DecidableEq Outcome] (reference candidate : PMF Outcome)
  (utility : Outcome → ℝ) (klWeight : WithTop ℝ) =>
  (∃ (finiteWeight : ℝ),
    klWeight = ↑finiteWeight ∧
    0 ≤ finiteWeight ∧
    ∀ (other : PMF Outcome),
      AppliedModelingLib.pmfExp other utility -
        finiteWeight * AppliedModelingLib.finiteKLDivergence other reference ≤
        AppliedModelingLib.pmfExp candidate utility -
        finiteWeight * AppliedModelingLib.finiteKLDivergence candidate reference) ∧
    klWeight = T ∧ candidate = reference
```

Direct retained dependencies

- [AppliedModelingLib.finiteKLDivergence](#)
- [AppliedModelingLib.pmfExp](#)

AppliedModelingLib.Learning.HumanFeedback.contextAveragedPolicyKLDivergence

Declaration location: reusable library

Lean declaration body

```
type:
{Context : Type u_1} →
{Response : Type u_2} →
[Fintype Context] →
[DecidableEq Context] →
[Fintype Response] →
[DecidableEq Response] →
PMF Context →
AppliedModelingLib.Learning.HumanFeedback.FinitePolicy Context Response →
AppliedModelingLib.Learning.HumanFeedback.FinitePolicy Context Response → ℝ

value:
fun {Context : Type u_1} {Response : Type u_2} [Fintype Context] [DecidableEq Context] [Fintype Response]
  [DecidableEq Response] (contextLaw : PMF Context)
  (first second : AppliedModelingLib.Learning.HumanFeedback.FinitePolicy Context Response) =>
  AppliedModelingLib.pmfExp contextLaw
  (AppliedModelingLib.Learning.HumanFeedback.pointwisePolicyKLDivergence first second)
```

Direct retained dependencies

- [AppliedModelingLib.Learning.HumanFeedback.FinitePolicy](#)
- [AppliedModelingLib.Learning.HumanFeedback.pointwisePolicyKLDivergence](#)
- [AppliedModelingLib.pmfExp](#)

AppliedModelingLib.Learning.HumanFeedback.policyExpectedScore

Declaration location: reusable library

Lean declaration body

```
type:
{Context : Type u_1} →
{Response : Type u_2} →
[Fintype Context] →
[DecidableEq Context] →
[Fintype Response] →
[DecidableEq Response] →
PMF Context →
AppliedModelingLib.Learning.HumanFeedback.FinitePolicy Context Response → (Context → Response → ℝ) → ℝ

value:
fun {Context : Type u_1} {Response : Type u_2} [Fintype Context] [DecidableEq Context] [Fintype Response]
  [DecidableEq Response] (contextLaw : PMF Context)
  (policy : AppliedModelingLib.Learning.HumanFeedback.FinitePolicy Context Response)
  (score : Context → Response → ℝ) =>
  AppliedModelingLib.pmfExp contextLaw fun (context : Context) =>
  AppliedModelingLib.pmfExp (policy context) (score context)
```

Direct retained dependencies

- [AppliedModelingLib.Learning.HumanFeedback.FinitePolicy](#)
- [AppliedModelingLib.pmfExp](#)

AppliedModelingLib.pmfPairExp

Declaration location: reusable library

Lean declaration body

```
type:
{α : Type u_1} →
{β : Type u_2} → [Fintype α] → [DecidableEq α] → [Fintype β] → [DecidableEq β] → PMF α → PMF β → (α → β → ℝ) → ℝ

value:
fun {α : Type u_1} {β : Type u_2} [Fintype α] [DecidableEq α] [Fintype β] [DecidableEq β] (μ : PMF α) (ν : PMF β)
(f : α → β → ℝ) =>
AppliedModelingLib.pmfExp μ fun (a : α) => AppliedModelingLib.pmfExp ν fun (b : β) => f a b
```

Direct retained dependencies

- [AppliedModelingLib.pmfExp](#)

AppliedModelingLib.Alignment.Welfare.populationBradleyTerryPolicyMargin

Declaration location: reusable library

Lean declaration body

```
type:
{User : Type u_1} →
{Alternative : Type u_2} →
[Fintype User] →
[DecidableEq User] →
[Fintype Alternative] →
[DecidableEq Alternative] →
PMF User →
AppliedModelingLib.Alignment.Welfare.FiniteUtilityProfile User Alternative →
ℝ → PMF Alternative → PMF Alternative → ℝ

value:
fun {User : Type u_1} {Alternative : Type u_2} [Fintype User] [DecidableEq User] [Fintype Alternative]
[DecidableEq Alternative] (population : PMF User)
(utility : AppliedModelingLib.Alignment.Welfare.FiniteUtilityProfile User Alternative) (btScale : ℝ)
(firstPolicy secondPolicy : PMF Alternative) =>
AppliedModelingLib.pmfPairExp firstPolicy secondPolicy fun (first second : Alternative) =>
(AppliedModelingLib.Alignment.Welfare.populationBradleyTerryPreference population utility btScale).prob PUnit.unit
first second -
1 / 2
```

Direct retained dependencies

- [AppliedModelingLib.Alignment.Welfare.FiniteUtilityProfile](#)
- [AppliedModelingLib.Alignment.Welfare.populationBradleyTerryPreference](#)
- [AppliedModelingLib.Learning.HumanFeedback.PairwisePreference](#)
- [AppliedModelingLib.pmfPairExp](#)

AppliedModelingLib.Learning.HumanFeedback.policyPreference

Declaration location: reusable library

Lean declaration body

```
type:
{Context : Type u_1} →
{Response : Type u_2} →
[Fintype Context] →
[DecidableEq Context] →
[Fintype Response] →
[DecidableEq Response] →
PMF Context →
AppliedModelingLib.Learning.HumanFeedback.PairwisePreference Context Response →
AppliedModelingLib.Learning.HumanFeedback.FinitePolicy Context Response →
AppliedModelingLib.Learning.HumanFeedback.FinitePolicy Context Response → ℝ

value:
fun {Context : Type u_1} {Response : Type u_2} [Fintype Context] [DecidableEq Context] [Fintype Response]
[DecidableEq Response] (contextLaw : PMF Context)
(preference : AppliedModelingLib.Learning.HumanFeedback.PairwisePreference Context Response)
(first second : AppliedModelingLib.Learning.HumanFeedback.FinitePolicy Context Response) =>
AppliedModelingLib.pmfExp contextLaw fun (context : Context) =>
AppliedModelingLib.pmfPairExp (first context) (second context) (preference.prob context)
```

Direct retained dependencies

- [AppliedModelingLib.Learning.HumanFeedback.FinitePolicy](#)
- [AppliedModelingLib.Learning.HumanFeedback.PairwisePreference](#)
- [AppliedModelingLib.pmfExp](#)
- [AppliedModelingLib.pmfPairExp](#)

AppliedModelingLib.GameTheory.PreferenceGame.preferenceGamePayoff

Declaration location: reusable library

Lean declaration body

```
type:
{Context : Type u_1} →
{Response : Type u_2} →
[Fintype Context] →
[DecidableEq Context] →
[Fintype Response] →
[DecidableEq Response] →
PMF Context →
AppliedModelingLib.Learning.HumanFeedback.PairwisePreference Context Response →
AppliedModelingLib.Learning.HumanFeedback.FinitePolicy Context Response →
AppliedModelingLib.Learning.HumanFeedback.FinitePolicy Context Response → ℝ

value:
fun {Context : Type u_1} {Response : Type u_2} [Fintype Context] [DecidableEq Context] [Fintype Response]
[DecidableEq Response] (contextLaw : PMF Context)
(preference : AppliedModelingLib.Learning.HumanFeedback.PairwisePreference Context Response)
(first second : AppliedModelingLib.Learning.HumanFeedback.FinitePolicy Context Response) =>
AppliedModelingLib.Learning.HumanFeedback.policyPreference contextLaw preference first second
```

Direct retained dependencies

- [AppliedModelingLib.Learning.HumanFeedback.FinitePolicy](#)
- [AppliedModelingLib.Learning.HumanFeedback.PairwisePreference](#)
- [AppliedModelingLib.Learning.HumanFeedback.policyPreference](#)

AppliedModelingLib.GameTheory.PreferenceGame.centeredPreferenceGamePayoff

Declaration location: reusable library

Lean declaration body

```
type:
{Context : Type u_1} →
{Response : Type u_2} →
[Fintype Context] →
[DecidableEq Context] →
[Fintype Response] →
[DecidableEq Response] →
PMF Context →
AppliedModelingLib.Learning.HumanFeedback.PairwisePreference Context Response →
AppliedModelingLib.Learning.HumanFeedback.FinitePolicy Context Response →
AppliedModelingLib.Learning.HumanFeedback.FinitePolicy Context Response → ℝ

value:
fun {Context : Type u_1} {Response : Type u_2} [Fintype Context] [DecidableEq Context] [Fintype Response]
[DecidableEq Response] (contextLaw : PMF Context)
(preference : AppliedModelingLib.Learning.HumanFeedback.PairwisePreference Context Response)
(first second : AppliedModelingLib.Learning.HumanFeedback.FinitePolicy Context Response) =>
AppliedModelingLib.GameTheory.PreferenceGame.preferenceGamePayoff contextLaw preference first second - 1 / 2
```

Direct retained dependencies

- [AppliedModelingLib.GameTheory.PreferenceGame.preferenceGamePayoff](#)
- [AppliedModelingLib.Learning.HumanFeedback.FinitePolicy](#)
- [AppliedModelingLib.Learning.HumanFeedback.PairwisePreference](#)

AppliedModelingLib.GameTheory.PreferenceGame.contextFreeCenteredPreferencePayoff

Declaration location: reusable library

Lean declaration body

```
type:
{Response : Type u_1} →
[Fintype Response] →
[DecidableEq Response] →
  AppliedModelingLib.Learning.HumanFeedback.PairwisePreference PUnit.{u_2 + 1} Response →
  PMF Response → PMF Response → ℝ

value:
fun {Response : Type u_1} [Fintype Response] [DecidableEq Response]
  (preference : AppliedModelingLib.Learning.HumanFeedback.PairwisePreference PUnit.{u_2 + 1} Response)
  (first second : PMF Response) =>
  AppliedModelingLib.GameTheory.PreferenceGame.centeredPreferenceGamePayoff (PMF.pure PUnit.unit) preference
  (fun (x : PUnit.{u_2 + 1}) => first) fun (x : PUnit.{u_2 + 1}) => second
```

Direct retained dependencies

- [AppliedModelingLib.GameTheory.PreferenceGame.centeredPreferenceGamePayoff](#)
- [AppliedModelingLib.Learning.HumanFeedback.PairwisePreference](#)

AppliedModelingLib.GameTheory.PreferenceGame.IsContextFreeExtendedRegularizedPreferenceGameEquilibrium

Declaration location: reusable library

Lean declaration body

```
type:
{Response : Type u_1} →
[Fintype Response] →
[DecidableEq Response] →
  AppliedModelingLib.Learning.HumanFeedback.PairwisePreference PUnit.{u_2 + 1} Response →
  PMF Response → PMF Response → WithTop ℝ → Prop

value:
fun {Response : Type u_1} [Fintype Response] [DecidableEq Response]
  (preference : AppliedModelingLib.Learning.HumanFeedback.PairwisePreference PUnit.{u_2 + 1} Response)
  (reference policy : PMF Response) (klWeight : WithTop ℝ) =>
  (∃ (finiteWeight : ℝ),
    klWeight = ↑finiteWeight ∧
    0 ≤ finiteWeight ∧
    ∀ (opponent : PMF Response),
      0 ≤
        AppliedModelingLib.GameTheory.PreferenceGame.contextFreeCenteredPreferencePayoff preference policy
        opponent +
        finiteWeight *
        (AppliedModelingLib.finiteKLDivergence opponent reference -
         AppliedModelingLib.finiteKLDivergence policy reference)) ∧
    klWeight = T ∧ policy = reference
```

Direct retained dependencies

- [AppliedModelingLib.GameTheory.PreferenceGame.contextFreeCenteredPreferencePayoff](#)
- [AppliedModelingLib.Learning.HumanFeedback.PairwisePreference](#)
- [AppliedModelingLib.finiteKLDivergence](#)

AppliedModelingLib.GameTheory.PreferenceGame.IsFiniteKLPreferenceMaximin

Declaration location: reusable library

Lean declaration body

```
type:
{Response : Type u_1} →
[Fintype Response] →
[DecidableEq Response] →
  AppliedModelingLib.Learning.HumanFeedback.PairwisePreference PUnit.{u_2 + 1} Response →
  PMF Response → ℝ → PMF Response → Prop

value:
fun {Response : Type u_1} [Fintype Response] [DecidableEq Response]
  (preference : AppliedModelingLib.Learning.HumanFeedback.PairwisePreference PUnit.{u_2 + 1} Response)
  (reference : PMF Response) (klBudget : ℝ) (policy : PMF Response) =>
  AppliedModelingLib.finiteKLDivergence policy reference ≤ klBudget ∧
  ∃ (worst : PMF Response),
    AppliedModelingLib.finiteKLDivergence worst reference ≤ klBudget ∧
    (∀ (opponent : PMF Response),
      AppliedModelingLib.finiteKLDivergence opponent reference ≤ klBudget →
      AppliedModelingLib.GameTheory.PreferenceGame.contextFreeCenteredPreferencePayoff preference policy worst ≤
      AppliedModelingLib.GameTheory.PreferenceGame.contextFreeCenteredPreferencePayoff preference policy
      opponent) ∧
    ∀ (candidate : PMF Response),
```

AppliedModelingLib.finiteKLDivergence candidate reference $\leq$ klBudget $\rightarrow$
AppliedModelingLib.GameTheory.PreferenceGame.contextFreeCenteredPreferencePayoff preference candidate
worst $\leq$
AppliedModelingLib.GameTheory.PreferenceGame.contextFreeCenteredPreferencePayoff preference policy worst

Direct retained dependencies

- [AppliedModelingLib.GameTheory.PreferenceGame.contextFreeCenteredPreferencePayoff](#)
- [AppliedModelingLib.Learning.HumanFeedback.PairwisePreference](#)
- [AppliedModelingLib.finiteKLDivergence](#)

AppliedModelingLib.Alignment.Welfare.IsConstrainedPopulationBradleyTerryMaximin

Declaration location: reusable library

Lean declaration body

```
type:
{User : Type u_1} →
{Alternative : Type u_2} →
[Fintype User] →
[DecidableEq User] →
[Fintype Alternative] →
[DecidableEq Alternative] →
PMF User →
  AppliedModelingLib.Alignment.Welfare.FiniteUtilityProfile User Alternative →
  ℝ → PMF Alternative → ℝ → PMF Alternative → Prop

value:
fun {User : Type u_1} {Alternative : Type u_2} [Fintype User] [DecidableEq User] [Fintype Alternative]
  [DecidableEq Alternative] (population : PMF User)
  (utility : AppliedModelingLib.Alignment.Welfare.FiniteUtilityProfile User Alternative) (btScale : ℝ)
  (reference : PMF Alternative) (klBudget : ℝ) (policy : PMF Alternative) =>
  AppliedModelingLib.GameTheory.PreferenceGame.IsFiniteKLPreferenceMaximin
  (have this : AppliedModelingLib.Learning.HumanFeedback.PairwisePreference PUnit.{1} Alternative :=
    AppliedModelingLib.Alignment.Welfare.populationBradleyTerryPreference population utility btScale;
    this)
  reference klBudget policy
```

Direct retained dependencies

- [AppliedModelingLib.Alignment.Welfare.FiniteUtilityProfile](#)
- [AppliedModelingLib.Alignment.Welfare.populationBradleyTerryPreference](#)
- [AppliedModelingLib.GameTheory.PreferenceGame.IsFiniteKLPreferenceMaximin](#)
- [AppliedModelingLib.Learning.HumanFeedback.PairwisePreference](#)

AppliedModelingLib.GameTheory.PreferenceGame.regularizedPreferenceGamePayoff

Declaration location: reusable library

Lean declaration body

```
type:
{Context : Type u_1} →
{Response : Type u_2} →
[Fintype Context] →
[DecidableEq Context] →
[Fintype Response] →
[DecidableEq Response] →
PMF Context →
  AppliedModelingLib.Learning.HumanFeedback.PairwisePreference Context Response →
  AppliedModelingLib.Learning.HumanFeedback.FinitePolicy Context Response →
  AppliedModelingLib.Learning.HumanFeedback.FinitePolicy Context Response →
  AppliedModelingLib.Learning.HumanFeedback.FinitePolicy Context Response → ℝ → ℝ

value:
fun {Context : Type u_1} {Response : Type u_2} [Fintype Context] [DecidableEq Context] [Fintype Response]
  [DecidableEq Response] (contextLaw : PMF Context)
  (preference : AppliedModelingLib.Learning.HumanFeedback.PairwisePreference Context Response)
  (reference first second : AppliedModelingLib.Learning.HumanFeedback.FinitePolicy Context Response)
  (klRegularization : ℝ) =>
  AppliedModelingLib.GameTheory.PreferenceGame.preferenceGamePayoff contextLaw preference first second -
  klRegularization *
  AppliedModelingLib.Learning.HumanFeedback.contextAveragedPolicyKLDivergence contextLaw first reference +
  klRegularization *
  AppliedModelingLib.Learning.HumanFeedback.contextAveragedPolicyKLDivergence contextLaw second reference
```

Direct retained dependencies

- [AppliedModelingLib.GameTheory.PreferenceGame.preferenceGamePayoff](#)
- [AppliedModelingLib.Learning.HumanFeedback.FinitePolicy](#)
- [AppliedModelingLib.Learning.HumanFeedback.PairwisePreference](#)
- [AppliedModelingLib.Learning.HumanFeedback.contextAveragedPolicyKLDivergence](#)

AppliedModelingLib.GameTheory.PreferenceGame.IsRegularizedPreferenceGameEquilibrium

Declaration location: reusable library

Lean declaration body

```
type:
{Context : Type u_1} →
{Response : Type u_2} →
[Fintype Context] →
[DecidableEq Context] →
[Fintype Response] →
[DecidableEq Response] →
PMF Context →
  AppliedModelingLib.Learning.HumanFeedback.PairwisePreference Context Response →
  AppliedModelingLib.Learning.HumanFeedback.FinitePolicy Context Response →
  AppliedModelingLib.Learning.HumanFeedback.FinitePolicy Context Response → ℝ → Prop

value:
fun {Context : Type u_1} {Response : Type u_2} [Fintype Context] [DecidableEq Context] [Fintype Response]
[DecidableEq Response] (contextLaw : PMF Context)
(preference : AppliedModelingLib.Learning.HumanFeedback.PairwisePreference Context Response)
(reference policy : AppliedModelingLib.Learning.HumanFeedback.FinitePolicy Context Response)
(klRegularization : ℝ) =>
  ∀ (alternative : AppliedModelingLib.Learning.HumanFeedback.FinitePolicy Context Response),
  1 / 2 ≤
    AppliedModelingLib.GameTheory.PreferenceGame.regularizedPreferenceGamePayoff contextLaw preference reference
    policy alternative klRegularization
```

Direct retained dependencies

- [AppliedModelingLib.GameTheory.PreferenceGame.regularizedPreferenceGamePayoff](#)
- [AppliedModelingLib.Learning.HumanFeedback.FinitePolicy](#)
- [AppliedModelingLib.Learning.HumanFeedback.PairwisePreference](#)

AppliedModelingLib.pmfProb

Declaration location: reusable library

Lean declaration body

```
type:
{α : Type u_1} → [Fintype α] → PMF α → (p : α → Prop) → [DecidablePred p] → ℝ

value:
fun {α : Type u_1} [Fintype α] (μ : PMF α) (p : α → Prop) [DecidablePred p] =>
  AppliedModelingLib.pmfExp μ fun (a : α) => if p a then 1 else 0
```

Direct retained dependencies

- [AppliedModelingLib.pmfExp](#)

AppliedModelingLib.pmfProduct

Declaration location: reusable library

Lean declaration body

```
type:
(ι : Type u_1) → (α : Type u_2) → [Fintype ι] → [DecidableEq ι] → [Fintype α] → PMF α → PMF (ι → α)

value:
fun (ι : Type u_1) (α : Type u_2) [Fintype ι] [DecidableEq ι] [Fintype α] (μ : PMF α) =>
  PMF.offintype (fun (f : ι → α) => ∏ i : ι, μ (f i)) (AppliedModelingLib.pmfProduct._proof_1 ι α μ)
```

AppliedModelingLib.uniformPMF

Declaration location: reusable library

Lean declaration body

```
type:
( $\alpha$  : Type u_1) → [Fintype  $\alpha$ ] → [Nonempty  $\alpha$ ] → PMF  $\alpha$ 

value:
fun ( $\alpha$  : Type u_1) [Fintype  $\alpha$ ] [Nonempty  $\alpha$ ] =>
  PMF.ofFintype (fun (x :  $\alpha$ ) => (1 (Fintype.card  $\alpha$ ))-1) (AppliedModelingLib.uniformPMF_proof_1  $\alpha$ )
```

GolzHaghtalabYang2025Distortion.lemma15DropAdd

Declaration location: paper-local

Lean declaration body

```
type:
(Fin 3 →  $\mathbb{R}$ ) → Fin 3 →  $\mathbb{R}$  → Fin 3 →  $\mathbb{R}$ 

value:
fun (utility : Fin 3 →  $\mathbb{R}$ ) (peak : Fin 3) (increment :  $\mathbb{R}$ ) (a : Fin 3) => if a = peak then 0 else utility a + increment
```

GolzHaghtalabYang2025Distortion.lemma15Increment

Declaration location: paper-local

Lean declaration body

```
type:
 $\mathbb{R}$  →  $\mathbb{R}$ 

value:
fun (delta :  $\mathbb{R}$ ) => Real.log ((Real.exp (delta / 2) + 1) ^ 3 / (2 * (Real.exp delta + 3)))
```

GolzHaghtalabYang2025Distortion.lemma15Peak

Declaration location: paper-local

Lean declaration body

```
type:
(Fin 3 →  $\mathbb{R}$ ) → Fin 3

value:
fun (utility : Fin 3 →  $\mathbb{R}$ ) => Classical.choose (GolzHaghtalabYang2025Distortion.lemma15Peak._proof_1 utility)
```

GolzHaghtalabYang2025Distortion.lemma15Population

Declaration location: paper-local

Lean declaration body

```
type:
PMF (Fin 3)

value:
AppliedModelingLib.uniformPMF (Fin 3)
```

Direct retained dependencies

- [AppliedModelingLib.uniformPMF](#)

GolzHaghtalabYang2025Distortion.lemma15Step

Declaration location: paper-local

Lean declaration body

```
type:
 $\mathbb{R}$  → (Fin 3 →  $\mathbb{R}$ ) → Fin 3 →  $\mathbb{R}$ 

value:
fun (beta :  $\mathbb{R}$ ) (utility : Fin 3 →  $\mathbb{R}$ ) (a : Fin 3) =>
  (have peak : Fin 3 := GolzHaghtalabYang2025Distortion.lemma15Peak utility;
   GolzHaghtalabYang2025Distortion.lemma15DropAdd utility peak
   (GolzHaghtalabYang2025Distortion.lemma15Increment (beta * utility peak) / beta))
  a
```

Direct retained dependencies

- [GolzhaghtalabYang2025Distortion.lemma15DropAdd](#)
- [GolzhaghtalabYang2025Distortion.lemma15Increment](#)
- [GolzhaghtalabYang2025Distortion.lemma15Peak](#)

GolzhaghtalabYang2025Distortion.lemma15Utility

Declaration location: paper-local

Lean declaration body

```
type:
ℝ → ℕ → Fin 3 → ℝ

value:
fun (beta : ℝ) (a : ℕ) (a_1 : Fin 3) =>
  Nat.brecOn (motive := fun (x : ℕ) => Fin 3 → ℝ) a (GolzhaghtalabYang2025Distortion.lemma15Utility_f beta) a_1
```

Direct retained dependencies

- [GolzhaghtalabYang2025Distortion.lemma15Step](#)

GolzhaghtalabYang2025Distortion.sourceDpoBinaryReportLoss

Declaration location: paper-local

Lean declaration body

```
type:
{Alternative : Type u_1} →
[Fintype Alternative] →
[DecidableEq Alternative] →
{horizon : ℕ} →
(Fin horizon → AppliedModelingLib.Learning.HumanFeedback.BinaryPairwiseReport Alternative) →
PMF Alternative → PMF Alternative → ℝ → ℝ

value:
fun {Alternative : Type u_1} [Fintype Alternative] [DecidableEq Alternative] {horizon : ℕ}
  (sample : Fin horizon → AppliedModelingLib.Learning.HumanFeedback.BinaryPairwiseReport Alternative)
  (reference policy : PMF Alternative) (klWeight : ℝ) =>
  -(AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset.ofBinaryReports sample).pairwiseLogLikelihood
  Real.sigmoid fun (alternative : Alternative) =>
    AppliedModelingLib.Learning.HumanFeedback.dpoImplicitReward
    (AppliedModelingLib.Alignment.Welfare.contextFreeAlternativePolicy policy)
    (AppliedModelingLib.Alignment.Welfare.contextFreeAlternativePolicy reference) klWeight PUnit.unit alternative
```

Direct retained dependencies

- [AppliedModelingLib.Alignment.Welfare.contextFreeAlternativePolicy](#)
- [AppliedModelingLib.Learning.HumanFeedback.BinaryPairwiseReport](#)
- [AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset.ofBinaryReports](#)
- [AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset.pairwiseLogLikelihood](#)
- [AppliedModelingLib.Learning.HumanFeedback.dpoImplicitReward](#)

GolzhaghtalabYang2025Distortion.IsSourceDpoLossMinimizer

Declaration location: paper-local

Lean declaration body

```
type:
{Alternative : Type u_1} →
[Fintype Alternative] →
[DecidableEq Alternative] →
{horizon : ℕ} →
(Fin horizon → AppliedModelingLib.Learning.HumanFeedback.BinaryPairwiseReport Alternative) →
PMF Alternative → ℝ → PMF Alternative → Prop

value:
fun {Alternative : Type u_1} [Fintype Alternative] [DecidableEq Alternative] {horizon : ℕ}
  (sample : Fin horizon → AppliedModelingLib.Learning.HumanFeedback.BinaryPairwiseReport Alternative)
  (reference : PMF Alternative) (klWeight : ℝ) (policy : PMF Alternative) =>
  AppliedModelingLib.PMFFullSupport policy ∧
  ∀ (candidate : PMF Alternative),
  AppliedModelingLib.PMFFullSupport candidate →
  GolzhaghtalabYang2025Distortion.sourceDpoBinaryReportLoss sample reference policy klWeight ≤
  GolzhaghtalabYang2025Distortion.sourceDpoBinaryReportLoss sample reference candidate klWeight
```

Direct retained dependencies

- [AppliedModelingLib.Learning.HumanFeedback.BinaryPairwiseReport](#)
- [AppliedModelingLib.PMFFullSupport](#)
- [GolzhaghtalabYang2025Distortion.sourceDpoBinaryReportLoss](#)

GolzHaghtalabYang2025Distortion.sourceDpoNormalizedMLEReward

Declaration location: paper-local

Lean declaration body

```
type:
{Alternative : Type u_1} →
[Fintype Alternative] →
[DecidableEq Alternative] → Alternative → PMF Alternative → PMF Alternative → ℝ → Alternative → ℝ

value:
fun {Alternative : Type u_1} [Fintype Alternative] [DecidableEq Alternative] (scoreReference : Alternative)
  (reference policy : PMF Alternative) (klWeight : ℝ) (a : Alternative) =>
  AppliedModelingLib.Learning.HumanFeedback.dpoImplicitReward
  (AppliedModelingLib.Alignment.Welfare.contextFreeAlternativePolicy policy)
  (AppliedModelingLib.Alignment.Welfare.contextFreeAlternativePolicy reference) klWeight PUnit.unit a -
  AppliedModelingLib.Learning.HumanFeedback.dpoImplicitReward
  (AppliedModelingLib.Alignment.Welfare.contextFreeAlternativePolicy policy)
  (AppliedModelingLib.Alignment.Welfare.contextFreeAlternativePolicy reference) klWeight PUnit.unit scoreReference
```

Direct retained dependencies

- [AppliedModelingLib.Alignment.Welfare.contextFreeAlternativePolicy](#)
- [AppliedModelingLib.Learning.HumanFeedback.dpoImplicitReward](#)

GolzHaghtalabYang2025Distortion.sourceDpoPolicy

Declaration location: paper-local

Lean declaration body

```
type:
{Alternative : Type u_1} →
[Fintype Alternative] →
[DecidableEq Alternative] →
PMF Alternative →
AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset.ScoreVector Alternative → ℝ → PMF Alternative

value:
fun {Alternative : Type u_1} [Fintype Alternative] [DecidableEq Alternative] (reference : PMF Alternative)
  (reward : AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset.ScoreVector Alternative) (klWeight : ℝ) =>
  AppliedModelingLib.Learning.HumanFeedback.dpoOptimalPolicy
  (AppliedModelingLib.Alignment.Welfare.contextFreeAlternativePolicy reference) (fun (x : PUnit.{1}) => reward)
  klWeight PUnit.unit
```

Direct retained dependencies

- [AppliedModelingLib.Alignment.Welfare.contextFreeAlternativePolicy](#)
- [AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset.ScoreVector](#)
- [AppliedModelingLib.Learning.HumanFeedback.dpoOptimalPolicy](#)

GolzHaghtalabYang2025Distortion.sourceRlhfbBinaryReportLoss

Declaration location: paper-local

Lean declaration body

```
type:
{Alternative : Type u_1} →
[Fintype Alternative] →
[DecidableEq Alternative] →
{horizon : ℕ} →
(Fin horizon → AppliedModelingLib.Learning.HumanFeedback.BinaryPairwiseReport Alternative) →
AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset.ScoreVector Alternative → ℝ

value:
fun {Alternative : Type u_1} [Fintype Alternative] [DecidableEq Alternative] {horizon : ℕ}
  (sample : Fin horizon → AppliedModelingLib.Learning.HumanFeedback.BinaryPairwiseReport Alternative)
  (reward : AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset.ScoreVector Alternative) =>
  -(AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset.ofBinaryReports sample).pairwiseLogLikelihood
  Real.sigmoid reward
```

Direct retained dependencies

- [AppliedModelingLib.Learning.HumanFeedback.BinaryPairwiseReport](#)
- [AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset.ScoreVector](#)
- [AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset.ofBinaryReports](#)
- [AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset.pairwiseLogLikelihood](#)

GolzHaghtalabYang2025Distortion.sourceRlhPolicy

Declaration location: paper-local

Lean declaration body

```
type:
{Alternative : Type u_1} →
[Fintype Alternative] →
[DecidableEq Alternative] →
PMF Alternative →
AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset.ScoreVector Alternative → ℝ → PMF Alternative

value:
fun {Alternative : Type u_1} [Fintype Alternative] [DecidableEq Alternative] (reference : PMF Alternative)
(reward : AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset.ScoreVector Alternative) (klWeight : ℝ) =>
AppliedModelingLib.Learning.HumanFeedback.contextExponentialTiltPolicy
(AppliedModelingLib.Alignment.Welfare.contextFreeAlternativePolicy reference) (fun (x : PUnit.{1}) => reward)
klWeight⁻¹ PUnit.unit
```

Direct retained dependencies

- [AppliedModelingLib.Alignment.Welfare.contextFreeAlternativePolicy](#)
- [AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset.ScoreVector](#)
- [AppliedModelingLib.Learning.HumanFeedback.contextExponentialTiltPolicy](#)

GolzHaghtalabYang2025Distortion.theorem10SourceRate

Declaration location: paper-local

Lean declaration body

```
type:
ℕ → ℕ → ℝ → ℝ → ℝ

value:
fun (users comparisonsPerUser : ℕ) (minimumMass failureBudget : ℝ) =>
8 * √(Real.log (4 / failureBudget) / (↑users * min 1 (↑comparisonsPerUser * minimumMass ^ 2))) +
16 * Real.log (4 / failureBudget) / (↑users * minimumMass ^ 2)
```

GolzHaghtalabYang2025Distortion.theorem10UnorderedPairIndicator

Declaration location: paper-local

Lean declaration body

```
type:
{Alternative : Type u_1} → [DecidableEq Alternative] → Alternative → Alternative → Alternative × Alternative → ℝ

value:
fun {Alternative : Type u_1} [DecidableEq Alternative] (first second : Alternative) (a : Alternative × Alternative) =>
match a with
| (left, right) => if left = first ∧ right = second ∨ left = second ∧ right = first then 1 else 0
```

GolzHaghtalabYang2025Distortion.theorem10UserUnorderedPairCount

Declaration location: paper-local

Lean declaration body

```
type:
{Alternative : Type u_1} →
[DecidableEq Alternative] →
{comparisonsPerUser : ℕ} → Alternative → Alternative → (Fin comparisonsPerUser → Alternative × Alternative) → ℝ

value:
fun {Alternative : Type u_1} [DecidableEq Alternative] {comparisonsPerUser : ℕ} (first second : Alternative)
(labels : Fin comparisonsPerUser → Alternative × Alternative) =>
∑ position : Fin comparisonsPerUser,
GolzHaghtalabYang2025Distortion.theorem10UnorderedPairIndicator first second (labels position)
```

Direct retained dependencies

- [GolzhaghtalabYang2025Distortion.theorem10UnorderedPairIndicator](#)

GolzHaghtalabYang2025Distortion.theorem12BernoulliReport

Declaration location: paper-local

Lean declaration body

```
type:
(probability : ℝ) → 0 ≤ probability → probability ≤ 1 → PMF Bool

value:
fun (probability : ℝ) (hnonneg : 0 ≤ probability) (hle_one : probability ≤ 1) =>
  PMF.bernoulli (probability, hnonneg)
  (GolzhaghtalabYang2025Distortion.theorem12BernoulliReport._proof_1 probability hnonneg hle_one)
```

GolzHaghtalabYang2025Distortion.theorem12CloneCollapse

Declaration location: paper-local

Lean declaration body

```
type:
{Alternative : Type v} → Alternative → (clones : ℕ) → Alternative ⊕ Fin clones → Alternative

value:
fun {Alternative : Type v} (split : Alternative) (clones : ℕ) (a : Alternative ⊕ Fin clones) =>
  match a with
  | Sum.inl alternative => alternative
  | Sum.inr val => split
```

GolzHaghtalabYang2025Distortion.theorem12EmpiricalBordaIncidences

Declaration location: paper-local

Lean declaration body

```
type:
{Report : Type u_1} →
{Alternative : Type u_2} →
[Fintype Alternative] → (Alternative → Report → ℝ) → Alternative → {horizon : ℕ} → (Fin horizon → Report) → ℝ

value:
fun {Report : Type u_1} {Alternative : Type u_2} [Fintype Alternative] (incidences : Alternative → Report → ℝ)
  (alternative : Alternative) {horizon : ℕ} (sample : Fin horizon → Report) =>
  AppliedModelingLib.Probability.finitelidScoreSum (incidences alternative) sample
```

Direct retained dependencies

- [AppliedModelingLib.Probability.finitelidScoreSum](#)

GolzHaghtalabYang2025Distortion.theorem12EmpiricalBordaWins

Declaration location: paper-local

Lean declaration body

```
type:
{Report : Type u_1} →
{Alternative : Type u_2} →
[Fintype Alternative] → (Alternative → Report → ℝ) → Alternative → {horizon : ℕ} → (Fin horizon → Report) → ℝ

value:
fun {Report : Type u_1} {Alternative : Type u_2} [Fintype Alternative] (wins : Alternative → Report → ℝ)
  (alternative : Alternative) {horizon : ℕ} (sample : Fin horizon → Report) =>
  AppliedModelingLib.Probability.finitelidScoreSum (wins alternative) sample
```

Direct retained dependencies

- [AppliedModelingLib.Probability.finitelidScoreSum](#)

GolzHaghtalabYang2025Distortion.theorem12EmpiricalBordaScore

Declaration location: paper-local

Lean declaration body

```
type:
{Report : Type u_1} →
{Alternative : Type u_2} →
[Fintype Alternative] →
(Alternative → Report → ℝ) → (Alternative → Report → ℝ) → Alternative → {horizon : ℕ} → (Fin horizon → Report) → ℝ

value:
fun {Report : Type u_1} {Alternative : Type u_2} [Fintype Alternative] (wins incidences : Alternative → Report → ℝ)
  (alternative : Alternative) {horizon : ℕ} (sample : Fin horizon → Report) =>
if GolzHaghtalabYang2025Distortion.theorem12EmpiricalBordaIncidences incidences alternative sample = 0 then 0
else
  GolzHaghtalabYang2025Distortion.theorem12EmpiricalBordaWins wins alternative sample /
  GolzHaghtalabYang2025Distortion.theorem12EmpiricalBordaIncidences incidences alternative sample
```

Direct retained dependencies

- [GolzHaghtalabYang2025Distortion.theorem12EmpiricalBordaIncidences](#)
- [GolzHaghtalabYang2025Distortion.theorem12EmpiricalBordaWins](#)

GolzHaghtalabYang2025Distortion.theorem12Eq10Coefficient

Declaration location: paper-local

Lean declaration body

```
type:
ℝ → ℝ → ℝ

value:
fun (beta gamma : ℝ) =>
  beta / 4 / (beta.sigmoid - 1 / 2) * (1 - gamma + ((beta * gamma).sigmoid - 1 / 2) / (beta.sigmoid - 1 / 2))
```

GolzHaghtalabYang2025Distortion.theorem12lidBordaReportBatchLaw

Declaration location: paper-local

Lean declaration body

```
type:
{Report : Type u_1} → [Fintype Report] → [DecidableEq Report] → PMF Report → (horizon : ℕ) → PMF (Fin horizon → Report)

value:
fun {Report : Type u_1} [Fintype Report] [DecidableEq Report] (reportLaw : PMF Report) (horizon : ℕ) =>
  AppliedModelingLib.pmfProduct (Fin horizon) Report reportLaw
```

Direct retained dependencies

- [AppliedModelingLib.pmfProduct](#)

GolzHaghtalabYang2025Distortion.theorem12lidEmpiricalBordaFailure

Declaration location: paper-local

Lean declaration body

```
type:
{Report : Type u_1} →
{Alternative : Type u_2} →
[Fintype Alternative] →
[DecidableEq Alternative] →
(Alternative → Report → ℝ) →
(Alternative → Report → ℝ) → Alternative → {horizon : ℕ} → (Fin horizon → Report) → Prop

value:
fun {Report : Type u_1} {Alternative : Type u_2} [Fintype Alternative] [DecidableEq Alternative]
  (wins incidences : Alternative → Report → ℝ) (winner : Alternative) {horizon : ℕ} (sample : Fin horizon → Report) =>
  ∃ (ordinary : { ordinary : Alternative // ordinary ≠ winner }),
  GolzHaghtalabYang2025Distortion.theorem12EmpiricalBordaScore wins incidences winner sample ≤
  GolzHaghtalabYang2025Distortion.theorem12EmpiricalBordaScore wins incidences (↑ ordinary) sample
```

Direct retained dependencies

- [GolzHaghtalabYang2025Distortion.theorem12EmpiricalBordaScore](#)

GolzHaghtalabYang2025Distortion.theorem12LogChoice

Declaration location: paper-local

Lean declaration body

```
type:
 $\mathbb{R} \rightarrow \mathbb{R}$ 

value:
fun (beta :  $\mathbb{R}$ ) => Real.log (beta + 1) / beta
```

GolzHaghtalabYang2025Distortion.theorem12OneComparisonIncidences

Declaration location: paper-local

Lean declaration body

```
type:
{Alternative : Type u_1} → [DecidableEq Alternative] → Alternative → (Alternative × Alternative) × Bool →  $\mathbb{R}$ 

value:
fun {Alternative : Type u_1} [DecidableEq Alternative] (alternative : Alternative)
  (a : (Alternative × Alternative) × Bool) =>
  match a with
  | ((first, second), snd) => (if first = alternative then 1 else 0) + if second = alternative then 1 else 0
```

GolzHaghtalabYang2025Distortion.theorem12OneComparisonReportLaw

Declaration location: paper-local

Lean declaration body

```
type:
{Alternative : Type u_1} →
[Fintype Alternative] →
[DecidableEq Alternative] →
PMF Alternative →
AppliedModelingLib.Learning.HumanFeedback.PairwisePreference PUnit.{u_2 + 1} Alternative →
PMF ((Alternative × Alternative) × Bool)

value:
fun {Alternative : Type u_1} [Fintype Alternative] [DecidableEq Alternative] (sampling : PMF Alternative)
  (preference : AppliedModelingLib.Learning.HumanFeedback.PairwisePreference PUnit.{u_2 + 1} Alternative) =>
sampling.bind fun (first : Alternative) =>
sampling.bind fun (second : Alternative) =>
PMF.map (fun (outcome : Bool) => ((first, second), outcome))
  (GolzHaghtalabYang2025Distortion.theorem12BernoulliReport (preference.prob PUnit.unit first second)
    (preference.nonneg PUnit.unit first second) (preference.le_one PUnit.unit first second))
```

Direct retained dependencies

- [AppliedModelingLib.Learning.HumanFeedback.PairwisePreference](#)
- [GolzHaghtalabYang2025Distortion.theorem12BernoulliReport](#)

GolzHaghtalabYang2025Distortion.theorem12OneComparisonWins

Declaration location: paper-local

Lean declaration body

```
type:
{Alternative : Type u_1} → [DecidableEq Alternative] → Alternative → (Alternative × Alternative) × Bool →  $\mathbb{R}$ 

value:
fun {Alternative : Type u_1} [DecidableEq Alternative] (alternative : Alternative)
  (a : (Alternative × Alternative) × Bool) =>
  match a with
  | ((first, second), outcome) =>
    (if first = alternative ∧ outcome = true then 1 else 0) + if second = alternative ∧ ¬outcome = true then 1 else 0
```

GolzHaghtalabYang2025Distortion.theorem12SplitUtility

Declaration location: paper-local

Lean declaration body

```
type:
{User : Type u} →
{OldAlternative : Type v} →
{NewAlternative : Type w} →
AppliedModelingLib.Alignment.Welfare.FiniteUtilityProfile User OldAlternative →
(NewAlternative → OldAlternative) → User → NewAlternative →  $\mathbb{R}$ 
```

```

value:
fun {User : Type u} {OldAlternative : Type v} {NewAlternative : Type w}
  (utility : AppliedModelingLib.Alignment.Welfare.FiniteUtilityProfile User OldAlternative)
  (collapse : NewAlternative → OldAlternative) (a : User) (a_1 : NewAlternative) =>
  utility a (collapse a_1)

```

Direct retained dependencies

- [AppliedModelingLib.Alignment.Welfare.FiniteUtilityProfile](#)

GolzHaghtalabYang2025Distortion.theorem12Utility

Declaration location: paper-local

Lean declaration body

```

type:
ℝ → ℝ → ℝ → Fin 3 → Fin 3 → ℝ

value:
fun (epsilon epsilonZero gamma : ℝ) (a a_1 : Fin 3) =>
  match a, a_1 with
  | 0, 0 => 1 - gamma
  | 0, 1 => 1
  | 0, 2 => 0
  | 1, 0 => 1
  | 1, 1 => 0
  | 1, 2 => epsilon
  | 2, 0 => 0
  | 2, 1 => 0
  | 2, 2 => epsilon + epsilonZero

```

GolzHaghtalabYang2025Distortion.theorem2SourceRate

Declaration location: paper-local

Lean declaration body

```

type:
ℕ → ℕ → ℕ → ℝ → ℝ → ℝ

value:
fun (alternativeCount users comparisonsPerUser : ℕ) (minimumMass delta : ℝ) =>
  have logLevel : ℝ := Real.log (8 * ↑alternativeCount ^ 2 / delta);
  20 * √((↑users * logLevel / min 1 (↑comparisonsPerUser * minimumMass ^ 2)) / ↑users +
    8 * (↑alternativeCount + 1) * logLevel / (↑users * minimumMass))

```

GolzHaghtalabYang2025Distortion.theorem2UserBatchReport

Declaration location: paper-local

Lean declaration body

```

type:
Type u_1 → ℕ → Type (max 0 u_1)

value:
fun (Alternative : Type u_1) (comparisonsPerUser : ℕ) => Fin comparisonsPerUser → (Alternative × Alternative) × Bool

```

GolzHaghtalabYang2025Distortion.theorem2UserBatchBordaIncidences

Declaration location: paper-local

Lean declaration body

```

type:
{Alternative : Type u_1} →
[DecidableEq Alternative] →
{comparisonsPerUser : ℕ} →
Alternative → GolzHaghtalabYang2025Distortion.theorem2UserBatchReport Alternative comparisonsPerUser → ℝ

value:
fun {Alternative : Type u_1} [DecidableEq Alternative] {comparisonsPerUser : ℕ} (alternative : Alternative)
  (report : GolzHaghtalabYang2025Distortion.theorem2UserBatchReport Alternative comparisonsPerUser) =>
  ∑ position : Fin comparisonsPerUser,
  GolzHaghtalabYang2025Distortion.theorem12OneComparisonIncidence alternative (report position)

```

Direct retained dependencies

- [GolzhaghtalabYang2025Distortion.theorem12OneComparisonIncidences](#)
- [GolzhaghtalabYang2025Distortion.theorem2UserBatchReport](#)

GolzhaghtalabYang2025Distortion.theorem2UserBatchBordaWins

Declaration location: paper-local

Lean declaration body

```
type:
{Alternative : Type u_1} →
[DecidableEq Alternative] →
{comparisonsPerUser : ℕ} →
Alternative → GolzhaghtalabYang2025Distortion.theorem2UserBatchReport Alternative comparisonsPerUser → ℝ

value:
fun {Alternative : Type u_1} [DecidableEq Alternative] {comparisonsPerUser : ℕ} (alternative : Alternative)
(report : GolzhaghtalabYang2025Distortion.theorem2UserBatchReport Alternative comparisonsPerUser) =>
  ∑ position : Fin comparisonsPerUser,
    GolzhaghtalabYang2025Distortion.theorem12OneComparisonWins alternative (report position)
```

Direct retained dependencies

- [GolzhaghtalabYang2025Distortion.theorem12OneComparisonWins](#)
- [GolzhaghtalabYang2025Distortion.theorem2UserBatchReport](#)

GolzhaghtalabYang2025Distortion.theorem2UserPairLabelLaw

Declaration location: paper-local

Lean declaration body

```
type:
{Alternative : Type u_1} → [Fintype Alternative] → PMF Alternative → PMF (Alternative × Alternative)

value:
fun {Alternative : Type u_1} [Fintype Alternative] (sampling : PMF Alternative) =>
  AppliedModelingLib.pmfProd sampling sampling
```

Direct retained dependencies

- [AppliedModelingLib.pmfProd](#)

GolzhaghtalabYang2025Distortion.theorem2UserResponseTable

Declaration location: paper-local

Lean declaration body

```
type:
Type u_1 → Type u_1

value:
fun (Alternative : Type u_1) => Alternative → Alternative → Bool
```

GolzhaghtalabYang2025Distortion.theorem10IidUserUnorderedPairCount

Declaration location: paper-local

Lean declaration body

```
type:
{Alternative : Type u_1} →
[DecidableEq Alternative] →
{users comparisonsPerUser : ℕ} →
Alternative →
Alternative →
(Fin users →
  GolzhaghtalabYang2025Distortion.theorem2UserResponseTable Alternative ×
  (Fin comparisonsPerUser → Alternative × Alternative)) →
  ℝ

value:
fun {Alternative : Type u_1} [DecidableEq Alternative] {users comparisonsPerUser : ℕ} (first second : Alternative)
(sample :
  Fin users →
  GolzhaghtalabYang2025Distortion.theorem2UserResponseTable Alternative ×
  (Fin comparisonsPerUser → Alternative × Alternative)) =>
  ∑ user : Fin users, GolzhaghtalabYang2025Distortion.theorem10UserUnorderedPairCount first second (sample user).2
```

Direct retained dependencies

- [GolzhaghtalabYang2025Distortion.theorem10UserUnorderedPairCount](#)
- [GolzhaghtalabYang2025Distortion.theorem2UserResponseTable](#)

GolzhaghtalabYang2025Distortion.theorem10UserUnorderedPairWins

Declaration location: paper-local

Lean declaration body

```
type:
{Alternative : Type u_1} →
[DecidableEq Alternative] →
{comparisonsPerUser : ℕ} →
Alternative →
Alternative →
GolzhaghtalabYang2025Distortion.theorem2UserResponseTable Alternative →
(Fin comparisonsPerUser → Alternative × Alternative) → ℝ

value:
fun {Alternative : Type u_1} [DecidableEq Alternative] {comparisonsPerUser : ℕ} (first second : Alternative)
(response : GolzhaghtalabYang2025Distortion.theorem2UserResponseTable Alternative)
(labels : Fin comparisonsPerUser → Alternative × Alternative) =>
Σ position : Fin comparisonsPerUser,
GolzhaghtalabYang2025Distortion.theorem10UnorderedPairIndicator first second (labels position) *
GolzhaghtalabYang2025Distortion.theorem12OneComparisonWins first
(labels position, response (labels position).1 (labels position).2)
```

Direct retained dependencies

- [GolzhaghtalabYang2025Distortion.theorem10UnorderedPairIndicator](#)
- [GolzhaghtalabYang2025Distortion.theorem12OneComparisonWins](#)
- [GolzhaghtalabYang2025Distortion.theorem2UserResponseTable](#)

GolzhaghtalabYang2025Distortion.theorem10lidUserUnorderedPairWins

Declaration location: paper-local

Lean declaration body

```
type:
{Alternative : Type u_1} →
[DecidableEq Alternative] →
{users comparisonsPerUser : ℕ} →
Alternative →
Alternative →
(Fin users →
GolzhaghtalabYang2025Distortion.theorem2UserResponseTable Alternative ×
(Fin comparisonsPerUser → Alternative × Alternative)) →
ℝ

value:
fun {Alternative : Type u_1} [DecidableEq Alternative] {users comparisonsPerUser : ℕ} (first second : Alternative)
(sample :
Fin users →
GolzhaghtalabYang2025Distortion.theorem2UserResponseTable Alternative ×
(Fin comparisonsPerUser → Alternative × Alternative)) =>
Σ user : Fin users,
GolzhaghtalabYang2025Distortion.theorem10UserUnorderedPairWins first second (sample user).1 (sample user).2
```

Direct retained dependencies

- [GolzhaghtalabYang2025Distortion.theorem10UserUnorderedPairWins](#)
- [GolzhaghtalabYang2025Distortion.theorem2UserResponseTable](#)

GolzhaghtalabYang2025Distortion.theorem10lidUserEmpiricalWinRate

Declaration location: paper-local

Lean declaration body

```
type:
{Alternative : Type u_1} →
[DecidableEq Alternative] →
{users comparisonsPerUser : ℕ} →
Alternative →
Alternative →
(Fin users →
GolzhaghtalabYang2025Distortion.theorem2UserResponseTable Alternative ×
(Fin comparisonsPerUser → Alternative × Alternative)) →
ℝ

value:
fun {Alternative : Type u_1} [DecidableEq Alternative] {users comparisonsPerUser : ℕ} (first second : Alternative)
(sample :
Fin users →
GolzhaghtalabYang2025Distortion.theorem2UserResponseTable Alternative ×
(Fin comparisonsPerUser → Alternative × Alternative)) =>
if GolzhaghtalabYang2025Distortion.theorem10lidUserUnorderedPairCount first second sample = 0 then 0
else
GolzhaghtalabYang2025Distortion.theorem10lidUserUnorderedPairWins first second sample /
GolzhaghtalabYang2025Distortion.theorem10lidUserUnorderedPairCount first second sample
```

Direct retained dependencies

- [GolzhaghtalabYang2025Distortion.theorem10lidUserUnorderedPairCount](#)
- [GolzhaghtalabYang2025Distortion.theorem10lidUserUnorderedPairWins](#)
- [GolzhaghtalabYang2025Distortion.theorem2UserResponseTable](#)

GolzhaghtalabYang2025Distortion.theorem10lidUserPaperWinRate

Declaration location: paper-local

Lean declaration body

```
type:
{Alternative : Type u_1} →
[DecidableEq Alternative] →
{users comparisonsPerUser : ℕ} →
Alternative →
Alternative →
(Fin users →
GolzhaghtalabYang2025Distortion.theorem2UserResponseTable Alternative ×
(Fin comparisonsPerUser → Alternative × Alternative)) →
ℝ

value:
fun {Alternative : Type u_1} [DecidableEq Alternative] {users comparisonsPerUser : ℕ} (first second : Alternative)
(sample :
Fin users →
GolzhaghtalabYang2025Distortion.theorem2UserResponseTable Alternative ×
(Fin comparisonsPerUser → Alternative × Alternative)) =>
if first = second then 1 / 2 else GolzhaghtalabYang2025Distortion.theorem10lidUserEmpiricalWinRate first second sample
```

Direct retained dependencies

- [GolzhaghtalabYang2025Distortion.theorem10lidUserEmpiricalWinRate](#)
- [GolzhaghtalabYang2025Distortion.theorem2UserResponseTable](#)

GolzhaghtalabYang2025Distortion.theorem11lidUserLatentBordaIncidences

Declaration location: paper-local

Lean declaration body

```
type:
{Alternative : Type u_1} →
[DecidableEq Alternative] →
{users comparisonsPerUser : ℕ} →
Alternative →
Alternative →
(Fin users →
GolzhaghtalabYang2025Distortion.theorem2UserResponseTable Alternative ×
(Fin comparisonsPerUser → Alternative × Alternative)) →
ℝ

value:
fun {Alternative : Type u_1} [DecidableEq Alternative] {users comparisonsPerUser : ℕ} (alternative : Alternative)
(sample :
Fin users →
GolzhaghtalabYang2025Distortion.theorem2UserResponseTable Alternative ×
(Fin comparisonsPerUser → Alternative × Alternative)) =>
Σ user : Fin users,
GolzhaghtalabYang2025Distortion.theorem2UserBatchBordaIncidences alternative
fun (position : Fin comparisonsPerUser) =>
((sample user).2 position, (sample user).1 ((sample user).2 position).1 ((sample user).2 position).2)
```

Direct retained dependencies

- [GolzhaghtalabYang2025Distortion.theorem2UserBatchBordaIncidences](#)
- [GolzhaghtalabYang2025Distortion.theorem2UserResponseTable](#)

GolzHaghtalabYang2025Distortion.theorem11lidUserLatentBordaWins

Declaration location: paper-local

Lean declaration body

```
type:
{Alternative : Type u_1} →
[DecidableEq Alternative] →
{users comparisonsPerUser : ℕ} →
Alternative →
(Fin users →
  GolzHaghtalabYang2025Distortion.theorem2UserResponseTable Alternative ×
  (Fin comparisonsPerUser → Alternative × Alternative)) →
ℝ

value:
fun {Alternative : Type u_1} [DecidableEq Alternative] {users comparisonsPerUser : ℕ} (alternative : Alternative)
  (sample :
    Fin users →
      GolzHaghtalabYang2025Distortion.theorem2UserResponseTable Alternative ×
      (Fin comparisonsPerUser → Alternative × Alternative)) =>
  ∑ user : Fin users,
    GolzHaghtalabYang2025Distortion.theorem2UserBatchBordaWins alternative fun (position : Fin comparisonsPerUser) =>
      ((sample user).2 position, (sample user).1 ((sample user).2 position).1 ((sample user).2 position).2)
```

Direct retained dependencies

- [GolzhaghtalabYang2025Distortion.theorem2UserBatchBordaWins](#)
- [GolzhaghtalabYang2025Distortion.theorem2UserResponseTable](#)

GolzHaghtalabYang2025Distortion.theorem11lidUserLatentBordaScore

Declaration location: paper-local

Lean declaration body

```
type:
{Alternative : Type u_1} →
[DecidableEq Alternative] →
{users comparisonsPerUser : ℕ} →
Alternative →
(Fin users →
  GolzHaghtalabYang2025Distortion.theorem2UserResponseTable Alternative ×
  (Fin comparisonsPerUser → Alternative × Alternative)) →
ℝ

value:
fun {Alternative : Type u_1} [DecidableEq Alternative] {users comparisonsPerUser : ℕ} (alternative : Alternative)
  (sample :
    Fin users →
      GolzHaghtalabYang2025Distortion.theorem2UserResponseTable Alternative ×
      (Fin comparisonsPerUser → Alternative × Alternative)) =>
  if GolzHaghtalabYang2025Distortion.theorem11lidUserLatentBordaIncidences alternative sample = 0 then 0
  else
    GolzHaghtalabYang2025Distortion.theorem11lidUserLatentBordaWins alternative sample /
    GolzHaghtalabYang2025Distortion.theorem11lidUserLatentBordaIncidences alternative sample
```

Direct retained dependencies

- [GolzhaghtalabYang2025Distortion.theorem11lidUserLatentBordaIncidences](#)
- [GolzhaghtalabYang2025Distortion.theorem11lidUserLatentBordaWins](#)
- [GolzhaghtalabYang2025Distortion.theorem2UserResponseTable](#)

GolzHaghtalabYang2025Distortion.theorem2UserResponseCalibrated

Declaration location: paper-local

Lean declaration body

```
type:
{Alternative : Type u_1} →
[Fintype Alternative] →
[DecidableEq Alternative] →
PMF (GolzhaghtalabYang2025Distortion.theorem2UserResponseTable Alternative) →
AppliedModelingLib.Learning.HumanFeedback.PairwisePreference PUnit.{u_2 + 1} Alternative → Prop

value:
fun {Alternative : Type u_1} [Fintype Alternative] [DecidableEq Alternative]
(responseLaw : PMF (GolzhaghtalabYang2025Distortion.theorem2UserResponseTable Alternative))
(preference : AppliedModelingLib.Learning.HumanFeedback.PairwisePreference PUnit.{u_2 + 1} Alternative) =>
∀ (first second : Alternative),
(AppliedModelingLib.pmfExp responseLaw
fun (response : GolzhaghtalabYang2025Distortion.theorem2UserResponseTable Alternative) =>
if response first second = true then 1 else 0) =
preference.prob PUnit.unit first second
```

Direct retained dependencies

- [AppliedModelingLib.Learning.HumanFeedback.PairwisePreference](#)
- [AppliedModelingLib.pmfExp](#)
- [GolzhaghtalabYang2025Distortion.theorem2UserResponseTable](#)

GolzhaghtalabYang2025Distortion.theorem3BalanceDenominator

Declaration location: paper-local

Lean declaration body

```
type:
 $\mathbb{R} \rightarrow \mathbb{R} \rightarrow \mathbb{R}$ 

value:
fun (beta epsilon :  $\mathbb{R}$ ) => beta.sigmoid + (beta * epsilon).sigmoid - 1
```

GolzhaghtalabYang2025Distortion.theorem3D2DiagonalEpsilon

Declaration location: paper-local

Lean declaration body

```
type:
 $\mathbb{N} \rightarrow \mathbb{R}$ 

value:
fun (n :  $\mathbb{N}$ ) => ( $\uparrow(n + 3)$ )-1
```

GolzhaghtalabYang2025Distortion.theorem3D2DiagonalXi

Declaration location: paper-local

Lean declaration body

```
type:
 $\mathbb{N} \rightarrow \mathbb{R}$ 

value:
fun (n :  $\mathbb{N}$ ) => 1 + GolzhaghtalabYang2025Distortion.theorem3D2DiagonalEpsilon n
```

Direct retained dependencies

- [GolzhaghtalabYang2025Distortion.theorem3D2DiagonalEpsilon](#)

GolzhaghtalabYang2025Distortion.theorem3DiagonalEpsilon

Declaration location: paper-local

Lean declaration body

```
type:
 $\mathbb{N} \rightarrow \mathbb{R}$ 

value:
fun (n :  $\mathbb{N}$ ) => ( $\uparrow(n + 2)$ )-1
```

GolzHaghtalabYang2025Distortion.theorem3lidEmpiricalStrictCondorcetLoser

Declaration location: paper-local

Lean declaration body

```
type:
{Report : Type u_1} →
{Alternative : Type u_2} →
[Fintype Alternative] →
[DecidableEq Alternative] →
(Alternative → Report → ℝ) → Alternative → {horizon : ℕ} → (Fin horizon → Report) → Prop

value:
fun {Report : Type u_1} {Alternative : Type u_2} [Fintype Alternative] [DecidableEq Alternative]
  (margin : Alternative → Report → ℝ) (special : Alternative) {horizon : ℕ} (sample : Fin horizon → Report) =>
  ∀ (ordinary : Alternative),
  ordinary ≠ special → 0 < AppliedModelingLib.Probability.finiteIdScoreSum (margin ordinary) sample
```

Direct retained dependencies

- [AppliedModelingLib.Probability.finiteIdScoreSum](#)

GolzHaghtalabYang2025Distortion.theorem3lidUserReportBatchLaw

Declaration location: paper-local

Lean declaration body

```
type:
{Report : Type u_1} → [Fintype Report] → [DecidableEq Report] → PMF Report → (horizon : ℕ) → PMF (Fin horizon → Report)

value:
fun {Report : Type u_1} [Fintype Report] [DecidableEq Report] (reportLaw : PMF Report) (horizon : ℕ) =>
  AppliedModelingLib.pmfProduct (Fin horizon) Report reportLaw
```

Direct retained dependencies

- [AppliedModelingLib.pmfProduct](#)

GolzHaghtalabYang2025Distortion.theorem3ManyAlternativeUtility

Declaration location: paper-local

Lean declaration body

```
type:
{Alternative : Type u_1} → [DecidableEq Alternative] → ℝ → ℝ → Alternative → Bool → Alternative → ℝ

value:
fun {Alternative : Type u_1} [DecidableEq Alternative] (epsilon xi : ℝ) (special : Alternative) (a : Bool)
  (a_1 : Alternative) =>
  match a, a_1 with
  | true, alternative => if alternative = special then 1 else 0
  | false, alternative => if alternative = special then 0 else xi * epsilon
```

GolzHaghtalabYang2025Distortion.theorem3ObservationSelectionLaw

Declaration location: paper-local

Lean declaration body

```
type:
{Observation : Type u_1} →
{Alternative : Type u_2} → PMF Observation → (Observation → PMF Alternative) → PMF Alternative

value:
fun {Observation : Type u_1} {Alternative : Type u_2} (observationLaw : PMF Observation)
  (rule : Observation → PMF Alternative) =>
  observationLaw.bind rule
```

GolzHaghtalabYang2025Distortion.theorem3OrdinarySpecialComparisonMargin

Declaration location: paper-local

Lean declaration body

```
type:
{Alternative : Type u_1} →
[DecidableEq Alternative] → Alternative → Alternative → (Alternative × Alternative) × Bool → ℝ

value:
fun {Alternative : Type u_1} [DecidableEq Alternative] (special ordinary : Alternative)
(a : (Alternative × Alternative) × Bool) =>
  match a with
  | ((first, second), outcome) =>
    if first = ordinary ∧ second = special then if outcome = true then 1 else -1
    else if first = special ∧ second = ordinary then if outcome = true then -1 else 1 else 0
```

GolzHaghtalabYang2025Distortion.theorem3ProbabilisticCondorcetLoserCriterion

Declaration location: paper-local

Lean declaration body

```
type:
{Observation : Type u_1} →
{Alternative : Type u_2} →
[Fintype Alternative] →
[DecidableEq Alternative] → (Observation → PMF Alternative) → (Observation → Alternative → Prop) → Prop

value:
fun {Observation : Type u_1} {Alternative : Type u_2} [Fintype Alternative] [DecidableEq Alternative]
(rule : Observation → PMF Alternative) (isCondorcetLoser : Observation → Alternative → Prop) =>
  ∀ (observation : Observation) (alternative : Alternative),
    isCondorcetLoser observation alternative → ((rule observation) alternative).toReal ≤ (↑ (Fintype.card Alternative))-1
```

GolzHaghtalabYang2025Distortion.theorem3ReportedOrdinarySpecialMargin

Declaration location: paper-local

Lean declaration body

```
type:
{Alternative : Type u_1} →
[DecidableEq Alternative] → {d : ℕ} → Alternative → Alternative → (Fin d → (Alternative × Alternative) × Bool) → ℝ

value:
fun {Alternative : Type u_1} [DecidableEq Alternative] {d : ℕ} (special ordinary : Alternative)
(report : Fin d → (Alternative × Alternative) × Bool) =>
  Σ comparisonIndex : Fin d,
    GolzHaghtalabYang2025Distortion.theorem3OrdinarySpecialComparisonMargin special ordinary (report comparisonIndex)
```

Direct retained dependencies

- [GolzHaghtalabYang2025Distortion.theorem3OrdinarySpecialComparisonMargin](#)

GolzHaghtalabYang2025Distortion.theorem3SpecialTypeMass

Declaration location: paper-local

Lean declaration body

```
type:
ℝ → ℝ → ℝ

value:
fun (beta epsilon : ℝ) =>
  ((beta * epsilon).sigmoid - 1 / 2) / GolzHaghtalabYang2025Distortion.theorem3BalanceDenominator beta epsilon
```

Direct retained dependencies

- [GolzHaghtalabYang2025Distortion.theorem3BalanceDenominator](#)

GolzHaghtalabYang2025Distortion.theorem12TypeAMass

Declaration location: paper-local

Lean declaration body

```
type:
ℝ → ℝ → ℝ

value:
fun (beta epsilon : ℝ) => GolzHaghtalabYang2025Distortion.theorem3SpecialTypeMass beta epsilon
```

Direct retained dependencies

- [GolzhaghtalabYang2025Distortion.theorem3SpecialTypeMass](#)

GolzhaghtalabYang2025Distortion.theorem12TypeBMass

Declaration location: paper-local

Lean declaration body

```
type:
 $\mathbb{R} \rightarrow \mathbb{R} \rightarrow \mathbb{R} \rightarrow \mathbb{R}$ 

value:
fun (beta epsilon gamma :  $\mathbb{R}$ ) =>
  GolzhaghtalabYang2025Distortion.theorem12TypeAMass beta epsilon * ((beta * gamma).sigmoid - 1 / 2) /
    (beta.sigmoid - 1 / 2)
```

Direct retained dependencies

- [GolzhaghtalabYang2025Distortion.theorem12TypeAMass](#)

GolzhaghtalabYang2025Distortion.theorem12TypeWeight

Declaration location: paper-local

Lean declaration body

```
type:
 $\mathbb{R} \rightarrow \mathbb{R} \rightarrow \mathbb{R} \rightarrow \text{Fin } 3 \rightarrow \mathbb{R}$ 

value:
fun (beta epsilon gamma :  $\mathbb{R}$ ) (a : Fin 3) =>
  match a with
  | 0 => GolzhaghtalabYang2025Distortion.theorem12TypeAMass beta epsilon
  | 1 => GolzhaghtalabYang2025Distortion.theorem12TypeBMass beta epsilon gamma
  | 2 =>
    1 - GolzhaghtalabYang2025Distortion.theorem12TypeAMass beta epsilon -
      GolzhaghtalabYang2025Distortion.theorem12TypeBMass beta epsilon gamma
```

Direct retained dependencies

- [GolzhaghtalabYang2025Distortion.theorem12TypeAMass](#)
- [GolzhaghtalabYang2025Distortion.theorem12TypeBMass](#)

GolzhaghtalabYang2025Distortion.theorem12Population

Declaration location: paper-local

Lean declaration body

```
type:
(beta epsilon gamma :  $\mathbb{R}$ )  $\rightarrow$  0 < beta  $\rightarrow$  0 < epsilon  $\rightarrow$  epsilon < 1  $\rightarrow$  0 < gamma  $\rightarrow$  gamma < 1  $\rightarrow$  PMF (Fin 3)

value:
fun (beta epsilon gamma :  $\mathbb{R}$ ) (hbeta : 0 < beta) (hepsilon : 0 < epsilon) (hepsilon_lt_one : epsilon < 1)
  (hgamma : 0 < gamma) (hgamma_lt_one : gamma < 1) =>
  AppliedModelingLib.finiteWeightedPMF (GolzhaghtalabYang2025Distortion.theorem12TypeWeight beta epsilon gamma)
    (GolzhaghtalabYang2025Distortion.theorem12TypeWeight_nonneg hbeta hepsilon hepsilon_lt_one hgamma hgamma_lt_one)
    (GolzhaghtalabYang2025Distortion.theorem12Population._proof_1 beta epsilon gamma)
```

Direct retained dependencies

- [AppliedModelingLib.finiteWeightedPMF](#)
- [GolzhaghtalabYang2025Distortion.theorem12TypeWeight](#)

GolzhaghtalabYang2025Distortion.theorem3TwoTypePopulation

Declaration location: paper-local

Lean declaration body

```
type:
(beta epsilon : ℝ) → 0 < beta → 0 < epsilon → PMF Bool

value:
fun (beta epsilon : ℝ) (hbeta : 0 < beta) (hepsilon : 0 < epsilon) =>
  PMF.bernoulli
    (GolzhaghtalabYang2025Distortion.theorem3SpecialTypeMass beta epsilon,
     GolzhaghtalabYang2025Distortion.theorem3TwoTypePopulation._proof_1 beta epsilon hbeta hepsilon)
    (GolzhaghtalabYang2025Distortion.theorem3TwoTypePopulation._proof_2 beta epsilon hbeta hepsilon)
```

Direct retained dependencies

- [GolzhaghtalabYang2025Distortion.theorem3SpecialTypeMass](#)

GolzhaghtalabYang2025Distortion.theorem3ManyAlternativeOneComparisonLaw

Declaration location: paper-local

Lean declaration body

```
type:
{Alternative : Type u_1} →
[Fintype Alternative] →
[DecidableEq Alternative] →
PMF (Alternative × Alternative) →
(beta epsilon : ℝ) → 0 < beta → 0 < epsilon → Alternative → PMF ((Alternative × Alternative) × Bool)

value:
fun {Alternative : Type u_1} [Fintype Alternative] [DecidableEq Alternative]
  (pairSampling : PMF (Alternative × Alternative)) (beta epsilon : ℝ) (hbeta : 0 < beta) (hepsilon : 0 < epsilon)
  (special : Alternative) =>
  pairSampling.bind fun (pair : Alternative × Alternative) =>
    let preference : AppliedModelingLib.Learning.HumanFeedback.PairwisePreference PUnit.{1} Alternative :=
      AppliedModelingLib.Alignment.Welfare.populationBradleyTerryPreference
        (GolzhaghtalabYang2025Distortion.theorem3TwoTypePopulation beta epsilon hbeta hepsilon)
        (GolzhaghtalabYang2025Distortion.theorem3ManyAlternativeUtility epsilon 1 special) beta;
    PMF.map (fun (outcome : Bool) => (pair, outcome))
      (PMF.bernoulli
        (preference.prob PUnit.unit pair.1 pair.2,
         GolzhaghtalabYang2025Distortion.theorem3ManyAlternativeOneComparisonLaw._proof_1 beta epsilon hbeta hepsilon
          special pair)
        (GolzhaghtalabYang2025Distortion.theorem3ManyAlternativeOneComparisonLaw._proof_2 beta epsilon hbeta hepsilon
          special pair))
```

Direct retained dependencies

- [AppliedModelingLib.Alignment.Welfare.populationBradleyTerryPreference](#)
- [AppliedModelingLib.Learning.HumanFeedback.PairwisePreference](#)
- [GolzhaghtalabYang2025Distortion.theorem3ManyAlternativeUtility](#)
- [GolzhaghtalabYang2025Distortion.theorem3TwoTypePopulation](#)

GolzhaghtalabYang2025Distortion.theorem3ManyAlternativeObservationLaw

Declaration location: paper-local

Lean declaration body

```
type:
{Alternative : Type u_1} →
[Fintype Alternative] →
[DecidableEq Alternative] →
PMF (Alternative × Alternative) →
(beta epsilon : ℝ) →
0 < beta → 0 < epsilon → Alternative → (horizon : ℕ) → PMF (Fin horizon → (Alternative × Alternative) × Bool)

value:
fun {Alternative : Type u_1} [Fintype Alternative] [DecidableEq Alternative]
  (pairSampling : PMF (Alternative × Alternative)) (beta epsilon : ℝ) (hbeta : 0 < beta) (hepsilon : 0 < epsilon)
  (special : Alternative) (horizon : ℕ) =>
  AppliedModelingLib.pmfProduct (Fin horizon) ((Alternative × Alternative) × Bool)
    (GolzhaghtalabYang2025Distortion.theorem3ManyAlternativeOneComparisonLaw pairSampling beta epsilon hbeta hepsilon
     special)
```

Direct retained dependencies

- [AppliedModelingLib.pmfProduct](#)
- [GolzhaghtalabYang2025Distortion.theorem3ManyAlternativeOneComparisonLaw](#)

GolzHaghtalabYang2025Distortion.theorem3PopulationComparisonOutcomeLaw

Declaration location: paper-local

Lean declaration body

```
type:
{Alternative : Type u_1} →
[Fintype Alternative] →
[DecidableEq Alternative] →
(beta epsilon : ℝ) → ℝ → 0 < beta → 0 < epsilon → Alternative → Alternative × Alternative → PMF Bool

value:
fun {Alternative : Type u_1} [Fintype Alternative] [DecidableEq Alternative] (beta epsilon xi : ℝ) (hbeta : 0 < beta)
(hepsilon : 0 < epsilon) (special : Alternative) (pair : Alternative × Alternative) =>
let preference : AppliedModelingLib.Learning.HumanFeedback.PairwisePreference PUnit.{1} Alternative :=
AppliedModelingLib.Alignment.Welfare.populationBradleyTerryPreference
(GolzHaghtalabYang2025Distortion.theorem3TwoTypePopulation beta epsilon hbeta hepsilon)
(GolzHaghtalabYang2025Distortion.theorem3ManyAlternativeUtility epsilon xi special) beta;
PMF.bernoulli
(preference.prob PUnit.unit pair.1 pair.2,
GolzHaghtalabYang2025Distortion.theorem3PopulationComparisonOutcomeLaw._proof_1 beta epsilon xi hbeta hepsilon
special pair)
(GolzHaghtalabYang2025Distortion.theorem3PopulationComparisonOutcomeLaw._proof_2 beta epsilon xi hbeta hepsilon
special pair)
```

Direct retained dependencies

- [AppliedModelingLib.Alignment.Welfare.populationBradleyTerryPreference](#)
- [AppliedModelingLib.Learning.HumanFeedback.PairwisePreference](#)
- [GolzHaghtalabYang2025Distortion.theorem3ManyAlternativeUtility](#)
- [GolzHaghtalabYang2025Distortion.theorem3TwoTypePopulation](#)

GolzHaghtalabYang2025Distortion.theorem3ManyAlternativeOneComparisonLawXi

Declaration location: paper-local

Lean declaration body

```
type:
{Alternative : Type u_1} →
[Fintype Alternative] →
[DecidableEq Alternative] →
PMF (Alternative × Alternative) →
(beta epsilon : ℝ) → ℝ → 0 < beta → 0 < epsilon → Alternative → PMF ((Alternative × Alternative) × Bool)

value:
fun {Alternative : Type u_1} [Fintype Alternative] [DecidableEq Alternative]
(pairSampling : PMF (Alternative × Alternative)) (beta epsilon xi : ℝ) (hbeta : 0 < beta) (hepsilon : 0 < epsilon)
(special : Alternative) =>
pairSampling.bind fun (pair : Alternative × Alternative) =>
PMF.map (fun (outcome : Bool) => (pair, outcome))
(GolzHaghtalabYang2025Distortion.theorem3PopulationComparisonOutcomeLaw beta epsilon xi hbeta hepsilon special
pair)
```

Direct retained dependencies

- [GolzHaghtalabYang2025Distortion.theorem3PopulationComparisonOutcomeLaw](#)

GolzHaghtalabYang2025Distortion.theorem6Alternative

Declaration location: paper-local

Lean declaration body

```
type:
ℕ → Type

value:
fun (copies : ℕ) => Fin 2 ⊕ Fin copies
```

GolzHaghtalabYang2025Distortion.theorem6A

Declaration location: paper-local

Lean declaration body

```
type:
{copies : ℕ} → GolzHaghtalabYang2025Distortion.theorem6Alternative copies

value:
fun {copies : ℕ} => Sum.inl 0
```

Direct retained dependencies

- [GolzHaghtalabYang2025Distortion.theorem6Alternative](#)

GolzHaghtalabYang2025Distortion.theorem6Epsilon

Declaration location: paper-local

Lean declaration body

```
type:
ℝ → ℝ

value:
fun (eta : ℝ) => Real.exp (-2 / eta)
```

GolzHaghtalabYang2025Distortion.theorem6MinorityMass

Declaration location: paper-local

Lean declaration body

```
type:
ℝ → ℝ

value:
fun (beta : ℝ) => 10 / (10 + Real.exp beta)
```

GolzHaghtalabYang2025Distortion.theorem6CopyCount

Declaration location: paper-local

Lean declaration body

```
type:
ℝ → ℕ

value:
fun (beta : ℝ) => ⌈2 / (GolzHaghtalabYang2025Distortion.theorem6MinorityMass beta * (1 / 3 - Real.exp 1 / 10))⌋ +
```

Direct retained dependencies

- [GolzHaghtalabYang2025Distortion.theorem6MinorityMass](#)

GolzHaghtalabYang2025Distortion.theorem6ReferenceWeight

Declaration location: paper-local

Lean declaration body

```
type:
ℝ → {copies : ℕ} → GolzHaghtalabYang2025Distortion.theorem6Alternative copies → ℝ

value:
fun (epsilon : ℝ) {copies : ℕ} (a : GolzHaghtalabYang2025Distortion.theorem6Alternative copies) =>
  match a with
  | Sum.inl val => (1 - epsilon) / 2
  | Sum.inr val => epsilon / ↑copies
```

Direct retained dependencies

- [GolzHaghtalabYang2025Distortion.theorem6Alternative](#)

GolzHaghtalabYang2025Distortion.theorem6Reference

Declaration location: paper-local

Lean declaration body

```
type:
(epsilon : ℝ) →
{copies : ℕ} →
0 < epsilon → epsilon < 1 → 0 < copies → PMF (GolzhaghtalabYang2025Distortion.theorem6Alternative copies)

value:
fun (epsilon : ℝ) {copies : ℕ} (hepsilon_pos : 0 < epsilon) (hepsilon_lt_one : epsilon < 1)
(hcopies_pos : 0 < copies) =>
AppliedModelingLib.finiteWeightedPMF (GolzhaghtalabYang2025Distortion.theorem6ReferenceWeight epsilon)
(GolzhaghtalabYang2025Distortion.theorem6Reference._proof_1 epsilon hepsilon_pos hepsilon_lt_one hcopies_pos)
(GolzhaghtalabYang2025Distortion.theorem6Reference._proof_2 epsilon hcopies_pos)
```

Direct retained dependencies

- [AppliedModelingLib.finiteWeightedPMF](#)
- [GolzhaghtalabYang2025Distortion.theorem6Alternative](#)
- [GolzhaghtalabYang2025Distortion.theorem6ReferenceWeight](#)

GolzhaghtalabYang2025Distortion.theorem6TypeWeight

Declaration location: paper-local

Lean declaration body

```
type:
ℝ → Fin 2 → ℝ

value:
fun (beta : ℝ) (a : Fin 2) =>
match a with
| 0 => GolzhaghtalabYang2025Distortion.theorem6MinorityMass beta
| 1 => 1 - GolzhaghtalabYang2025Distortion.theorem6MinorityMass beta
```

Direct retained dependencies

- [GolzhaghtalabYang2025Distortion.theorem6MinorityMass](#)

GolzhaghtalabYang2025Distortion.theorem6Population

Declaration location: paper-local

Lean declaration body

```
type:
ℝ → PMF (Fin 2)

value:
fun (beta : ℝ) =>
AppliedModelingLib.finiteWeightedPMF (GolzhaghtalabYang2025Distortion.theorem6TypeWeight beta)
(GolzhaghtalabYang2025Distortion.theorem6Population._proof_1 beta)
(GolzhaghtalabYang2025Distortion.theorem6Population._proof_2 beta)
```

Direct retained dependencies

- [AppliedModelingLib.finiteWeightedPMF](#)
- [GolzhaghtalabYang2025Distortion.theorem6TypeWeight](#)

GolzhaghtalabYang2025Distortion.theorem6Utility

Declaration location: paper-local

Lean declaration body

```
type:
ℝ → {copies : ℕ} → Fin 2 → GolzhaghtalabYang2025Distortion.theorem6Alternative copies → ℝ

value:
fun (beta : ℝ) {copies : ℕ} (a : Fin 2) (a_1 : GolzhaghtalabYang2025Distortion.theorem6Alternative copies) =>
match a, a_1 with
| 0, Sum.inl 0 => 0
| 0, Sum.inl 1 => 1
| 0, Sum.inr val => 0
| 1, Sum.inl 0 => 1 / beta
| 1, Sum.inl 1 => 0
| 1, Sum.inr val => 1
```

Direct retained dependencies

- [GolzhaghtalabYang2025Distortion.theorem6Alternative](#)

GolzhaghtalabYang2025Distortion.theorem6Preference

Declaration location: paper-local

Lean declaration body

```
type:
  ℝ →
  {copies : ℕ} →
  AppliedModelingLib.Learning.HumanFeedback.PairwisePreference PUnit.{1}
  (GolzhaghtalabYang2025Distortion.theorem6Alternative copies)

value:
fun (beta : ℝ) {copies : ℕ} =>
  AppliedModelingLib.Alignment.Welfare.populationBradleyTerryPreference
  (GolzhaghtalabYang2025Distortion.theorem6Population beta) (GolzhaghtalabYang2025Distortion.theorem6Utility beta)
  beta
```

Direct retained dependencies

- [AppliedModelingLib.Alignment.Welfare.populationBradleyTerryPreference](#)
- [AppliedModelingLib.Learning.HumanFeedback.PairwisePreference](#)
- [GolzhaghtalabYang2025Distortion.theorem6Alternative](#)
- [GolzhaghtalabYang2025Distortion.theorem6Population](#)
- [GolzhaghtalabYang2025Distortion.theorem6Utility](#)

GolzhaghtalabYang2025Distortion.theorem6Welfare

Declaration location: paper-local

Lean declaration body

```
type:
  ℝ → {copies : ℕ} → GolzhaghtalabYang2025Distortion.theorem6Alternative copies → ℝ

value:
fun (beta : ℝ) {copies : ℕ} (a : GolzhaghtalabYang2025Distortion.theorem6Alternative copies) =>
  AppliedModelingLib.Alignment.Welfare.populationAverageUtility
  (GolzhaghtalabYang2025Distortion.theorem6Population beta) (GolzhaghtalabYang2025Distortion.theorem6Utility beta) a
```

Direct retained dependencies

- [AppliedModelingLib.Alignment.Welfare.populationAverageUtility](#)
- [GolzhaghtalabYang2025Distortion.theorem6Alternative](#)
- [GolzhaghtalabYang2025Distortion.theorem6Population](#)
- [GolzhaghtalabYang2025Distortion.theorem6Utility](#)

GolzhaghtalabYang2025Distortion.theorem7EmpiricalPreference

Declaration location: paper-local

Lean declaration body

```
type:
  {Alternative : Type u_1} →
  (empiricalRate : Alternative → Alternative → ℝ) →
  (∀ (first second : Alternative), 0 ≤ empiricalRate first second) →
  (∀ (first second : Alternative), empiricalRate first second ≤ 1) →
  AppliedModelingLib.Learning.HumanFeedback.PairwisePreference PUnit.{1} Alternative

value:
fun {Alternative : Type u_1} (empiricalRate : Alternative → Alternative → ℝ)
  (hempirical_nonneg : ∀ (first second : Alternative), 0 ≤ empiricalRate first second)
  (hempirical_le_one : ∀ (first second : Alternative), empiricalRate first second ≤ 1) =>
  {
    prob := fun (x : PUnit.{1}) (first second : Alternative) =>
      1 / 2 + (empiricalRate first second - empiricalRate second first) / 2,
    nonneg :=
      GolzhaghtalabYang2025Distortion.theorem7EmpiricalPreference._proof_2 empiricalRate hempirical_nonneg
      hempirical_le_one,
    le_one :=
      GolzhaghtalabYang2025Distortion.theorem7EmpiricalPreference._proof_3 empiricalRate hempirical_nonneg
      hempirical_le_one,
    complementary := GolzhaghtalabYang2025Distortion.theorem7EmpiricalPreference._proof_4 empiricalRate }
  }
```

Direct retained dependencies

- [AppliedModelingLib.Learning.HumanFeedback.PairwisePreference](#)

GolzHaghtalabYang2025Distortion.theorem7lidUserBatchLaw

Declaration location: paper-local

Lean declaration body

```
type:
{Alternative : Type u_1} →
[Fintype Alternative] →
[DecidableEq Alternative] →
PMF (GolzHaghtalabYang2025Distortion.theorem2UserResponseTable Alternative) →
PMF Alternative →
(users comparisonsPerUser : ℕ) →
PMF
(Fin users →
GolzHaghtalabYang2025Distortion.theorem2UserResponseTable Alternative ×
(Fin comparisonsPerUser → Alternative × Alternative))

value:
fun {Alternative : Type u_1} [Fintype Alternative] [DecidableEq Alternative]
(responseLaw : PMF (GolzHaghtalabYang2025Distortion.theorem2UserResponseTable Alternative))
(sampling : PMF Alternative) (users comparisonsPerUser : ℕ) =>
AppliedModelingLib.pmfProduct (Fin users)
(GolzHaghtalabYang2025Distortion.theorem2UserResponseTable Alternative ×
(Fin comparisonsPerUser → Alternative × Alternative))
(AppliedModelingLib.pmfProd responseLaw
(AppliedModelingLib.pmfProduct (Fin comparisonsPerUser) (Alternative × Alternative)
(GolzHaghtalabYang2025Distortion.theorem2UserPairLabelLaw sampling)))
```

Direct retained dependencies

- [AppliedModelingLib.pmfProd](#)
- [AppliedModelingLib.pmfProduct](#)
- [GolzHaghtalabYang2025Distortion.theorem2UserPairLabelLaw](#)
- [GolzHaghtalabYang2025Distortion.theorem2UserResponseTable](#)

GolzHaghtalabYang2025Distortion.theorem7lidUserEmpiricalPolicy

Declaration location: paper-local

Lean declaration body

```
type:
{Alternative : Type u_1} →
[Fintype Alternative] →
[DecidableEq Alternative] →
{users comparisonsPerUser : ℕ} →
PMF Alternative →
(klBudget : ℝ) →
0 ≤ klBudget →
(Fin users →
GolzHaghtalabYang2025Distortion.theorem2UserResponseTable Alternative ×
(Fin comparisonsPerUser → Alternative × Alternative)) →
PMF Alternative

value:
fun {Alternative : Type u_1} [Fintype Alternative] [DecidableEq Alternative] {users comparisonsPerUser : ℕ}
(reference : PMF Alternative) (klBudget : ℝ) (hbudget : 0 ≤ klBudget)
(sample :
Fin users →
GolzHaghtalabYang2025Distortion.theorem2UserResponseTable Alternative ×
(Fin comparisonsPerUser → Alternative × Alternative)) =>
Classical.choose
(GolzHaghtalabYang2025Distortion.theorem7lidUserEmpiricalPolicy._proof_1 reference klBudget hbudget sample)
```

Direct retained dependencies

- [AppliedModelingLib.GameTheory.PreferenceGame.IsFiniteKLPreferenceMaximin](#)
- [GolzHaghtalabYang2025Distortion.theorem10lidUserPaperWinRate](#)
- [GolzHaghtalabYang2025Distortion.theorem2UserResponseTable](#)
- [GolzHaghtalabYang2025Distortion.theorem7EmpiricalPreference](#)

GolzHaghtalabYang2025Distortion.theorem9OffPathPairs

Declaration location: paper-local

Lean declaration body

```

type:
(m : ℕ) → Finset (Fin m × Fin m)

value:
fun (m : ℕ) => {pair : Fin m × Fin m | pair.1 ≠ pair.2 ∧ ↑pair.1 + 1 ≠ ↑pair.2 ∧ ↑pair.2 + 1 ≠ ↑pair.1}

```

GolzHaghtalabYang2025Distortion.theorem9OffPathPair

Declaration location: paper-local

Lean declaration body

```

type:
ℕ → Type

value:
fun (m : ℕ) => ↑(GolzHaghtalabYang2025Distortion.theorem9OffPathPairs m)

```

Direct retained dependencies

- [GolzHaghtalabYang2025Distortion.theorem9OffPathPairs](#)

GolzHaghtalabYang2025Distortion.theorem9OffPathSampling

Declaration location: paper-local

Lean declaration body

```

type:
(m : ℕ) → 3 ≤ m → AppliedModelingLib.Learning.HumanFeedback.CorrelatedPairSampling (Fin m)

value:
fun (m : ℕ) (hm : 3 ≤ m) =>
  PMF.map Subtype.val (AppliedModelingLib.uniformPMF (GolzHaghtalabYang2025Distortion.theorem9OffPathPair m))

```

Direct retained dependencies

- [AppliedModelingLib.Learning.HumanFeedback.CorrelatedPairSampling](#)
- [AppliedModelingLib.uniformPMF](#)
- [GolzHaghtalabYang2025Distortion.theorem9OffPathPair](#)
- [GolzHaghtalabYang2025Distortion.theorem9OffPathPairs](#)

GolzHaghtalabYang2025Distortion.theorem9PathOrientation

Declaration location: paper-local

Lean declaration body

```

type:
{m : ℕ} → 2 ≤ m → Fin (m - 1) × Bool → Fin m × Fin m

value:
fun {m : ℕ} (hm : 2 ≤ m) (edge : Fin (m - 1) × Bool) =>
  have lower : Fin m := (↑edge.1, GolzHaghtalabYang2025Distortion.theorem9PathOrientation._proof_3 hm edge);
  have upper : Fin m := (↑edge.1 + 1, GolzHaghtalabYang2025Distortion.theorem9PathOrientation._proof_4 hm edge);
  if edge.2 = true then (lower, upper) else (upper, lower)

```

GolzHaghtalabYang2025Distortion.theorem9PathSampling

Declaration location: paper-local

Lean declaration body

```

type:
(m : ℕ) → 2 ≤ m → AppliedModelingLib.Learning.HumanFeedback.CorrelatedPairSampling (Fin m)

value:
fun (m : ℕ) (hm : 2 ≤ m) =>
  PMF.map (GolzHaghtalabYang2025Distortion.theorem9PathOrientation hm)
    (AppliedModelingLib.uniformPMF (Fin (m - 1) × Bool))

```

Direct retained dependencies

- [AppliedModelingLib.Learning.HumanFeedback.CorrelatedPairSampling](#)
- [AppliedModelingLib.uniformPMF](#)
- [GolzHaghtalabYang2025Distortion.theorem9PathOrientation](#)

GolzHaghtalabYang2025Distortion.theorem9ComparisonSampling

Declaration location: paper-local

Lean declaration body

```
type:
(m : ℕ) → 3 ≤ m → NNReal → AppliedModelingLib.Learning.HumanFeedback.CorrelatedPairSampling (Fin m)

value:
fun (m : ℕ) (hm : 3 ≤ m) (epsilon : NNReal) =>
  AppliedModelingLib.binaryMixturePMF (1 - epsilon)
    (GolzHaghtalabYang2025Distortion.theorem9ComparisonSampling._proof_1 epsilon)
    (GolzHaghtalabYang2025Distortion.theorem9PathSampling m
      (GolzHaghtalabYang2025Distortion.theorem9ComparisonSampling._proof_2 m hm))
    (GolzHaghtalabYang2025Distortion.theorem9OffPathSampling m hm)
```

Direct retained dependencies

- [AppliedModelingLib.Learning.HumanFeedback.CorrelatedPairSampling](#)
- [AppliedModelingLib.binaryMixturePMF](#)
- [GolzHaghtalabYang2025Distortion.theorem9OffPathSampling](#)
- [GolzHaghtalabYang2025Distortion.theorem9PathSampling](#)

GolzHaghtalabYang2025Distortion.theorem9PrefixProfile

Declaration location: paper-local

Lean declaration body

```
type:
ℝ → (m : ℕ) → Fin 3 → Fin m → ℝ

value:
fun (beta : ℝ) (m : ℕ) (a : Fin 3) (a_1 : Fin m) => GolzHaghtalabYang2025Distortion.lemma15Utility beta (↑ a_1) a
```

Direct retained dependencies

- [GolzHaghtalabYang2025Distortion.lemma15Utility](#)

GolzHaghtalabYang2025Distortion.theorem9PrefixPreference

Declaration location: paper-local

Lean declaration body

```
type:
ℝ → (m : ℕ) → AppliedModelingLib.Learning.HumanFeedback.PairwisePreference PUnit.{1} (Fin m)

value:
fun (beta : ℝ) (m : ℕ) =>
  AppliedModelingLib.Alignment.Welfare.populationBradleyTerryPreference
    (GolzHaghtalabYang2025Distortion.lemma15Population (GolzHaghtalabYang2025Distortion.theorem9PrefixProfile beta m)
     beta)
```

Direct retained dependencies

- [AppliedModelingLib.Alignment.Welfare.populationBradleyTerryPreference](#)
- [AppliedModelingLib.Learning.HumanFeedback.PairwisePreference](#)
- [GolzHaghtalabYang2025Distortion.lemma15Population](#)
- [GolzHaghtalabYang2025Distortion.theorem9PrefixProfile](#)

GolzHaghtalabYang2025Distortion.theorem9OneComparisonReportLaw

Declaration location: paper-local

Lean declaration body

```
type:
 $\mathbb{R} \rightarrow (m : \mathbb{N}) \rightarrow 3 \leq m \rightarrow \text{NNReal} \rightarrow \text{PMF} (\text{AppliedModelingLib.Learning.HumanFeedback.BinaryPairwiseReport} (\text{Fin } m))$ 

value:
fun (beta :  $\mathbb{R}$ ) (m :  $\mathbb{N}$ ) (hm :  $3 \leq m$ ) (epsilon :  $\text{NNReal}$ ) =>
  AppliedModelingLib.Learning.HumanFeedback.correlatedOneComparisonReportLaw
    (GolzHaghtalabYang2025Distortion.theorem9ComparisonSampling m hm epsilon)
    (GolzHaghtalabYang2025Distortion.theorem9PrefixPreference beta m)
```

Direct retained dependencies

- [AppliedModelingLib.Learning.HumanFeedback.BinaryPairwiseReport](#)
- [AppliedModelingLib.Learning.HumanFeedback.correlatedOneComparisonReportLaw](#)
- [GolzHaghtalabYang2025Distortion.theorem9ComparisonSampling](#)
- [GolzHaghtalabYang2025Distortion.theorem9PrefixPreference](#)

GolzHaghtalabYang2025Distortion.theorem9UniformReference

Declaration location: paper-local

Lean declaration body

```
type:
 $(m : \mathbb{N}) \rightarrow 1 \leq m \rightarrow \text{PMF} (\text{Fin } m)$ 

value:
fun (m :  $\mathbb{N}$ ) (hm :  $1 \leq m$ ) => AppliedModelingLib.uniformPMF (Fin m)
```

Direct retained dependencies

- [AppliedModelingLib.uniformPMF](#)

Material library prerequisites

AppliedModelingLib.Alignment.Welfare.FiniteUtilityProfile

Source locator: cited publication, lines 174-179

Verbatim paper-source connection

Let $A = \{1, \dots, m\}$ be a finite set of alternatives. The population of users is described by a probability distribution D over utility vectors $u = (u(1), \dots, u(m))$, whose entries $0 \leq u(x) \leq 1$ indicate a user's utility for alternative x .³ The objective is to find an alternative x such that its average utility $\text{AvgUtil}(x) := \mathbb{E}_{u \sim D} [u(x)]$ across the user population (also known as the utilitarian social welfare) is as high as possible. We extend this notation to probability distributions π over alternatives by setting $\text{AvgUtil}(\pi) := \mathbb{E}_{x \sim \pi} [\text{AvgUtil}(x)]$.

Lean-expanded library semantic target

```
type:
Type u_1 → Type u_2 → Type (max u_1 u_2)

value:
fun (User : Type u_1) (Alternative : Type u_2) => User → Alternative → ℝ
```

Exact Lean library declaration

```
/-- A user's cardinal utility for each finite alternative. -/
abbrev FiniteUtilityProfile (User Alternative : Type*) := User → Alternative → ℝ
```

Recorded source-to-library screening Verdict: matches

Reason: This represents a cardinal utility vector as user to alternative to real; finiteness of alternatives is imposed at source-facing use sites.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

AppliedModelingLib.finiteKLDivergence

Source locator: cited publication, lines 237-253

Verbatim paper-source connection

Alignment Setting. The alignment setting generalizes the social choice setting in two ways: first, user utilities $u_i(y \mid x)$ may depend on a state x ; second, the goal in determining a policy π is not purely to maximize the reward $\text{AvgUtil}(\pi \mid x)$, but a trade-off between this reward and the goal of remaining close to a reference policy $\pi_{\text{ref}}(\cdot \mid x) \in \Delta(A)$ in terms of the KL divergence $\text{DKL}(\cdot \mid \pi_{\text{ref}})$. For theoretical tractability, we focus our analysis on a single state x , which we from here on omit from the notation. Conceptually, this treatment of alignment on a state-by-state basis corresponds to an assumption that our policy class is expressive enough so that it can take the optimal distribution of actions at each state and abstracts from the generalization problem of estimating the population's preference between a pair of alternatives at the given state x based on preferences in similar states x_0 . Having set aside the dependency between states, we focus on how the regularization with respect to a reference policy impacts the ability to optimize the average utility of three alignment methods: RLHF, DPO, and NLHF. In addition to the pairwise comparisons, these methods take in a reference policy $\pi_{\text{ref}} \in \Delta(A)$ and a KL bound $\tau \geq 0$, and map these inputs to a policy π in the KL-ball $B_\tau(\pi_{\text{ref}}) := \{\pi \in \Delta(A) \mid \text{DKL}(\pi \mid \pi_{\text{ref}}) \leq \tau\}$ around the reference policy. RLHF, DPO, and NLHF are typically implemented with a KL regularization rather than our KL-constrained formulation, which we adopt to enable a comparison on equal terms. In Section 5, we show that these perspectives are equivalent, and that distortion upper bounds carry over to regularized alignment methods.

[Next verbatim source excerpt]

If $\tau = 0$, the only feasible policy is π_{ref} , and the claim clearly holds for $\lambda = \infty$. We also assume that $\pi_{\text{ref}}(a) > 0$ for all $a \in M$. Otherwise if $\pi_{\text{ref}}(a) = 0$ for some $a \in M$, then both the regularized and constrained versions forbid any policy to put nonzero probability on a , which leads to an effectively smaller candidate set.

Lean-expanded library semantic target

```
type:
{α : Type u_1} → [Fintype α] → [DecidableEq α] → PMF α → PMF α → ℝ

value:
fun {α : Type u_1} [Fintype α] [DecidableEq α] (first second : PMF α) =>
  ∑ outcome : α, (first outcome).toReal * (Real.log (first outcome).toReal - Real.log (second outcome).toReal)
```

Exact Lean library declaration

```
-- The finite real-valued KL expression between two probability mass functions. -/
noncomputable def finiteKLDivergence {α : Type*} [Fintype α] [DecidableEq α]
  (first second : PMF α) : ℝ :=
  ∑ outcome : α,
    (first outcome).toReal *
      (Real.log (first outcome).toReal - Real.log (second outcome).toReal)
```

Recorded source-to-library screening Verdict: matches

Reason: This is the finite-sum KL expression; bundled full support makes it the ordinary finite KL divergence and zero first-policy mass contributes zero.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

AppliedModelingLib.Alignment.Welfare.InFiniteKLBall

Source locator: cited publication, lines 237-253

Verbatim paper-source connection

Alignment Setting. The alignment setting generalizes the social choice setting in two ways: first, user utilities $u_i(y \mid x)$ may depend on a state x ; second, the goal in determining a policy π is not purely to maximize the reward $\text{AvgUtil}(\pi \mid x)$, but a trade-off between this reward and the goal of remaining close to a reference policy $\pi_{\text{ref}}(\cdot \mid x) \in \Delta(A)$ in terms of the KL divergence $\text{DKL}(\cdot \mid \pi_{\text{ref}})$. For theoretical tractability, we focus our analysis on a single state x , which we from here on omit from the notation. Conceptually, this treatment of alignment on a state-by-state basis corresponds to an assumption that our policy class is expressive enough so that it can take the optimal distribution of actions at each state⁷ and abstracts from the generalization problem of estimating the population's preference between a pair of alternatives at the given state x based on preferences in similar states x_0 . Having set aside the dependency between states, we focus on how the regularization with respect to a reference policy impacts the ability to optimize the average utility of three alignment methods: RLHF, DPO, and NLHF. In addition to the pairwise comparisons, these methods take in a reference policy $\pi_{\text{ref}} \in \Delta(A)$ and a KL bound $\tau \geq 0$, and map these inputs to a policy π in the KL-ball $B_\tau(\pi_{\text{ref}}) := \{\pi \in \Delta(A) \mid \text{DKL}(\pi \mid \pi_{\text{ref}}) \leq \tau\}$ around the reference policy. RLHF, DPO, and NLHF are typically implemented with a KL regularization rather than our KL-constrained formulation, which we adopt to enable a comparison on equal terms. In Section 5, we show that these perspectives are equivalent, and that distortion upper bounds carry over to regularized alignment methods.

[Next verbatim source excerpt]

If $\tau = 0$, the only feasible policy is π_{ref} , and the claim clearly holds for $\lambda = \infty$. We also assume that $\pi_{\text{ref}}(a) > 0$ for all $a \in M$. Otherwise if $\pi_{\text{ref}}(a) = 0$ for some $a \in M$, then both the regularized and constrained versions forbid any policy to put nonzero probability on a , which leads to an effectively smaller candidate set.

Lean-expanded library semantic target

```
type:
{Alternative : Type u_1} →
[FIntype Alternative] → [DecidableEq Alternative] → PMF Alternative → ℝ → PMF Alternative → Prop

value:
fun {Alternative : Type u_1} [FIntype Alternative] [DecidableEq Alternative] (reference : PMF Alternative)
(klBudget : ℝ) (policy : PMF Alternative) =>
AppliedModelingLib.finiteKLDivergence policy reference ≤ klBudget
```

Exact Lean library declaration

```
/-- A finite policy belongs to the source KL ball around `reference`. -/
def InFiniteKLBall {Alternative : Type*} [FIntype Alternative] [DecidableEq Alternative]
(reference : PMF Alternative) (klBudget : ℝ) (policy : PMF Alternative) : Prop :=
finiteKLDivergence policy reference ≤ klBudget
```

Direct retained dependencies

- [AppliedModelingLib.finiteKLDivergence](#)

Recorded source-to-library screening Verdict: matches

Reason: This exactly expresses the source normalization $0 \leq u(x) \leq 1$ for every user and alternative.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

AppliedModelingLib.Alignment.Welfare.ReferenceAlternativePolicy

Source locator: cited publication, lines 237-253

Verbatim paper-source connection

Alignment Setting. The alignment setting generalizes the social choice setting in two ways: first, user utilities $u_i(y \mid x)$ may depend on a state x ; second, the goal in determining a policy π is not purely to maximize the reward $\text{AvgUtil}(\pi \mid x)$, but a trade-off between this reward and the goal of remaining close to a reference policy $\pi_{\text{ref}}(\cdot \mid x) \in \Delta(A)$ in terms of the KL divergence $\text{DKL}(\pi \mid \pi_{\text{ref}})$. For theoretical tractability, we focus our analysis on a single state x , which we from here on omit from the notation. Conceptually, this treatment of alignment on a state-by-state basis corresponds to an assumption that our policy class is expressive enough so that it can take the optimal distribution of actions at each state⁷ and abstracts from the generalization problem of estimating the population's preference between a pair of alternatives at the given state x based on preferences in similar states x_0 . Having set aside the dependency between states, we focus on how the regularization with respect to a reference policy impacts the ability to optimize the average utility of three alignment methods: RLHF, DPO, and NLHF. In addition to the pairwise comparisons, these methods take in a reference policy $\pi_{\text{ref}} \in \Delta(A)$ and a KL bound $\tau \geq 0$, and map these inputs to a policy π in the KL-ball $B_\tau(\pi_{\text{ref}}) := \{\pi \in \Delta(A) \mid \text{DKL}(\pi \mid \pi_{\text{ref}}) \leq \tau\}$ around the reference policy. RLHF, DPO, and NLHF are typically implemented with a KL regularization rather than our KL-constrained formulation, which we adopt to enable a comparison on equal terms. In Section 5, we show that these perspectives are equivalent, and that distortion upper bounds carry over to regularized alignment methods.

[Next verbatim source excerpt]

If $\tau = 0$, the only feasible policy is π_{ref} , and the claim clearly holds for $\lambda = \infty$. We also assume that $\pi_{\text{ref}}(a) > 0$ for all $a \in M$. Otherwise if $\pi_{\text{ref}}(a) = 0$ for some $a \in M$, then both the regularized and constrained versions forbid any policy to put nonzero probability on a , which leads to an effectively smaller candidate set.

Lean-expanded library semantic target

```
type:
Type u_1 → Type u_1

value:
fun (Alternative : Type u_1) => PMF Alternative
```

Exact Lean library declaration

```
-- A context-free reference policy over alternatives, kept distinct from comparison sampling. -/
abbrev ReferenceAlternativePolicy (Alternative : Type*) := PMF Alternative
```

Recorded source-to-library screening Verdict: matches

Reason: The finite-population PMF specialization is precisely expectation of $u(x)$ under the population law.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

AppliedModelingLib.Alignment.Welfare.UnitIntervalUtilityProfile

Source locator: cited publication, lines 174-179

Verbatim paper-source connection

Let $A = \{1, \dots, m\}$ be a finite set of alternatives. The population of users is described by a probability distribution D over utility vectors $u = (u(1), \dots, u(m))$, whose entries $0 \leq u(x) \leq 1$ indicate a user's utility for alternative x .³ The objective is to find an alternative x such that its average utility $\text{AvgUtil}(x) := \mathbb{E}_{u \sim D} [u(x)]$ across the user population (also known as the utilitarian social welfare) is as high as possible. We extend this notation to probability distributions π over alternatives by setting $\text{AvgUtil}(\pi) := \mathbb{E}_{x \sim \pi} [\text{AvgUtil}(x)]$.

Lean-expanded library semantic target

```
type:
{User : Type u_1} →
{Alternative : Type u_2} → AppliedModelingLib.Alignment.Welfare.FiniteUtilityProfile User Alternative → Prop

value:
fun {User : Type u_1} {Alternative : Type u_2}
  (utility : AppliedModelingLib.Alignment.Welfare.FiniteUtilityProfile User Alternative) =>
  ∀ (user : User) (alternative : Alternative), 0 ≤ utility user alternative ∧ utility user alternative ≤ 1
```

Exact Lean library declaration

```
-- The source distortion model's utility normalization to the unit interval. -/
def UnitIntervalUtilityProfile {User Alternative : Type*}
  (utility : FiniteUtilityProfile User Alternative) : Prop :=
  ∀ user alternative, 0 ≤ utility user alternative ∧ utility user alternative ≤ 1
```

Direct retained dependencies

- [AppliedModelingLib.Alignment.Welfare.FiniteUtilityProfile](#)

Recorded source-to-library screening Verdict: matches

Reason: This exactly nests the alternative-policy expectation over population average utilities.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

AppliedModelingLib.Alignment.Welfare.populationAverageUtility

Source locator: cited publication, lines 174-179

Verbatim paper-source connection

Let $A = \{1, \dots, m\}$ be a finite set of alternatives. The population of users is described by a probability distribution D over utility vectors $u = (u(1), \dots, u(m))$, whose entries $0 \leq u(x) \leq 1$ indicate a user's utility for alternative x .³ The objective is to find an alternative x such that its average utility $\text{AvgUtil}(x) := \mathbb{E}_{u \sim D} [u(x)]$ across the user population (also known as the utilitarian social welfare) is as high as possible. We extend this notation to probability distributions π over alternatives by setting $\text{AvgUtil}(\pi) := \mathbb{E}_{x \sim \pi} [\text{AvgUtil}(x)]$.

Lean-expanded library semantic target

```
type:
{User : Type u_1} →
{Alternative : Type u_2} →
[Fintype User] →
[DecidableEq User] →
PMF User → AppliedModelingLib.Alignment.Welfare.FiniteUtilityProfile User Alternative → Alternative → ℝ

value:
fun {User : Type u_1} {Alternative : Type u_2} [Fintype User] [DecidableEq User] (population : PMF User)
  (utility : AppliedModelingLib.Alignment.Welfare.FiniteUtilityProfile User Alternative)
  (alternative : Alternative) =>
  AppliedModelingLib.pmfExp population fun (user : User) => utility user alternative
```

Exact Lean library declaration

```
/-- The utilitarian average utility of one alternative over a user population. -/
noncomputable def populationAverageUtility
  {User Alternative : Type*} [Fintype User] [DecidableEq User]
  (population : PMF User) (utility : FiniteUtilityProfile User Alternative)
  (alternative : Alternative) : ℝ :=
  pmfExp population (fun user => utility user alternative)
```

Direct retained dependencies

- [AppliedModelingLib.Alignment.Welfare.FiniteUtilityProfile](#)
- [AppliedModelingLib.pmfExp](#)

Recorded source-to-library screening Verdict: matches

Reason: This is the appropriate reusable interface for a valid pairwise marginal probability, with range and complementarity; the preceding constructor supplies the Bradley-Terry formula.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

AppliedModelingLib.Alignment.Welfare.policyAverageUtility

Source locator: cited publication, lines 174-179

Verbatim paper-source connection

Let $A = \{1, \dots, m\}$ be a finite set of alternatives. The population of users is described by a probability distribution D over utility vectors $u = (u(1), \dots, u(m))$, whose entries $0 \leq u(x) \leq 1$ indicate a user's utility for alternative x .³ The objective is to find an alternative x such that its average utility $\text{AvgUtil}(x) := \mathbb{E}_{u \sim D} [u(x)]$ across the user population (also known as the utilitarian social welfare) is as high as possible. We extend this notation to probability distributions π over alternatives by setting $\text{AvgUtil}(\pi) := \mathbb{E}_{x \sim \pi} [\text{AvgUtil}(x)]$.

Lean-expanded library semantic target

```
type:
{User : Type u_1} →
{Alternative : Type u_2} →
[Fintype User] →
[DecidableEq User] →
[Fintype Alternative] →
[DecidableEq Alternative] →
PMF User → AppliedModelingLib.Alignment.Welfare.FiniteUtilityProfile User Alternative → PMF Alternative → ℝ

value:
fun {User : Type u_1} {Alternative : Type u_2} [Fintype User] [DecidableEq User] [Fintype Alternative]
[DecidableEq Alternative] (population : PMF User)
(utility : AppliedModelingLib.Alignment.Welfare.FiniteUtilityProfile User Alternative) (policy : PMF Alternative) =>
AppliedModelingLib.pmfExp policy (AppliedModelingLib.Alignment.Welfare.populationAverageUtility population utility)
```

Exact Lean library declaration

```
/-- The average utility of a randomized alternative policy. -/
noncomputable def policyAverageUtility
{User Alternative : Type*} [Fintype User] [DecidableEq User]
[Fintype Alternative] [DecidableEq Alternative]
(population : PMF User) (utility : FiniteUtilityProfile User Alternative)
(policy : PMF Alternative) : ℝ :=
pmfExp policy (populationAverageUtility population utility)
```

Direct retained dependencies

- [AppliedModelingLib.Alignment.Welfare.FiniteUtilityProfile](#)
- [AppliedModelingLib.Alignment.Welfare.populationAverageUtility](#)
- [AppliedModelingLib.pmfExp](#)

Recorded source-to-library screening Verdict: matches

Reason: This computes the population expectation of the Bradley-Terry sigmoid of beta times the utility gap; the source-facing theorems retain positive beta.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

AppliedModelingLib.Learning.HumanFeedback.PairwisePreference

Source locator: cited publication, lines 180-200

Verbatim paper-source connection

Both voting rules and alignment methods observe comparisons from n users. We model each $i = 1, \dots, n$ as a fresh user with independently drawn utility vector $u_i \sim D$. For exposition, we assume that each user i provides an equal number $d \geq 1$ of pairwise comparisons. For each i , and for $j = 1, \dots, d$, we independently draw alternatives x_{ji}, y_{ji} from a fixed distribution μ over

3

Our assumption that utilities be in $[0, 1]$ is weaker than any of the three assumptions — unit-sum, unit-range, or approval [23] — made in classic distortion to avoid trivial infinite lower bounds.

3

[form-feed in source extraction]

alternatives, in which the minimum probability mass $\mu_{\min} := \min_{x \in A} \mu(x)$ is positive.⁴ User i then compares each pair $\{x_{ji}, y_{ji}\}$ (for $j = 1, \dots, d$) through a Bradley-Terry model based on i 's utilities: i prefers x_{ji} over y_{ji} (written “ x_{ji} [U+001F control character] y_{ji} ”) with probability $\sigma \beta \cdot (u_i(x_{ji}) - u_i(y_{ji}))$, where [U+0001 control character]

$\sigma(t) := 1/(1+e^{-t})$ is the logistic sigmoid function and $\beta > 0$ is a temperature parameter, and prefers y_{ji} over x_{ji} (“ y_{ji} [U+001F control character] x_{ji} ”) otherwise.⁵ Whereas this specifies the marginal probability of each pairwise comparison, we make no assumption about the correlation between i 's choices. For example, i might derive the pairwise comparisons from a Plackett-Luce ranking, ensuring that [U+0002 control character] the user's comparisons [U+0001 control character] $\hookrightarrow$ character][U+0003 control character] are always consistent.⁶ We set $p(x$ [U+001F control character] $y)$ for the expected win rate $E_{u \sim D} \sigma \beta \cdot (u(x) - u(y))$.

Lean declaration type (metadata)

```
field prob:
{Context : Type u_1} →
{Response : Type u_2} →
AppliedModelingLib.Learning.HumanFeedback.PairwisePreference Context Response → Context → Response → Response → ℝ
```

```
field nonneg:
∀ {Context : Type u_1} {Response : Type u_2}
(self : AppliedModelingLib.Learning.HumanFeedback.PairwisePreference Context Response) (context : Context)
(first second : Response), 0 ≤ self.prob context first second
```

```
field le_one:
∀ {Context : Type u_1} {Response : Type u_2}
(self : AppliedModelingLib.Learning.HumanFeedback.PairwisePreference Context Response) (context : Context)
(first second : Response), self.prob context first second ≤ 1
```

```
field complementary:
∀ {Context : Type u_1} {Response : Type u_2}
(self : AppliedModelingLib.Learning.HumanFeedback.PairwisePreference Context Response) (context : Context)
(first second : Response), self.prob context first second + self.prob context second first = 1
```

Exact Lean library declaration

```
/-- A valid probability that one response is preferred to another in a context. -/
structure PairwisePreference (Context Response : Type*) where
  /- Probability that the first response is preferred to the second. -/
  prob : Context → Response → Response → ℝ
  nonneg : ∀ context first second, 0 ≤ prob context first second
  le_one : ∀ context first second, prob context first second ≤ 1
  complementary : ∀ context first second,
    prob context first second + prob context second first = 1
```

Recorded source-to-library screening Verdict: matches

Reason: This is exactly the context-free reference policy in the finite alternative simplex, distinct from comparison sampling as in the source.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

AppliedModelingLib.Alignment.Welfare.populationBradleyTerryPreference

Source locator: cited publication, lines 180-200

Verbatim paper-source connection

Both voting rules and alignment methods observe comparisons from n users. We model each $i = 1, \dots, n$ as a fresh user with independently drawn utility vector $u_i \sim D$. For exposition, we assume that each user i provides an equal number $d \geq 1$ of pairwise comparisons. For each i , and for $j = 1, \dots, d$, we independently draw alternatives x_{ji}, y_{ji} from a fixed distribution μ over

3

Our assumption that utilities be in $[0, 1]$ is weaker than any of the three assumptions — unit-sum, unit-range, or approval [23] — made in classic distortion to avoid trivial infinite lower bounds.

3

[form-feed in source extraction]

alternatives, in which the minimum probability mass $\mu_{\min} := \min_{x \in A} \mu(x)$ is positive.⁴ User i then compares each pair $\{x_{ji}, y_{ji}\}$ (for $j = 1, \dots, d$) through a Bradley-Terry model based on i 's utilities: i prefers x_{ji} over y_{ji} (written “ x_{ji} [U+001F control character] y_{ji} ”) with probability $\sigma \beta \cdot (u_i(x_{ji}) - u_i(y_{ji}))$, where [U+0001 control character]

$\sigma(t) := 1/(1+e^{-t})$ is the logistic sigmoid function and $\beta > 0$ is a temperature parameter, and prefers y_{ji} over x_{ji} (“ y_{ji} [U+001F control character] x_{ji} ”) otherwise.⁵ Whereas this specifies the marginal probability of each pairwise comparison, we make no assumption about the correlation between i 's choices. For example, i might derive the pairwise comparisons from a Plackett-Luce ranking, ensuring that [U+0002 control character] the user's comparisons [U+0001 control character] $\leftrightarrow$ character][U+0003 control character] are always consistent.⁶ We set $p(x$ [U+001F control character] $y)$ for the expected win rate $E u \sim D \sigma \beta \cdot (u(x) - u(y))$.

Lean-expanded library semantic target

```
type:
{User : Type u_1} →
{Alternative : Type u_2} →
[Fintype User] →
[DecidableEq User] →
PMF User →
AppliedModelingLib.Alignment.Welfare.FiniteUtilityProfile User Alternative →
ℝ → AppliedModelingLib.Learning.HumanFeedback.PairwisePreference PUnit.{u_3 + 1} Alternative

value:
fun {User : Type u_1} {Alternative : Type u_2} [Fintype User] [DecidableEq User] (population : PMF User)
  (utility : AppliedModelingLib.Alignment.Welfare.FiniteUtilityProfile User Alternative) (btScale : ℝ) =>
{
  prob := fun (x : PUnit.{u_3 + 1}) (first second : Alternative) =>
    AppliedModelingLib.pmfExp population fun (user : User) =>
      (btScale * (utility user first - utility user second)).sigmoid,
  nonneg := AppliedModelingLib.Alignment.Welfare.populationBradleyTerryPreference._proof_1 population utility btScale,
  le_one := AppliedModelingLib.Alignment.Welfare.populationBradleyTerryPreference._proof_2 population utility btScale,
  complementary :=
    AppliedModelingLib.Alignment.Welfare.populationBradleyTerryPreference._proof_3 population utility btScale }
```

Exact Lean library declaration

```
/-- Aggregate independent user Bradley-Terry comparisons into one valid preference model. -/
noncomputable def populationBradleyTerryPreference
  {User Alternative : Type*} [Fintype User] [DecidableEq User]
  (population : PMF User) (utility : FiniteUtilityProfile User Alternative)
  (btScale : ℝ) : PairwisePreference PUnit Alternative where
prob _ first second :=
  pmfExp population fun user =>
    Real.sigmoid (btScale * (utility user first - utility user second))
nonneg _ first second :=
  pmfExp nonneg_of_forall_nonneg population _ fun user => Real.sigmoid_nonneg _
le_one _ first second :=
  pmfExp_le_of_forall_le population _ 1 fun user => Real.sigmoid_le_one _
complementary := by
  intro _ first second
  rw [← pmfExp_add, ← pmfExp_const population 1]
  refine pmfExp_congr population fun user => ?
  have hneg : btScale * (utility user second - utility user first) =
    -(btScale * (utility user first - utility user second)) := by ring
  rw [hneg, Real.sigmoid_neg]
  ring
```

Direct retained dependencies

- [AppliedModelingLib.Alignment.Welfare.FiniteUtilityProfile](#)
- [AppliedModelingLib.Learning.HumanFeedback.PairwisePreference](#)
- [AppliedModelingLib.pmfExp](#)

Recorded source-to-library screening **Verdict:** matches

Reason: This is the exact independent-product construction used for independently sampled displayed alternatives, instantiated with the source sampling law in both factors.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

AppliedModelingLib.PMFFullSupport

Source locator: cited publication, lines 237-253

Verbatim paper-source connection

Alignment Setting. The alignment setting generalizes the social choice setting in two ways: first, user utilities $u_i(y \mid x)$ may depend on a state x ; second, the goal in determining a policy π is not purely to maximize the reward $\text{AvgUtil}(\pi \mid x)$, but a trade-off between this reward and the goal of remaining close to a reference policy $\pi_{\text{ref}}(\cdot \mid x) \in \Delta(A)$ in terms of the KL divergence $\text{DKL}(\pi \mid \pi_{\text{ref}})$. For theoretical tractability, we focus our analysis on a single state x , which we from here on omit from the notation. Conceptually, this treatment of alignment on a state-by-state basis corresponds to an assumption that our policy class is expressive enough so that it can take the optimal distribution of actions at each state⁷ and abstracts from the generalization problem of estimating the population’s preference between a pair of alternatives at the given state x based on preferences in similar states x_0 . Having set aside the dependency between states, we focus on how the regularization with respect to a reference policy impacts the ability to optimize the average utility of three alignment methods: RLHF, DPO, and NLHF. In addition to the pairwise comparisons, these methods take in a reference policy $\pi_{\text{ref}} \in \Delta(A)$ and a KL bound $\tau \geq 0$, and map these inputs to a policy π in the KL-ball $B_\tau(\pi_{\text{ref}}) := \{\pi \in \Delta(A) \mid \text{DKL}(\pi \mid \pi_{\text{ref}}) \leq \tau\}$ around the reference policy. RLHF, DPO, and NLHF are typically implemented with a KL regularization rather than our KL-constrained formulation, which we adopt to enable a comparison on equal terms. In Section 5, we show that these perspectives are equivalent, and that distortion upper bounds carry over to regularized alignment methods.

[Next verbatim source excerpt]

If $\tau = 0$, the only feasible policy is π_{ref} , and the claim clearly holds for $\lambda = \infty$. We also assume that $\pi_{\text{ref}}(a) > 0$ for all $a \in M$. Otherwise if $\pi_{\text{ref}}(a) = 0$ for some $a \in M$, then both the regularized and constrained versions forbid any policy to put nonzero probability on a , which leads to an effectively smaller candidate set.

Lean-expanded library semantic target

```
type:
{α : Type u_1} → PMF α → Prop

value:
fun {α : Type u_1} (law : PMF α) => ∀ (outcome : α), 0 < (law outcome).toReal
```

Exact Lean library declaration

```
-- Every atom of a finite probability mass function has strictly positive real mass. -/
def PMFFullSupport {α : Type*} (law : PMF α) : Prop :=
  ∀ outcome, 0 < (law outcome).toReal
```

Recorded source-to-library screening Verdict: matches

Reason: This is the source clarification’s exact positive-mass condition for every candidate under the reference policy.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

AppliedModelingLib.pmfProd

Source locator: cited publication, lines 180-200

Verbatim paper-source connection

Both voting rules and alignment methods observe comparisons from n users. We model each $i = 1, \dots, n$ as a fresh user with independently drawn utility vector $u_i \sim D$. For exposition, we assume that each user i provides an equal number $d \geq 1$ of pairwise comparisons. For each i , and for $j = 1, \dots, d$, we independently draw alternatives x_{ji} , y_{ji} from a fixed distribution μ over

3

Our assumption that utilities be in $[0, 1]$ is weaker than any of the three assumptions — unit-sum, unit-range, or approval [23] — made in classic distortion to avoid trivial infinite lower bounds.

3

[form-feed in source extraction]

alternatives, in which the minimum probability mass $\mu_{\min} := \min_{x \in A} \mu(x)$ is positive.⁴ User i then compares each pair $\{x_{ji}, y_{ji}\}$ (for $j = 1, \dots, d$) through a Bradley-Terry model based on i 's utilities: i prefers x_{ji} over y_{ji} (written “ x_{ji} [U+001F control character] y_{ji} ”) with probability $\sigma \beta \cdot (u_i(x_{ji}) - u_i(y_{ji}))$, where [U+0001 control character]

$\sigma(t) := 1/(1+e^{-t})$ is the logistic sigmoid function and $\beta > 0$ is a temperature parameter, and prefers y_{ji} over x_{ji} (“ y_{ji} [U+001F control character] x_{ji} ”) otherwise.⁵ Whereas this specifies the marginal probability of each pairwise comparison, we make no assumption about the correlation between i 's choices. For example, i might derive the pairwise comparisons from a Plackett-Luce ranking, ensuring that [U+0002 control character] the user's comparisons [U+0001 control character] [U+0003 control character] are always consistent.⁶ We set $p(x$ [U+001F control character] $y)$ for the expected win rate $E_{u \sim D} \sigma \beta \cdot (u(x) - u(y))$.

Lean-expanded library semantic target

type:
{ α : Type u_1} → { β : Type u_2} → [Fintype α] → [Fintype β] → PMF α → PMF β → PMF ($\alpha \times \beta$)

value:
fun { α : Type u_1} { β : Type u_2} [Fintype α] [Fintype β] (μ : PMF α) (ν : PMF β) =>
PMF.ofFintype (fun (p : $\alpha \times \beta$) => $\mu.p.1 * \nu.p.2$) (AppliedModelingLib.pmfProd._proof_1 μ ν)

Exact Lean library declaration

```
/-- Independent product PMF on `α × β`. -/
noncomputable def pmfProd {α β : Type*} [Fintype α] [Fintype β]
  (μ : PMF α) (ν : PMF β) : PMF (α × β) :=
  PMF.ofFintype (fun p : α × β => μ.p.1 * ν.p.2) (by
    classical
    have hμ : ∑ a : α, μ a = 1 := by
      rw [← PMF.tsum_coe μ, tsum_fintype]
    have hν : ∑ b : β, ν b = 1 := by
      rw [← PMF.tsum_coe ν, tsum_fintype]
    calc
      ∑ p : α × β, μ p.1 * ν p.2
      = ∑ a : α, ∑ b : β, μ a * ν b := by
        rw [Fintype.sum_prod_type]
      = ∑ a : α, μ a * (∑ b : β, ν b) := by
        simp [Finset.mul_sum]
      = 1 := by
        simp [hμ, hν])
```

Recorded source-to-library screening Verdict: matches

Reason: This exactly defines the KL ball around the reference policy. Nonnegative radius and full-support reference remain explicit surrounding source-model assumptions.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Paper-specific semantic prerequisites

GolzHaghtalabYang2025Distortion.IsEmpiricalMaximalLottery

Paper-local declaration:

GolzHaghtalabYang2025Distortion.IsEmpiricalMaximalLottery

Source locator: cited publication, lines 230-236

Verbatim paper-source connection

the winner uniformly among all alternatives with maximum Borda score. The Maximal Lotteries voting rule first computes the margin matrix $M \in \mathbb{R}^{m \times m}$, where $M_{x,y} = \#(x[U+001F \text{ control character}]y) - \#(y[U+001F \text{ control character}]x)$. It then considers a symmetric two-player zero-sum game in which player 1 selects alternative x_1 , player 2 selects alternative x_2 , and the payoffs are M_{x_1, x_2} for player 1 and $M_{x_2, x_1} = -M_{x_1, x_2}$ for player 2. The maximal lotteries rule returns a distribution $\pi \in \arg\max_{\pi_1 \in \Delta(A)} \min_{\pi_2 \in \Delta(A)} \mathbb{E}_{x_1 \sim \pi_1, x_2 \sim \pi_2} [M_{x_1, x_2}]$, i.e., a mixed strategy in Nash equilibrium. When several such π exist, our results hold for any choice.

Lean-expanded paper semantic target

```
type:
{Alternative : Type u_1} →
[FIntype Alternative] → [DecidableEq Alternative] → (Alternative → Alternative → ℝ) → PMF Alternative → Prop

value:
fun {Alternative : Type u_1} [FIntype Alternative] [DecidableEq Alternative]
(empiricalRate : Alternative → Alternative → ℝ) (lottery : PMF Alternative) =>
  ∀ (opponent : PMF Alternative),
    0 ≤
      AppliedModelingLib.pmfPairExp lottery opponent fun (first second : Alternative) =>
        empiricalRate first second - empiricalRate second first
```

Direct retained dependencies

- [AppliedModelingLib.pmfPairExp](#)

Recorded source-to-declaration screening Verdict: matches

Reason: This is the exact finite zero-sum empirical-equilibrium condition against every opponent. Its unobserved-pair rule is separately disclosed.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

GolzHaghtalabYang2025Distortion.IsSourceAlignmentBenchmark

Paper-local declaration:

GolzHaghtalabYang2025Distortion.IsSourceAlignmentBenchmark

Source locator: cited publication, lines 281-285

Verbatim paper-source connection

To generalize the definition of distortion to alignment methods, we set the maximum average utility of any policy in the KL-ball as the benchmark. For fixed m, D, β, d, μ and correlation between pairwise comparisons, the distortion of alignment method f is

$$\text{dist}(f) = \sup_{D, \pi_{\text{ref}}, \tau} \limsup_{n \rightarrow \infty} \frac{\max_{\pi \in B_{\tau}(\pi_{\text{ref}})} \text{AvgUtil}(\pi)}{\text{AvgUtil}_n(f(\cdot, \pi_{\text{ref}}, \tau))}.$$

Lean-expanded paper semantic target

type:

```
{Alternative : Type u_1} →  
[Fintype Alternative] → [DecidableEq Alternative] → (Alternative → ℝ) → PMF Alternative → PMF Alternative → ℝ → Prop
```

value:

```
fun {Alternative : Type u_1} [Fintype Alternative] [DecidableEq Alternative] (averageUtility : Alternative → ℝ)  
  (reference benchmark : PMF Alternative) (klBudget : ℝ) =>  
  AppliedModelingLib.Alignment.Welfare.InFiniteKLBall reference klBudget benchmark ^  
    (policy : PMF Alternative),  
    AppliedModelingLib.Alignment.Welfare.InFiniteKLBall reference klBudget policy →  
    AppliedModelingLib.pmfExp policy averageUtility ≤ AppliedModelingLib.pmfExp benchmark averageUtility
```

Direct retained dependencies

- [AppliedModelingLib.Alignment.Welfare.InFiniteKLBall](#)
- [AppliedModelingLib.pmfExp](#)

Recorded source-to-declaration screening Verdict: matches

Reason: This selects a welfare-maximizing policy in the stated KL ball.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Paper-local declaration:

GolzHaghtalabYang2025Distortion.sourceMaximalLotteryMargin

Source locator: cited publication, lines 230-236

Verbatim paper-source connection

the winner uniformly among all alternatives with maximum Borda score. The Maximal Lotteries voting rule first computes the margin matrix $M \in \mathbb{R}^{m \times m}$, where $M_{x,y} = \#(x[U+001F \text{ control character}]y) - \#(y[U+001F \text{ control character}]x)$. It then considers a symmetric two-player zero-sum game in which player 1 selects alternative x_1 , player 2 selects alternative x_2 , and the payoffs are M_{x_1,x_2} for player 1 and $M_{x_2,x_1} = -M_{x_1,x_2}$ for player 2. The maximal lotteries rule returns a distribution $\pi \in \arg\max_{\pi_1 \in \Delta(A)} \min_{\pi_2 \in \Delta(A)} \mathbb{E}_{x_1 \sim \pi_1, x_2 \sim \pi_2} [M_{x_1,x_2}]$, i.e., a mixed strategy in Nash equilibrium. When several such π exist, our results hold for any choice.

Lean-expanded paper semantic target

```
type:
{Alternative : Type u_1} →
(wins : Alternative → Alternative → ℕ) → (first second : Alternative) → wins first second + wins second first ≠ 0 → ℝ

value:
fun {Alternative : Type u_1} (wins : Alternative → Alternative → ℕ) (first second : Alternative)
  (hdenominator : wins first second + wins second first ≠ 0) =>
  (↑(wins first second) - ↑(wins second first)) / (↑(wins first second) + ↑(wins second first))
```

Recorded source-to-declaration screening Verdict: matches

Reason: This is the literal normalized win-loss margin, explicitly restricted to its source nonzero-denominator domain.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

GolzHaghtalabYang2025Distortion.sourceMaximalLotteryValue

Paper-local declaration:

GolzHaghtalabYang2025Distortion.sourceMaximalLotteryValue

Source locator: cited publication, lines 230-236

Verbatim paper-source connection

the winner uniformly among all alternatives with maximum Borda score. The Maximal Lotteries voting rule first computes the margin matrix $M \in \mathbb{R}^{m \times m}$, where $M_{x,y} = \#(x[U+001F \text{ control character}]y) - \#(y[U+001F \text{ control character}]x)$. It then considers a symmetric two-player zero-sum game in which player 1 selects alternative x_1 , player 2 selects alternative x_2 , and the payoffs are M_{x_1, x_2} for player 1 and $M_{x_2, x_1} = -M_{x_1, x_2}$ for player 2. The maximal lotteries rule returns a distribution $\pi \in \arg\max_{\pi_1 \in \Delta(A)} \min_{\pi_2 \in \Delta(A)} \mathbb{E}_{x_1 \sim \pi_1, x_2 \sim \pi_2} [M_{x_1, x_2}]$, i.e., a mixed strategy in Nash equilibrium. When several such π exist, our results hold for any choice.

Lean-expanded paper semantic target

```
type:
{Alternative : Type u_1} →
[Fintype Alternative] →
[DecidableEq Alternative] →
(wins : Alternative → Alternative → ℕ) →
(∀ (first second : Alternative), first ≠ second → wins first second + wins second first ≠ 0) →
PMF Alternative → ℝ

value:
fun {Alternative : Type u_1} [Fintype Alternative] [DecidableEq Alternative] (wins : Alternative → Alternative → ℕ)
(hdenominator : ∀ (first second : Alternative), first ≠ second → wins first second + wins second first ≠ 0)
(candidate : PMF Alternative) =>
sinf
(Set.range fun (opponent : PMF Alternative) =>
AppliedModelingLib.pmfExp candidate fun (first : Alternative) =>
AppliedModelingLib.pmfExp opponent fun (second : Alternative) =>
if h : first = second then 0
else
GolzHaghtalabYang2025Distortion.sourceMaximalLotteryMargin wins first second (hdenominator first second h))
```

Direct retained dependencies

- [AppliedModelingLib.pmfExp](#)
- [GolzHaghtalabYang2025Distortion.sourceMaximalLotteryMargin](#)

Recorded source-to-declaration screening Verdict: matches

Reason: This is the minimum expected source margin over opponent lotteries, with diagonal zero and the same off-diagonal domain.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

GolzHaghtalabYang2025Distortion.IsSourceMaximalLottery

Paper-local declaration:

GolzHaghtalabYang2025Distortion.IsSourceMaximalLottery

Source locator: cited publication, lines 230-236

Verbatim paper-source connection

the winner uniformly among all alternatives with maximum Borda score. The Maximal Lotteries voting rule first computes the margin matrix $M \in \mathbb{R}^{m \times m}$, where $M_{x,y} = \#(x[U+001F \text{ control character}]y) - \#(y[U+001F \text{ control character}]x)$. It then considers a symmetric two-player zero-sum game in which player 1 selects alternative x_1 , player 2 selects alternative x_2 , and the payoffs are M_{x_1, x_2} for player 1 and $M_{x_2, x_1} = -M_{x_1, x_2}$ for player 2. The maximal lotteries rule returns a distribution $\pi \in \arg\max_{\pi_1 \in \Delta(A)} \min_{\pi_2 \in \Delta(A)} \mathbb{E}_{x_1 \sim \pi_1, x_2 \sim \pi_2} [M_{x_1, x_2}]$, i.e., a mixed strategy in Nash equilibrium. When several such π exist, our results hold for any choice.

Lean-expanded paper semantic target

```
type:
{Alternative : Type u_1} →
[Fintype Alternative] →
[DecidableEq Alternative] →
(wins : Alternative → Alternative → ℕ) →
(∀ (first second : Alternative), first ≠ second → wins first second + wins second first ≠ 0) →
PMF Alternative → Prop

value:
fun {Alternative : Type u_1} [Fintype Alternative] [DecidableEq Alternative] (wins : Alternative → Alternative → ℕ)
(hdenominator : ∀ (first second : Alternative), first ≠ second → wins first second + wins second first ≠ 0)
(lottery : PMF Alternative) =>
∀ (candidate : PMF Alternative),
GolzHaghtalabYang2025Distortion.sourceMaximalLotteryValue wins hdenominator candidate ≤
GolzHaghtalabYang2025Distortion.sourceMaximalLotteryValue wins hdenominator lottery
```

Direct retained dependencies

- [GolzHaghtalabYang2025Distortion.sourceMaximalLotteryValue](#)

Recorded source-to-declaration screening Verdict: matches

Reason: This is the source maximin condition on its raw nonzero-denominator margin domain.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Paper-local declaration:

GolzHaghtalabYang2025Distortion.IsSourceSocialChoiceOptimalAverageUtility

Source locator: cited publication, lines 202-215

Verbatim paper-source connection

Social Choice Setting. A voting rule f observes the sampled pairwise comparisons $\{x_{ji} \mid i \in [n], j \in [d]\}$ and maps them to a probability distribution over alternatives. For some m, D, β, d, μ , and ϵ , the correlation between comparisons, the average utility of f for n samples is $\text{AvgUtil}_n(f) := \mathbb{E} \text{AvgUtil } f(\{x_{ji} \mid i \in [n], j \in [d]\})$, where the expectation is taken over the pairwise comparisons. The distortion of f on D is the competitive ratio between $\text{AvgUtil}_n(f)$ and the optimal average utility $\max_{x \in A} \text{AvgUtil}(x)$ in the limit of $n \rightarrow \infty$ samples, and the distortion of f the worst-case distortion over all D :

$$\text{dist}(f, D) := \limsup_{n \rightarrow \infty} \frac{\max_{x \in A} \text{AvgUtil}(x)}{\text{AvgUtil}_n(f)}, \quad \text{dist}(f) = \sup_D \text{dist}(f, D).$$

Lean-expanded paper semantic target

```
type:
{Alternative : Type u_1} → [Fintype Alternative] → (Alternative → ℝ) → ℝ → Prop

value:
fun {Alternative : Type u_1} [Fintype Alternative] (averageUtility : Alternative → ℝ) (optimalAverageUtility : ℝ) =>
(∃ (alternative : Alternative), averageUtility alternative = optimalAverageUtility) ∧
∀ (alternative : Alternative), averageUtility alternative ≤ optimalAverageUtility
```

Recorded source-to-declaration screening Verdict: matches

Reason: This captures attainment and dominance of the finite optimal social-choice welfare benchmark.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Paper-local declaration:

GolzHaghtalabYang2025Distortion.corollary4lidUserEmpiricalMaximalLottery

Source locator: cited publication, lines 230-236

Verbatim paper-source connection

the winner uniformly among all alternatives with maximum Borda score. The Maximal Lotteries voting rule first computes the margin matrix $M \in \mathbb{R}^{m \times m}$, where $M_{x,y} = \#(x[U+001F \text{ control character}]y) - \#(y[U+001F \text{ control character}]x)$. It then considers a symmetric two-player zero-sum game in which player 1 selects alternative x_1 , player 2 selects alternative x_2 , and the payoffs are M_{x_1, x_2} for player 1 and $M_{x_2, x_1} = -M_{x_1, x_2}$ for player 2. The maximal lotteries rule returns a distribution $\pi \in \arg\max_{\pi_1 \in \Delta(A)} \min_{\pi_2 \in \Delta(A)} \mathbb{E}_{x_1 \sim \pi_1, x_2 \sim \pi_2} [M_{x_1, x_2}]$, i.e., a mixed strategy in Nash equilibrium. When several such π exist, our results hold for any choice.

Lean-expanded paper semantic target

```
type:
{Alternative : Type u_1} →
[Fintype Alternative] →
[DecidableEq Alternative] →
[Nonempty Alternative] →
{users comparisonsPerUser : ℕ} →
(Fin users →
  GolzHaghtalabYang2025Distortion.theorem2UserResponseTable Alternative ×
  (Fin comparisonsPerUser → Alternative × Alternative)) →
PMF Alternative

value:
fun {Alternative : Type u_1} [Fintype Alternative] [DecidableEq Alternative] [Nonempty Alternative]
  {users comparisonsPerUser : ℕ}
  (sample :
    Fin users →
      GolzHaghtalabYang2025Distortion.theorem2UserResponseTable Alternative ×
      (Fin comparisonsPerUser → Alternative × Alternative)) =>
GolzHaghtalabYang2025Distortion.theorem7lidUserEmpiricalPolicy (AppliedModelingLib.uniformPMF Alternative)
(Real.log ↑ (Fintype.card Alternative))
(GolzHaghtalabYang2025Distortion.corollary4_uniformFullSimplexBudget_nonneg Alternative) sample
```

Direct retained dependencies

- [AppliedModelingLib.uniformPMF](#)
- [GolzHaghtalabYang2025Distortion.theorem2UserResponseTable](#)
- [GolzHaghtalabYang2025Distortion.theorem7lidUserEmpiricalPolicy](#)

Recorded source-to-declaration screening Verdict: matches

Reason: The uniform reference with radius $\log |A|$ contains the simplex; observed pairs recover the paper ratios and unobserved off-diagonal pairs have the explicitly disclosed neutral 0/0-to-zero margin.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

GolzHaghtalabYang2025Distortion.empiricalBordaDefinitionSpec

Paper-local declaration:

GolzHaghtalabYang2025Distortion.empiricalBordaDefinitionSpec

Source locator: cited publication, lines 220-230

Verbatim paper-source connection

For alternatives x, y , let $\#(x \text{ [U+001F control character] } y) := \{(i, j) \mid x_{ji} = x, y_{ij} = y\}$ denote the number of pairwise comparisons in which x beat y . The (normalized) Borda score [45] of alternative x is

$$BC(x) := \frac{1}{P} \sum_{y \in A} \#(x \text{ [U+001F control character] } y) - \frac{1}{P} \sum_{y \in A} \#(y \text{ [U+001F control character] } x)$$

i.e., the fraction of pairwise comparisons involving x in which it wins. The Borda voting rule chooses the winner uniformly among all alternatives with maximum Borda score. The Maximal Lotteries vot-

Lean-expanded paper semantic target

```
type:
{Report : Type u_1} →
{Alternative : Type u_2} →
[Fintype Alternative] →
(Alternative → Report → ℝ) →
(Alternative → Report → ℝ) → ((horizon : ℕ) → (Fin horizon → Report) → PMF Alternative) → Prop

value:
fun {Report : Type u_1} {Alternative : Type u_2} [Fintype Alternative] (wins incidences : Alternative → Report → ℝ)
  (rule : (horizon : ℕ) → (Fin horizon → Report) → PMF Alternative) =>
  ∀ (horizon : ℕ) (sample : Fin horizon → Report),
  have score : Alternative → ℝ := fun (alternative : Alternative) =>
    if AppliedModelingLib.Probability.finitelidScoreSum (incidences alternative) sample = 0 then 0
    else
      AppliedModelingLib.Probability.finitelidScoreSum (wins alternative) sample /
        AppliedModelingLib.Probability.finitelidScoreSum (incidences alternative) sample;
  (∀ (alternative : Alternative),
    0 < ((rule horizon sample) alternative).toReal ↔ ∀ (other : Alternative), score other ≤ score alternative) ∧
  (∀ (first second : Alternative),
    (∀ (other : Alternative), score other ≤ score first) →
    (∀ (other : Alternative), score other ≤ score second) →
    ((rule horizon sample) first).toReal = ((rule horizon sample) second).toReal)
```

Direct retained dependencies

- [AppliedModelingLib.Probability.finitelidScoreSum](#)

Recorded source-to-declaration screening Verdict: matches

Reason: This retains the source positive-incidence quotient and uniform maximizer selection. Score zero is limited to the otherwise undefined zero-incidence evaluator branch and is not a model premise.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Paper-local declaration:

GolzHaghtalabYang2025Distortion.nlhfDefinitionSpec

Source locator: cited publication, lines 277-280

Verbatim paper-source connection

The alignment method NLHF was inspired in part by a desire to better align with the preferences of a heterogeneous group [36]. NLHF naturally adapts the definition of maximal lotteries by constraining both players' mixed strategies to the KL-ball, i.e., $\pi_{NLHF} := \arg \max_{\pi_1 \in B_{\tau}(\pi_{ref})} \min_{\pi_2 \in B_{\tau}(\pi_{ref})} \mathbb{E}_{x_1 \sim \pi_1, x_2 \sim \pi_2} [M_{x_1, x_2}]$.

[Next verbatim source excerpt]

If $\tau = 0$, the only feasible policy is π_{ref} , and the claim clearly holds for $\lambda = \infty$. We also assume that $\pi_{ref}(a) > 0$ for all $a \in M$. Otherwise if $\pi_{ref}(a) = 0$ for some $a \in M$, then both the regularized and constrained versions forbid any policy to put nonzero probability on a , which leads to an effectively smaller candidate set.

Lean-expanded paper semantic target

```
type:
{User Alternative : Type} →
[Fintype User] →
[DecidableEq User] →
[Fintype Alternative] →
[DecidableEq Alternative] →
PMF User →
AppliedModelingLib.Alignment.Welfare.FiniteUtilityProfile User Alternative →
ℝ → PMF Alternative → ℝ → PMF Alternative → Prop

value:
fun {User Alternative : Type} [Fintype User] [DecidableEq User] [Fintype Alternative] [DecidableEq Alternative]
(population : PMF User) (utility : AppliedModelingLib.Alignment.Welfare.FiniteUtilityProfile User Alternative)
(btScale : ℝ) (reference : PMF Alternative) (klBudget : ℝ) (policy : PMF Alternative) =>
AppliedModelingLib.PMFFullSupport reference ∧
AppliedModelingLib.finiteKLDivergence policy reference ≤ klBudget ∧
∃ (worst : PMF Alternative),
AppliedModelingLib.finiteKLDivergence worst reference ≤ klBudget ∧
(∀ (opponent : PMF Alternative),
AppliedModelingLib.finiteKLDivergence opponent reference ≤ klBudget →
AppliedModelingLib.GameTheory.PreferenceGame.contextFreeCenteredPreferencePayoff
(AppliedModelingLib.Alignment.Welfare.populationBradleyTerryPreference population utility btScale)
policy worst ≤
AppliedModelingLib.GameTheory.PreferenceGame.contextFreeCenteredPreferencePayoff
(AppliedModelingLib.Alignment.Welfare.populationBradleyTerryPreference population utility btScale)
policy opponent) ∧
∀ (candidate : PMF Alternative),
AppliedModelingLib.finiteKLDivergence candidate reference ≤ klBudget →
AppliedModelingLib.GameTheory.PreferenceGame.contextFreeCenteredPreferencePayoff
(AppliedModelingLib.Alignment.Welfare.populationBradleyTerryPreference population utility btScale)
candidate worst ≤
AppliedModelingLib.GameTheory.PreferenceGame.contextFreeCenteredPreferencePayoff
(AppliedModelingLib.Alignment.Welfare.populationBradleyTerryPreference population utility btScale)
policy worst
```

Direct retained dependencies

- [AppliedModelingLib.Alignment.Welfare.FiniteUtilityProfile](#)
- [AppliedModelingLib.Alignment.Welfare.populationBradleyTerryPreference](#)
- [AppliedModelingLib.GameTheory.PreferenceGame.contextFreeCenteredPreferencePayoff](#)
- [AppliedModelingLib.PMFFullSupport](#)
- [AppliedModelingLib.finiteKLDivergence](#)

Recorded source-to-declaration screening Verdict: matches

Reason: Both players use the same KL ball and centered population Bradley-Terry payoff; reference-support scope is covered by the source-supported positive-support/reduced-candidate-set clarification.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Paper-local declaration:

GolzHaghtalabYang2025Distortion.populationBordaDefinitionSpec

Source locator: cited publication, lines 372-375

Verbatim paper-source connection

for $n \rightarrow \infty$, assuming
 Proof sketch (full proof in Appendix E.1). For exposition, we sketch this proofP
 that each alternative's Borda count has converged to its expectation $BC?(x) := \sum_{y \in A} \mu(y) \cdot p(x \succ y)$.
 Let $\hat{x} = \operatorname{argmax}_{x \in A} BC?(x)$ be the Borda winner in the limit, and $x^* = \operatorname{argmax}_{x \in A} \operatorname{AvgUtil}(x)$ be

Lean-expanded paper semantic target

```
type:
{Alternative : Type u_1} →
[Fintype Alternative] →
[DecidableEq Alternative] →
PMF Alternative →
AppliedModelingLib.Learning.HumanFeedback.PairwisePreference PUnit.{1} Alternative → Alternative → Prop

value:
fun {Alternative : Type u_1} [Fintype Alternative] [DecidableEq Alternative] (sampling : PMF Alternative)
  (preference : AppliedModelingLib.Learning.HumanFeedback.PairwisePreference PUnit.{1} Alternative)
  (winner : Alternative) =>
  ∀ (alternative : Alternative),
  (AppliedModelingLib.pmfExp sampling fun (opponent : Alternative) =>
    preference.prob PUnit.unit alternative opponent) ≤
  AppliedModelingLib.pmfExp sampling fun (opponent : Alternative) => preference.prob PUnit.unit winner opponent
```

Direct retained dependencies

- [AppliedModelingLib.Learning.HumanFeedback.PairwisePreference](#)
- [AppliedModelingLib.pmfExp](#)

Recorded source-to-declaration screening Verdict: matches

Reason: This uses the source sampling-weighted population pairwise score and a maximizing alternative.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Paper-local declaration:

GolzHaghtalabYang2025Distortion.rlhfDefinitionSpec

Source locator: cited publication, lines 269-276

Verbatim paper-source connection

The RLHF method first estimates rewards for each alternative, using maximum likelihood estimation assuming that comparisons were generated by a single Bradley-Terry model:

$$r := \underset{P}{\operatorname{argmax}} \sum_{1 \leq i \leq n, 1 \leq j \leq d} \log \frac{\sigma(r(x_{ji}) - r(y_{ij}))}{[U + 0001 \text{ control character}]} .8$$

Next, RLHF uses PPO [44] to compute the policy $\pi_{\text{RLHF}} := \operatorname{argmax}_{\pi \in \mathcal{B}\tau} (\pi_{\text{ref}}) \mathbb{E}_{x \sim \pi} [r(x)]$ with maximum expected reward within the KL-ball. In our setup, DPO is equivalent to RLHF (see Appendix F.3) and thus has the same distortion.

[Next verbatim source excerpt]

If $\tau = 0$, the only feasible policy is π_{ref} , and the claim clearly holds for $\lambda = \infty$. We also assume that $\pi_{\text{ref}}(a) > 0$ for all $a \in M$. Otherwise if $\pi_{\text{ref}}(a) = 0$ for some $a \in M$, then both the regularized and constrained versions forbid any policy to put nonzero probability on a , which leads to an effectively smaller candidate set.

Lean-expanded paper semantic target

```
type:
{Alternative : Type u_1} →
[Fintype Alternative] →
[DecidableEq Alternative] →
{horizon : ℕ} →
(Fin horizon → AppliedModelingLib.Learning.HumanFeedback.BinaryPairwiseReport Alternative) →
Alternative →
PMF Alternative →
PMF Alternative →
ℝ → AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset.ScoreVector Alternative → Prop

value:
fun {Alternative : Type u_1} [Fintype Alternative] [DecidableEq Alternative] {horizon : ℕ}
(sample : Fin horizon → AppliedModelingLib.Learning.HumanFeedback.BinaryPairwiseReport Alternative)
(scoreReference : Alternative) (referencePolicy : PMF Alternative) (klBudget : ℝ)
(score : AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset.ScoreVector Alternative) =>
AppliedModelingLib.PMFFullSupport reference ∧
(score scoreReference = 0 ∧
  IsMaxOn
    ((AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset.ofBinaryReports sample).pairwiseLogLikelihood
     Real.sigmoid)
    {candidate : AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset.ScoreVector Alternative |
     candidate scoreReference = 0}
    score) ∧
AppliedModelingLib.Learning.HumanFeedback.contextAveragedPolicyKLDivergence (PMF.pure PUnit.unit)
  (AppliedModelingLib.Alignment.Welfare.contextFreeAlternativePolicy policy)
  (AppliedModelingLib.Alignment.Welfare.contextFreeAlternativePolicy reference) ≤
klBudget ∧
∀ (other : AppliedModelingLib.Learning.HumanFeedback.FinitePolicy PUnit.{1} Alternative),
AppliedModelingLib.Learning.HumanFeedback.contextAveragedPolicyKLDivergence (PMF.pure PUnit.unit) other
  (AppliedModelingLib.Alignment.Welfare.contextFreeAlternativePolicy reference) ≤
klBudget →
(AppliedModelingLib.Learning.HumanFeedback.policyExpectedScore (PMF.pure PUnit.unit) other
 fun (x : PUnit.{1}) => score) ≤
AppliedModelingLib.Learning.HumanFeedback.policyExpectedScore (PMF.pure PUnit.unit)
  (AppliedModelingLib.Alignment.Welfare.contextFreeAlternativePolicy policy) fun (x : PUnit.{1}) => score
```

Direct retained dependencies

- [AppliedModelingLib.Alignment.Welfare.contextFreeAlternativePolicy](#)
- [AppliedModelingLib.Learning.HumanFeedback.BinaryPairwiseReport](#)
- [AppliedModelingLib.Learning.HumanFeedback.FinitePolicy](#)
- [AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset.ScoreVector](#)
- [AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset.ofBinaryReports](#)
- [AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset.pairwiseLogLikelihood](#)
- [AppliedModelingLib.Learning.HumanFeedback.contextAveragedPolicyKLDivergence](#)
- [AppliedModelingLib.Learning.HumanFeedback.policyExpectedScore](#)
- [AppliedModelingLib.PMFFullSupport](#)

Recorded source-to-declaration screening **Verdict:** matches

Reason: This faithfully specifies finite logistic MLE up to harmless reference-score normalization, followed by reward maximization in the same KL ball.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

GolzHaghtalabYang2025Distortion.sourceAlignmentAverageUtility

Paper-local declaration:

GolzHaghtalabYang2025Distortion.sourceAlignmentAverageUtility

Source locator: cited publication, lines 281-285

Verbatim paper-source connection

To generalize the definition of distortion to alignment methods, we set the maximum average utility of any policy in the KL-ball as the benchmark. For fixed m, D, β, d, μ and correlation between pairwise comparisons, the distortion of alignment method f is

$$\text{dist}(f) = \sup_{D, \pi_{\text{ref}}, \tau} \limsup_{n \rightarrow \infty} \max_{\pi \in B_{\tau}(\pi_{\text{ref}})} \text{AvgUtil}(\pi) - \text{AvgUtil}_n(f(\cdot, \pi_{\text{ref}}, \tau)).$$

Lean-expanded paper semantic target

```
type:
{Alternative : Type u_1} →
{Observation : Type u_2} →
[Fintype Alternative] →
[Fintype Observation] →
(Alternative → ℝ) →
((horizon : ℕ) → PMF (Fin horizon → Observation)) →
((horizon : ℕ) → (Fin horizon → Observation) → PMF Alternative → ℝ → PMF Alternative) →
PMF Alternative → ℝ → ℕ → ℝ

value:
fun {Alternative : Type u_1} {Observation : Type u_2} [Fintype Alternative] [Fintype Observation]
  (averageUtility : Alternative → ℝ) (observationLaw : (horizon : ℕ) → PMF (Fin horizon → Observation))
  (method : (horizon : ℕ) → (Fin horizon → Observation) → PMF Alternative → ℝ → PMF Alternative)
  (reference : PMF Alternative) (klBudget : ℝ) (horizon : ℕ) =>
AppliedModelingLib.pmfExp (observationLaw horizon) fun (observations : Fin horizon → Observation) =>
AppliedModelingLib.pmfExp (method horizon observations reference klBudget) averageUtility
```

Direct retained dependencies

- [AppliedModelingLib.pmfExp](#)

Recorded source-to-declaration screening Verdict: matches

Reason: This is expected policy welfare over the full report law, parameterized by reference and KL budget.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

GolzHaghtalabYang2025Distortion.sourceAlignmentDistortion

Paper-local declaration:

GolzHaghtalabYang2025Distortion.sourceAlignmentDistortion

Source locator: cited publication, lines 281-285

Verbatim paper-source connection

To generalize the definition of distortion to alignment methods, we set the maximum average utility of any policy in the KL-ball as the benchmark. For fixed m, D, β, d, μ and correlation between pairwise comparisons, the distortion of alignment method f is

$$\text{dist}(f) = \sup_{D, \pi_{\text{ref}}, \tau} \limsup_{n \rightarrow \infty} \frac{\max_{\pi \in B_{\tau}(\pi_{\text{ref}})} \text{AvgUtil}(\pi)}{\text{AvgUtil}_n(f(\cdot, \pi_{\text{ref}}, \tau))}.$$

Lean-expanded paper semantic target

```
type:
{Alternative : Type u_1} →
{Observation : Type u_2} →
[Fintype Alternative] →
[DecidableEq Alternative] →
[Fintype Observation] →
(Alternative → ℝ) →
ℝ →
((horizon : ℕ) → PMF (Fin horizon → Observation)) →
((horizon : ℕ) → (Fin horizon → Observation) → PMF Alternative → ℝ → PMF Alternative) →
PMF Alternative → ℝ → ℝ

value:
fun {Alternative : Type u_1} {Observation : Type u_2} [Fintype Alternative] [DecidableEq Alternative]
[Fintype Observation] (averageUtility : Alternative → ℝ) (benchmarkUtility : ℝ)
(observationLaw : (horizon : ℕ) → PMF (Fin horizon → Observation))
(method : (horizon : ℕ) → (Fin horizon → Observation) → PMF Alternative → ℝ → PMF Alternative)
(reference : PMF Alternative) (klBudget : ℝ) =>
Filter.limsup
(fun (horizon : ℕ) =>
benchmarkUtility /
GolzHaghtalabYang2025Distortion.sourceAlignmentAverageUtility averageUtility observationLaw method reference
klBudget horizon)
Filter.atTop
```

Direct retained dependencies

- [GolzHaghtalabYang2025Distortion.sourceAlignmentAverageUtility](#)

Recorded source-to-declaration screening Verdict: matches

Reason: This is the fixed-instance limsup of the KL-ball welfare benchmark over method welfare.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

GolzHaghtalabYang2025Distortion.sourceSocialChoiceAverageUtility

Paper-local declaration:

GolzHaghtalabYang2025Distortion.sourceSocialChoiceAverageUtility

Source locator: cited publication, lines 202-215

Verbatim paper-source connection

Social Choice Setting. A voting rule f observes the sampled pairwise comparisons $\{x_{ji} \mid i \in [n], j \in [d]\}$ and maps them to a probability distribution over alternatives. For some m, D, β, d, μ , and $\mathbb{E} \text{AvgUtil}(f) = \mathbb{E} \text{AvgUtil}(f(\{x_{ji} \mid i \in [n], j \in [d]\}))$, where the expectation is taken over the pairwise comparisons. The distortion of f on D is the competitive ratio between $\text{AvgUtil}(f)$ and the optimal average utility $\max_{x \in A} \text{AvgUtil}(x)$ in the limit of $n \rightarrow \infty$ samples, and the distortion of f the worst-case distortion over all D :

$$\text{dist}(f, D) := \limsup_{n \rightarrow \infty} \max_{x \in A} \frac{\text{AvgUtil}(f)}{\text{AvgUtil}(x)}, \quad \text{dist}(f) = \sup_D \text{dist}(f, D).$$

Lean-expanded paper semantic target

```
type:
{Alternative : Type u_1} →
{Observation : Type u_2} →
[Fintype Alternative] →
[Fintype Observation] →
(Alternative → ℝ) →
((horizon : ℕ) → PMF (Fin horizon → Observation)) →
((horizon : ℕ) → (Fin horizon → Observation) → PMF Alternative) → ℕ → ℝ

value:
fun {Alternative : Type u_1} {Observation : Type u_2} [Fintype Alternative] [Fintype Observation]
  (averageUtility : Alternative → ℝ) (observationLaw : (horizon : ℕ) → PMF (Fin horizon → Observation))
  (rule : (horizon : ℕ) → (Fin horizon → Observation) → PMF Alternative) (horizon : ℕ) =>
  AppliedModelingLib.pmfExp (observationLaw horizon) fun (observations : Fin horizon → Observation) =>
  AppliedModelingLib.pmfExp (rule horizon observations) averageUtility
```

Direct retained dependencies

- [AppliedModelingLib.pmfExp](#)

Recorded source-to-declaration screening Verdict: matches

Reason: This is the exact expected social-choice welfare definition and preserves the source's lack of a false within-user comparison-independence assumption.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

GolzHaghtalabYang2025Distortion.sourceSocialChoiceDistortion

Paper-local declaration:

GolzHaghtalabYang2025Distortion.sourceSocialChoiceDistortion

Source locator: cited publication, lines 202-215

Verbatim paper-source connection

Social Choice Setting. A voting rule f observes the sampled pairwise comparisons $\{x_{ji} \mid i \in [n], j \in [d]\}$ and maps them to a probability distribution over alternatives. For some m, D, β, d, μ , and ϵ , the correlation between comparisons, the average utility of f for n samples is $\text{AvgUtil}_n(f) := \mathbb{E} \text{AvgUtil } f(\{x_{ji} \mid i \in [n], j \in [d]\})$, where the expectation is taken over the pairwise comparisons. The distortion of f on D is the competitive ratio between $\text{AvgUtil}_n(f)$ and the optimal average utility $\max_{x \in A} \text{AvgUtil}(x)$ in the limit of $n \rightarrow \infty$ samples, and the distortion of f the worst-case distortion over all D :

$$\text{dist}(f, D) := \limsup_{n \rightarrow \infty} \max_{x \in A} \frac{\text{AvgUtil}_n(f)}{\text{AvgUtil}(x)}, \quad \text{dist}(f) = \sup_D \text{dist}(f, D).$$

Lean-expanded paper semantic target

```
type:
{Alternative : Type u_1} →
{Observation : Type u_2} →
[Fintype Alternative] →
[Fintype Observation] →
(Alternative → ℝ) →
ℝ →
((horizon : ℕ) → PMF (Fin horizon → Observation)) →
((horizon : ℕ) → (Fin horizon → Observation) → PMF Alternative) → ℝ

value:
fun {Alternative : Type u_1} {Observation : Type u_2} [Fintype Alternative] [Fintype Observation]
  (averageUtility : Alternative → ℝ) (optimalAverageUtility : ℝ)
  (observationLaw : (horizon : ℕ) → PMF (Fin horizon → Observation))
  (rule : (horizon : ℕ) → (Fin horizon → Observation) → PMF Alternative) =>
  Filter.limsup
    (fun (horizon : ℕ) =>
      optimalAverageUtility /
      GolzHaghtalabYang2025Distortion.sourceSocialChoiceAverageUtility averageUtility observationLaw rule horizon)
  Filter.atTop
```

Direct retained dependencies

- [GolzHaghtalabYang2025Distortion.sourceSocialChoiceAverageUtility](#)

Recorded source-to-declaration screening Verdict: matches

Reason: This is the fixed-instance limsup of optimal alternative welfare over expected rule welfare.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

GolzHaghtalabYang2025Distortion.sourceWorstCaseAlignmentDistortion

Paper-local declaration:

GolzHaghtalabYang2025Distortion.sourceWorstCaseAlignmentDistortion

Source locator: cited publication, lines 281-285

Verbatim paper-source connection

To generalize the definition of distortion to alignment methods, we set the maximum average utility of any policy in the KL-ball as the benchmark. For fixed m , D , β , d , μ and correlation between pairwise comparisons, the distortion of alignment method f is

$$\text{dist}(f) = \sup_{D, \pi_{\text{ref}}, \tau} \limsup_{n \rightarrow \infty} \frac{\max_{\pi \in B_{\tau}(\pi_{\text{ref}})} \text{AvgUtil}(\pi)}{\text{AvgUtil}_n(f(\cdot, \pi_{\text{ref}}, \tau))}.$$

Lean-expanded paper semantic target

type:

{Instance : Type u_1} → (Instance → ℝ) → ℝ

value:

fun {Instance : Type u_1} (distortion : Instance → ℝ) => sSup (Set.range distortion)

Recorded source-to-declaration screening Verdict: matches

Reason: This takes the supremum over the source alignment-instance scope, including utility distribution, reference, and KL budget.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Paper-local declaration:
GolzhaghtalabYang2025Distortion.sourceWorstCaseSocialChoiceDistortion
Source locator: cited publication, lines 202-215

Verbatim paper-source connection

Social Choice Setting. A voting rule f observes the sampled pairwise comparisons $\{x_{ji} \mid i \in [n], j \in [d]\}$ and maps them to a probability distribution over alternatives. For some m, D, β, d, μ , and ϵ , the correlation between comparisons, the average utility of f for n samples is $\text{AvgUtil}_n(f) := \mathbb{E} \text{AvgUtil}_n(f(\{x_{ji} \mid i \in [n], j \in [d]\}))$, where the expectation is taken over the pairwise comparisons. The distortion of f on D is the competitive ratio between $\text{AvgUtil}_n(f)$ and the optimal average utility $\max_{x \in A} \text{AvgUtil}_n(x)$ in the limit of $n \rightarrow \infty$ samples, and the distortion of f the worst-case distortion over all D :

$$\text{dist}(f, D) := \limsup_{n \rightarrow \infty} \frac{\max_{x \in A} \text{AvgUtil}_n(x)}{\text{AvgUtil}_n(f)}, \quad \text{dist}(f) = \sup_D \text{dist}(f, D).$$

Lean-expanded paper semantic target

```
type:
{Instance : Type u_1} → (Instance → ℝ) → ℝ

value:
fun {Instance : Type u_1} (distortion : Instance → ℝ) => sSup (Set.range distortion)
```

Recorded source-to-declaration screening Verdict: matches
Reason: This takes the supremum over the source social-choice instance scope.
Reviewer check: ☐ Matches source input
If left unchecked, explain the discrepancy or uncertainty below.
Reviewer annotation

Source claims

Review row 1

Source locator: cited publication, lines 338-344

Verbatim source input

Lemma 1 (Linearization of Expected Win-Rates). Let $L := \sigma(0) = 1/4$ and $\beta := \frac{1}{2\beta \cdot 1 + e - \beta}$. For any pair of alternatives $x, y \in A$, we have

$$\beta \cdot (\beta \cdot \text{AvgUtil}(x) - L \cdot \text{AvgUtil}(y)) \leq p(x \text{ [U+001F control character] } y) - L \leq \beta \cdot (L \cdot \text{AvgUtil}(x) - \beta \cdot \text{AvgUtil}(y)).$$

3.1 Upper Bound on Borda Distortion

[Next verbatim source excerpt]

Let $A = \{1, \dots, m\}$ be a finite set of alternatives. The population of users is described by a probability distribution D over utility vectors $u = (u(1), \dots, u(m))$, whose entries $0 \leq u(x) \leq 1$ indicate a user's utility for alternative x .³ The objective is to find an alternative x such that its average utility $\text{AvgUtil}(x) := \mathbb{E}_{u \sim D} [u(x)]$ across the user population (also known as the utilitarian social welfare) is as high as possible. We extend this notation to probability distributions π over alternatives by setting $\text{AvgUtil}(\pi) := \mathbb{E}_{x \sim \pi} [\text{AvgUtil}(x)]$. Both voting rules and alignment methods observe comparisons from n users. We model each $i = 1, \dots, n$ as a fresh user with independently drawn utility vector $u_i \sim D$. For exposition, we assume that each user i provides an equal number $d \geq 1$ of pairwise comparisons. For each i , and for $j = 1, \dots, d$, we independently draw alternatives x_{ji}, y_{ji} from a fixed distribution μ over

³ Our assumption that utilities be in $[0, 1]$ is weaker than any of the three assumptions — unit-sum, unit-range, or approval [23] — made in classic distortion to avoid trivial infinite lower bounds.

3

[form-feed in source extraction]

alternatives, in which the minimum probability mass $\mu_{\min} := \min_{x \in A} \mu(x)$ is positive.⁴ User i then compares each pair $\{x_{ji}, y_{ji}\}$ (for $j = 1, \dots, d$) through a Bradley-Terry model based on i 's utilities: i prefers x_{ji} over y_{ji} (written " x_{ji} [U+001F control character] y_{ji} ") with probability $\sigma(\beta \cdot (u_i(x_{ji}) - u_i(y_{ji})))$, where [U+0001 control character]

$\sigma(t) := 1/(1+e^{-t})$ is the logistic sigmoid function and $\beta > 0$ is a temperature parameter, and prefers y_{ji} over x_{ji} (" y_{ji} [U+001F control character] x_{ji} ") otherwise.⁵ Whereas this specifies the marginal probability of each pairwise comparison, we make no assumption about the correlation between i 's choices. For example, i might derive the pairwise comparisons from a Plackett-Luce ranking, ensuring that [U+0002 control character] the user's comparisons [U+0001 control character] [U+0003 control character] are always consistent.⁶ We set $p(x \text{ [U+001F control character] } y)$ for the expected win rate $\mathbb{E}_{u \sim D} \sigma(\beta \cdot (u(x) - u(y)))$.

Expanded PaperInterface specification

```
∀ {User Alternative : Type} [inst : Fintype User] [inst_1 : DecidableEq User] (population : PMF User)
  (utility : AppliedModelingLib.Alignment.Welfare.FiniteUtilityProfile User Alternative),
  AppliedModelingLib.Alignment.Welfare.UnitIntervalUtilityProfile utility →
  ∀ {btScale : ℝ},
  0 < btScale →
  ∀ (first second : Alternative),
  btScale *
    (AppliedModelingLib.Alignment.Welfare.sigmoidChordSlope btScale *
      AppliedModelingLib.Alignment.Welfare.populationAverageUtility population utility first -
      1 / 4 * AppliedModelingLib.Alignment.Welfare.populationAverageUtility population utility second) ≤
    (AppliedModelingLib.Alignment.Welfare.populationBradleyTerryPreference population utility btScale).prob
    PUnit.unit first second -
    1 / 2 ∧
  (AppliedModelingLib.Alignment.Welfare.populationBradleyTerryPreference population utility btScale).prob
    PUnit.unit first second -
    1 / 2 ≤
    btScale *
      (1 / 4 * AppliedModelingLib.Alignment.Welfare.populationAverageUtility population utility first -
        AppliedModelingLib.Alignment.Welfare.sigmoidChordSlope btScale *
        AppliedModelingLib.Alignment.Welfare.populationAverageUtility population utility second)
```

Retained governing declarations

- [AppliedModelingLib.Alignment.Welfare.FiniteUtilityProfile](#)
- [AppliedModelingLib.Alignment.Welfare.UnitIntervalUtilityProfile](#)
- [AppliedModelingLib.Alignment.Welfare.populationAverageUtility](#)
- [AppliedModelingLib.Alignment.Welfare.populationBradleyTerryPreference](#)
- [AppliedModelingLib.Alignment.Welfare.sigmoidChordSlope](#)
- [AppliedModelingLib.Learning.HumanFeedback.PairwisePreference](#)

Lean-elaborated claim roles

```
1. parameter: Type
2. parameter: Type
3. parameter: Fintype User
4. parameter: DecidableEq User
5. parameter: PMF User
6. parameter: AppliedModelingLib.Alignment.Welfare.FiniteUtilityProfile User Alternative
7. assumption: AppliedModelingLib.Alignment.Welfare.UnitIntervalUtilityProfile utility
8. parameter: ℝ
9. assumption: 0 < btScale
10. parameter: Alternative
11. parameter: Alternative
12. conclusion: btScale *
    (AppliedModelingLib.Alignment.Welfare.sigmoidChordSlope btScale *
      AppliedModelingLib.Alignment.Welfare.populationAverageUtility population utility first -
      1 / 4 * AppliedModelingLib.Alignment.Welfare.populationAverageUtility population utility second) ≤
    (AppliedModelingLib.Alignment.Welfare.populationBradleyTerryPreference population utility btScale).prob PUnit.unit
    first second -
    1 / 2 ≤
    (AppliedModelingLib.Alignment.Welfare.populationBradleyTerryPreference population utility btScale).prob PUnit.unit
    first second -
    1 / 2 ≤
    btScale *
    (1 / 4 * AppliedModelingLib.Alignment.Welfare.populationAverageUtility population utility first -
      AppliedModelingLib.Alignment.Welfare.sigmoidChordSlope btScale *
      AppliedModelingLib.Alignment.Welfare.populationAverageUtility population utility second)
```

Lean proof endpoint (built separately)

GolzHaghtalabYang2025Distortion.lemma1_linearization

Recorded source-to-Spec screening Verdict: matches

Reason: This is the source two-sided Bradley-Terry linearization using $L = 1/4$ and the source chord coefficient.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 2

Source locator: cited publication, lines 1358-1370

Verbatim source input

Lemma 10. For any instance D with any number of alternatives m , any distribution μ over alternatives with $\mu_{\min} = \min_{x \in A} \mu(x)$, and n i.i.d. users sampled from D where each user labels d comparison pairs following the Bradley-Terry[U+0011 control character] model with temperature β , we have that with probability

at least $1 - \delta$ where $\delta \geq m \exp - 8$, the empirical win rates $p_n(x [U+001F control character] y) := \#(x[U+001F control character]y) + \#(y[U+001F control character]x)$ satisfies that:

$$\forall x, y \in A, |p_n(x [U+001F control character] y) - p(x [U+001F control character] y)| \leq O\left(\frac{\log(m/\delta)}{n \cdot \min\{1, d \cdot \mu_{2\min}\}} + \frac{\log(m/\delta)}{n \mu_{2\min}}\right).$$

[Next verbatim source excerpt]

Let $A = \{1, \dots, m\}$ be a finite set of alternatives. The population of users is described by a probability distribution D over utility vectors $u = (u(1), \dots, u(m))$, whose entries $0 \leq u(x) \leq 1$ indicate a user's utility for alternative x .³ The objective is to find an alternative x such that its average utility $\text{AvgUtil}(x) := \mathbb{E}_{u \sim D} [u(x)]$ across the user population (also known as the utilitarian social welfare) is as high as possible. We extend this notation to probability distributions π over alternatives by setting $\text{AvgUtil}(\pi) := \mathbb{E}_{x \sim \pi} [\text{AvgUtil}(x)]$.

Both voting rules and alignment methods observe comparisons from n users. We model each $i = 1, \dots, n$ as a fresh user with independently drawn utility vector $u_i \sim D$. For exposition, we assume that each user i provides an equal number $d \geq 1$ of pairwise comparisons. For each i , and for $j = 1, \dots, d$, we independently draw alternatives x_{ji}, y_{ji} from a fixed distribution μ over

³ Our assumption that utilities be in $[0, 1]$ is weaker than any of the three assumptions — unit-sum, unit-range, or approval [23] — made in classic distortion to avoid trivial infinite lower bounds.

3

[form-feed in source extraction]

alternatives, in which the minimum probability mass $\mu_{\min} := \min_{x \in A} \mu(x)$ is positive.⁴ User i then compares each pair $\{x_{ji}, y_{ji}\}$ (for $j = 1, \dots, d$) through a Bradley-Terry model based on i 's utilities: i prefers x_{ji} over y_{ji} (written " x_{ji} [U+001F control character] y_{ji} ") with probability $\sigma \beta \cdot (u_i(x_{ji}) - u_i(y_{ji}))$, where [U+0001 control character]

$\sigma(t) := 1/(1+e^{-t})$ is the logistic sigmoid function and $\beta > 0$ is a temperature parameter, and prefers y_{ji} over x_{ji} (" y_{ji} [U+001F control character] x_{ji} ") otherwise.⁵ Whereas this specifies the marginal probability of each pairwise comparison, we make no assumption about the correlation between i 's choices. For example, i might derive the pairwise comparisons from a Plackett-Luce ranking, ensuring that [U+0002 control character] the user's comparisons [U+0001 control character] [U+0003 control character] are always consistent.⁶ We set $p(x [U+001F control character] y)$ for the expected win rate $\mathbb{E}_{u \sim D} \sigma \beta \cdot (u(x) - u(y))$.

Expanded PaperInterface specification

```

∀ {Alternative : Type} [inst : Fintype Alternative] [inst_1 : DecidableEq Alternative] [Nonempty Alternative]
{responseLaw : PMF (GolzHaghtalabYang2025Distortion.theorem2UserResponseTable Alternative)}
{preference : AppliedModelingLib.Learning.HumanFeedback.PairwisePreference PUnit.{1} Alternative},
GolzHaghtalabYang2025Distortion.theorem2UserResponseCalibrated responseLaw preference →
  ∀ (sampling : PMF Alternative) (users comparisonsPerUser : ℕ) (minimumMass delta : ℝ),
    0 < users →
      0 < comparisonsPerUser →
        0 < minimumMass →
          (∀ (alternative : Alternative), minimumMass ≤ (sampling alternative).toReal) →
            0 < delta →
              delta ≤ 1 →
                ↑(Fintype.card Alternative) ^ 2 * Real.exp (-↑(users * comparisonsPerUser) * minimumMass ^ 2 / 8) ≤
                  delta →
                    have failureBudget : ℝ := delta / ↑(Fintype.card Alternative) ^ 2;
                    (AppliedModelingLib.pmfProb
                      (AppliedModelingLib.pmfProduct (Fin users)
                        (GolzHaghtalabYang2025Distortion.theorem2UserResponseTable Alternative ×
                          (Fin comparisonsPerUser → Alternative × Alternative))
                        (AppliedModelingLib.pmfProd responseLaw
                          (AppliedModelingLib.pmfProduct (Fin comparisonsPerUser) (Alternative × Alternative)
                            (GolzHaghtalabYang2025Distortion.theorem2UserPairLabelLaw sampling))))))
                    fun
                      (sample :
                        Fin users →
                          GolzHaghtalabYang2025Distortion.theorem2UserResponseTable Alternative ×
                            (Fin comparisonsPerUser → Alternative × Alternative)) =>
                        ∃ (pair : Alternative × Alternative),
                          GolzHaghtalabYang2025Distortion.theorem10SourceRate users comparisonsPerUser minimumMass
                            failureBudget <
                              |GolzHaghtalabYang2025Distortion.theorem10lidUserPaperWinRate pair.1 pair.2 sample -
                                preference.prob PUnit.unit pair.1 pair.2| ≤
                                delta

```

Retained governing declarations

- [AppliedModelingLib.Learning.HumanFeedback.PairwisePreference](#)

- [AppliedModelingLib.pmfProb](#)
- [AppliedModelingLib.pmfProd](#)
- [AppliedModelingLib.pmfProduct](#)
- [GolzhaghtalabYang2025Distortion.theorem10IidUserPaperWinRate](#)
- [GolzhaghtalabYang2025Distortion.theorem10SourceRate](#)
- [GolzhaghtalabYang2025Distortion.theorem2UserPairLabelLaw](#)
- [GolzhaghtalabYang2025Distortion.theorem2UserResponseCalibrated](#)
- [GolzhaghtalabYang2025Distortion.theorem2UserResponseTable](#)

Lean-elaborated claim roles

```

1. parameter: Type
2. parameter: Fintype Alternative
3. parameter: DecidableEq Alternative
4. assumption: Nonempty Alternative
5. parameter: PMF (GolzhaghtalabYang2025Distortion.theorem2UserResponseTable Alternative)
6. parameter: AppliedModelingLib.Learning.HumanFeedback.PairwisePreference PUnit.{1} Alternative
7. assumption: GolzhaghtalabYang2025Distortion.theorem2UserResponseCalibrated responseLaw preference
8. parameter: PMF Alternative
9. parameter: ℕ
10. parameter: ℕ
11. parameter: ℝ
12. parameter: ℝ
13. assumption: 0 < users
14. assumption: 0 < comparisonsPerUser
15. assumption: 0 < minimumMass
16. assumption: ∀ (alternative : Alternative), minimumMass ≤ (sampling alternative).toReal
17. assumption: 0 < delta
18. assumption: delta ≤ 1
19. assumption: ↑(Fintype.card Alternative) ^ 2 * Real.exp (- ↑(users * comparisonsPerUser) * minimumMass ^ 2 / 8) ≤ delta
20. conclusion: (AppliedModelingLib.pmfProb
  (AppliedModelingLib.pmfProduct (Fin users)
    (GolzhaghtalabYang2025Distortion.theorem2UserResponseTable Alternative ×
      (Fin comparisonsPerUser → Alternative × Alternative))
    (AppliedModelingLib.pmfProd responseLaw
      (AppliedModelingLib.pmfProduct (Fin comparisonsPerUser) (Alternative × Alternative)
        (GolzhaghtalabYang2025Distortion.theorem2UserPairLabelLaw sampling))))))
fun
  (sample :
    Fin users →
      GolzhaghtalabYang2025Distortion.theorem2UserResponseTable Alternative ×
        (Fin comparisonsPerUser → Alternative × Alternative)) =>
  ∃ (pair : Alternative × Alternative),
    GolzhaghtalabYang2025Distortion.theorem10SourceRate users comparisonsPerUser minimumMass
      (delta / ↑(Fintype.card Alternative) ^ 2) <
      |GolzhaghtalabYang2025Distortion.theorem10IidUserPaperWinRate pair.1 pair.2 sample -
        preference.prob PUnit.unit pair.1 pair.2|) ≤
delta

```

Lean proof endpoint (built separately)

GolzhaghtalabYang2025Distortion.lemma10_empirical_win_rate_concentration

Recorded source-to-Spec screening Verdict: matches

Reason: This is the source uniform empirical pairwise-rate concentration statement under iid users, positive sampling mass, and its stated exponential confidence regime.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 3

Source locator: cited publication, lines 1464-1478

Verbatim source input

Lemma 11. For any instance D with any number of alternatives m , any distribution μ over alternatives with $\mu_{\min} = \min_{x \in A} \mu(x)$, and n i.i.d. users sampled from D where each user labels d comparison pairs, the normalized Borda score $BC_n(x)$ of any alternative $x \in A$ satisfies that with probability at least $1 - \delta$ where $\delta \geq 2m \exp(-\frac{n \mu_{\min}}{8 \log(m/\delta)})$,

$$\forall x \in A, |BC_n(x) - BC^*(x)| \leq \frac{\log(m/\delta)}{n \cdot \min\{1, d \cdot \mu_{\min}\}} + \frac{m \log(m/\delta)}{n \mu_{\min}}, \quad (5)$$

where $BC^*(x)$ is the limiting normalized Borda score of candidate x , defined as

$$BC^*(x) := \sum_{y \in A} \mu(y) \cdot p(x \text{ [U+001F control character] } y) = \mu(x) + \sum_{y \neq x} \mu(y) \cdot p(x \text{ [U+001F control character] } y). \quad (6)$$

[Next verbatim source excerpt]

Let $A = \{1, \dots, m\}$ be a finite set of alternatives. The population of users is described by a probability distribution D over utility vectors $u = (u(1), \dots, u(m))$, whose entries $0 \leq u(x) \leq 1$ indicate a user's utility for alternative x .³ The objective is to find an alternative x such that its average utility $\text{AvgUtil}(x) := \mathbb{E}_{u \sim D} [u(x)]$ across the user population (also known as the utilitarian social welfare) is as high as possible. We extend this notation to probability distributions π over alternatives by setting $\text{AvgUtil}(\pi) := \mathbb{E}_{x \sim \pi} [\text{AvgUtil}(x)]$. Both voting rules and alignment methods observe comparisons from n users. We model each $i = 1, \dots, n$ as a fresh user with independently drawn utility vector $u_i \sim D$. For exposition, we assume that each user i provides an equal number $d \geq 1$ of pairwise comparisons. For each i , and for $j = 1, \dots, d$, we independently draw alternatives x_{ji}, y_{ji} from a fixed distribution μ over

³ Our assumption that utilities be in $[0, 1]$ is weaker than any of the three assumptions — unit-sum, unit-range, or approval [23] — made in classic distortion to avoid trivial infinite lower bounds.

3

[form-feed in source extraction]

alternatives, in which the minimum probability mass $\mu_{\min} := \min_{x \in A} \mu(x)$ is positive.⁴ User i then compares each pair $\{x_{ji}, y_{ji}\}$ (for $j = 1, \dots, d$) through a Bradley-Terry model based on i 's utilities: i prefers x_{ji} over y_{ji} (written “ x_{ji} [U+001F control character] y_{ji} ”) with probability $\sigma(\beta \cdot (u_i(x_{ji}) - u_i(y_{ji})))$, where

$\sigma(t) := 1/(1+e^{-t})$ is the logistic sigmoid function and $\beta > 0$ is a temperature parameter, and prefers y_{ji} over x_{ji} (“ y_{ji} [U+001F control character] x_{ji} ”) otherwise.⁵ Whereas this specifies the marginal probability of each pairwise comparison, we make no assumption about the correlation between i 's choices. For example, i might derive the pairwise comparisons from a Plackett-Luce ranking, ensuring that [U+0002 control character] the user's comparisons [U+0001 control character] [U+0003 control character] are always consistent.⁶ We set $p(x \text{ [U+001F control character] } y)$ for the expected win rate $\mathbb{E}_{u \sim D} \sigma(\beta \cdot (u(x) - u(y)))$.

[Next verbatim source excerpt]

Social Choice Setting. A voting rule f observes the sampled pairwise comparisons $\{x_{ji} \text{ [U+001F control character] } y_{ji} \mid i \in [n], j \in [d]\}$ and maps them to a probability distribution over alternatives. For some m, D, β, d, μ , and [U+0002 control character] the correlation between comparisons, the average utility of f for n samples is $\text{AvgUtil}_n(f) := \mathbb{E} \text{AvgUtil}(f(\{x_{ji} \text{ [U+001F control character] } y_{ji} \}_{i,j}))$, where the expectation is taken over the pairwise comparisons. The [U+0001 control character] [U+0003 control character]

distortion of f on D is the competitive ratio between $\text{AvgUtil}_n(f)$ and the optimal average utility $\max_{x \in A} \text{AvgUtil}(x)$ in the limit of $n \rightarrow \infty$ samples, and the distortion of f the worst-case distortion over all D :

$$\text{dist}(f, D) := \limsup_{n \rightarrow \infty} \frac{\max_{x \in A} \text{AvgUtil}(x)}{\text{AvgUtil}_n(f)}, \quad \text{dist}(f) = \sup_D \text{dist}(f, D).$$

For alternatives x, y , let $\#(x \text{ [U+001F control character] } y) := \{(i, j) \mid x_{ji} = x, y_{ji} = y\}$ denote the number of pairwise comparisons in which x beat y . The (normalized) Borda score [45] of alternative x is

$$BC(x) := \frac{1}{P} \sum_{y \in A} \#(x \text{ [U+001F control character] } y) - \frac{1}{P} \sum_{y \in A} \#(y \text{ [U+001F control character] } x),$$

i.e., the fraction of pairwise comparisons involving x in which it wins. The Borda voting rule chooses the winner uniformly among all alternatives with maximum Borda score. The Maximal Lotteries voting rule first computes the margin matrix $M \in \mathbb{R}^{m \times m}$, where $M_{x,y} = \#(x \text{ [U+001F control character] } y) - \#(y \text{ [U+001F control character] } x)$. It then considers a symmetric two-player zero-sum game in which player 1 selects alternative x_1 , player 2 selects alternative x_2 , and the payoffs are M_{x_1, x_2} for player 1 and $M_{x_2, x_1} = -M_{x_1, x_2}$ for player 2. The maximal lotteries rule returns a distribution $\pi \in \Delta(A)$ such that $\pi_1, \pi_2 \in \Delta(A)$ and $\mathbb{E}_{x_1 \sim \pi_1, x_2 \sim \pi_2} [M_{x_1, x_2}] = 0$, i.e., a mixed strategy in Nash equilibrium. When several such π exist, our results hold for any choice.

Expanded PaperInterface specification

$\forall \{ \text{Alternative} : \text{Type} \} [\text{inst} : \text{Fintype Alternative}] [\text{inst}_1 : \text{DecidableEq Alternative}] [\text{Nonempty Alternative}]$

```

(responseLaw : PMF (GolzhaghtalabYang2025Distortion.theorem2UserResponseTable Alternative))
(sampling : PMF Alternative)
(preference : AppliedModelingLib.Learning.HumanFeedback.PairwisePreference PUnit.{1} Alternative)
(users comparisonsPerUser : ℕ),
0 < users →
0 < comparisonsPerUser →
GolzhaghtalabYang2025Distortion.theorem2UserResponseCalibrated responseLaw preference →
  ∀ (minimumMass delta : ℝ),
    0 < minimumMass →
      (∀ (alternative : Alternative), minimumMass ≤ (sampling alternative).toReal) →
        0 < delta →
          delta ≤ 1 →
            2 * ↑(Fintype.card Alternative) * Real.exp (-↑(users * comparisonsPerUser) * minimumMass / 8) ≤
              delta →
                (AppliedModelingLib.pmfProb
                  (AppliedModelingLib.pmfProduct (Fin users)
                    (GolzhaghtalabYang2025Distortion.theorem2UserResponseTable Alternative ×
                      (Fin comparisonsPerUser → Alternative × Alternative))
                    (AppliedModelingLib.pmfProd responseLaw
                      (AppliedModelingLib.pmfProduct (Fin comparisonsPerUser) (Alternative × Alternative)
                        (GolzhaghtalabYang2025Distortion.theorem2UserPairLabelLaw sampling))))))
                fun
                  (sample :
                    Fin users →
                      GolzhaghtalabYang2025Distortion.theorem2UserResponseTable Alternative ×
                        (Fin comparisonsPerUser → Alternative × Alternative)) =>
                    have logLevel : ℝ := Real.log (8 * ↑(Fintype.card Alternative) ^ 2 / delta);
                    ∃ (alternative : Alternative),
                      20 * √(↑users * logLevel / min 1 (↑comparisonsPerUser * minimumMass ^ 2)) / ↑users +
                        8 * (↑(Fintype.card Alternative) + 1) * logLevel / (↑users * minimumMass) <
                          |GolzhaghtalabYang2025Distortion.theorem11IidUserLatentBordaScore alternative sample -
                            AppliedModelingLib.Alignment.Welfare.pairwiseBordaScore sampling preference
                            alternative|) ≤
                  delta
                delta

```

Retained governing declarations

- [AppliedModelingLib.Alignment.Welfare.pairwiseBordaScore](#)
- [AppliedModelingLib.Learning.HumanFeedback.PairwisePreference](#)
- [AppliedModelingLib.pmfProb](#)
- [AppliedModelingLib.pmfProd](#)
- [AppliedModelingLib.pmfProduct](#)
- [GolzhaghtalabYang2025Distortion.theorem11IidUserLatentBordaScore](#)
- [GolzhaghtalabYang2025Distortion.theorem2UserPairLabelLaw](#)
- [GolzhaghtalabYang2025Distortion.theorem2UserResponseCalibrated](#)
- [GolzhaghtalabYang2025Distortion.theorem2UserResponseTable](#)

Lean-elaborated claim roles

```

1. parameter: Type
2. parameter: Fintype Alternative
3. parameter: DecidableEq Alternative
4. assumption: Nonempty Alternative
5. parameter: PMF (GolzhaghtalabYang2025Distortion.theorem2UserResponseTable Alternative)
6. parameter: PMF Alternative
7. parameter: AppliedModelingLib.Learning.HumanFeedback.PairwisePreference PUnit.{1} Alternative
8. parameter: ℕ
9. parameter: ℕ
10. assumption: 0 < users
11. assumption: 0 < comparisonsPerUser
12. assumption: GolzhaghtalabYang2025Distortion.theorem2UserResponseCalibrated responseLaw preference
13. parameter: ℝ
14. parameter: ℝ
15. assumption: 0 < minimumMass
16. assumption: ∀ (alternative : Alternative), minimumMass ≤ (sampling alternative).toReal
17. assumption: 0 < delta
18. assumption: delta ≤ 1
19. assumption: 2 * ↑(Fintype.card Alternative) * Real.exp (-↑(users * comparisonsPerUser) * minimumMass / 8) ≤ delta
20. conclusion: (AppliedModelingLib.pmfProb
  (AppliedModelingLib.pmfProduct (Fin users)
    (GolzhaghtalabYang2025Distortion.theorem2UserResponseTable Alternative ×
      (Fin comparisonsPerUser → Alternative × Alternative))
    (AppliedModelingLib.pmfProd responseLaw
      (AppliedModelingLib.pmfProduct (Fin comparisonsPerUser) (Alternative × Alternative)
        (GolzhaghtalabYang2025Distortion.theorem2UserPairLabelLaw sampling))))))
  fun
    (sample :
      Fin users →
        GolzhaghtalabYang2025Distortion.theorem2UserResponseTable Alternative ×
          (Fin comparisonsPerUser → Alternative × Alternative)) =>
      have logLevel : ℝ := Real.log (8 * ↑(Fintype.card Alternative) ^ 2 / delta);
      ∃ (alternative : Alternative),
        20 * √(↑users * logLevel / min 1 (↑comparisonsPerUser * minimumMass ^ 2)) / ↑users +
          8 * (↑(Fintype.card Alternative) + 1) * logLevel / (↑users * minimumMass) <
            |GolzhaghtalabYang2025Distortion.theorem11IidUserLatentBordaScore alternative sample -
              AppliedModelingLib.Alignment.Welfare.pairwiseBordaScore sampling preference
              alternative|) ≤
      delta
    delta

```

Lean proof endpoint (built separately)

GolzHaghtalabYang2025Distortion.lemma11_borda_concentration

Recorded source-to-Spec screening Verdict: matches

Reason: This retains the simultaneous normalized-Borda concentration claim with the stated rate dependence and confidence regime.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 4

Source locator: cited publication, lines 2312-2324

Verbatim source input

Lemma 15. For any $\beta > 0$, there is an infinite sequence $a_1, a_2, \dots$ of alternatives, and a distribution D of utility functions over these alternatives such that

- $\text{AvgUtil}(a_1) = 1/3$,
- for all $t \geq 2$, $0 < \text{AvgUtil}(a_t) \leq \text{AvgUtil}(a_{t-1}) - 32\beta \log 1 + \tanh \beta/4 \cdot \text{AvgUtil}(a_{t-1}) < \text{AvgUtil}(a_{t-1})$, and
- for all $t \geq 2$, $p(a_t | a_{t-1}) > 1/2$.

Proof. Our population D will be a uniform distribution over three utility vectors, u_1, u_2 , and u_3 . We define the sequence of alternatives and prove the claim by induction over $t \geq 1$.

[Next verbatim source excerpt]

Let $A = \{1, \dots, m\}$ be a finite set of alternatives. The population of users is described by a probability distribution D over utility vectors $u = (u(1), \dots, u(m))$, whose entries $0 \leq u(x) \leq 1$ indicate a user's utility for alternative x .³ The objective is to find an alternative x such that its average utility $\text{AvgUtil}(x) := \mathbb{E}_{u \sim D} [u(x)]$ across the user population (also known as the utilitarian social welfare) is as high as possible. We extend this notation to probability distributions π over alternatives by setting $\text{AvgUtil}(\pi) := \mathbb{E}_{x \sim \pi} [\text{AvgUtil}(x)]$. Both voting rules and alignment methods observe comparisons from n users. We model each $i = 1, \dots, n$ as a fresh user with independently drawn utility vector $u_i \sim D$. For exposition, we assume that each user i provides an equal number $d \geq 1$ of pairwise comparisons. For each i , and for $j = 1, \dots, d$, we independently draw alternatives x_{ji}, y_{ji} from a fixed distribution μ over

³ Our assumption that utilities be in $[0, 1]$ is weaker than any of the three assumptions — unit-sum, unit-range, or approval [23] — made in classic distortion to avoid trivial infinite lower bounds.

3

[form-feed in source extraction]
alternatives, in which the minimum probability mass $\mu_{\min} := \min_{x \in A} \mu(x)$ is positive.⁴ User i then compares each pair $\{x_{ji}, y_{ji}\}$ (for $j = 1, \dots, d$) through a Bradley-Terry model based on i 's utilities: i prefers x_{ji} over y_{ji} (written " $x_{ji} [U+001F \text{ control character}] y_{ji}$ ") with probability $\sigma \beta \cdot (u_i(x_{ji}) - u_i(y_{ji}))$, where

$\sigma(t) := 1/(1+e^{-t})$ is the logistic sigmoid function and $\beta > 0$ is a temperature parameter, and prefers y_{ji} over x_{ji} (" $y_{ji} [U+001F \text{ control character}] x_{ji}$ ") otherwise.⁵ Whereas this specifies the marginal probability of each pairwise comparison, we make no assumption about the correlation between i 's choices. For example, i might derive the pairwise comparisons from a Plackett-Luce ranking, ensuring that $[U+0002 \text{ control character}]$ the user's comparisons $[U+0001 \text{ control character}] [U+0003 \text{ control character}]$ are always consistent.⁶ We set $p(x [U+001F \text{ control character}] y)$ for the expected win rate $\mathbb{E}_{u \sim D} \sigma \beta \cdot (u(x) - u(y))$.

Expanded PaperInterface specification

```

∀ {beta : ℝ},
0 < beta →
  ∃ (population : PMF (Fin 3)) (utility : AppliedModelingLib.Alignment.Welfare.FiniteUtilityProfile (Fin 3) ℕ),
    AppliedModelingLib.Alignment.Welfare.UnitIntervalUtilityProfile utility ∧
    AppliedModelingLib.Alignment.Welfare.populationAverageUtility population utility 0 = 1 / 3 ∧
    ∀ (n : ℕ),
      0 < AppliedModelingLib.Alignment.Welfare.populationAverageUtility population utility (n + 1) ∧
      AppliedModelingLib.Alignment.Welfare.populationAverageUtility population utility (n + 1) ≤
        AppliedModelingLib.Alignment.Welfare.populationAverageUtility population utility n -
          2 / (3 * beta) *
            Real.log
              (1 +
                Real.tanh
                  (beta *
                    4) ^
                    3)
              )
    AppliedModelingLib.Alignment.Welfare.populationAverageUtility population utility (n + 1) <
      AppliedModelingLib.Alignment.Welfare.populationAverageUtility population utility n ∧
      1 / 2 <
        (AppliedModelingLib.Alignment.Welfare.populationBradleyTerryPreference population utility beta).prob
          PUnit.unit (n + 1) n

```

Retained governing declarations

- [AppliedModelingLib.Alignment.Welfare.FiniteUtilityProfile](#)
- [AppliedModelingLib.Alignment.Welfare.UnitIntervalUtilityProfile](#)
- [AppliedModelingLib.Alignment.Welfare.populationAverageUtility](#)
- [AppliedModelingLib.Alignment.Welfare.populationBradleyTerryPreference](#)
- [AppliedModelingLib.Learning.HumanFeedback.PairwisePreference](#)

Lean-elaborated claim roles

```

1. parameter: ℝ
2. assumption: 0 < beta
3. conclusion: ∃ (population : PMF (Fin 3)) (utility : AppliedModelingLib.Alignment.Welfare.FiniteUtilityProfile (Fin 3) ℕ),
  AppliedModelingLib.Alignment.Welfare.UnitIntervalUtilityProfile utility ∧
  AppliedModelingLib.Alignment.Welfare.populationAverageUtility population utility 0 = 1 / 3 ∧
  ∀ (n : ℕ),
    0 < AppliedModelingLib.Alignment.Welfare.populationAverageUtility population utility (n + 1) ∧
    AppliedModelingLib.Alignment.Welfare.populationAverageUtility population utility (n + 1) ≤
      AppliedModelingLib.Alignment.Welfare.populationAverageUtility population utility n -
        2 / (3 * beta) *
          Real.log
            (1 +
              Real.tanh
                (beta * AppliedModelingLib.Alignment.Welfare.populationAverageUtility population utility n /
                  4) ^
                3) ∧
    AppliedModelingLib.Alignment.Welfare.populationAverageUtility population utility (n + 1) <
      AppliedModelingLib.Alignment.Welfare.populationAverageUtility population utility n ∧
    1 / 2 <
      (AppliedModelingLib.Alignment.Welfare.populationBradleyTerryPreference population utility beta).prob
      PUnit.unit (n + 1) n

```

Lean proof endpoint (built separately)

GolzHaghtalabYang2025Distortion.lemma15_infinite_sequence

Recorded source-to-Spec screening Verdict: matches

Reason: The infinite sequence, initial welfare one-third, stated welfare decay, and adjacent win-rate strictly above one-half are all retained.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 5

Source locator: cited publication, lines 352-372

Verbatim source input

Theorem 2 (Borda Distortion Upper Bound). For any instance D with any number of alternatives m ,

distribution μ over alternatives, and temperature β , Borda has at most distortion $\beta^2 \cdot 1 - e^{-\frac{1+e-\beta^2}{2\beta}}$.

In the finite-sample regime, we have that

$$\text{AvgUtiln}(\text{Borda}) \geq \beta^2 \cdot 1 - e^{-\frac{1+e-\beta^2}{2\beta}} \cdot \max_{x \in A} \text{AvgUtil}(x) - O\left(\frac{1}{\sqrt{n}}\right)$$

for $n \rightarrow \infty$, assuming

[Next verbatim source excerpt]

Let $A = \{1, \dots, m\}$ be a finite set of alternatives. The population of users is described by a probability distribution D over utility vectors $u = (u(1), \dots, u(m))$, whose entries $0 \leq u(x) \leq 1$ indicate a user's utility for alternative x . The objective is to find an alternative x such that its average utility $\text{AvgUtil}(x) := \mathbb{E}_{u \sim D} [u(x)]$ across the user population (also known as the utilitarian social welfare) is as high as possible. We extend this notation to probability distributions π over alternatives by setting $\text{AvgUtil}(\pi) := \mathbb{E}_{x \sim \pi} [\text{AvgUtil}(x)]$. Both voting rules and alignment methods observe comparisons from n users. We model each $i = 1, \dots, n$ as a fresh user with independently drawn utility vector $u_i \sim D$. For exposition, we assume that each user i provides an equal number $d \geq 1$ of pairwise comparisons. For each i , and for $j = 1, \dots, d$, we independently draw alternatives x_{ji}, y_{ji} from a fixed distribution μ over

alternatives, in which the minimum probability mass $\mu_{\min} := \min_{x \in A} \mu(x)$ is positive. User i then compares each pair $\{x_{ji}, y_{ji}\}$ (for $j = 1, \dots, d$) through a Bradley-Terry model based on i 's utilities: i prefers x_{ji} over y_{ji} (written " $x_{ji} \succ y_{ji}$ ") with probability $\sigma(\beta \cdot (u_i(x_{ji}) - u_i(y_{ji})))$, where

3

[form-feed in source extraction]

$\sigma(t) := 1/(1+e^{-t})$ is the logistic sigmoid function and $\beta > 0$ is a temperature parameter, and prefers y_{ij} over x_{ji} (" $y_{ij} \succ x_{ji}$ ") otherwise. Whereas this specifies the marginal probability of each pairwise comparison, we make no assumption about the correlation between i 's choices. For example, i might derive the pairwise comparisons from a Plackett-Luce ranking, ensuring that the user's comparisons are always consistent. We set $p(x \succ y)$ for the expected win rate $\mathbb{E}_{u \sim D} [\sigma(\beta \cdot (u(x) - u(y)))]$.

Our assumption that utilities be in $[0, 1]$ is weaker than any of the three assumptions — unit-sum, unit-range, or approval [23] — made in classic distortion to avoid trivial infinite lower bounds.

[Next verbatim source excerpt]

Social Choice Setting. A voting rule f observes the sampled pairwise comparisons $\{x_{ji} \succ y_{ji}\}_{i \in [n], j \in [d]}$ and maps them to a probability distribution over alternatives. For some m, D, β, d, μ , and

the correlation between comparisons, the average utility of f for n samples is $\text{AvgUtiln}(f) := \mathbb{E} \text{AvgUtil}(f(\{x_{ji} \succ y_{ji}\}_{i \in [n], j \in [d]}))$, where the expectation is taken over the pairwise comparisons. The

distortion of f on D is the competitive ratio between $\text{AvgUtiln}(f)$ and the optimal average utility $\max_{x \in A} \text{AvgUtil}(x)$ in the limit of $n \rightarrow \infty$ samples, and the distortion of f the worst-case distortion over all D :

$$\text{dist}(f, D) := \limsup_{n \rightarrow \infty} \frac{\text{AvgUtiln}(f)}{\max_{x \in A} \text{AvgUtil}(x)}, \quad \text{dist}(f) = \sup_D \text{dist}(f, D).$$

For alternatives x, y , let $\#(x \succ y) := \#\{(i, j) \mid x_{ji} = x, y_{ji} = y\}$ denote the number of pairwise comparisons in which x beat y . The (normalized) Borda score [45] of alternative x is

$$\text{BC}(x) := \frac{1}{P} \sum_{y \in A} \#(x \succ y) - \frac{1}{P} \sum_{y \in A} \#(y \succ x)$$

i.e., the fraction of pairwise comparisons involving x in which it wins. The Borda voting rule chooses the winner uniformly among all alternatives with maximum Borda score. The Maximal Lotteries voting rule first computes the margin matrix $M \in \mathbb{R}^{m \times m}$, where $M_{x,y} = \#(x \succ y) - \#(y \succ x)$. It then considers a symmetric two-player zero-sum game in which player 1 selects alternative x_1 , player 2 selects alternative x_2 , and the payoffs are M_{x_1, x_2} for player 1 and $-M_{x_1, x_2}$ for player 2. The maximal lotteries rule returns a distribution $\pi \in \Delta(A)$ such that $\pi_1, \pi_2 \in \Delta(A)$ and $\pi_1 \succsim \pi_2$ for M , i.e., a mixed strategy in Nash equilibrium. When several such π exist, our results hold for any choice.

Expanded PaperInterface specification

```

(∀ {User Alternative : Type} [inst : Fintype User] [inst_1 : DecidableEq User] [inst_2 : Fintype Alternative]
[inst_3 : DecidableEq Alternative] (population : PMF User)
(utility : AppliedModelingLib.Alignment.Welfare.FiniteUtilityProfile User Alternative),
AppliedModelingLib.Alignment.Welfare.UnitIntervalUtilityProfile utility →
  ∀ (sampling : PMF Alternative) {btScale : ℝ},
  0 < btScale →
  ∀ (winner : Alternative),
  (∀ (alternative : Alternative),
    AppliedModelingLib.Alignment.Welfare.pairwiseBordaScore sampling
    (AppliedModelingLib.Alignment.Welfare.populationBradleyTerryPreference population utility btScale)
    alternative ≤
    AppliedModelingLib.Alignment.Welfare.pairwiseBordaScore sampling
    (AppliedModelingLib.Alignment.Welfare.populationBradleyTerryPreference population utility btScale)
    winner) →
  ∀ (alternative : Alternative),
  (AppliedModelingLib.Alignment.Welfare.sigmoidChordSlope btScale / (1 / 4)) ^ 2 *
  AppliedModelingLib.Alignment.Welfare.populationAverageUtility population utility alternative ≤
  AppliedModelingLib.Alignment.Welfare.populationAverageUtility population utility winner) ∧
  ∀ {User Alternative : Type} [inst : Fintype User] [inst_1 : DecidableEq User] [inst_2 : Fintype Alternative]
[inst_3 : DecidableEq Alternative] [Nonempty Alternative] (population : PMF User)
(utility : AppliedModelingLib.Alignment.Welfare.FiniteUtilityProfile User Alternative),
AppliedModelingLib.Alignment.Welfare.UnitIntervalUtilityProfile utility →
  ∀ (sampling : PMF Alternative) {btScale : ℝ},
  0 < btScale →
  ∀ (responseLaw : PMF (GolzhaghtalabYang2025Distortion.theorem2UserResponseTable Alternative)),
  GolzhaghtalabYang2025Distortion.theorem2UserResponseCalibrated responseLaw
  (AppliedModelingLib.Alignment.Welfare.populationBradleyTerryPreference population utility btScale) →
  ∀ (users comparisonsPerUser : ℕ),
  0 < users →
  0 < comparisonsPerUser →
  ∀ (minimumMass delta : ℝ),
  0 < minimumMass →
  (∀ (alternative : Alternative), minimumMass ≤ (sampling alternative).toReal) →
  0 < delta →
  delta ≤ 1 →
  2 * ↑(Fintype.card Alternative) *
  Real.exp (-↑(users * comparisonsPerUser) * minimumMass / 8) ≤
  delta →
  ∀
  (winner :
    (Fin users →
      GolzhaghtalabYang2025Distortion.theorem2UserResponseTable Alternative ×
      (Fin comparisonsPerUser → Alternative × Alternative)) →
    Alternative),
  (∀
    (sample :
      Fin users →
      GolzhaghtalabYang2025Distortion.theorem2UserResponseTable Alternative ×
      (Fin comparisonsPerUser → Alternative × Alternative))
    (alternative : Alternative),
    GolzhaghtalabYang2025Distortion.theorem11lidUserLatentBordaScore alternative
    sample ≤
    GolzhaghtalabYang2025Distortion.theorem11lidUserLatentBordaScore (winner sample)
    sample) →
  ∀ (benchmark : Alternative),
  (1 - delta) *
  ((AppliedModelingLib.Alignment.Welfare.sigmoidChordSlope btScale / (1 / 4)) ^
  2 *
  AppliedModelingLib.Alignment.Welfare.populationAverageUtility population
  utility benchmark -
  2 *
  GolzhaghtalabYang2025Distortion.theorem2SourceRate
  (Fintype.card Alternative) users comparisonsPerUser minimumMass
  delta /
  (btScale * (1 / 4))) ≤
  AppliedModelingLib.pmfExp
  (AppliedModelingLib.pmfProduct (Fin users)
  (GolzhaghtalabYang2025Distortion.theorem2UserResponseTable Alternative ×
  (Fin comparisonsPerUser → Alternative × Alternative))
  (AppliedModelingLib.pmfProd responseLaw
  (AppliedModelingLib.pmfProduct (Fin comparisonsPerUser)
  (Alternative × Alternative)
  (GolzhaghtalabYang2025Distortion.theorem2UserPairLabelLaw sampling))))
  fun
  (sample :
    Fin users →
    GolzhaghtalabYang2025Distortion.theorem2UserResponseTable Alternative ×
    (Fin comparisonsPerUser → Alternative × Alternative)) =>
  AppliedModelingLib.Alignment.Welfare.populationAverageUtility population
  utility (winner sample)

```

Retained governing declarations

- [AppliedModelingLib.Alignment.Welfare.FiniteUtilityProfile](#)
- [AppliedModelingLib.Alignment.Welfare.UnitIntervalUtilityProfile](#)
- [AppliedModelingLib.Alignment.Welfare.pairwiseBordaScore](#)
- [AppliedModelingLib.Alignment.Welfare.populationAverageUtility](#)
- [AppliedModelingLib.Alignment.Welfare.populationBradleyTerryPreference](#)
- [AppliedModelingLib.Alignment.Welfare.sigmoidChordSlope](#)

- [AppliedModelingLib.pmfExp](#)
- [AppliedModelingLib.pmfProd](#)
- [AppliedModelingLib.pmfProduct](#)
- [GolzhaghtalabYang2025Distortion.theorem11iidUserLatentBordaScore](#)
- [GolzhaghtalabYang2025Distortion.theorem2SourceRate](#)
- [GolzhaghtalabYang2025Distortion.theorem2UserPairLabelLaw](#)
- [GolzhaghtalabYang2025Distortion.theorem2UserResponseCalibrated](#)
- [GolzhaghtalabYang2025Distortion.theorem2UserResponseTable](#)

Lean-elaborated claim roles

```

1. conclusion: (∀ {User Alternative : Type} [inst : Fintype User] [inst_1 : DecidableEq User] [inst_2 : Fintype Alternative]
[inst_3 : DecidableEq Alternative] (population : PMF User)
(utility : AppliedModelingLib.Alignment.Welfare.FiniteUtilityProfile User Alternative),
AppliedModelingLib.Alignment.Welfare.UnitIntervalUtilityProfile utility →
  ∀ (sampling : PMF Alternative) {btScale : ℝ},
  0 < btScale →
    ∀ (winner : Alternative),
    (∀ (alternative : Alternative),
      AppliedModelingLib.Alignment.Welfare.pairwiseBordaScore sampling
      (AppliedModelingLib.Alignment.Welfare.populationBradleyTerryPreference population utility btScale)
      alternative ≤
      AppliedModelingLib.Alignment.Welfare.pairwiseBordaScore sampling
      (AppliedModelingLib.Alignment.Welfare.populationBradleyTerryPreference population utility btScale)
      winner) →
      ∀ (alternative : Alternative),
      (AppliedModelingLib.Alignment.Welfare.sigmoidChordSlope btScale / (1 / 4)) ^ 2 *
      AppliedModelingLib.Alignment.Welfare.populationAverageUtility population utility alternative ≤
      AppliedModelingLib.Alignment.Welfare.populationAverageUtility population utility winner) ∧
  ∀ {User Alternative : Type} [inst : Fintype User] [inst_1 : DecidableEq User] [inst_2 : Fintype Alternative]
[inst_3 : DecidableEq Alternative] [Nonempty Alternative] (population : PMF User)
(utility : AppliedModelingLib.Alignment.Welfare.FiniteUtilityProfile User Alternative),
AppliedModelingLib.Alignment.Welfare.UnitIntervalUtilityProfile utility →
  ∀ (sampling : PMF Alternative) {btScale : ℝ},
  0 < btScale →
    ∀ (responseLaw : PMF (GolzhaghtalabYang2025Distortion.theorem2UserResponseTable Alternative)),
    GolzhaghtalabYang2025Distortion.theorem2UserResponseCalibrated responseLaw
    (AppliedModelingLib.Alignment.Welfare.populationBradleyTerryPreference population utility btScale) →
    ∀ (users comparisonsPerUser : ℕ),
    0 < users →
    0 < comparisonsPerUser →
    ∀ (minimumMass delta : ℝ),
    0 < minimumMass →
    (∀ (alternative : Alternative), minimumMass ≤ (sampling alternative).toReal) →
    0 < delta →
    delta ≤ 1 →
    2 * ↑(Fintype.card Alternative) *
    Real.exp (-↑(users * comparisonsPerUser) * minimumMass / 8) ≤
    delta →
    ∀
    (winner :
      (Fin users →
        GolzhaghtalabYang2025Distortion.theorem2UserResponseTable Alternative ×
        (Fin comparisonsPerUser → Alternative × Alternative)) →
        Alternative),
    (∀
      (sample :
        (Fin users →
          GolzhaghtalabYang2025Distortion.theorem2UserResponseTable Alternative ×
          (Fin comparisonsPerUser → Alternative × Alternative))
        (alternative : Alternative),
        GolzhaghtalabYang2025Distortion.theorem11iidUserLatentBordaScore alternative
        sample ≤
        GolzhaghtalabYang2025Distortion.theorem11iidUserLatentBordaScore (winner sample)
        sample) →
      (benchmark : Alternative),
      (1 - delta) *
      ((AppliedModelingLib.Alignment.Welfare.sigmoidChordSlope btScale / (1 / 4)) ^
        2 *
        AppliedModelingLib.Alignment.Welfare.populationAverageUtility population
        utility benchmark -
        2 *
        GolzhaghtalabYang2025Distortion.theorem2SourceRate
        (Fintype.card Alternative) users comparisonsPerUser minimumMass
        delta /
        (btScale * (1 / 4))) ≤
      AppliedModelingLib.pmfExp
      (AppliedModelingLib.pmfProduct (Fin users)
      (GolzhaghtalabYang2025Distortion.theorem2UserResponseTable Alternative ×
      (Fin comparisonsPerUser → Alternative × Alternative))
      (AppliedModelingLib.pmfProd responseLaw
      (AppliedModelingLib.pmfProduct (Fin comparisonsPerUser)
      (Alternative × Alternative)
      (GolzhaghtalabYang2025Distortion.theorem2UserPairLabelLaw sampling))))
    fun
    (sample :
      (Fin users →
        GolzhaghtalabYang2025Distortion.theorem2UserResponseTable Alternative ×
        (Fin comparisonsPerUser → Alternative × Alternative)) =>
      AppliedModelingLib.Alignment.Welfare.populationAverageUtility population
      utility (winner sample)

```

Lean proof endpoint (built separately)

GolzHaghtalabYang2025Distortion.theorem2_borda_distortion

Recorded source-to-Spec screening Verdict: matches

Reason: The population squared chord/tangent Borda factor and literal finite-user expected-welfare source rate are retained. The empirical-Borda zero-incidence branch is now an explicit neutral-score finite-rule convention, not a model premise.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 6

Source locator: cited publication, lines 431-436

Verbatim source input

Theorem 3 (Voting Rule-Independent Distortion Lower Bound). Fix any $\beta > 0$. If each user provides $d = 1 + e^{-\beta}$ comparison, no voting rule can guarantee distortion better than $\beta^2 \cdot 1 - e^{-\beta}$ for large m . If each user reports $d \geq 2$ pairwise comparisons, any voting rule that puts at most $1/m$ probability mass on a Condorcet loser must have at least the above distortion.

[Next verbatim source excerpt]

Let $A = \{1, \dots, m\}$ be a finite set of alternatives. The population of users is described by a probability distribution D over utility vectors $u = (u(1), \dots, u(m))$, whose entries $0 \leq u(x) \leq 1$ indicate a user's utility for alternative x .³ The objective is to find an alternative x such that its average utility $\text{AvgUtil}(x) := \mathbb{E}_{u \sim D} [u(x)]$ across the user population (also known as the utilitarian social welfare) is as high as possible. We extend this notation to probability distributions π over alternatives by setting $\text{AvgUtil}(\pi) := \mathbb{E}_{x \sim \pi} [\text{AvgUtil}(x)]$. Both voting rules and alignment methods observe comparisons from n users. We model each $i = 1, \dots, n$ as a fresh user with independently drawn utility vector $u_i \sim D$. For exposition, we assume that each user i provides an equal number $d \geq 1$ of pairwise comparisons. For each i , and for $j = 1, \dots, d$, we independently draw alternatives x_{ji}, y_{ji} from a fixed distribution μ over

³ Our assumption that utilities be in $[0, 1]$ is weaker than any of the three assumptions — unit-sum, unit-range, or approval [23] — made in classic distortion to avoid trivial infinite lower bounds.

3

[form-feed in source extraction]

alternatives, in which the minimum probability mass $\mu_{\min} := \min_{x \in A} \mu(x)$ is positive.⁴ User i then compares each pair $\{x_{ji}, y_{ji}\}$ (for $j = 1, \dots, d$) through a Bradley-Terry model based on i 's utilities: i prefers x_{ji} over y_{ji} (written “ x_{ji} [U+001F control character] y_{ji} ”) with probability $\sigma(\beta \cdot (u_i(x_{ji}) - u_i(y_{ji})))$, where $\sigma(t) := 1/(1 + e^{-t})$ is the logistic sigmoid function and $\beta > 0$ is a temperature parameter, and prefers

y_{ji} over x_{ji} (“ y_{ji} [U+001F control character] x_{ji} ”) otherwise.⁵ Whereas this specifies the marginal probability of each pairwise comparison, we make no assumption about the correlation between i 's choices. For example, i might derive the pairwise comparisons from a Plackett-Luce ranking, ensuring that [U+0002 control character] the user's comparisons [U+0001 control character] [U+0003 control character] are always consistent.⁶ We set $p(x [U+001F control character] y)$ for the expected win rate $\mathbb{E}_{u \sim D} \sigma(\beta \cdot (u(x) - u(y)))$.

[Next verbatim source excerpt]

Social Choice Setting. A voting rule f observes the sampled pairwise comparisons $\{x_{ji} [U+001F control character] y_{ji} \mid i \in [n], j \in [d]\}$ and maps them to a probability distribution over alternatives. For some m, D, β, d, μ , and

[U+0002 control character] the correlation between comparisons, [U+0001 control character] the average utility of f for n samples is $\text{AvgUtil}_n(f) := \mathbb{E} \text{AvgUtil}(f(\{x_{ji} [U+001F control character] y_{ji} \mid i, j\}))$, where the expectation is taken over the pairwise comparisons. The [U+0001 control character] [U+0003 control character]

distortion of f on D is the competitive ratio between $\text{AvgUtil}_n(f)$ and the optimal average utility $\max_{x \in A} \text{AvgUtil}(x)$ in the limit of $n \rightarrow \infty$ samples, and the distortion of f the worst-case distortion over all D :

$$\text{dist}(f, D) := \limsup_{n \rightarrow \infty} \max_{x \in A} \frac{\text{AvgUtil}_n(f)}{\text{AvgUtil}(x)}, \quad \text{dist}(f) = \sup_D \text{dist}(f, D).$$

For alternatives x, y , let $\#(x [U+001F control character] y) := \{(i, j) \mid x_{ji} = x, y_{ji} = y\}$ denote the number of pairwise comparisons in which x beat y . The (normalized) Borda score [45] of alternative x is

$$\text{BC}(x) := \frac{1}{P} \sum_{y \in A} \#(x [U+001F control character] y) - \frac{1}{P} \sum_{y \in A} \#(y [U+001F control character] x)$$

i.e., the fraction of pairwise comparisons involving x in which it wins. The Borda voting rule chooses the winner uniformly among all alternatives with maximum Borda score. The Maximal Lotteries voting rule first computes the margin matrix $M \in \mathbb{R}^{m \times m}$, where $M_{x,y} = \#(x [U+001F control character] y) - \#(y [U+001F control character] x)$.

[U+0001 control character] [U+0003 control character] . It then considers a symmetric two-player zero-sum game in which player 1 selects alternative x_1 , player 2 selects alternative x_2 , and the payoffs are M_{x_1, x_2} for player 1 and $M_{x_2, x_1} = -M_{x_1, x_2}$ for player 2. The maximal lotteries rule returns a distribution $\pi \in \Delta(A)$ such that $\pi \in \arg \max_{\pi} \mathbb{E}_{x_1 \sim \pi} \mathbb{E}_{x_2 \sim \pi} [M_{x_1, x_2}]$, i.e., a mixed strategy in Nash equilibrium. When several such π exist, our results hold for any choice.

Expanded PaperInterface specification

```
(∀ {beta : ℝ} (hbeta : 0 < beta) (pairSampling : (n : ℕ) → PMF (Fin (n + 2) × Fin (n + 2))))
(rule : (n horizon : ℕ) → (Fin horizon → (Fin (n + 2) × Fin (n + 2)) × Bool) → PMF (Fin (n + 2))) (delta : ℝ),
0 < delta →
  ∃ (N : ℕ),
  ∀ n ≥ N,
  ∃ (special : Fin (n + 2)),
  {horizon : ℕ |
    have policy : PMF (Fin (n + 2)) :=
      GolzHaghtalabYang2025Distortion.theorem3ObservationSelectionLaw
```

```

(GolzHaghtalabYang2025Distortion.theorem3ManyAlternativeObservationLaw (pairSampling n) beta
(GolzHaghtalabYang2025Distortion.theorem3DiagonalEpsilon n) hbeta
(GolzHaghtalabYang2025Distortion.theorem3DiagonalEpsilon_pos n) special horizon)
(rule n horizon);
(∀ (alternative : Fin (n + 2)),
AppliedModelingLib.Alignment.Welfare.populationAverageUtility
(GolzHaghtalabYang2025Distortion.theorem3TwoTypePopulation beta
(GolzHaghtalabYang2025Distortion.theorem3DiagonalEpsilon n) hbeta
(GolzHaghtalabYang2025Distortion.theorem3DiagonalEpsilon_pos n))
(GolzHaghtalabYang2025Distortion.theorem3ManyAlternativeUtility
(GolzHaghtalabYang2025Distortion.theorem3DiagonalEpsilon n) 1 special)
alternative ≤
AppliedModelingLib.Alignment.Welfare.populationAverageUtility
(GolzHaghtalabYang2025Distortion.theorem3TwoTypePopulation beta
(GolzHaghtalabYang2025Distortion.theorem3DiagonalEpsilon n) hbeta
(GolzHaghtalabYang2025Distortion.theorem3DiagonalEpsilon_pos n))
(GolzHaghtalabYang2025Distortion.theorem3ManyAlternativeUtility
(GolzHaghtalabYang2025Distortion.theorem3DiagonalEpsilon n) 1 special)
special) ∧
beta / 2 * (1 + Real.exp (-beta)) / (1 - Real.exp (-beta)) - delta <
GolzHaghtalabYang2025Distortion.theorem3SpecialTypeMass beta
(GolzHaghtalabYang2025Distortion.theorem3DiagonalEpsilon n) /
AppliedModelingLib.Alignment.Welfare.policyAverageUtility
(GolzHaghtalabYang2025Distortion.theorem3TwoTypePopulation beta
(GolzHaghtalabYang2025Distortion.theorem3DiagonalEpsilon n) hbeta
(GolzHaghtalabYang2025Distortion.theorem3DiagonalEpsilon_pos n))
(GolzHaghtalabYang2025Distortion.theorem3ManyAlternativeUtility
(GolzHaghtalabYang2025Distortion.theorem3DiagonalEpsilon n) 1 special)
policy).Infinite) ∧
∀ {beta : ℝ} (hbeta : 0 < beta) (d : ℕ),
2 ≤ d →
∀ (pairSampling : (n : ℕ) → PMF (Fin (n + 3) × Fin (n + 3))),
(∀ (n : ℕ) (ordinary : Fin (n + 3)), ordinary ≠ 0 → 0 < ((pairSampling n) (ordinary, 0)).toReal) →
∀ (reportLaw : (n : ℕ) → PMF (Fin d → (Fin (n + 3) × Fin (n + 3)) × Bool)),
(∀ (n : ℕ) (comparisonIndex : Fin d),
PMF.map (fun (report : Fin d → (Fin (n + 3) × Fin (n + 3)) × Bool) => report comparisonIndex)
(reportLaw n) =
GolzHaghtalabYang2025Distortion.theorem3ManyAlternativeOneComparisonLawXi (pairSampling n) beta
(GolzHaghtalabYang2025Distortion.theorem3D2DiagonalEpsilon n)
(GolzHaghtalabYang2025Distortion.theorem3D2DiagonalXi n) hbeta
(GolzHaghtalabYang2025Distortion.theorem3D2DiagonalEpsilon_pos n) 0) →
∀
(rule :
(n horizon : ℕ) → (Fin horizon → Fin d → (Fin (n + 3) × Fin (n + 3)) × Bool) → PMF (Fin (n + 3))),
(∀ (n horizon : ℕ),
GolzHaghtalabYang2025Distortion.theorem3ProbabilisticCondorcetLoserCriterion (rule n horizon)
fun (observation : Fin horizon → Fin d → (Fin (n + 3) × Fin (n + 3)) × Bool)
(alternative : Fin (n + 3)) =>
GolzHaghtalabYang2025Distortion.theorem3IidEmpiricalStrictCondorcetLoser
(fun (ordinary : Fin (n + 3)) =>
GolzHaghtalabYang2025Distortion.theorem3ReportedOrdinarySpecialMargin alternative ordinary)
alternative observation) →
∀ (delta : ℝ),
0 < delta →
∃ (N : ℕ),
∀ n ≥ N,
∀r (horizon : ℕ) in Filter.atTop,
beta / 2 * (1 + Real.exp (-beta)) / (1 - Real.exp (-beta)) - delta <
GolzHaghtalabYang2025Distortion.theorem3SpecialTypeMass beta
(GolzHaghtalabYang2025Distortion.theorem3D2DiagonalEpsilon n) /
AppliedModelingLib.Alignment.Welfare.policyAverageUtility
(GolzHaghtalabYang2025Distortion.theorem3TwoTypePopulation beta
(GolzHaghtalabYang2025Distortion.theorem3D2DiagonalEpsilon n) hbeta
(GolzHaghtalabYang2025Distortion.theorem3D2DiagonalEpsilon_pos n))
(GolzHaghtalabYang2025Distortion.theorem3ManyAlternativeUtility
(GolzHaghtalabYang2025Distortion.theorem3D2DiagonalEpsilon n)
(GolzHaghtalabYang2025Distortion.theorem3D2DiagonalXi n) 0)
(GolzHaghtalabYang2025Distortion.theorem3ObservationSelectionLaw
(GolzHaghtalabYang2025Distortion.theorem3IidUserReportBatchLaw (reportLaw n)
horizon)
(rule n horizon))

```

Retained governing declarations

- [AppliedModelingLib.Alignment.Welfare.policyAverageUtility](#)
- [AppliedModelingLib.Alignment.Welfare.populationAverageUtility](#)
- [GolzHaghtalabYang2025Distortion.theorem3D2DiagonalEpsilon](#)
- [GolzHaghtalabYang2025Distortion.theorem3D2DiagonalXi](#)
- [GolzHaghtalabYang2025Distortion.theorem3DiagonalEpsilon](#)
- [GolzHaghtalabYang2025Distortion.theorem3IidEmpiricalStrictCondorcetLoser](#)
- [GolzHaghtalabYang2025Distortion.theorem3IidUserReportBatchLaw](#)
- [GolzHaghtalabYang2025Distortion.theorem3ManyAlternativeObservationLaw](#)
- [GolzHaghtalabYang2025Distortion.theorem3ManyAlternativeOneComparisonLawXi](#)
- [GolzHaghtalabYang2025Distortion.theorem3ManyAlternativeUtility](#)
- [GolzHaghtalabYang2025Distortion.theorem3ObservationSelectionLaw](#)

- [GolzhaghtalabYang2025Distortion.theorem3ProbabilisticCondorcetLoserCriterion](#)
- [GolzhaghtalabYang2025Distortion.theorem3ReportedOrdinarySpecialMargin](#)
- [GolzhaghtalabYang2025Distortion.theorem3SpecialTypeMass](#)
- [GolzhaghtalabYang2025Distortion.theorem3TwoTypePopulation](#)

Lean-elaborated claim roles

```

1. conclusion: (∀ {beta : ℝ} (hbeta : 0 < beta) (pairSampling : (n : ℕ) → PMF (Fin (n + 2) × Fin (n + 2)))
(rule : (n horizon : ℕ) → (Fin horizon → (Fin (n + 2) × Fin (n + 2)) × Bool) → PMF (Fin (n + 2))) (delta : ℝ),
0 < delta →
  ∃ (N : ℕ),
  ∀ n ≥ N,
  ∃ (special : Fin (n + 2)),
  { horizon : ℕ |
    have policy : PMF (Fin (n + 2)) :=
      GolzhaghtalabYang2025Distortion.theorem3ObservationSelectionLaw
        (GolzhaghtalabYang2025Distortion.theorem3ManyAlternativeObservationLaw (pairSampling n) beta
        (GolzhaghtalabYang2025Distortion.theorem3DiagonalEpsilon n) hbeta
        (GolzhaghtalabYang2025Distortion.theorem3DiagonalEpsilon_pos n) special horizon)
      (rule n horizon);
    (∀ (alternative : Fin (n + 2)),
      AppliedModelingLib.Alignment.Welfare.populationAverageUtility
        (GolzhaghtalabYang2025Distortion.theorem3TwoTypePopulation beta
        (GolzhaghtalabYang2025Distortion.theorem3DiagonalEpsilon n) hbeta
        (GolzhaghtalabYang2025Distortion.theorem3DiagonalEpsilon_pos n))
        (GolzhaghtalabYang2025Distortion.theorem3ManyAlternativeUtility
        (GolzhaghtalabYang2025Distortion.theorem3DiagonalEpsilon n) 1 special)
        alternative ≤
      AppliedModelingLib.Alignment.Welfare.populationAverageUtility
        (GolzhaghtalabYang2025Distortion.theorem3TwoTypePopulation beta
        (GolzhaghtalabYang2025Distortion.theorem3DiagonalEpsilon n) hbeta
        (GolzhaghtalabYang2025Distortion.theorem3DiagonalEpsilon_pos n))
        (GolzhaghtalabYang2025Distortion.theorem3ManyAlternativeUtility
        (GolzhaghtalabYang2025Distortion.theorem3DiagonalEpsilon n) 1 special)
        special) ∧
      beta / 2 * (1 + Real.exp (-beta)) / (1 - Real.exp (-beta)) - delta <
      GolzhaghtalabYang2025Distortion.theorem3SpecialTypeMass beta
        (GolzhaghtalabYang2025Distortion.theorem3DiagonalEpsilon n) /
      AppliedModelingLib.Alignment.Welfare.policyAverageUtility
        (GolzhaghtalabYang2025Distortion.theorem3TwoTypePopulation beta
        (GolzhaghtalabYang2025Distortion.theorem3DiagonalEpsilon n) hbeta
        (GolzhaghtalabYang2025Distortion.theorem3DiagonalEpsilon_pos n))
        (GolzhaghtalabYang2025Distortion.theorem3ManyAlternativeUtility
        (GolzhaghtalabYang2025Distortion.theorem3DiagonalEpsilon n) 1 special)
        policy).Infinite) ∧
    policy}.Infinite) ∧
  2 ≤ d →
    ∀ (pairSampling : (n : ℕ) → PMF (Fin (n + 3) × Fin (n + 3))),
    (∀ (n : ℕ) (ordinary : Fin (n + 3)), ordinary ≠ 0 → 0 < ((pairSampling n) (ordinary, 0)).toReal) →
    ∀ (reportLaw : (n : ℕ) → PMF (Fin d → (Fin (n + 3) × Fin (n + 3)) × Bool)),
    (∀ (n : ℕ) (comparisonIndex : Fin d),
      PMF.map (fun (report : Fin d → (Fin (n + 3) × Fin (n + 3)) × Bool) => report comparisonIndex)
        (reportLaw n) =
      GolzhaghtalabYang2025Distortion.theorem3ManyAlternativeOneComparisonLawXi (pairSampling n) beta
        (GolzhaghtalabYang2025Distortion.theorem3D2DiagonalEpsilon n)
        (GolzhaghtalabYang2025Distortion.theorem3D2DiagonalXi n) hbeta
        (GolzhaghtalabYang2025Distortion.theorem3D2DiagonalEpsilon_pos n) 0) →
    ∀
    (rule :
      (n horizon : ℕ) → (Fin horizon → Fin d → (Fin (n + 3) × Fin (n + 3)) × Bool) → PMF (Fin (n + 3))),
    (∀ (n horizon : ℕ),
      GolzhaghtalabYang2025Distortion.theorem3ProbabilisticCondorcetLoserCriterion (rule n horizon)
      fun (observation : Fin horizon → Fin d → (Fin (n + 3) × Fin (n + 3)) × Bool)
        (alternative : Fin (n + 3)) =>
      GolzhaghtalabYang2025Distortion.theorem3IidEmpiricalStrictCondorcetLoser
        (fun (ordinary : Fin (n + 3)) =>
      GolzhaghtalabYang2025Distortion.theorem3ReportedOrdinarySpecialMargin alternative ordinary)
        alternative observation) →
    ∀ (delta : ℝ),
    0 < delta →
    ∃ (N : ℕ),
    ∀ n ≥ N,
    ∀f (horizon : ℕ) in Filter.atTop,
    beta / 2 * (1 + Real.exp (-beta)) / (1 - Real.exp (-beta)) - delta <
      GolzhaghtalabYang2025Distortion.theorem3SpecialTypeMass beta
        (GolzhaghtalabYang2025Distortion.theorem3D2DiagonalEpsilon n) /
      AppliedModelingLib.Alignment.Welfare.policyAverageUtility
        (GolzhaghtalabYang2025Distortion.theorem3TwoTypePopulation beta
        (GolzhaghtalabYang2025Distortion.theorem3D2DiagonalEpsilon n) hbeta
        (GolzhaghtalabYang2025Distortion.theorem3D2DiagonalEpsilon_pos n))
        (GolzhaghtalabYang2025Distortion.theorem3ManyAlternativeUtility
        (GolzhaghtalabYang2025Distortion.theorem3D2DiagonalEpsilon n)
        (GolzhaghtalabYang2025Distortion.theorem3D2DiagonalXi n) 0)
        (GolzhaghtalabYang2025Distortion.theorem3ObservationSelectionLaw
        (GolzhaghtalabYang2025Distortion.theorem3IidUserReportBatchLaw (reportLaw n)
        horizon)
        (rule n horizon))

```

Lean proof endpoint (built separately)

GolzhaghtalabYang2025Distortion.theorem3_voting_rule_lower_bound

Recorded source-to-Spec screening **Verdict:** matches

Reason: Both the $d = 1$ arbitrary-rule and $d \geq 2$ Condorcet-loser lower-bound branches are covered.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 7

Source locator: cited publication, lines 450-454

Verbatim source input

Theorem 5 (Borda Distortion Lower Bound, Informally). For any $\beta > 0$ and $m \geq 3$, the distortion $\frac{1}{1+e^{-\beta}}$ guaranteed by Borda (and, hence, RLHF) is greater than and bounded away from $\beta^2 \frac{1-e^{-\beta}}{2}$. In particular, as $\beta \rightarrow \infty$, this distortion guarantee is at least $(1 - o(1)) \cdot \beta$.

[Next verbatim source excerpt]

Let $A = \{1, \dots, m\}$ be a finite set of alternatives. The population of users is described by a probability distribution D over utility vectors $u = (u(1), \dots, u(m))$, whose entries $0 \leq u(x) \leq 1$ indicate a user's utility for alternative x .³ The objective is to find an alternative x such that its average utility $\text{AvgUtil}(x) := \mathbb{E}_{u \sim D} [u(x)]$ across the user population (also known as the utilitarian social welfare) is as high as possible. We extend this notation to probability distributions π over alternatives by setting $\text{AvgUtil}(\pi) := \mathbb{E}_{x \sim \pi} [\text{AvgUtil}(x)]$. Both voting rules and alignment methods observe comparisons from n users. We model each $i = 1, \dots, n$ as a fresh user with independently drawn utility vector $u_i \sim D$. For exposition, we assume that each user i provides an equal number $d \geq 1$ of pairwise comparisons. For each i , and for $j = 1, \dots, d$, we independently draw alternatives x_{ji}, y_{ji} from a fixed distribution μ over

³ Our assumption that utilities be in $[0, 1]$ is weaker than any of the three assumptions — unit-sum, unit-range, or approval [23] — made in classic distortion to avoid trivial infinite lower bounds.

3

[form-feed in source extraction]

alternatives, in which the minimum probability mass $\mu_{\min} := \min_{x \in A} \mu(x)$ is positive.⁴ User i then compares each pair $\{x_{ji}, y_{ji}\}$ (for $j = 1, \dots, d$) through a Bradley-Terry model based on i 's utilities: i prefers x_{ji} over y_{ji} (written “ $x_{ji} \text{ [U+001F control character] } y_{ji}$ ”) with probability $\sigma(\beta \cdot (u_i(x_{ji}) - u_i(y_{ji})))$, where $\sigma(t) := 1/(1+e^{-t})$ is the logistic sigmoid function and $\beta > 0$ is a temperature parameter, and prefers y_{ji} over x_{ji} (“ $y_{ji} \text{ [U+001F control character] } x_{ji}$ ”) otherwise.⁵ Whereas this specifies the marginal probability of each pairwise comparison, we make no assumption about the correlation between i 's choices. For example, i might derive the pairwise comparisons from a Plackett-Luce ranking, ensuring that $\text{[U+0002 control character]}$ the user's comparisons $\text{[U+0001 control character]}$ are always consistent.⁶ We set $p(x \text{ [U+001F control character] } y)$ for the expected win rate $\mathbb{E}_{u \sim D} \sigma(\beta \cdot (u(x) - u(y)))$.

[Next verbatim source excerpt]

Social Choice Setting. A voting rule f observes the sampled pairwise comparisons $\{x_{ji} \text{ [U+001F control character] } y_{ji} \mid i \in [n], j \in [d]\}$ and maps them to a probability distribution over alternatives. For some m, D, β, d, μ , and $\text{[U+0002 control character]}$ the correlation between comparisons, the average utility of f for n samples is $\text{AvgUtil}_n(f) := \mathbb{E} \text{AvgUtil}(f(\{x_{ji} \text{ [U+001F control character] } y_{ji} \}_{i,j}))$, where the expectation is taken over the pairwise comparisons. The $\text{[U+0001 control character]}$ distortion of f on D is the competitive ratio between $\text{AvgUtil}_n(f)$ and the optimal average utility $\max_{x \in A} \text{AvgUtil}(x)$ in the limit of $n \rightarrow \infty$ samples, and the distortion of f the worst-case distortion over all D :

$$\text{dist}(f, D) := \limsup_{n \rightarrow \infty} \max_{x \in A} \frac{\text{AvgUtil}_n(f)}{\text{AvgUtil}(x)}, \quad \text{dist}(f) = \sup_D \text{dist}(f, D).$$

For alternatives x, y , let $\#(x \text{ [U+001F control character] } y) := \{(i, j) \mid x_{ji} = x, y_{ji} = y\}$ denote the number of pairwise comparisons in which x beat y . The (normalized) Borda score [45] of alternative x is

$$\text{BC}(x) := \frac{\sum_{y \in A} \#(x \text{ [U+001F control character] } y)}{\sum_{y \in A} \#(y \text{ [U+001F control character] } x)}$$

i.e., the fraction of pairwise comparisons involving x in which it wins. The Borda voting rule chooses the winner uniformly among all alternatives with maximum Borda score. The Maximal Lotteries voting rule first computes the margin matrix $M \in \mathbb{R}^{m \times m}$, where $M_{x,y} = \#(x \text{ [U+001F control character] } y) - \#(y \text{ [U+001F control character] } x)$. It then considers a symmetric two-player zero-sum game in which player 1 selects alternative x_1 , player 2 selects alternative x_2 , and the payoffs are M_{x_1, x_2} for player 1 and $M_{x_2, x_1} = -M_{x_1, x_2}$ for player 2. The maximal lotteries rule returns a distribution $\pi \in \Delta(A)$ such that $\pi_1, \pi_2 \in \Delta(A)$ $\mathbb{E}_{x_1 \sim \pi_1, x_2 \sim \pi_2} [M_{x_1, x_2}] = 0$, i.e., a mixed strategy in Nash equilibrium. When several such π exist, our results hold for any choice.

Expanded PaperInterface specification

```
(∀ {beta : ℝ} (hbeta : 0 < beta),
have gamma : ℝ := GolzHaghtalabYang2025Distortion.theorem12LogChoice beta;
∃ (hgamma : 0 < gamma) (hgamma_lt_one : gamma < 1),
beta / 2 * (1 + Real.exp (-beta)) / (1 - Real.exp (-beta)) <
GolzHaghtalabYang2025Distortion.theorem12Eq10Coefficient beta gamma ∧
∀ q < GolzHaghtalabYang2025Distortion.theorem12Eq10Coefficient beta gamma,
∀ (m : ℕ),
3 ≤ m →
∃ (epsilon : ℝ) (hepsilon : 0 < epsilon) (hepsilon_lt_half : epsilon < 1 / 2) (sampling :
PMF (Fin 3 ⊕ Fin (m - 3))),
```

```

epsilon ^ 2 < 1 - epsilon ∧
Fintype.card (Fin 3 ⊗ Fin (m - 3)) = m ∧
have collapse : Fin 3 ⊗ Fin (m - 3) → Fin 3 :=
  GolzHaghtalabYang2025Distortion.theorem12CloneCollapse 1 (m - 3);
have splitUtility :
  AppliedModelingLib.Alignment.Welfare.FiniteUtilityProfile (Fin 3) (Fin 3 ⊗ Fin (m - 3)) :=
  GolzHaghtalabYang2025Distortion.theorem12SplitUtility
  (GolzHaghtalabYang2025Distortion.theorem12Utility epsilon (epsilon ^ 2) gamma) collapse;
have population : PMF (Fin 3) :=
  GolzHaghtalabYang2025Distortion.theorem12Population beta epsilon gamma hbeta hepsilon
  (GolzHaghtalabYang2025Distortion.theorem5Spec._proof_2 epsilon hepsilon hepsilon_lt_half) hgamma
  hgamma_lt_one;
have preference :
  AppliedModelingLib.Learning.HumanFeedback.PairwisePreference PUnit.{1} (Fin 3 ⊗ Fin (m - 3)) :=
  AppliedModelingLib.Alignment.Welfare.populationBradleyTerryPreference population splitUtility
  beta;
AppliedModelingLib.Alignment.Welfare.UnitIntervalUtilityProfile splitUtility ∧
AppliedModelingLib.PMFFullSupport sampling ∧
q <
  AppliedModelingLib.Alignment.Welfare.populationAverageUtility population splitUtility
  (Sum.inl 0) /
  AppliedModelingLib.Alignment.Welfare.populationAverageUtility population splitUtility
  (Sum.inl 2) ∧
Filter.Tendsto
  (fun (horizon : ℕ) =>
    AppliedModelingLib.pmfProb
      (GolzHaghtalabYang2025Distortion.theorem12IidBordaReportBatchLaw
        (GolzHaghtalabYang2025Distortion.theorem12OneComparisonReportLaw sampling preference)
        horizon)
    fun (sample : Fin horizon → ((Fin 3 ⊗ Fin (m - 3)) × (Fin 3 ⊗ Fin (m - 3))) × Bool) =>
      GolzHaghtalabYang2025Distortion.theorem12IidEmpiricalBordaFailure
        GolzHaghtalabYang2025Distortion.theorem12OneComparisonWins
        GolzHaghtalabYang2025Distortion.theorem12OneComparisonIncidences (Sum.inl 2) sample)
    Filter.atTop (nhds 0)) ∧
Filter.Tendsto
  (fun (beta : ℝ) =>
    GolzHaghtalabYang2025Distortion.theorem12Eq10Coefficient beta
    (GolzHaghtalabYang2025Distortion.theorem12LogChoice beta) /
    beta)
  Filter.atTop (nhds 1)

```

Retained governing declarations

- [AppliedModelingLib.Alignment.Welfare.FiniteUtilityProfile](#)
- [AppliedModelingLib.Alignment.Welfare.UnitIntervalUtilityProfile](#)
- [AppliedModelingLib.Alignment.Welfare.populationAverageUtility](#)
- [AppliedModelingLib.Alignment.Welfare.populationBradleyTerryPreference](#)
- [AppliedModelingLib.Learning.HumanFeedback.PairwisePreference](#)
- [AppliedModelingLib.PMFFullSupport](#)
- [AppliedModelingLib.pmfProb](#)
- [GolzHaghtalabYang2025Distortion.theorem12CloneCollapse](#)
- [GolzHaghtalabYang2025Distortion.theorem12EmpiricalBordaScore](#)
- [GolzHaghtalabYang2025Distortion.theorem12Eq10Coefficient](#)
- [GolzHaghtalabYang2025Distortion.theorem12IidBordaReportBatchLaw](#)
- [GolzHaghtalabYang2025Distortion.theorem12IidEmpiricalBordaFailure](#)
- [GolzHaghtalabYang2025Distortion.theorem12LogChoice](#)
- [GolzHaghtalabYang2025Distortion.theorem12OneComparisonIncidences](#)
- [GolzHaghtalabYang2025Distortion.theorem12OneComparisonReportLaw](#)
- [GolzHaghtalabYang2025Distortion.theorem12OneComparisonWins](#)
- [GolzHaghtalabYang2025Distortion.theorem12Population](#)
- [GolzHaghtalabYang2025Distortion.theorem12SplitUtility](#)
- [GolzHaghtalabYang2025Distortion.theorem12Utility](#)

Lean-elaborated claim roles

```
1. conclusion: (∀ {beta : ℝ} (hbeta : 0 < beta),
have gamma : ℝ := GolzHaghtalabYang2025Distortion.theorem12LogChoice beta;
∃ (hgamma : 0 < gamma) (hgamma_lt_one : gamma < 1),
beta / 2 * (1 + Real.exp (-beta)) / (1 - Real.exp (-beta)) <
  GolzHaghtalabYang2025Distortion.theorem12Eq10Coefficient beta gamma ∧
  ∀ q < GolzHaghtalabYang2025Distortion.theorem12Eq10Coefficient beta gamma,
    ∀ (m : ℕ),
      3 ≤ m →
        ∃ (epsilon : ℝ) (hepsilon : 0 < epsilon) (hepsilon_lt_half : epsilon < 1 / 2) (sampling :
          PMF (Fin 3 ⊕ Fin (m - 3))),
          epsilon ^ 2 < 1 - epsilon ∧
          Fintype.card (Fin 3 ⊕ Fin (m - 3)) = m ∧
          have collapse : Fin 3 ⊕ Fin (m - 3) → Fin 3 :=
            GolzHaghtalabYang2025Distortion.theorem12CloneCollapse 1 (m - 3);
          have splitUtility :
            AppliedModelingLib.Alignment.Welfare.FiniteUtilityProfile (Fin 3) (Fin 3 ⊕ Fin (m - 3)) :=
              GolzHaghtalabYang2025Distortion.theorem12SplitUtility
              (GolzHaghtalabYang2025Distortion.theorem12Utility epsilon (epsilon ^ 2) gamma) collapse;
          have population : PMF (Fin 3) :=
            GolzHaghtalabYang2025Distortion.theorem12Population beta epsilon gamma hbeta hepsilon
            (GolzHaghtalabYang2025Distortion.theorem5Spec._proof_2 epsilon hepsilon hepsilon_lt_half) hgamma
            hgamma_lt_one;
          have preference :
            AppliedModelingLib.Learning.HumanFeedback.PairwisePreference PUnit.{1} (Fin 3 ⊕ Fin (m - 3)) :=
              AppliedModelingLib.Alignment.Welfare.populationBradleyTerryPreference population splitUtility
              beta;
          AppliedModelingLib.Alignment.Welfare.UnitIntervalUtilityProfile splitUtility ∧
          AppliedModelingLib.PMFFullSupport sampling ∧
          q <
            AppliedModelingLib.Alignment.Welfare.populationAverageUtility population splitUtility
              (Sum.inl 0) /
            AppliedModelingLib.Alignment.Welfare.populationAverageUtility population splitUtility
              (Sum.inl 2) ∧
          Filter.Tendsto
            (fun (horizon : ℕ) =>
              AppliedModelingLib.pmfProb
                (GolzHaghtalabYang2025Distortion.theorem12lidBordaReportBatchLaw
                  (GolzHaghtalabYang2025Distortion.theorem12OneComparisonReportLaw sampling preference)
                  horizon)
                fun (sample : Fin horizon → ((Fin 3 ⊕ Fin (m - 3)) × (Fin 3 ⊕ Fin (m - 3))) × Bool) =>
                  GolzHaghtalabYang2025Distortion.theorem12lidEmpiricalBordaFailure
                    GolzHaghtalabYang2025Distortion.theorem12OneComparisonWins
                    GolzHaghtalabYang2025Distortion.theorem12OneComparisonIncidences (Sum.inl 2) sample)
              Filter.atTop (nhds 0)) ∧
          Filter.Tendsto
            (fun (beta : ℝ) =>
              GolzHaghtalabYang2025Distortion.theorem12Eq10Coefficient beta
              (GolzHaghtalabYang2025Distortion.theorem12LogChoice beta) /
              beta)
            Filter.atTop (nhds 1))
```

Lean proof endpoint (built separately)

GolzHaghtalabYang2025Distortion.theorem5_borda_lower_bound

Recorded source-to-Spec screening Verdict: matches

Reason: This realizes the source Borda/RLHF lower bound strictly above the universal coefficient and asymptotic to beta.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 8

Source locator: cited publication, lines 1728-1734

Verbatim source input

Theorem 12 (Lower Bound for Borda; Formal Version of Theorem 5). For any $\beta > 0$ and $m \geq 3$, the Borda voting rule (and, hence, RHLF) cannot guarantee a distortion better than $\max_{0 < \gamma < 1} \frac{1 + e^{-\beta}}{2} \frac{\sigma(\beta\gamma) - 1/2}{\sigma(\beta) - 1/2}$. This bound is strictly higher than the voting-rule independent lower bound $\frac{\beta}{2} \frac{1 + e^{-\beta}}{1 - e^{-\beta}}$ for all β and is at least $(1 - o(1)) \beta$ as $\beta \rightarrow \infty$.

[Next verbatim source excerpt]

Let $A = \{1, \dots, m\}$ be a finite set of alternatives. The population of users is described by a probability distribution D over utility vectors $u = (u(1), \dots, u(m))$, whose entries $0 \leq u(x) \leq 1$ indicate a user's utility for alternative x . The objective is to find an alternative x such that its average utility $\text{AvgUtil}(x) := \mathbb{E}_{u \sim D} [u(x)]$ across the user population (also known as the utilitarian social welfare) is as high as possible. We extend this notation to probability distributions π over alternatives by setting $\text{AvgUtil}(\pi) := \mathbb{E}_{x \sim \pi} [\text{AvgUtil}(x)]$. Both voting rules and alignment methods observe comparisons from n users. We model each $i = 1, \dots, n$ as a fresh user with independently drawn utility vector $u_i \sim D$. For exposition, we assume that each user i provides an equal number $d \geq 1$ of pairwise comparisons. For each i , and for $j = 1, \dots, d$, we independently draw alternatives x_{ji}, y_{ji} from a fixed distribution μ over

alternatives, in which the minimum probability mass $\mu_{\min} := \min_{x \in A} \mu(x)$ is positive. User i then compares each pair $\{x_{ji}, y_{ji}\}$ (for $j = 1, \dots, d$) through a Bradley-Terry model based on i 's utilities: i prefers x_{ji} over y_{ji} (written " x_{ji} [U+001F control character] y_{ji} ") with probability $\sigma(\beta \cdot (u_i(x_{ji}) - u_i(y_{ji})))$, where

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[form-feed in source extraction]

$\sigma(t) := 1/(1 + e^{-t})$ is the logistic sigmoid function and $\beta > 0$ is a temperature parameter, and prefers y_{ji} over x_{ji} (" y_{ji} [U+001F control character] x_{ji} ") otherwise. Whereas this specifies the marginal probability of each pairwise comparison, we make no assumption about the correlation between i 's choices. For example, i might derive the pairwise comparisons from a Plackett-Luce ranking, ensuring that [U+0002 control character] the user's comparisons [U+0001 control character] are always consistent. We set $p(x$ [U+001F control character] $y)$ for the expected win rate $\mathbb{E}_{u \sim D} \sigma(\beta \cdot (u(x) - u(y)))$.

[Next verbatim source excerpt]

Social Choice Setting. A voting rule f observes the sampled pairwise comparisons $\{x_{ji} \text{ [U+001F control character] } y_{ji} \mid i \in [n], j \in [d]\}$ and maps them to a probability distribution over alternatives. For some m, D, β, d, μ , and [U+0002 control character] the correlation between comparisons, the average utility of f for n samples is $\text{AvgUtil}_n(f) := \mathbb{E} \text{AvgUtil}(f(\{x_{ji} \text{ [U+001F control character] } y_{ji} \}_{i,j}))$, where the expectation is taken over the pairwise comparisons. The [U+0001 control character] [U+0003 control character]

distortion of f on D is the competitive ratio between $\text{AvgUtil}_n(f)$ and the optimal average utility $\max_{x \in A} \text{AvgUtil}(x)$ in the limit of $n \rightarrow \infty$ samples, and the distortion of f the worst-case distortion over all D :

$$\text{dist}(f, D) := \limsup_{n \rightarrow \infty} \max_{x \in A} \frac{\text{AvgUtil}_n(f)}{\text{AvgUtil}(x)}, \quad \text{dist}(f) = \sup_D \text{dist}(f, D).$$

For alternatives x, y , let $\#(x \text{ [U+001F control character] } y) := \{(i, j) \mid x_{ji} = x, y_{ji} = y\}$ denote the number of pairwise comparisons in which x beat y . The (normalized) Borda score [45] of alternative x is

$$\text{BC}(x) := \frac{\sum_{y \in A} \#(x \text{ [U+001F control character] } y) - \sum_{y \in A} \#(y \text{ [U+001F control character] } x)}{\sum_{y \in A} \#(y \text{ [U+001F control character] } x)}$$

i.e., the fraction of pairwise comparisons involving x in which it wins. The Borda voting rule chooses the winner uniformly among all alternatives with maximum Borda score. The Maximal Lotteries voting rule first computes the margin matrix $M \in \mathbb{R}^{m \times m}$, where $M_{x,y} = \#(x \text{ [U+001F control character] } y) - \#(y \text{ [U+001F control character] } x)$. It then considers a symmetric two-player zero-sum game in which player 1 selects alternative x_1 , player 2 selects alternative x_2 , and the payoffs are M_{x_1, x_2} for player 1 and $M_{x_2, x_1} = -M_{x_1, x_2}$ for player 2. The maximal lotteries rule returns a distribution $\pi \in \arg\max_{\pi_1 \in \Delta(A), \pi_2 \in \Delta(A)} \mathbb{E}_{x_1 \sim \pi_1, x_2 \sim \pi_2} [M_{x_1, x_2}]$, i.e., a mixed strategy in Nash equilibrium. When several such π exist, our results hold for any choice.

Expanded PaperInterface specification

$\forall \{\beta, \gamma, q : \mathbb{R}\} \text{ (hbeta : } 0 < \beta \text{) (hgamma : } 0 < \gamma \text{) (hgamma_lt_one : } \gamma < 1 \text{),}$
 $q < \text{GolzhaghtalabYang2025Distortion.theorem12Eq10Coefficient } \beta, \gamma \rightarrow$
 $\forall (m : \mathbb{N}),$
 $3 \leq m \rightarrow$
 $\exists (\epsilon : \mathbb{R}) \text{ (hepsilon : } 0 < \epsilon \text{) (hepsilon_lt_half : } \epsilon < 1/2 \text{) (sampling :}$
 $\text{PMF (Fin 3 } \oplus \text{ Fin (m - 3))),}$
 $\epsilon^2 < 1 - \epsilon \wedge$
 $\text{Finite.card (Fin 3 } \oplus \text{ Fin (m - 3)) = m } \wedge$

```

have collapse : Fin 3 ⊕ Fin (m - 3) → Fin 3 :=
  GolzHaghtalabYang2025Distortion.theorem12CloneCollapse 1 (m - 3);
have splitUtility :
  AppliedModelingLib.Alignment.Welfare.FiniteUtilityProfile (Fin 3) (Fin 3 ⊕ Fin (m - 3)) :=
  GolzHaghtalabYang2025Distortion.theorem12SplitUtility
  (GolzHaghtalabYang2025Distortion.theorem12Utility epsilon (epsilon ^ 2) gamma) collapse;
have population : PMF (Fin 3) :=
  GolzHaghtalabYang2025Distortion.theorem12Population beta epsilon gamma hbeta hepsilon
  (GolzHaghtalabYang2025Distortion.theorem5Spec._proof_2 epsilon hepsilon hepsilon_lt_half) hgamma
  hgamma_lt_one;
have preference :
  AppliedModelingLib.Learning.HumanFeedback.PairwisePreference PUnit.{1} (Fin 3 ⊕ Fin (m - 3)) :=
  AppliedModelingLib.Alignment.Welfare.populationBradleyTerryPreference population splitUtility beta;
AppliedModelingLib.Alignment.Welfare.UnitIntervalUtilityProfile splitUtility ∧
AppliedModelingLib.PMFFullSupport sampling ∧
q <
  AppliedModelingLib.Alignment.Welfare.populationAverageUtility population splitUtility
  (Sum.inl 0) /
  AppliedModelingLib.Alignment.Welfare.populationAverageUtility population splitUtility
  (Sum.inl 2) ∧
Filter.Tendsto
  (fun (horizon : ℕ) =>
    AppliedModelingLib.pmfProb
    (GolzHaghtalabYang2025Distortion.theorem12lidBordaReportBatchLaw
      (GolzHaghtalabYang2025Distortion.theorem12OneComparisonReportLaw sampling preference)
      horizon)
    fun (sample : Fin horizon → ((Fin 3 ⊕ Fin (m - 3)) × (Fin 3 ⊕ Fin (m - 3))) × Bool) =>
      GolzHaghtalabYang2025Distortion.theorem12lidEmpiricalBordaFailure
      GolzHaghtalabYang2025Distortion.theorem12OneComparisonWins
      GolzHaghtalabYang2025Distortion.theorem12OneComparisonIncidences (Sum.inl 2) sample)
  Filter.atTop (nhds 0)) ∧
(∀ {beta gamma : ℝ},
  0 < beta →
  0 < gamma →
  gamma < 1 →
  beta / 2 * (1 + Real.exp (-beta)) / (1 - Real.exp (-beta)) <
    GolzHaghtalabYang2025Distortion.theorem12Eq10Coefficient beta gamma) ∧
Filter.Tendsto
  (fun (beta : ℝ) =>
    GolzHaghtalabYang2025Distortion.theorem12Eq10Coefficient beta
    (GolzHaghtalabYang2025Distortion.theorem12LogChoice beta) /
    beta)
  Filter.atTop (nhds 1)

```

Retained governing declarations

- [AppliedModelingLib.Alignment.Welfare.FiniteUtilityProfile](#)
- [AppliedModelingLib.Alignment.Welfare.UnitIntervalUtilityProfile](#)
- [AppliedModelingLib.Alignment.Welfare.populationAverageUtility](#)
- [AppliedModelingLib.Alignment.Welfare.populationBradleyTerryPreference](#)
- [AppliedModelingLib.Learning.HumanFeedback.PairwisePreference](#)
- [AppliedModelingLib.PMFFullSupport](#)
- [AppliedModelingLib.pmfProb](#)
- [GolzHaghtalabYang2025Distortion.theorem12CloneCollapse](#)
- [GolzHaghtalabYang2025Distortion.theorem12EmpiricalBordaScore](#)
- [GolzHaghtalabYang2025Distortion.theorem12Eq10Coefficient](#)
- [GolzHaghtalabYang2025Distortion.theorem12lidBordaReportBatchLaw](#)
- [GolzHaghtalabYang2025Distortion.theorem12lidEmpiricalBordaFailure](#)
- [GolzHaghtalabYang2025Distortion.theorem12LogChoice](#)
- [GolzHaghtalabYang2025Distortion.theorem12OneComparisonIncidences](#)
- [GolzHaghtalabYang2025Distortion.theorem12OneComparisonReportLaw](#)
- [GolzHaghtalabYang2025Distortion.theorem12OneComparisonWins](#)
- [GolzHaghtalabYang2025Distortion.theorem12Population](#)
- [GolzHaghtalabYang2025Distortion.theorem12SplitUtility](#)
- [GolzHaghtalabYang2025Distortion.theorem12Utility](#)

Lean-elaborated claim roles

```

1. conclusion: (∀ {beta gamma q : ℝ} (hbeta : 0 < beta) (hgamma : 0 < gamma) (hgamma_lt_one : gamma < 1),
  q < GolzHaghtalabYang2025Distortion.theorem12Eq10Coefficient beta gamma →
  ∀ (m : ℕ),
  3 ≤ m →
  ∃ (epsilon : ℝ) (hepsilon : 0 < epsilon) (hepsilon_lt_half : epsilon < 1 / 2) (sampling :
    PMF (Fin 3 ⊕ Fin (m - 3))),
  epsilon ^ 2 < 1 - epsilon ∧
  Fintype.card (Fin 3 ⊕ Fin (m - 3)) = m ∧
  have collapse : Fin 3 ⊕ Fin (m - 3) → Fin 3 :=
    GolzHaghtalabYang2025Distortion.theorem12CloneCollapse 1 (m - 3);
  have splitUtility :
    AppliedModelingLib.Alignment.Welfare.FiniteUtilityProfile (Fin 3) (Fin 3 ⊕ Fin (m - 3)) :=
      GolzHaghtalabYang2025Distortion.theorem12SplitUtility
        (GolzHaghtalabYang2025Distortion.theorem12Utility epsilon (epsilon ^ 2) gamma) collapse;
  have population : PMF (Fin 3) :=
    GolzHaghtalabYang2025Distortion.theorem12Population beta epsilon gamma hbeta hepsilon
      (GolzHaghtalabYang2025Distortion.theorem5Spec._proof_2 epsilon hepsilon hepsilon_lt_half) hgamma
      hgamma_lt_one;
  have preference :
    AppliedModelingLib.Learning.HumanFeedback.PairwisePreference PUnit.{1} (Fin 3 ⊕ Fin (m - 3)) :=
      AppliedModelingLib.Alignment.Welfare.populationBradleyTerryPreference population splitUtility beta;
  AppliedModelingLib.Alignment.Welfare.UnitIntervalUtilityProfile splitUtility ∧
  AppliedModelingLib.PMFFullSupport sampling ∧
  q <
    AppliedModelingLib.Alignment.Welfare.populationAverageUtility population splitUtility
      (Sum.inl 0) /
    AppliedModelingLib.Alignment.Welfare.populationAverageUtility population splitUtility
      (Sum.inl 2) ∧
  Filter.Tendsto
    (fun (horizon : ℕ) =>
      AppliedModelingLib.pmfProb
        (GolzHaghtalabYang2025Distortion.theorem12lidBordaReportBatchLaw
          (GolzHaghtalabYang2025Distortion.theorem12OneComparisonReportLaw sampling preference)
          horizon)
        fun (sample : Fin horizon → ((Fin 3 ⊕ Fin (m - 3)) × (Fin 3 ⊕ Fin (m - 3))) × Bool) =>
          GolzHaghtalabYang2025Distortion.theorem12lidEmpiricalBordaFailure
            GolzHaghtalabYang2025Distortion.theorem12OneComparisonWins
              GolzHaghtalabYang2025Distortion.theorem12OneComparisonIncidences (Sum.inl 2) sample)
      Filter.atTop (nhds 0)) ∧
  (∀ {beta gamma : ℝ},
  0 < beta →
  0 < gamma →
  gamma < 1 →
  beta / 2 * (1 + Real.exp (-beta)) / (1 - Real.exp (-beta)) <
    GolzHaghtalabYang2025Distortion.theorem12Eq10Coefficient beta gamma) ∧
  Filter.Tendsto
    (fun (beta : ℝ) =>
      GolzHaghtalabYang2025Distortion.theorem12Eq10Coefficient beta
        (GolzHaghtalabYang2025Distortion.theorem12LogChoice beta) /
        beta)
    Filter.atTop (nhds 1)

```

Lean proof endpoint (built separately)

GolzHaghtalabYang2025Distortion.theorem12_borda_lower_bound

Recorded source-to-Spec screening Verdict: matches

Reason: The all-m Borda lower-bound construction, its strict separation from the universal factor, and the asymptotic coefficient are retained.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 9

Source locator: cited publication, lines 494-496

Verbatim source input

Theorem 6 (RLHF Distortion Lower Bound). For $m \geq 3$, there is a sequence of alignment problems on which the distortion of RLHF scales as $e\Omega(\beta)$ in β .

[Next verbatim source excerpt]

Let $A = \{1, \dots, m\}$ be a finite set of alternatives. The population of users is described by a probability distribution D over utility vectors $u = (u(1), \dots, u(m))$, whose entries $0 \leq u(x) \leq 1$ indicate a user's utility for alternative x .³ The objective is to find an alternative x such that its average utility $\text{AvgUtil}(x) := \mathbb{E}_{u \sim D} [u(x)]$ across the user population (also known as the utilitarian social welfare) is as high as possible. We extend this notation to probability distributions π over alternatives by setting $\text{AvgUtil}(\pi) := \mathbb{E}_{x \sim \pi} [\text{AvgUtil}(x)]$.

Both voting rules and alignment methods observe comparisons from n users. We model each $i = 1, \dots, n$ as a fresh user with independently drawn utility vector $u_i \sim D$. For exposition, we assume that each user i provides an equal number $d \geq 1$ of pairwise comparisons. For each i , and for $j = 1, \dots, d$, we independently draw alternatives x_{ji}, y_{ji} from a fixed distribution μ over

3

Our assumption that utilities be in $[0, 1]$ is weaker than any of the three assumptions — unit-sum, unit-range, or approval [23] — made in classic distortion to avoid trivial infinite lower bounds.

3

[form-feed in source extraction]

alternatives, in which the minimum probability mass $\mu_{\min} := \min_{x \in A} \mu(x)$ is positive.⁴ User i then compares each pair $\{x_{ji}, y_{ji}\}$ (for $j = 1, \dots, d$) through a Bradley-Terry model based on i 's utilities: i prefers x_{ji} over y_{ji} (written “ x_{ji} [U+001F control character] y_{ji} ”) with probability $\sigma(\beta \cdot (u_i(x_{ji}) - u_i(y_{ji})))$, where [U+0001 control character]

$\sigma(t) := 1/(1+e^{-t})$ is the logistic sigmoid function and $\beta > 0$ is a temperature parameter, and prefers y_{ji} over x_{ji} (“ y_{ji} [U+001F control character] x_{ji} ”) otherwise.⁵ Whereas this specifies the marginal probability of each pairwise comparison, we make no assumption about the correlation between i 's choices. For example, i might derive the pairwise comparisons from a Plackett-Luce ranking, ensuring that [U+0002 control character] the user's comparisons [U+0001 control character] [U+0003 control character] are always consistent.⁶ We set $p(x$ [U+001F control character] $y)$ for the expected win rate $\mathbb{E}_{u \sim D} \sigma(\beta \cdot (u(x) - u(y)))$.

[Next verbatim source excerpt]

Alignment Setting. The alignment setting generalizes the social choice setting in two ways: first, user utilities $u_i(y | x)$ may depend on a state x ; second, the goal in determining a policy π is not purely to maximize the reward $\text{AvgUtil}(\pi | x)$, but a trade-off between this reward and the goal of remaining close to a reference policy $\pi_{\text{ref}}(\cdot | x) \in \Delta(A)$ in terms of the KL divergence $\text{DKL}(\cdot \| \pi_{\text{ref}})$. For theoretical tractability, we focus our analysis on a single state x , which we from here on omit from the notation. Conceptually, this treatment of alignment on a state-by-state basis corresponds to an assumption that our policy class is expressive enough so that it can take the optimal distribution of actions at each state⁷ and abstracts from the generalization problem of estimating the population's preference between a pair of alternatives at the given state x based on preferences in similar states x_0 . Having set aside the dependency between states, we focus on how the regularization with respect to a reference policy impacts the ability to optimize the average utility of three alignment methods: RLHF, DPO, and NLHF. In addition to the pairwise comparisons, these methods take in a reference policy $\pi_{\text{ref}} \in \Delta(A)$ and a KL bound $\tau \geq 0$, and map these inputs to a policy π in the KL-ball $\mathcal{B}_\tau(\pi_{\text{ref}}) := \{\pi \in \Delta(A) \mid \text{DKL}(\pi \| \pi_{\text{ref}}) \leq \tau\}$ around the reference policy. RLHF, DPO, and NLHF are typically implemented with a KL regularization rather than our KL-constrained formulation, which we adopt to enable a comparison on equal terms. In Section 5, we show that these perspectives are equivalent, and that distortion upper bounds carry over to regularized alignment methods.

4

In Section 5, we discuss how most of our results extend more general distributions over comparison pairs, which can, for example, capture k -wise comparisons.

5

Should we sample the same alternative $x = x_{ji} = y_{ji}$ twice for a pair, the user is not asked for a pairwise comparison. We record this as “ x [U+001F control character] x ”, in a slight abuse of notation.

6

In particular, if we sample the same unordered pair twice for a user, the answers can be perfectly correlated.

7

This assumption is common in the literature; see, for example, Rafailov et al. [43]'s application of first-order optimality conditions of PPO loss minimization.

4

[form-feed in source extraction]

The RLHF method first estimates rewards for each alternative, using maximum likelihood estimation assuming that comparisons were generated by a single Bradley-Terry model:

$$r := \arg\max_{\pi \in \Delta(A)} \sum_{1 \leq i \leq n, 1 \leq j \leq d} \log \sigma(r(x_{ji}) - r(y_{ji})).$$

[U+0001 control character]

Next, RLHF uses PPO [44] to compute the policy $\pi_{\text{RLHF}} := \arg\max_{\pi \in \mathcal{B}_\tau(\pi_{\text{ref}})} \mathbb{E}_{x \sim \pi} [r(x)]$ with maximum expected reward within the KL-ball. In our setup, DPO is equivalent to RLHF (see Appendix F.3) and thus has the same distortion.

The alignment method NLHF was inspired in part by a desire to better align with the preferences of a heterogeneous group [36]. NLHF naturally adapts the definition of maximal lotteries by constraining both players' mixed strategies to the KL-ball, i.e., $\pi_{\text{NLHF}} := \arg\max_{\pi_1 \in \mathcal{B}_\tau(\pi_{\text{ref}}), \pi_2 \in \mathcal{B}_\tau(\pi_{\text{ref}})} \mathbb{E}_{x_1 \sim \pi_1, x_2 \sim \pi_2} [M_{x_1, x_2}]$.

To generalize the definition of distortion to alignment methods, we set the maximum average utility of any policy in the KL-ball as the benchmark. For fixed m, D, β, d, μ and correlation between pairwise comparisons, the distortion of alignment method f is

$$\text{dist}(f) = \sup_{D, \pi_{\text{ref}}, \tau} \lim_{\sup n \rightarrow \infty} \frac{\max_{\pi \in \mathcal{B}_\tau(\pi_{\text{ref}})} \text{AvgUtil}(\pi)}{\text{AvgUtil}(f(\cdot, \pi_{\text{ref}}, \tau))}.$$

[Next verbatim source excerpt]

Theorem 6 (RLHF Distortion Lower Bound). For $m \geq 3$, there is a sequence of alignment problems on which the distortion of RLHF scales as $e\Omega(\beta)$ in β .

Proof of Theorem 6. Suppose that the instance has m alternatives $A = \{a, b, c_1, \dots, c_{m-2}\}$, where $m-2 \geq 4e\beta$. Let the data collection distribution be uniform over all candidates, i.e., $\mu = \text{Uniform}(A)$. We consider the following distribution D over utility vectors, such that the utility vector of a random agent $i \sim D$ satisfies

($u_i(a), u_i(b), u_i(c_1), \dots, u_i(c_{m-2})$) = [U+0011 control character] 1 (type I) with probability δ ,
 $\beta, 0, 1, \dots, 1$ (type II) with probability $1 - \delta$,

where $\delta = 10 + e^{-\beta} = \Theta(e^{-\beta})$. In other words, type I users have a strong preference for candidate b but only constitute a δ fraction of the population, while type II users have a strong preference for c and weak preference for a [U+001F control character] b and make up for a $1 - \delta$ fraction of the population. For the reference policy and the KL budget, we set $\pi_{\text{ref}}(a) = \pi_{\text{ref}}(b) = 1 - \epsilon$ and $\pi_{\text{ref}}(c_i) = \frac{\epsilon}{m-2}$ for

all $i \in [m - 2]$. We leave the choice of ϵ to be determined later. The KL budget τ is set to be $\tau = 1$.

32

[form-feed in source extraction]

Analysis of the MLE reward. Now we show that when $n \rightarrow \infty$, the MLE reward satisfies $r(b) - r(a) > 0$. According to [46, 42], it suffices to show that $\lim_{n \rightarrow \infty} \text{BC}_n(b) - \text{BC}_n(a) > 0$,

[Next verbatim source excerpt]

[form-feed in source extraction]

- For π_{RLHF} , we assume that the estimated reward is shifted such that $r(c) = 0$ (as a result, $r(a) < r(b) < 0$). Since $\pi_0 = (0, 1, 0)$ also satisfies the KL constraint, we have $r(\pi_{\text{RLHF}}) \geq r(\pi_0)$. Together with the fact that $\pi_{\text{RLHF}}(c) \leq \eta$, we have

Expanded PaperInterface specification

```

∀ {beta : ℝ},
  10 ≤ beta →
    ∃ (eta : ℝ) (heta_pos : 0 < eta) (eta_le : eta ≤ 1),
      3 ≤
        Fintype.card
          (GolzHaghtalabYang2025Distortion.theorem6Alternative
            (GolzHaghtalabYang2025Distortion.theorem6CopyCount beta)) ∧
        ∀m (path :
          ℕ →
            AppliedModelingLib.Learning.HumanFeedback.BinaryPairwiseReport
              (GolzHaghtalabYang2025Distortion.theorem6Alternative
                (GolzHaghtalabYang2025Distortion.theorem6CopyCount
                  beta))) →
              AppliedModelingLib.finitePMFIdPathMeasure
                (GolzHaghtalabYang2025Distortion.theorem12OneComparisonReportLaw
                  (AppliedModelingLib.uniformPMF
                    (GolzHaghtalabYang2025Distortion.theorem6Alternative
                      (GolzHaghtalabYang2025Distortion.theorem6CopyCount beta))))
                (GolzHaghtalabYang2025Distortion.theorem6Preference beta)),
        ∀r (horizon : ℕ) in Filter.atTop,
          ∀
            (reward :
              AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset.ScoreVector
                (GolzHaghtalabYang2025Distortion.theorem6Alternative
                  (GolzHaghtalabYang2025Distortion.theorem6CopyCount beta))),
            (AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset.ofBinaryReports
              (AppliedModelingLib.finitePMFIdPathPrefix path horizon)).isPairwiseMLE
              Real.sigmoid GolzHaghtalabYang2025Distortion.theorem6A reward →
            ∃ (rewardPolicy :
              PMF
                (GolzHaghtalabYang2025Distortion.theorem6Alternative
                  (GolzHaghtalabYang2025Distortion.theorem6CopyCount beta)))
              (welfarePolicy :
                PMF
                  (GolzHaghtalabYang2025Distortion.theorem6Alternative
                    (GolzHaghtalabYang2025Distortion.theorem6CopyCount beta))),
            AppliedModelingLib.finiteKLDivergence rewardPolicy
              (GolzHaghtalabYang2025Distortion.theorem6Reference
                (GolzHaghtalabYang2025Distortion.theorem6Epsilon eta)
                (GolzHaghtalabYang2025Distortion.theorem6Epsilon_pos eta)
                (GolzHaghtalabYang2025Distortion.theorem6Epsilon_lt_one heta_pos)
                (GolzHaghtalabYang2025Distortion.theorem6CopyCount_pos beta)) ≤
              1 ∧
            (∀
              (other :
                PMF
                  (GolzHaghtalabYang2025Distortion.theorem6Alternative
                    (GolzHaghtalabYang2025Distortion.theorem6CopyCount beta))),
              AppliedModelingLib.finiteKLDivergence other
                (GolzHaghtalabYang2025Distortion.theorem6Reference
                  (GolzHaghtalabYang2025Distortion.theorem6Epsilon eta)
                  (GolzHaghtalabYang2025Distortion.theorem6Epsilon_pos eta)
                  (GolzHaghtalabYang2025Distortion.theorem6Epsilon_lt_one heta_pos)
                  (GolzHaghtalabYang2025Distortion.theorem6CopyCount_pos beta)) ≤
                  1 →
                AppliedModelingLib.pmfExp other reward ≤ AppliedModelingLib.pmfExp rewardPolicy reward) ∧
            AppliedModelingLib.finiteKLDivergence welfarePolicy
              (GolzHaghtalabYang2025Distortion.theorem6Reference
                (GolzHaghtalabYang2025Distortion.theorem6Epsilon eta)
                (GolzHaghtalabYang2025Distortion.theorem6Epsilon_pos eta)
                (GolzHaghtalabYang2025Distortion.theorem6Epsilon_lt_one heta_pos)
                (GolzHaghtalabYang2025Distortion.theorem6CopyCount_pos beta))

```

```

(GolzHaghtalabYang2025Distortion.theorem6CopyCount_pos beta)) ≤
1 ∧
(∀
  (other :
    PMF
    (GolzHaghtalabYang2025Distortion.theorem6Alternative
      (GolzHaghtalabYang2025Distortion.theorem6CopyCount beta))),
    AppliedModelingLib.finiteKLDivergence other
    (GolzHaghtalabYang2025Distortion.theorem6Reference
      (GolzHaghtalabYang2025Distortion.theorem6Epsilon eta)
      (GolzHaghtalabYang2025Distortion.theorem6Epsilon_pos eta)
      (GolzHaghtalabYang2025Distortion.theorem6Epsilon_lt_one eta_pos)
      (GolzHaghtalabYang2025Distortion.theorem6CopyCount_pos beta)) ≤
    1 →
    AppliedModelingLib.pmfExp other (GolzHaghtalabYang2025Distortion.theorem6Welfare beta) ≤
    AppliedModelingLib.pmfExp welfarePolicy
      (GolzHaghtalabYang2025Distortion.theorem6Welfare beta)) ∧
    Real.exp beta / (44 * beta) ≤
    AppliedModelingLib.pmfExp welfarePolicy
      (GolzHaghtalabYang2025Distortion.theorem6Welfare beta) /
    AppliedModelingLib.pmfExp rewardPolicy
      (GolzHaghtalabYang2025Distortion.theorem6Welfare beta))

```

Retained governing declarations

- [AppliedModelingLib.Learning.HumanFeedback.BinaryPairwiseReport](#)
- [AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset.ScoreVector](#)
- [AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset.isPairwiseMLE](#)
- [AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset.ofBinaryReports](#)
- [AppliedModelingLib.finiteKLDivergence](#)
- [AppliedModelingLib.finitePMFIidPathMeasure](#)
- [AppliedModelingLib.finitePMFIidPathPrefix](#)
- [AppliedModelingLib.pmfExp](#)
- [AppliedModelingLib.uniformPMF](#)
- [GolzHaghtalabYang2025Distortion.theorem12OneComparisonReportLaw](#)
- [GolzHaghtalabYang2025Distortion.theorem6A](#)
- [GolzHaghtalabYang2025Distortion.theorem6Alternative](#)
- [GolzHaghtalabYang2025Distortion.theorem6CopyCount](#)
- [GolzHaghtalabYang2025Distortion.theorem6Epsilon](#)
- [GolzHaghtalabYang2025Distortion.theorem6Preference](#)
- [GolzHaghtalabYang2025Distortion.theorem6Reference](#)
- [GolzHaghtalabYang2025Distortion.theorem6Welfare](#)

Lean-elaborated claim roles

```

1. parameter: ℝ
2. assumption: 10 ≤ beta
3. conclusion: ∃ (eta : ℝ) (heta_pos : 0 < eta) ( _ : eta ≤ 1),
3 ≤
  Fintype.card
    (GolzHaghtalabYang2025Distortion.theorem6Alternative (GolzHaghtalabYang2025Distortion.theorem6CopyCount beta)) ∧
  ∀m (path :
    ℕ →
    AppliedModelingLib.Learning.HumanFeedback.BinaryPairwiseReport
      (GolzHaghtalabYang2025Distortion.theorem6Alternative
        (GolzHaghtalabYang2025Distortion.theorem6CopyCount
          beta))) → AppliedModelingLib.finitePMFIidPathMeasure
      (GolzHaghtalabYang2025Distortion.theorem12OneComparisonReportLaw
        (AppliedModelingLib.uniformPMF
          (GolzHaghtalabYang2025Distortion.theorem6Alternative
            (GolzHaghtalabYang2025Distortion.theorem6CopyCount beta))))
      (GolzHaghtalabYang2025Distortion.theorem6Preference beta)),
  ∀r (horizon : ℕ) in Filter.atTop,
  ∀
    (reward :
      AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset.ScoreVector
        (GolzHaghtalabYang2025Distortion.theorem6Alternative
          (GolzHaghtalabYang2025Distortion.theorem6CopyCount beta))),
    (AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset.ofBinaryReports
      (AppliedModelingLib.finitePMFIidPathPrefix path horizon)).isPairwiseMLE
      Real.sigmoid GolzHaghtalabYang2025Distortion.theorem6A reward →
    ∃ (rewardPolicy :
      PMF
      (GolzHaghtalabYang2025Distortion.theorem6Alternative
        (GolzHaghtalabYang2025Distortion.theorem6CopyCount beta)))

```

```

(welfarePolicy :
PMF
  (GolzhaghtalabYang2025Distortion.theorem6Alternative
    (GolzhaghtalabYang2025Distortion.theorem6CopyCount beta))),
AppliedModelingLib.finiteKLDivergence rewardPolicy
  (GolzhaghtalabYang2025Distortion.theorem6Reference
    (GolzhaghtalabYang2025Distortion.theorem6Epsilon eta)
    (GolzhaghtalabYang2025Distortion.theorem6Epsilon_pos eta)
    (GolzhaghtalabYang2025Distortion.theorem6Epsilon_lt_one heta_pos)
    (GolzhaghtalabYang2025Distortion.theorem6CopyCount_pos beta)) ≤
1 ∧
(∀
  (other :
    PMF
      (GolzhaghtalabYang2025Distortion.theorem6Alternative
        (GolzhaghtalabYang2025Distortion.theorem6CopyCount beta))),
    AppliedModelingLib.finiteKLDivergence other
      (GolzhaghtalabYang2025Distortion.theorem6Reference
        (GolzhaghtalabYang2025Distortion.theorem6Epsilon eta)
        (GolzhaghtalabYang2025Distortion.theorem6Epsilon_pos eta)
        (GolzhaghtalabYang2025Distortion.theorem6Epsilon_lt_one heta_pos)
        (GolzhaghtalabYang2025Distortion.theorem6CopyCount_pos beta)) ≤
1 →
    AppliedModelingLib.pmfExp other reward ≤ AppliedModelingLib.pmfExp rewardPolicy reward) ∧
AppliedModelingLib.finiteKLDivergence welfarePolicy
  (GolzhaghtalabYang2025Distortion.theorem6Reference
    (GolzhaghtalabYang2025Distortion.theorem6Epsilon eta)
    (GolzhaghtalabYang2025Distortion.theorem6Epsilon_pos eta)
    (GolzhaghtalabYang2025Distortion.theorem6Epsilon_lt_one heta_pos)
    (GolzhaghtalabYang2025Distortion.theorem6CopyCount_pos beta)) ≤
1 ∧
(∀
  (other :
    PMF
      (GolzhaghtalabYang2025Distortion.theorem6Alternative
        (GolzhaghtalabYang2025Distortion.theorem6CopyCount beta))),
    AppliedModelingLib.finiteKLDivergence other
      (GolzhaghtalabYang2025Distortion.theorem6Reference
        (GolzhaghtalabYang2025Distortion.theorem6Epsilon eta)
        (GolzhaghtalabYang2025Distortion.theorem6Epsilon_pos eta)
        (GolzhaghtalabYang2025Distortion.theorem6Epsilon_lt_one heta_pos)
        (GolzhaghtalabYang2025Distortion.theorem6CopyCount_pos beta)) ≤
1 →
    AppliedModelingLib.pmfExp other (GolzhaghtalabYang2025Distortion.theorem6Welfare beta) ≤
    AppliedModelingLib.pmfExp welfarePolicy
      (GolzhaghtalabYang2025Distortion.theorem6Welfare beta)) ∧
Real.exp beta / (44 * beta) ≤
AppliedModelingLib.pmfExp welfarePolicy (GolzhaghtalabYang2025Distortion.theorem6Welfare beta) /
AppliedModelingLib.pmfExp rewardPolicy (GolzhaghtalabYang2025Distortion.theorem6Welfare beta)

```

Lean proof endpoint (built separately)

GolzhaghtalabYang2025Distortion.theorem6_rlhf_distortion

Recorded source-to-Spec screening Verdict: matches

Reason: Matches the source-defined asymptotic reading: Theorem 6 asserts a sequence of finite alignment problems with $e^{\Omega(\beta)}$ distortion, and Appendix F.2 itself requires $m - 2 \geq 4 \exp(\beta)$. The expanded Spec makes that sequence explicit through $m(\beta)$, witnesses the asymptotic tail by $\beta \geq 10$, and derives the auxiliary $\exp(\beta) \geq 100$ regime internally; it does not add a fixed- m or extra main-text premise.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 10

Source locator: cited publication, lines 543-558

Verbatim source input

Theorem 7 (NLHF Distortion Upper Bound). For any instance D and any m , data distribution μ , temperature β of the Bradley-Terry model, and any reference policy π_{ref} and KL budget τ , we have

$\text{dist}(\text{NLHF}) \leq \beta^2 \cdot \frac{1+e^{-\beta}}{1-e^{-\beta}}$. In the finite-sample regime, we have

$$\text{AvgUtil}(\text{NLHF}) \geq \beta^2 \cdot \frac{1+e^{-\beta}}{1+e^{-\beta}} \cdot \max_{\pi \in \mathcal{B}_\tau(\pi_{\text{ref}})} \text{AvgUtil}(\pi) - \frac{1}{\beta} \cdot \min\{1, d \cdot \mu_2\} + \frac{n \cdot \beta \mu_2}{\min}$$

[Next verbatim source excerpt]

If $\tau = 0$, the only feasible policy is π_{ref} , and the claim clearly holds for $\lambda = \infty$. We also assume that $\pi_{\text{ref}}(a) > 0$ for all $a \in M$. Otherwise if $\pi_{\text{ref}}(a) = 0$ for some $a \in M$, then both the regularized and constrained versions forbid any policy to put nonzero probability on a , which leads to an effectively smaller candidate set.

[Next verbatim source excerpt]

Let $A = \{1, \dots, m\}$ be a finite set of alternatives. The population of users is described by a probability distribution D over utility vectors $u = (u(1), \dots, u(m))$, whose entries $0 \leq u(x) \leq 1$ indicate a user's utility for alternative x .³ The objective is to find an alternative x such that its average utility $\text{AvgUtil}(x) := \mathbb{E}_{u \sim D} [u(x)]$ across the user population (also known as the utilitarian social welfare) is as high as possible. We extend this notation to probability distributions π over alternatives by setting $\text{AvgUtil}(\pi) := \mathbb{E}_{x \sim \pi} [\text{AvgUtil}(x)]$.

Both voting rules and alignment methods observe comparisons from n users. We model each $i = 1, \dots, n$ as a fresh user with independently drawn utility vector $u_i \sim D$. For exposition, we assume that each user i provides an equal number $d \geq 1$ of pairwise comparisons. For each i , and for $j = 1, \dots, d$, we independently draw alternatives x_{ji} , y_{ji} from a fixed distribution μ over

³ Our assumption that utilities be in $[0, 1]$ is weaker than any of the three assumptions — unit-sum, unit-range, or approval [23] — made in classic distortion to avoid trivial infinite lower bounds.

3

[form-feed in source extraction]

alternatives, in which the minimum probability mass $\mu_{\min} := \min_{x \in A} \mu(x)$ is positive.⁴ User i then compares each pair $\{x_{ji}, y_{ji}\}$ (for $j = 1, \dots, d$) through a Bradley-Terry model based on i 's utilities: i prefers x_{ji} over y_{ji} (written " x_{ji} [U+001F control character] y_{ji} ") with probability $\sigma \beta \cdot (u_i(x_{ji}) - u_i(y_{ji}))$, where [U+0001 control character]

$\sigma(t) := 1/(1+e^{-t})$ is the logistic sigmoid function and $\beta > 0$ is a temperature parameter, and prefers y_{ji} over x_{ji} (" y_{ji} [U+001F control character] x_{ji} ") otherwise.⁵ Whereas this specifies the marginal probability of each pairwise comparison, we make no assumption about the correlation between i 's choices. For example, i might derive the pairwise comparisons from a Plackett-Luce ranking, ensuring that [U+0002 control character] the user's comparisons [U+0001 control character] [U+0003 control character] are always consistent.⁶ We set $p(x$ [U+001F control character] $y)$ for the expected win rate $\mathbb{E}_{u \sim D} \sigma \beta \cdot (u(x) - u(y))$.

[Next verbatim source excerpt]

Alignment Setting. The alignment setting generalizes the social choice setting in two ways: first, user utilities $u_i(y | x)$ may depend on a state x ; second, the goal in determining a policy π is not purely to maximize the reward $\text{AvgUtil}(\pi | x)$, but a trade-off between this reward and the goal of remaining close to a reference policy π_{ref} ($\cdot | x$) in terms of the KL divergence $\text{DKL}(\cdot \| \pi_{\text{ref}})$. For theoretical tractability, we focus our analysis on a single state x , which we from here on omit from the notation. Conceptually, this treatment of alignment on a state-by-state basis corresponds to an assumption that our policy class is expressive enough so that it can take the optimal distribution of actions at each state⁷ and abstracts from the generalization problem of estimating the population's preference between a pair of alternatives at the given state x based on preferences in similar states x_0 . Having set aside the dependency between states, we focus on how the regularization with respect to a reference policy impacts the ability to optimize the average utility of three alignment methods: RLHF, DPO, and NLHF. In addition to the pairwise comparisons, these methods take in a reference policy $\pi_{\text{ref}} \in \Delta(A)$ and a KL bound $\tau \geq 0$, and map these inputs to a policy π in the KL-ball $\mathcal{B}_\tau(\pi_{\text{ref}}) := \{\pi \in \Delta(A) \mid \text{DKL}(\pi \| \pi_{\text{ref}}) \leq \tau\}$ around the reference policy. RLHF, DPO, and NLHF are typically implemented with a KL regularization rather than our KL-constrained formulation, which we adopt to enable a comparison on equal terms. In Section 5, we show that these perspectives are equivalent, and that distortion upper bounds carry over to regularized alignment methods.

⁴ In Section 5, we discuss how most of our results extend more general distributions over comparison pairs, which can, for example, capture k -wise comparisons.

⁵ Should we sample the same alternative $x = x_{ji} = y_{ji}$ twice for a pair, the user is not asked for a pairwise comparison. We record this as " x [U+001F control character] x ", in a slight abuse of notation.

⁶ In particular, if we sample the same unordered pair twice for a user, the answers can be perfectly correlated.

⁷ This assumption is common in the literature; see, for example, Rafailov et al. [43]'s application of first-order optimality conditions of PPO loss minimization.

4

[form-feed in source extraction]

The RLHF method first estimates rewards for each alternative, using maximum likelihood estimation assuming that comparisons were generated by a single Bradley-Terry model:

$$r := \underset{p}{\operatorname{argmax}} \operatorname{Er}_{m \leq i \leq n, 1 \leq j \leq d} \log \frac{\sigma(r(x_{ji}) - r(y_{ij}))}{[U + 0.001 \text{ control character}]}$$

Next, RLHF uses PPO [44] to compute the policy $\pi_{\text{RLHF}} := \operatorname{argmax}_{\pi \in \mathcal{B}\tau} (\pi_{\text{ref}}) \operatorname{Ex} \sim \pi [r(x)]$ with maximum expected reward within the KL-ball. In our setup, DPO is equivalent to RLHF (see Appendix F.3) and thus has the same distortion.

The alignment method NLHF was inspired in part by a desire to better align with the preferences of a heterogeneous group [36]. NLHF naturally adapts the definition of maximal lotteries by constraining both players' mixed strategies to the KL-ball, i.e., $\pi_{\text{NLHF}} := \operatorname{argmax}_{\pi_1 \in \mathcal{B}\tau} (\pi_{\text{ref}}) \min_{\pi_2 \in \mathcal{B}\tau} (\pi_{\text{ref}}) \operatorname{Ex}_1 \sim \pi_1, x_2 \sim \pi_2 [Mx_1, x_2]$.

To generalize the definition of distortion to alignment methods, we set the maximum average utility of any policy in the KL-ball as the benchmark. For fixed m, D, β, d, μ and correlation between pairwise comparisons, the distortion of alignment method f is

$$\operatorname{dist}(f) = \sup_{D, \pi_{\text{ref}}, \tau} \limsup_{n \rightarrow \infty} \frac{\max_{\pi \in \mathcal{B}\tau} (\pi_{\text{ref}}) \operatorname{AvgUtil}(\pi)}{\operatorname{AvgUtil}(f(\cdot, \pi_{\text{ref}}, \tau))}.$$

Expanded PaperInterface specification

```
(∀ {User Alternative : Type} [inst : Fintype User] [inst_1 : DecidableEq User] [inst_2 : Fintype Alternative]
[inst_3 : DecidableEq Alternative] (population : PMF User)
(utility : AppliedModelingLib.Alignment.Welfare.FiniteUtilityProfile User Alternative),
AppliedModelingLib.Alignment.Welfare.UnitIntervalUtilityProfile utility →
  ∀ {btScale : ℝ},
  0 < btScale →
    ∀ (reference : PMF Alternative) (klBudget : ℝ) (nlhfPolicy benchmark : PMF Alternative),
    0 ≤ klBudget →
      AppliedModelingLib.PMFFullSupport reference →
        AppliedModelingLib.Alignment.Welfare.IsConstrainedPopulationBradleyTerryMaximin population utility
        btScale reference klBudget nlhfPolicy →
          AppliedModelingLib.Alignment.Welfare.InFiniteKLBall reference klBudget benchmark →
            AppliedModelingLib.Alignment.Welfare.sigmoidChordSlope btScale / (1 / 4) *
              AppliedModelingLib.Alignment.Welfare.policyAverageUtility population utility benchmark ≤
                AppliedModelingLib.Alignment.Welfare.policyAverageUtility population utility nlhfPolicy) ∧
  ∀ {User Alternative : Type} [inst : Fintype User] [inst_1 : DecidableEq User] [inst_2 : Fintype Alternative]
[inst_3 : DecidableEq Alternative] [Nonempty Alternative] {users comparisonsPerUser : ℕ} (population : PMF User)
(utility : AppliedModelingLib.Alignment.Welfare.FiniteUtilityProfile User Alternative),
AppliedModelingLib.Alignment.Welfare.UnitIntervalUtilityProfile utility →
  ∀ {btScale : ℝ},
  0 < btScale →
    ∀ (responseLaw : PMF (GolzHaghtalabYang2025Distortion.theorem2UserResponseTable Alternative))
    (preference : AppliedModelingLib.Learning.HumanFeedback.PairwisePreference PUnit.{1} Alternative),
    GolzHaghtalabYang2025Distortion.theorem2UserResponseCalibrated responseLaw preference →
      ∀ (sampling : PMF Alternative),
      0 < users →
        0 < comparisonsPerUser →
          ∀ (minimumMass : ℝ),
          0 < minimumMass →
            (∀ (alternative : Alternative), minimumMass ≤ (sampling alternative).toReal) →
              ∀ (reference : PMF Alternative) (klBudget : ℝ) (hbudget : 0 ≤ klBudget),
                AppliedModelingLib.PMFFullSupport reference →
                  ∀ (delta : ℝ),
                  0 < delta →
                    delta ≤ 1 →
                      1 (Fintype.card Alternative) ^ 2 *
                        Real.exp (-↑(users * comparisonsPerUser) * minimumMass ^ 2 / 8) ≤
                          delta →
                            preference =
                              AppliedModelingLib.Alignment.Welfare.populationBradleyTerryPreference
                              population utility btScale →
                                have failureBudget : ℝ := delta / ↑(Fintype.card Alternative) ^ 2;
                                ∃ (benchmark : PMF Alternative),
                                  AppliedModelingLib.finiteKLDivergence benchmark reference ≤ klBudget ∧
                                    (∀ (other : PMF Alternative),
                                      AppliedModelingLib.finiteKLDivergence other reference ≤ klBudget →
                                        AppliedModelingLib.Alignment.Welfare.policyAverageUtility population
                                        utility other ≤
                                          AppliedModelingLib.Alignment.Welfare.policyAverageUtility population
                                          utility benchmark) ∧
                                      (1 - delta) *
                                        (AppliedModelingLib.Alignment.Welfare.sigmoidChordSlope btScale /
                                          (1 / 4) *
                                            AppliedModelingLib.Alignment.Welfare.policyAverageUtility
                                            population utility benchmark -
                                              GolzHaghtalabYang2025Distortion.theorem10SourceRate users
                                              comparisonsPerUser minimumMass failureBudget /
                                              (btScale * (1 / 4))) ≤
                                                AppliedModelingLib.pmfExp
                                                (GolzHaghtalabYang2025Distortion.theorem7lidUserBatchLaw responseLaw
                                                  sampling users comparisonsPerUser)
                                      fun
                                        (sample :
                                          Fin users →
                                            GolzHaghtalabYang2025Distortion.theorem2UserResponseTable
                                              Alternative ×
                                              (Fin comparisonsPerUser → Alternative × Alternative)) =>
                                          AppliedModelingLib.Alignment.Welfare.policyAverageUtility population
                                          utility
                                          (GolzHaghtalabYang2025Distortion.theorem7lidUserEmpiricalPolicy
                                            reference klBudget hbudget sample))
```

Retained governing declarations

- [AppliedModelingLib.Alignment.Welfare.FiniteUtilityProfile](#)

- [AppliedModelingLib.Alignment.Welfare.InFiniteKLBall](#)
- [AppliedModelingLib.Alignment.Welfare.IsConstrainedPopulationBradleyTerryMaximin](#)
- [AppliedModelingLib.Alignment.Welfare.UnitIntervalUtilityProfile](#)
- [AppliedModelingLib.Alignment.Welfare.policyAverageUtility](#)
- [AppliedModelingLib.Alignment.Welfare.populationBradleyTerryPreference](#)
- [AppliedModelingLib.Alignment.Welfare.sigmoidChordSlope](#)
- [AppliedModelingLib.Learning.HumanFeedback.PairwisePreference](#)
- [AppliedModelingLib.PMFFullSupport](#)
- [AppliedModelingLib.finiteKLDivergence](#)
- [AppliedModelingLib.pmfExp](#)
- [GolzHaghtalabYang2025Distortion.theorem10SourceRate](#)
- [GolzHaghtalabYang2025Distortion.theorem2UserResponseCalibrated](#)
- [GolzHaghtalabYang2025Distortion.theorem2UserResponseTable](#)
- [GolzHaghtalabYang2025Distortion.theorem7IidUserBatchLaw](#)
- [GolzHaghtalabYang2025Distortion.theorem7IidUserEmpiricalPolicy](#)

Lean-elaborated claim roles

```

1. conclusion: (∀ {User Alternative : Type} [inst : Fintype User] [inst_1 : DecidableEq User] [inst_2 : Fintype Alternative]
[inst_3 : DecidableEq Alternative] (population : PMF User)
(utility : AppliedModelingLib.Alignment.Welfare.FiniteUtilityProfile User Alternative),
AppliedModelingLib.Alignment.Welfare.UnitIntervalUtilityProfile utility →
  ∀ {btScale : ℝ},
  0 < btScale →
    ∀ (reference : PMF Alternative) (klBudget : ℝ) (nlhfPolicy benchmark : PMF Alternative),
    0 ≤ klBudget →
      AppliedModelingLib.PMFFullSupport reference →
        AppliedModelingLib.Alignment.Welfare.IsConstrainedPopulationBradleyTerryMaximin population utility
        btScale reference klBudget nlhfPolicy →
          AppliedModelingLib.Alignment.Welfare.InFiniteKLBall reference klBudget benchmark →
            AppliedModelingLib.Alignment.Welfare.sigmoidChordSlope btScale / (1 / 4) *
            AppliedModelingLib.Alignment.Welfare.policyAverageUtility population utility benchmark ≤
            AppliedModelingLib.Alignment.Welfare.policyAverageUtility population utility nlhfPolicy) ∧
  ∀ {User Alternative : Type} [inst : Fintype User] [inst_1 : DecidableEq User] [inst_2 : Fintype Alternative]
[inst_3 : DecidableEq Alternative] [Nonempty Alternative] {users comparisonsPerUser : ℕ} (population : PMF User)
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AppliedModelingLib.Alignment.Welfare.UnitIntervalUtilityProfile utility →
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  0 < btScale →
    ∀ (responseLaw : PMF (GolzHaghtalabYang2025Distortion.theorem2UserResponseTable Alternative))
(preference : AppliedModelingLib.Learning.HumanFeedback.PairwisePreference PUnit.{1} Alternative),
GolzHaghtalabYang2025Distortion.theorem2UserResponseCalibrated responseLaw preference →
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      0 < minimumMass →
        (∀ (alternative : Alternative), minimumMass ≤ (sampling alternative).toReal) →
          ∀ (reference : PMF Alternative) (klBudget : ℝ) (hbudget : 0 ≤ klBudget),
          AppliedModelingLib.PMFFullSupport reference →
            ∀ (delta : ℝ),
            0 < delta →
              delta ≤ 1 →
                ↑(Fintype.card Alternative) ^ 2 *
                Real.exp (-↑(users * comparisonsPerUser) * minimumMass ^ 2 / 8) ≤
                delta →
                  preference =
                    AppliedModelingLib.Alignment.Welfare.populationBradleyTerryPreference
                    population utility btScale →
                      have failureBudget : ℝ := delta / ↑(Fintype.card Alternative) ^ 2;
                      ∃ (benchmark : PMF Alternative),
                        AppliedModelingLib.finiteKLDivergence benchmark reference ≤ klBudget ∧
                        (∀ (other : PMF Alternative),
                          AppliedModelingLib.finiteKLDivergence other reference ≤ klBudget →
                            AppliedModelingLib.Alignment.Welfare.policyAverageUtility population
                            utility other ≤
                            AppliedModelingLib.Alignment.Welfare.policyAverageUtility population
                            utility benchmark) ∧
                        (1 - delta) *
                        (AppliedModelingLib.Alignment.Welfare.sigmoidChordSlope btScale /
                        (1 / 4) *
                        AppliedModelingLib.Alignment.Welfare.policyAverageUtility
                        population utility benchmark -
                        GolzHaghtalabYang2025Distortion.theorem10SourceRate users
                        comparisonsPerUser minimumMass failureBudget /
                        (btScale * (1 / 4))) ≤
                        AppliedModelingLib.pmfExp
                        (GolzHaghtalabYang2025Distortion.theorem7IidUserBatchLaw responseLaw
                        sampling users comparisonsPerUser)
                        fun

```

```

(sample :
  Fin users →
    GolzHaghtalabYang2025Distortion.theorem2UserResponseTable
      Alternative ×
      (Fin comparisonsPerUser → Alternative × Alternative)) =>
AppliedModelingLib.Alignment.Welfare.policyAverageUtility population
utility
(GolzHaghtalabYang2025Distortion.theorem7lidUserEmpiricalPolicy
reference klBudget hbudget sample)

```

Lean proof endpoint (built separately)

GolzHaghtalabYang2025Distortion.theorem7_nlhf_distortion

Recorded source-to-Spec screening Verdict: matches

Reason: The constrained-NLHF population factor and finite-sample expected-welfare guarantee over the KL ball are retained.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 11

Source locator: cited publication, lines 444-449

Verbatim source input

Corollary 4 (of Theorem 7). The Maximal Lotteries voting rule has a distortion of $\beta 2 \cdot 1 - e^{1+\beta}$.

The Borda rule, in contrast, does not match this optimal distortion bound, as shown by the following bound that we prove and state in full detail in Appendix E.2.

[Next verbatim source excerpt]

Let $A = \{1, \dots, m\}$ be a finite set of alternatives. The population of users is described by a probability distribution D over utility vectors $u = (u(1), \dots, u(m))$, whose entries $0 \leq u(x) \leq 1$ indicate a user's utility for alternative x .³ The objective is to find an alternative x such that its average utility $\text{AvgUtil}(x) := \mathbb{E}_{u \sim D} [u(x)]$ across the user population (also known as the utilitarian social welfare) is as high as possible. We extend this notation to probability distributions π over alternatives by setting $\text{AvgUtil}(\pi) := \mathbb{E}_{x \sim \pi} [\text{AvgUtil}(x)]$. Both voting rules and alignment methods observe comparisons from n users. We model each $i = 1, \dots, n$ as a fresh user with independently drawn utility vector $u_i \sim D$. For exposition, we assume that each user i provides an equal number $d \geq 1$ of pairwise comparisons. For each i , and for $j = 1, \dots, d$, we independently draw alternatives x_{ji}, y_{ji} from a fixed distribution μ over

³ Our assumption that utilities be in $[0, 1]$ is weaker than any of the three assumptions — unit-sum, unit-range, or approval [23] — made in classic distortion to avoid trivial infinite lower bounds.

3

[form-feed in source extraction]

alternatives, in which the minimum probability mass $\mu_{\min} := \min_{x \in A} \mu(x)$ is positive.⁴ User i then compares each pair $\{x_{ji}, y_{ji}\}$ (for $j = 1, \dots, d$) through a Bradley-Terry model based on i 's utilities: i prefers x_{ji} over y_{ji} (written “ x_{ji} [U+001F control character] y_{ji} ”) with probability $\sigma(\beta \cdot (u_i(x_{ji}) - u_i(y_{ji})))$, where

$\sigma(t) := 1/(1+e^{-t})$ is the logistic sigmoid function and $\beta > 0$ is a temperature parameter, and prefers y_{ji} over x_{ji} (“ y_{ji} [U+001F control character] x_{ji} ”) otherwise.⁵ Whereas this specifies the marginal probability of each pairwise comparison, we make no assumption about the correlation between i 's choices. For example, i might derive the pairwise comparisons from a Plackett-Luce ranking, ensuring that [U+0002 control character] the user's comparisons [U+0001 control character] [U+0003 control character] are always consistent.⁶ We set $p(x [U+001F control character] y)$ for the expected win rate $\mathbb{E}_{u \sim D} \sigma(\beta \cdot (u(x) - u(y)))$.

[Next verbatim source excerpt]

Social Choice Setting. A voting rule f observes the sampled pairwise comparisons $\{x_{ji} [U+001F control character] y_{ji} \mid i \in [n], j \in [d]\}$ and maps them to a probability distribution over alternatives. For some m, D, β, d, μ , and [U+0002 control character] the correlation between comparisons, the average utility of f for n samples is $\text{AvgUtil}_n(f) := \mathbb{E} \text{AvgUtil}(f(\{x_{ji} [U+001F control character] y_{ji} \mid i, j\}))$, where the expectation is taken over the pairwise comparisons. The [U+0001 control character][U+0003 control character]

distortion of f on D is the competitive ratio between $\text{AvgUtil}_n(f)$ and the optimal average utility $\max_{x \in A} \text{AvgUtil}(x)$ in the limit of $n \rightarrow \infty$ samples, and the distortion of f the worst-case distortion over all D :

$$\text{dist}(f, D) := \limsup_{n \rightarrow \infty} \max_{x \in A} \frac{\text{AvgUtil}_n(f)}{\text{AvgUtil}(x)}, \quad \text{dist}(f) = \sup_D \text{dist}(f, D).$$

For alternatives x, y , let $\#(x [U+001F control character] y) := \{(i, j) \mid x_{ji} = x, y_{ji} = y\}$ denote the number of pairwise comparisons in which x beat y . The (normalized) Borda score [45] of alternative x is

$$\text{BC}(x) := \frac{1}{P} \sum_{y \in A} \#(x [U+001F control character] y) - \frac{1}{P} \sum_{y \in A} \#(y [U+001F control character] x)$$

i.e., the fraction of pairwise comparisons involving x in which it wins. The Borda voting rule chooses the winner uniformly among all alternatives with maximum Borda score. The Maximal Lotteries voting rule first computes the margin matrix $M \in \mathbb{R}^{m \times m}$, where $M_{x,y} = \#(x [U+001F control character] y) - \#(y [U+001F control character] x)$. It then considers a symmetric two-player zero-sum game in which player 1 selects alternative x_1 , player 2 selects alternative x_2 , and the payoffs are M_{x_1, x_2} for player 1 and $M_{x_2, x_1} = -M_{x_1, x_2}$ for player 2. The maximal lotteries rule returns a distribution $\pi \in \arg\max_{\pi \in \Delta(A)} \min_{\pi_2 \in \Delta(A)} \mathbb{E}_{x_1 \sim \pi_1, x_2 \sim \pi_2} [M_{x_1, x_2}]$, i.e., a mixed strategy in Nash equilibrium. When several such π exist, our results hold for any choice.

Expanded PaperInterface specification

```
(∀ {User Alternative : Type} [inst : Fintype User] [inst_1 : DecidableEq User] [inst_2 : Fintype Alternative]
 [inst_3 : DecidableEq Alternative] (population : PMF User)
 (utility : AppliedModelingLib.Alignment.Welfare.FiniteUtilityProfile User Alternative),
 AppliedModelingLib.Alignment.Welfare.UnitIntervalUtilityProfile utility →
  ∀ {btScale : ℝ},
  0 < btScale →
  ∀ (maximalLottery benchmark : PMF Alternative),
  (∀ (opponent : PMF Alternative),
   0 ≤
```

```

AppliedModelingLib.Alignment.Welfare.populationBradleyTerryPolicyMargin population utility btScale
maximalLottery opponent) →
AppliedModelingLib.Alignment.Welfare.sigmoidChordSlope btScale / (1 / 4) *
AppliedModelingLib.Alignment.Welfare.policyAverageUtility population utility benchmark ≤
AppliedModelingLib.Alignment.Welfare.policyAverageUtility population utility maximalLottery) ∧
∀ {User Alternative : Type} [inst : Fintype User] [inst_1 : DecidableEq User] [inst_2 : Fintype Alternative]
[inst_3 : DecidableEq Alternative] [inst_4 : Nonempty Alternative] {users comparisonsPerUser : ℕ}
(population : PMF User) (utility : AppliedModelingLib.Alignment.Welfare.FiniteUtilityProfile User Alternative),
AppliedModelingLib.Alignment.Welfare.UnitIntervalUtilityProfile utility →
  ∀ {btScale : ℝ},
  0 < btScale →
  ∀ (responseLaw : PMF (GolzHaghtalabYang2025Distortion.theorem2UserResponseTable Alternative))
  (preference : AppliedModelingLib.Learning.HumanFeedback.PairwisePreference PUnit.{1} Alternative),
  GolzHaghtalabYang2025Distortion.theorem2UserResponseCalibrated responseLaw preference →
  ∀ (sampling : PMF Alternative),
  0 < users →
  0 < comparisonsPerUser →
  ∀ (minimumMass : ℝ),
  0 < minimumMass →
  (∀ (alternative : Alternative), minimumMass ≤ (sampling alternative).toReal) →
  ∀ (delta : ℝ),
  0 < delta →
  delta ≤ 1 →
  ↑(Fintype.card Alternative) ^ 2 *
  Real.exp (-↑(users * comparisonsPerUser) * minimumMass ^ 2 / 8) ≤
  delta →
  preference =
  AppliedModelingLib.Alignment.Welfare.populationBradleyTerryPreference population
  utility btScale →
  (∀
  (sample :
  Fin users →
  GolzHaghtalabYang2025Distortion.theorem2UserResponseTable Alternative ×
  (Fin comparisonsPerUser → Alternative × Alternative)),
  GolzHaghtalabYang2025Distortion.IsEmpiricalMaximalLottery
  (fun (first second : Alternative) =>
  GolzHaghtalabYang2025Distortion.theorem10IidUserPaperWinRate first second
  sample)
  (GolzHaghtalabYang2025Distortion.corollary4IidUserEmpiricalMaximalLottery
  sample)) ∧
  have failureBudget : ℝ := delta / ↑(Fintype.card Alternative) ^ 2;
  ∃ (benchmark : PMF Alternative),
  (∀ (other : PMF Alternative),
  AppliedModelingLib.Alignment.Welfare.policyAverageUtility population utility
  other ≤
  AppliedModelingLib.Alignment.Welfare.policyAverageUtility population
  utility benchmark) ∧
  (1 - delta) *
  (AppliedModelingLib.Alignment.Welfare.sigmoidChordSlope btScale /
  (1 / 4) *
  AppliedModelingLib.Alignment.Welfare.policyAverageUtility population
  utility benchmark -
  GolzHaghtalabYang2025Distortion.theorem10SourceRate users
  comparisonsPerUser minimumMass failureBudget /
  (btScale * (1 / 4))) ≤
  AppliedModelingLib.pmfExp
  (GolzHaghtalabYang2025Distortion.theorem7IidUserBatchLaw responseLaw
  sampling users comparisonsPerUser)
  fun
  (sample :
  Fin users →
  GolzHaghtalabYang2025Distortion.theorem2UserResponseTable
  Alternative ×
  (Fin comparisonsPerUser → Alternative × Alternative)) =>
  AppliedModelingLib.Alignment.Welfare.policyAverageUtility population
  utility
  (GolzHaghtalabYang2025Distortion.corollary4IidUserEmpiricalMaximalLottery
  sample)

```

Retained governing declarations

- [AppliedModelingLib.Alignment.Welfare.FiniteUtilityProfile](#)
- [AppliedModelingLib.Alignment.Welfare.UnitIntervalUtilityProfile](#)
- [AppliedModelingLib.Alignment.Welfare.policyAverageUtility](#)
- [AppliedModelingLib.Alignment.Welfare.populationBradleyTerryPolicyMargin](#)
- [AppliedModelingLib.Alignment.Welfare.populationBradleyTerryPreference](#)
- [AppliedModelingLib.Alignment.Welfare.sigmoidChordSlope](#)
- [AppliedModelingLib.Learning.HumanFeedback.PairwisePreference](#)
- [AppliedModelingLib.pmfExp](#)
- [GolzHaghtalabYang2025Distortion.IsEmpiricalMaximalLottery](#)
- [GolzHaghtalabYang2025Distortion.corollary4IidUserEmpiricalMaximalLottery](#)
- [GolzHaghtalabYang2025Distortion.theorem10IidUserPaperWinRate](#)
- [GolzHaghtalabYang2025Distortion.theorem10SourceRate](#)

- [GolzHaghtalabYang2025Distortion.theorem2UserResponseCalibrated](#)
- [GolzHaghtalabYang2025Distortion.theorem2UserResponseTable](#)
- [GolzHaghtalabYang2025Distortion.theorem7IidUserBatchLaw](#)

Lean-elaborated claim roles

```

1. conclusion: (∀ {User Alternative : Type} [inst : Fintype User] [inst_1 : DecidableEq User] [inst_2 : Fintype Alternative]
[inst_3 : DecidableEq Alternative] (population : PMF User)
(utility : AppliedModelingLib.Alignment.Welfare.FiniteUtilityProfile User Alternative),
AppliedModelingLib.Alignment.Welfare.UnitIntervalUtilityProfile utility →
  ∀ {btScale : ℝ},
  0 < btScale →
    ∀ (maximalLottery benchmark : PMF Alternative),
    (∀ (opponent : PMF Alternative),
      0 ≤
        AppliedModelingLib.Alignment.Welfare.populationBradleyTerryPolicyMargin population utility btScale
        maximalLottery opponent) →
      AppliedModelingLib.Alignment.Welfare.sigmoidChordSlope btScale / (1 / 4) *
        AppliedModelingLib.Alignment.Welfare.policyAverageUtility population utility benchmark ≤
        AppliedModelingLib.Alignment.Welfare.policyAverageUtility population utility maximalLottery) ∧
  ∀ {User Alternative : Type} [inst : Fintype User] [inst_1 : DecidableEq User] [inst_2 : Fintype Alternative]
[inst_3 : DecidableEq Alternative] [inst_4 : Nonempty Alternative] {users comparisonsPerUser : ℕ}
(population : PMF User) (utility : AppliedModelingLib.Alignment.Welfare.FiniteUtilityProfile User Alternative),
AppliedModelingLib.Alignment.Welfare.UnitIntervalUtilityProfile utility →
  ∀ {btScale : ℝ},
  0 < btScale →
    ∀ (responseLaw : PMF (GolzHaghtalabYang2025Distortion.theorem2UserResponseTable Alternative))
    (preference : AppliedModelingLib.Learning.HumanFeedback.PairwisePreference PUnit.{1} Alternative),
    GolzHaghtalabYang2025Distortion.theorem2UserResponseCalibrated responseLaw preference →
      ∀ (sampling : PMF Alternative),
      0 < users →
        0 < comparisonsPerUser →
          ∀ (minimumMass : ℝ),
          0 < minimumMass →
            (∀ (alternative : Alternative), minimumMass ≤ (sampling alternative).toReal) →
              ∀ (delta : ℝ),
              0 < delta →
                delta ≤ 1 →
                  ↑(Fintype.card Alternative) ^ 2 *
                    Real.exp (-↑(users * comparisonsPerUser) * minimumMass ^ 2 / 8) ≤
                      delta →
                        preference =
                          AppliedModelingLib.Alignment.Welfare.populationBradleyTerryPreference population
                          utility btScale →
                            (∀
                              (sample :
                                Fin users →
                                  GolzHaghtalabYang2025Distortion.theorem2UserResponseTable Alternative ×
                                  (Fin comparisonsPerUser → Alternative × Alternative)),
                                GolzHaghtalabYang2025Distortion.IsEmpiricalMaximalLottery
                                (fun (first second : Alternative) =>
                                  GolzHaghtalabYang2025Distortion.theorem10IidUserPaperWinRate first second
                                  sample)
                                (GolzHaghtalabYang2025Distortion.corollary4IidUserEmpiricalMaximalLottery
                                  sample)) ∧
                                have failureBudget : ℝ := delta / ↑(Fintype.card Alternative) ^ 2;
                                ∃ (benchmark : PMF Alternative),
                                (∀ (other : PMF Alternative),
                                  AppliedModelingLib.Alignment.Welfare.policyAverageUtility population utility
                                  other ≤
                                  AppliedModelingLib.Alignment.Welfare.policyAverageUtility population
                                  utility benchmark) ∧
                                  (1 - delta) *
                                  (AppliedModelingLib.Alignment.Welfare.sigmoidChordSlope btScale /
                                    (1 / 4) *
                                    AppliedModelingLib.Alignment.Welfare.policyAverageUtility population
                                    utility benchmark -
                                    GolzHaghtalabYang2025Distortion.theorem10SourceRate users
                                    comparisonsPerUser minimumMass failureBudget /
                                    (btScale * (1 / 4))) ≤
                                    AppliedModelingLib.pmfExp
                                    (GolzHaghtalabYang2025Distortion.theorem7IidUserBatchLaw responseLaw
                                    sampling users comparisonsPerUser)
                                fun
                                  (sample :
                                    Fin users →
                                      GolzHaghtalabYang2025Distortion.theorem2UserResponseTable
                                      Alternative ×
                                      (Fin comparisonsPerUser → Alternative × Alternative)) =>
                                    AppliedModelingLib.Alignment.Welfare.policyAverageUtility population
                                    utility
                                    (GolzHaghtalabYang2025Distortion.corollary4IidUserEmpiricalMaximalLottery
                                      sample)

```

Lean proof endpoint (built separately)

GolzHaghtalabYang2025Distortion.corollary4_maximal_lottery

Recorded source-to-Spec screening Verdict: matches

Reason: The Maximal-Lotteries factor and finite expected-welfare route are retained. Positive-count margins are the

printed ratios; the explicitly disclosed neutral zero margin totalizes only unobserved off-diagonal pairs and adds no stochastic premise.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 12

Source locator: cited publication, lines 2166-2178

Verbatim source input

Proposition 13 (Equivalence between Constrained and Regularized NLHF). Let π be the output of the τ -constrained NLHF method defined in Section 2, i.e.,

$$\pi = \operatorname{argmax}_{\pi_1 \in B_\tau(\pi_{\text{ref}})} \min_{\pi_2 \in B_\tau(\pi_{\text{ref}})} E_{x_1 \sim \pi_1, x_2 \sim \pi_2} [M_{x_1, x_2}], \quad (14)$$

and let $\tilde{\pi}$ be the output of the λ -regularized NLHF method [36], i.e.,

$$\tilde{\pi} = \operatorname{argmax}_{\pi_1 \in \Delta(M)} \min_{\pi_2 \in \Delta(M)} E_{x_1 \sim \pi_1, x_2 \sim \pi_2} [M_{x_1, x_2}] - \lambda \cdot \text{DKL}(\pi_1 \parallel \pi_{\text{ref}}) + \lambda \cdot \text{DKL}(\pi_2 \parallel \pi_{\text{ref}}). \quad (15)$$

Then, for each $\lambda \in [0, \infty]$ with solution $\tilde{\pi}$ for Eq. (15), we have that $\tilde{\pi}$ is also an optimal solution to the τ -constrained optimization problem in Eq. (14), where $\tau = \text{DKL}(\tilde{\pi} \parallel \pi_{\text{ref}})$. Conversely, for each $\tau \geq 0$ with solution π for Eq. (14), there exists $\lambda \in [0, \infty]$ such that π is also an optimal solution to the λ -regularized optimization problem in Eq. (15).

[Next verbatim source excerpt]

Let $A = \{1, \dots, m\}$ be a finite set of alternatives. The population of users is described by a probability distribution D over utility vectors $u = (u(1), \dots, u(m))$, whose entries $0 \leq u(x) \leq 1$ indicate a user's utility for alternative x .³ The objective is to find an alternative x such that its average utility $\text{AvgUtil}(x) := E_{u \sim D} [u(x)]$ across the user population (also known as the utilitarian social welfare) is as high as possible. We extend this notation to probability distributions π over alternatives by setting $\text{AvgUtil}(\pi) := E_{x \sim \pi} [\text{AvgUtil}(x)]$.

Both voting rules and alignment methods observe comparisons from n users. We model each $i = 1, \dots, n$ as a fresh user with independently drawn utility vector $u_i \sim D$. For exposition, we assume that each user i provides an equal number $d \geq 1$ of pairwise comparisons. For each i , and for $j = 1, \dots, d$, we independently draw alternatives x_{ji}, y_{ji} from a fixed distribution μ over

3

Our assumption that utilities be in $[0, 1]$ is weaker than any of the three assumptions — unit-sum, unit-range, or approval [23] — made in classic distortion to avoid trivial infinite lower bounds.

3

[form-feed in source extraction]

alternatives, in which the minimum probability mass $\mu_{\min} := \min_{x \in A} \mu(x)$ is positive.⁴ User i then compares each pair $\{x_{ji}, y_{ji}\}$ (for $j = 1, \dots, d$) through a Bradley-Terry model based on i 's utilities: i prefers x_{ji} over y_{ji} (written “ x_{ji} [U+001F control character] y_{ji} ”) with probability $\sigma(\beta \cdot (u_i(x_{ji}) - u_i(y_{ji})))$, where [U+0001 control character]

$\sigma(t) := 1/(1+e^{-t})$ is the logistic sigmoid function and $\beta > 0$ is a temperature parameter, and prefers y_{ji} over x_{ji} (“ y_{ji} [U+001F control character] x_{ji} ”) otherwise.⁵ Whereas this specifies the marginal probability of each pairwise comparison, we make no assumption about the correlation between i 's choices. For example, i might derive the pairwise comparisons from a Plackett-Luce ranking, ensuring that [U+0002 control character] the user's comparisons [U+0001 control character] [U+0003 control character] are always consistent.⁶ We set $p(x$ [U+001F control character] $y)$ for the expected win rate $E_{u \sim D} \sigma(\beta \cdot (u(x) - u(y)))$.

[Next verbatim source excerpt]

Alignment Setting. The alignment setting generalizes the social choice setting in two ways: first, user utilities $u_i(y \mid x)$ may depend on a state x ; second, the goal in determining a policy π is not purely to maximize the reward $\text{AvgUtil}(\pi \mid x)$, but a trade-off between this reward and the goal of remaining close to a reference policy $\pi_{\text{ref}}(\cdot \mid x) \in \Delta(A)$ in terms of the KL divergence $\text{DKL}(\pi \parallel \pi_{\text{ref}})$. For theoretical tractability, we focus our analysis on a single state x , which we from here on omit from the notation. Conceptually, this treatment of alignment on a state-by-state basis corresponds to an assumption that our policy class is expressive enough so that it can take the optimal distribution of actions at each state⁷ and abstracts from the generalization problem of estimating the population's preference between a pair of alternatives at the given state x based on preferences in similar states x_0 . Having set aside the dependency between states, we focus on how the regularization with respect to a reference policy impacts the ability to optimize the average utility of three alignment methods: RLHF, DPO, and NLHF. In addition to the pairwise comparisons, these methods take in a reference policy $\pi_{\text{ref}} \in \Delta(A)$ and a KL bound $\tau \geq 0$, and map these inputs to a policy π in the KL-ball $B_\tau(\pi_{\text{ref}}) := \{\pi \in \Delta(A) \mid \text{DKL}(\pi \parallel \pi_{\text{ref}}) \leq \tau\}$ around the reference policy. RLHF, DPO, and NLHF are typically implemented with a KL regularization rather than our KL-constrained formulation, which we adopt to enable a comparison on equal terms. In Section 5, we show that these perspectives are equivalent, and that distortion upper bounds carry over to regularized alignment methods.

4

In Section 5, we discuss how most of our results extend more general distributions over comparison pairs, which can, for example, capture k -wise comparisons.

5

Should we sample the same alternative $x = x_{ji} = y_{ji}$ twice for a pair, the user is not asked for a pairwise comparison. We record this as “ x [U+001F control character] x ”, in a slight abuse of notation.

6

In particular, if we sample the same unordered pair twice for a user, the answers can be perfectly correlated.

7

This assumption is common in the literature; see, for example, Rafailov et al. [43]'s application of first-order optimality conditions of PPO loss minimization.

4

[form-feed in source extraction]

The RLHF method first estimates rewards for each alternative, using maximum likelihood estimation assuming that comparisons were generated by a single Bradley-Terry model:

$$r := \operatorname{argmax}_{\pi} \sum_{1 \leq i \leq n, 1 \leq j \leq d} \log \sigma(r(x_{ji}) - r(y_{ji})). \quad [U+0001 \text{ control character}]$$

Next, RLHF uses PPO [44] to compute the policy $\pi_{\text{RLHF}} := \operatorname{argmax}_{\pi \in B_\tau(\pi_{\text{ref}})} E_{x \sim \pi} [r(x)]$ with maximum expected reward within the KL-ball. In our setup, DPO is equivalent to RLHF (see Appendix F.3) and thus has the same distortion.

The alignment method NLHF was inspired in part by a desire to better align with the preferences of a heterogeneous group [36]. NLHF naturally adapts the definition of maximal lotteries by constraining both players' mixed strategies to the KL-ball, i.e., $\pi_{NLHF} := \operatorname{argmax}_{\pi_1 \in \mathcal{B}_\tau(\pi_{ref})} \min_{\pi_2 \in \mathcal{B}_\tau(\pi_{ref})} \mathbb{E}_{x_1 \sim \pi_1, x_2 \sim \pi_2} [M_{x_1, x_2}]$. To generalize the definition of distortion to alignment methods, we set the maximum average utility of any policy in the KL-ball as the benchmark. For fixed m, D, β, d, μ and correlation between pairwise comparisons, the distortion of alignment method f is

$$\operatorname{dist}(f) = \sup_{D, \pi_{ref}, \tau} \limsup_{n \rightarrow \infty} \frac{\operatorname{AvgUtil}(\pi)}{\operatorname{AvgUtil}_n(f(\cdot, \pi_{ref}, \tau))}.$$

Expanded PaperInterface specification

```
(∀ {Alternative : Type} [inst : Fintype Alternative] [inst_1 : DecidableEq Alternative]
  (preference : AppliedModelingLib.Learning.HumanFeedback.PairwisePreference PUnit.{1} Alternative)
  (reference_policy : PMF Alternative) (klWeight : WithTop ℝ),
  AppliedModelingLib.PMFFullSupport reference →
  AppliedModelingLib.GameTheory.PreferenceGame.IsContextFreeExtendedRegularizedPreferenceGameEquilibrium preference
  reference_policy klWeight →
  AppliedModelingLib.GameTheory.PreferenceGame.IsFiniteKLPreferenceMaximin preference reference
  (AppliedModelingLib.finiteKLDivergence policy reference) policy) ∧
  ∀ {Alternative : Type} [inst : Fintype Alternative] [inst_1 : DecidableEq Alternative]
  (preference : AppliedModelingLib.Learning.HumanFeedback.PairwisePreference PUnit.{1} Alternative)
  (reference_policy : PMF Alternative) (klBudget : ℝ),
  0 ≤ klBudget →
  AppliedModelingLib.PMFFullSupport reference →
  AppliedModelingLib.GameTheory.PreferenceGame.IsFiniteKLPreferenceMaximin preference reference klBudget policy →
  ∃ (klWeight : WithTop ℝ),
  AppliedModelingLib.GameTheory.PreferenceGame.IsContextFreeExtendedRegularizedPreferenceGameEquilibrium
  preference reference_policy klWeight
```

Retained governing declarations

- [AppliedModelingLib.GameTheory.PreferenceGame.IsContextFreeExtendedRegularizedPreferenceGameEquilibrium](#)
- [AppliedModelingLib.GameTheory.PreferenceGame.IsFiniteKLPreferenceMaximin](#)
- [AppliedModelingLib.Learning.HumanFeedback.PairwisePreference](#)
- [AppliedModelingLib.PMFFullSupport](#)
- [AppliedModelingLib.finiteKLDivergence](#)

Lean-elaborated claim roles

```
1. conclusion: (∀ {Alternative : Type} [inst : Fintype Alternative] [inst_1 : DecidableEq Alternative]
  (preference : AppliedModelingLib.Learning.HumanFeedback.PairwisePreference PUnit.{1} Alternative)
  (reference_policy : PMF Alternative) (klWeight : WithTop ℝ),
  AppliedModelingLib.PMFFullSupport reference →
  AppliedModelingLib.GameTheory.PreferenceGame.IsContextFreeExtendedRegularizedPreferenceGameEquilibrium preference
  reference_policy klWeight →
  AppliedModelingLib.GameTheory.PreferenceGame.IsFiniteKLPreferenceMaximin preference reference
  (AppliedModelingLib.finiteKLDivergence policy reference) policy) ∧
  ∀ {Alternative : Type} [inst : Fintype Alternative] [inst_1 : DecidableEq Alternative]
  (preference : AppliedModelingLib.Learning.HumanFeedback.PairwisePreference PUnit.{1} Alternative)
  (reference_policy : PMF Alternative) (klBudget : ℝ),
  0 ≤ klBudget →
  AppliedModelingLib.PMFFullSupport reference →
  AppliedModelingLib.GameTheory.PreferenceGame.IsFiniteKLPreferenceMaximin preference reference klBudget policy →
  ∃ (klWeight : WithTop ℝ),
  AppliedModelingLib.GameTheory.PreferenceGame.IsContextFreeExtendedRegularizedPreferenceGameEquilibrium
  preference reference_policy klWeight
```

Lean proof endpoint (built separately)

GolzHaghtalabYang2025Distortion.proposition13_nlhf_equivalence

Recorded source-to-Spec screening Verdict: matches

Reason: Both regularized-to-constrained and constrained-to-regularized NLHF directions are formalized, including extended regularization weights.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 13

Source locator: cited publication, lines 607-636

Verbatim source input

Corollary 8. If λ -regularized NLHF (for any $\lambda \geq 0$) returns a policy $\tilde{\pi}_{\text{NLHF}}$, this policy's average utility is at least a $\frac{1-\beta}{1+e^{-\beta}}$ fraction of the optimal average utility of any policy π with $\text{DKL}(\pi \parallel \pi_{\text{ref}}) \leq \frac{1}{10}$ (minus finite-sample errors, see Theorem 7).
Why not define distortion by flexibly selecting the benchmark based on the output policy's KL divergence? For this definition, an alignment method that always returns the reference policy would spuriously achieve distortion 1: because its KL divergence is 0, we would benchmark it only against the reference policy, i.e., itself.

9

[form-feed in source extraction]
Sampling of Comparison Pairs. In our model, we assume that each voter provides d pairwise comparisons, where both members x_{ji}, y_{ij} of each comparison pair are sampled i.i.d. from μ . More generally, we can model $\{x_{ji}, y_{ij}\} \sim \nu$ where ν is a distribution over unordered alternative pairs, or even a distribution over d pairs of alternatives from which $\{x_{1i}, y_{i1}, \dots, x_{di}, y_{id}\}$ are sampled. (To keep the alignment methods well defined, we assume that each comparison pair has positive probability of [U+0001 control character] of these models can, for example, express k -wise (rather than pairwise) being sampled.) The latter comparisons, if $d = 2r$ are all pairs inside a randomly chosen set of r alternatives. Almost all of our results continue to hold in these general models: the lower bound for all alignment methods that satisfy the Condorcet loser criterion in the social choice setting (Theorem 3), the exponential lower bound for RLHF (Theorem 6), and the upper bound for NLHF/Maximal Lotteries (Theorem 7)¹¹. Given that our proofs continue to work out, the only “disadvantage” of these stronger models for sampling comparison pairs is that, without a distribution μ , the Borda voting rule is no longer defined (and we see no obvious way to generalize the Borda-MLE equivalence [46, 42]). It seems that RLHF does not only become harder to analyze under these comparison-pair models, but actually performs worse: we show in Appendix G that RLHF can have a distortion that is not bounded in β in these extended models, leading to an even clearer separation with NLHF.

[Next verbatim source excerpt]

If $\tau = 0$, the only feasible policy is π_{ref} , and the claim clearly holds for $\lambda = \infty$. We also assume that $\pi_{\text{ref}}(a) > 0$ for all $a \in M$. Otherwise if $\pi_{\text{ref}}(a) = 0$ for some $a \in M$, then both the regularized and constrained versions forbid any policy to put nonzero probability on a , which leads to an effectively smaller candidate set.

[Next verbatim source excerpt]

Let $A = \{1, \dots, m\}$ be a finite set of alternatives. The population of users is described by a probability distribution D over utility vectors $u = (u(1), \dots, u(m))$, whose entries $0 \leq u(x) \leq 1$ indicate a user's utility for alternative x .³ The objective is to find an alternative x such that its average utility $\text{AvgUtil}(x) := \mathbb{E}_{u \sim D} [u(x)]$ across the user population (also known as the utilitarian social welfare) is as high as possible. We extend this notation to probability distributions π over alternatives by setting $\text{AvgUtil}(\pi) := \mathbb{E}_{x \sim \pi} [\text{AvgUtil}(x)]$. Both voting rules and alignment methods observe comparisons from n users. We model each $i = 1, \dots, n$ as a fresh user with independently drawn utility vector $u_i \sim D$. For exposition, we assume that each user i provides an equal number $d \geq 1$ of pairwise comparisons. For each i , and for $j = 1, \dots, d$, we independently draw alternatives x_{ji}, y_{ij} from a fixed distribution μ over

³ Our assumption that utilities be in $[0, 1]$ is weaker than any of the three assumptions — unit-sum, unit-range, or approval [23] — made in classic distortion to avoid trivial infinite lower bounds.

3

[form-feed in source extraction]
alternatives, in which the minimum probability mass $\mu_{\min} := \min_{x \in A} \mu(x)$ is positive.⁴ User i then compares each pair $\{x_{ji}, y_{ij}\}$ (for $j = 1, \dots, d$) through a Bradley-Terry model based on i 's utilities: i prefers x_{ji} over y_{ij} (written “ x_{ji} [U+001F control character] y_{ij} ”) with probability $\sigma(\beta \cdot (u_i(x_{ji}) - u_i(y_{ij})))$, where [U+0001 control character]

$\sigma(t) := 1/(1+e^{-t})$ is the logistic sigmoid function and $\beta > 0$ is a temperature parameter, and prefers y_{ij} over x_{ji} (“ y_{ij} [U+001F control character] x_{ji} ”) otherwise.⁵ Whereas this specifies the marginal probability of each pairwise comparison, we make no assumption about the correlation between i 's choices. For example, i might derive the pairwise comparisons from a Plackett-Luce ranking, ensuring that [U+0002 control character] the user's comparisons [U+0001 control character] [U+0003 control character] are always consistent.⁶ We set $p(x$ [U+001F control character] $y)$ for the expected win rate $\mathbb{E}_{u \sim D} \sigma(\beta \cdot (u(x) - u(y)))$.

[Next verbatim source excerpt]

Alignment Setting. The alignment setting generalizes the social choice setting in two ways: first, user utilities $u_i(y \mid x)$ may depend on a state x ; second, the goal in determining a policy π is not purely to maximize the reward $\text{AvgUtil}(\pi \mid x)$, but a trade-off between this reward and the goal of remaining close to a reference policy $\pi_{\text{ref}}(\cdot \mid x) \in \Delta(A)$ in terms of the KL divergence $\text{DKL}(\pi \parallel \pi_{\text{ref}})$. For theoretical tractability, we focus our analysis on a single state x , which we from here on omit from the notation. Conceptually, this treatment of alignment on a state-by-state basis corresponds to an assumption that our policy class is expressive enough so that it can take the optimal distribution of actions at each state⁷ and abstracts from the generalization problem of estimating the population's preference between a pair of alternatives at the given state x based on preferences in similar states x_0 . Having set aside the dependency between states, we focus on how the regularization with respect to a reference policy impacts the ability to optimize the average utility of three alignment methods: RLHF, DPO, and NLHF. In addition to the pairwise comparisons, these methods take in a reference policy $\pi_{\text{ref}} \in \Delta(A)$ and a KL bound $\tau \geq 0$, and map these inputs to a policy π in the KL-ball $\mathcal{B}_\tau(\pi_{\text{ref}}) := \{\pi \in \Delta(A) \mid \text{DKL}(\pi \parallel \pi_{\text{ref}}) \leq \tau\}$ around the reference policy. RLHF, DPO, and NLHF are typically implemented with a KL regularization rather than our KL-constrained formulation, which we adopt to enable a comparison on equal terms. In Section 5, we show that these perspectives are equivalent, and that distortion upper bounds carry over to regularized alignment methods.

4 In Section 5, we discuss how most of our results extend more general distributions over comparison pairs, which can, for example, capture k-wise comparisons.

5 Should we sample the same alternative $x = x_{ji} = y_{ij}$ twice for a pair, the user is not asked for a pairwise comparison. We record this as “x [U+001F control character]i x”, in a slight abuse of notation.

6 In particular, if we sample the same unordered pair twice for a user, the answers can be perfectly correlated.

7 This assumption is common in the literature; see, for example, Rafailov et al. [43]’s application of first-order optimality conditions of PPO loss minimization.

4

[form-feed in source extraction]

The RLHF method first estimates rewards for each alternative, using maximum likelihood estimation assuming that comparisons were generated by a single Bradley-Terry model:

$$r := \underset{p}{\operatorname{argmax}}_{r \in \mathbb{R}^m} \sum_{1 \leq i \leq n, 1 \leq j \leq d} \log \sigma(r(x_{ji}) - r(y_{ij})) \cdot 8$$

[U+0001 control character]

Next, RLHF uses PPO [44] to compute the policy $\pi_{\text{RLHF}} := \operatorname{argmax}_{\pi \in \mathcal{B}\tau} (\pi_{\text{ref}}) \mathbb{E}_{x \sim \pi} [r(x)]$ with maximum expected reward within the KL-ball. In our setup, DPO is equivalent to RLHF (see Appendix F.3) and thus has the same distortion.

The alignment method NLHF was inspired in part by a desire to better align with the preferences of a heterogeneous group [36]. NLHF naturally adapts the definition of maximal lotteries by constraining both players’ mixed strategies to the KL-ball, i.e., $\pi_{\text{NLHF}} := \operatorname{argmax}_{\pi_1 \in \mathcal{B}\tau} (\pi_{\text{ref}}) \min_{\pi_2 \in \mathcal{B}\tau} (\pi_{\text{ref}}) \mathbb{E}_{x_1 \sim \pi_1, x_2 \sim \pi_2} [M_{x_1, x_2}]$.

To generalize the definition of distortion to alignment methods, we set the maximum average utility of any policy in the KL-ball as the benchmark. For fixed m, D, β, d, μ and correlation between pairwise comparisons, the distortion of alignment method f is

$$\operatorname{dist}(f) = \sup_{D, \pi_{\text{ref}}, \tau} \limsup_{n \rightarrow \infty} \frac{\operatorname{AvgUtil}(\pi)}{\operatorname{AvgUtil}(f(\cdot, \pi_{\text{ref}}, \tau))}.$$

Expanded PaperInterface specification

```

∀ {User Alternative : Type} [inst : Fintype User] [inst_1 : DecidableEq User] [inst_2 : Fintype Alternative]
[inst_3 : DecidableEq Alternative] (population : PMF User)
(utility : AppliedModelingLib.Alignment.Welfare.FiniteUtilityProfile User Alternative),
AppliedModelingLib.Alignment.Welfare.UnitIntervalUtilityProfile utility →
  ∀ {btScale : ℝ},
  0 < btScale →
    ∀ (reference policy : PMF Alternative) (klRegularization : ℝ),
    0 ≤ klRegularization →
      AppliedModelingLib.PMFFullSupport reference →
        AppliedModelingLib.GameTheory.PreferenceGame.IsRegularizedPreferenceGameEquilibrium (PMF.pure PUnit.unit)
        (AppliedModelingLib.Alignment.Welfare.populationBradleyTerryPreference population utility btScale)
        (AppliedModelingLib.Alignment.Welfare.contextFreeAlternativePolicy reference)
        (AppliedModelingLib.Alignment.Welfare.contextFreeAlternativePolicy policy) klRegularization →
      ∀ (benchmark : PMF Alternative),
        AppliedModelingLib.finiteKLDivergence benchmark reference ≤
          AppliedModelingLib.finiteKLDivergence policy reference →
            AppliedModelingLib.Alignment.Welfare.sigmoidChordSlope btScale / (1 / 4) *
            AppliedModelingLib.Alignment.Welfare.policyAverageUtility population utility benchmark ≤
            AppliedModelingLib.Alignment.Welfare.policyAverageUtility population utility policy

```

Retained governing declarations

- [AppliedModelingLib.Alignment.Welfare.FiniteUtilityProfile](#)
- [AppliedModelingLib.Alignment.Welfare.UnitIntervalUtilityProfile](#)
- [AppliedModelingLib.Alignment.Welfare.contextFreeAlternativePolicy](#)
- [AppliedModelingLib.Alignment.Welfare.policyAverageUtility](#)
- [AppliedModelingLib.Alignment.Welfare.populationBradleyTerryPreference](#)
- [AppliedModelingLib.Alignment.Welfare.sigmoidChordSlope](#)
- [AppliedModelingLib.GameTheory.PreferenceGame.IsRegularizedPreferenceGameEquilibrium](#)
- [AppliedModelingLib.PMFFullSupport](#)
- [AppliedModelingLib.finiteKLDivergence](#)

Lean-elaborated claim roles

1. parameter: Type
2. parameter: Type
3. parameter: Fintype User
4. parameter: DecidableEq User
5. parameter: Fintype Alternative
6. parameter: DecidableEq Alternative
7. parameter: PMF User
8. parameter: AppliedModelingLib.Alignment.Welfare.FiniteUtilityProfile User Alternative
9. assumption: AppliedModelingLib.Alignment.Welfare.UnitIntervalUtilityProfile utility
10. parameter: ℝ
11. assumption: 0 < btScale
12. parameter: PMF Alternative

13. parameter: PMF Alternative
 14. parameter: $\mathbb{R}$
 15. assumption: $0 \leq \text{klRegularization}$
 16. assumption: AppliedModelingLib.PMFFullSupport reference
 17. assumption: AppliedModelingLib.GameTheory.PreferenceGame.IsRegularizedPreferenceGameEquilibrium (PMF.pure PUnit.unit)
 (AppliedModelingLib.Alignment.Welfare.populationBradleyTerryPreference population utility btScale)
 (AppliedModelingLib.Alignment.Welfare.contextFreeAlternativePolicy reference)
 (AppliedModelingLib.Alignment.Welfare.contextFreeAlternativePolicy policy) klRegularization
 18. parameter: PMF Alternative
 19. assumption: AppliedModelingLib.finiteKLDivergence benchmark reference $\leq$ AppliedModelingLib.finiteKLDivergence policy reference
 20. conclusion: AppliedModelingLib.Alignment.Welfare.sigmoidChordSlope btScale / (1 / 4) *
 AppliedModelingLib.Alignment.Welfare.policyAverageUtility population utility benchmark $\leq$
 AppliedModelingLib.Alignment.Welfare.policyAverageUtility population utility policy

Lean proof endpoint (built separately)

GolzHaghtalabYang2025Distortion.corollary8_regularized_nlhf

Recorded source-to-Spec screening Verdict: matches

Reason: Regularized NLHF is compared with every policy of no greater KL divergence and receives the Theorem-7 welfare factor.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 14

Source locator: cited publication, lines 2263-2280

Verbatim source input

Proposition 14 (Equivalence between Constrained and Regularized RLHF). Let r be the MLE reward learned from the comparison data, and let π_τ be the output of the τ -constrained RLHF method defined in Section 2, i.e.,

$$\pi_\tau = \operatorname{argmax}_{\pi \in \mathcal{B}_\tau(\pi_{\text{ref}})} \mathbb{E}[r(x)], \quad (19)$$

and let $\tilde{\pi}_\lambda$ be the output of the λ -regularized RLHF method, i.e.,

$$\tilde{\pi}_\lambda = \operatorname{argmax}_{\pi \in \Delta(M)} \mathbb{E}[r(x)] - \lambda \cdot \text{DKL}(\pi \parallel \pi_{\text{ref}}). \quad (20)$$

Then, for each $\lambda \in [0, \infty]$ with solution $\tilde{\pi}_\lambda$ for Eq. (20), we have that $\tilde{\pi}_\lambda$ is also an optimal solution to the τ -constrained optimization problem in Eq. (19), where $\tau = \text{DKL}(\tilde{\pi}_\lambda \parallel \pi_{\text{ref}})$. Conversely, for each $\tau \geq 0$ with solution π_τ for Eq. (19), there exists $\lambda \in [0, \infty]$ such that π_τ is also an optimal solution to the λ -regularized optimization problem in Eq. (20).

¹⁴ If $\tau = 0$, the only feasible policy is π_{ref} , and the claim clearly holds for $\lambda = \infty$. We also assume that $\pi_{\text{ref}}(a) > 0$ for all $a \in M$. Otherwise if $\pi_{\text{ref}}(a) = 0$ for some $a \in M$, then both the regularized and constrained versions forbid any policy to put nonzero probability on a , which leads to an effectively smaller candidate set.

[Next verbatim source excerpt]

Let $A = \{1, \dots, m\}$ be a finite set of alternatives. The population of users is described by a probability distribution D over utility vectors $u = (u(1), \dots, u(m))$, whose entries $0 \leq u(x) \leq 1$ indicate a user's utility for alternative x .³ The objective is to find an alternative x such that its average utility $\text{AvgUtil}(x) := \mathbb{E}_{u \sim D}[u(x)]$ across the user population (also known as the utilitarian social welfare) is as high as possible. We extend this notation to probability distributions π over alternatives by setting $\text{AvgUtil}(\pi) := \mathbb{E}_{x \sim \pi}[\text{AvgUtil}(x)]$.

Both voting rules and alignment methods observe comparisons from n users. We model each $i = 1, \dots, n$ as a fresh user with independently drawn utility vector $u_i \sim D$. For exposition, we assume that each user i provides an equal number $d \geq 1$ of pairwise comparisons. For each i , and for $j = 1, \dots, d$, we independently draw alternatives x_{ji} , y_{ji} from a fixed distribution μ over

³ Our assumption that utilities be in $[0, 1]$ is weaker than any of the three assumptions — unit-sum, unit-range, or approval [23] — made in classic distortion to avoid trivial infinite lower bounds.

3

[form-feed in source extraction]

alternatives, in which the minimum probability mass $\mu_{\min} := \min_{x \in A} \mu(x)$ is positive.⁴ User i then compares each pair $\{x_{ji}, y_{ji}\}$ (for $j = 1, \dots, d$) through a Bradley-Terry model based on i 's utilities: i prefers x_{ji} over y_{ji} (written “ x_{ji} [U+001F control character] y_{ji} ”) with probability $\sigma(\beta \cdot (u_i(x_{ji}) - u_i(y_{ji})))$, where [U+0001 control character]

$\sigma(t) := 1/(1+e^{-t})$ is the logistic sigmoid function and $\beta > 0$ is a temperature parameter, and prefers y_{ji} over x_{ji} (“ y_{ji} [U+001F control character] x_{ji} ”) otherwise.⁵ Whereas this specifies the marginal probability of each pairwise comparison, we make no assumption about the correlation between i 's choices. For example, i might derive the pairwise comparisons from a Plackett-Luce ranking, ensuring that [U+0002 control character] the user's comparisons [U+0001 control character] [U+0003 control character] are always consistent.⁶ We set $p(x$ [U+001F control character] $y)$ for the expected win rate $\mathbb{E}_{u \sim D} \sigma(\beta \cdot (u(x) - u(y)))$.

[Next verbatim source excerpt]

Alignment Setting. The alignment setting generalizes the social choice setting in two ways: first, user utilities $u_i(y \parallel x)$ may depend on a state x ; second, the goal in determining a policy π is not purely to maximize the reward $\text{AvgUtil}(\pi \parallel x)$, but a trade-off between this reward and the goal of remaining close to a reference policy $\pi_{\text{ref}}(\cdot \parallel x) \in \Delta(A)$ in terms of the KL divergence $\text{DKL}(\pi \parallel \pi_{\text{ref}})$. For theoretical tractability, we focus our analysis on a single state x , which we from here on omit from the notation. Conceptually, this treatment of alignment on a state-by-state basis corresponds to an assumption that our policy class is expressive enough so that it can take the optimal distribution of actions at each state⁷ and abstracts from the generalization problem of estimating the population's preference between a pair of alternatives at the given state x based on preferences in similar states x_0 . Having set aside the dependency between states, we focus on how the regularization with respect to a reference policy impacts the ability to optimize the average utility of three alignment methods: RLHF, DPO, and NLHF. In addition to the pairwise comparisons, these methods take in a reference policy $\pi_{\text{ref}} \in \Delta(A)$ and a KL bound $\tau \geq 0$, and map these inputs to a policy π in the KL-ball $\mathcal{B}_\tau(\pi_{\text{ref}}) := \{\pi \in \Delta(A) \mid \text{DKL}(\pi \parallel \pi_{\text{ref}}) \leq \tau\}$ around the reference policy. RLHF, DPO, and NLHF are typically implemented with a KL regularization rather than our KL-constrained formulation, which we adopt to enable a comparison on equal terms. In Section 5, we show that these perspectives are equivalent, and that distortion upper bounds carry over to regularized alignment methods.

⁴ In Section 5, we discuss how most of our results extend more general distributions over comparison pairs, which can, for example, capture k -wise comparisons.

⁵ Should we sample the same alternative $x = x_{ji} = y_{ji}$ twice for a pair, the user is not asked for a pairwise comparison. We record this as “ x [U+001F control character] x ”, in a slight abuse of notation.

⁶ In particular, if we sample the same unordered pair twice for a user, the answers can be perfectly correlated.

⁷ This assumption is common in the literature; see, for example, Rafailov et al. [43]'s application of first-order optimality conditions of PPO loss minimization.

4

[form-feed in source extraction]

The RLHF method first estimates rewards for each alternative, using maximum likelihood estimation assuming that comparisons were generated by a single Bradley-Terry model:
 $r := \operatorname{argmax}_{r \in \mathbb{R}^m} \sum_{1 \leq i \leq n, 1 \leq j \leq d} \log \sigma(r(x_{ji}) - r(y_{ji}))$.⁸

Next, RLHF uses PPO [44] to compute the policy $\pi_{\text{RLHF}} := \arg\max_{\pi \in \mathcal{B}\tau} (\pi_{\text{ref}}) \mathbb{E}_{x \sim \pi} [r(x)]$ with maximum expected reward within the KL-ball. In our setup, DPO is equivalent to RLHF (see Appendix F.3) and thus has the same distortion. The alignment method NLHF was inspired in part by a desire to better align with the preferences of a heterogeneous group [36]. NLHF naturally adapts the definition of maximal lotteries by constraining both players' mixed strategies to the KL-ball, i.e., $\pi_{\text{NLHF}} := \arg\max_{\pi_1 \in \mathcal{B}\tau} (\pi_{\text{ref}}) \min_{\pi_2 \in \mathcal{B}\tau} (\pi_{\text{ref}}) \mathbb{E}_{x_1 \sim \pi_1, x_2 \sim \pi_2} [M_{x_1, x_2}]$. To generalize the definition of distortion to alignment methods, we set the maximum average utility of any policy in the KL-ball as the benchmark. For fixed m, D, β, d, μ and correlation between pairwise comparisons, the distortion of alignment method f is

$$\text{dist}(f) = \sup_{D, \pi_{\text{ref}}, \tau} \limsup_{n \rightarrow \infty} \frac{\text{AvgUtil}(\pi)}{\text{AvgUtil}(f(\cdot, \pi_{\text{ref}}, \tau))}.$$

Expanded PaperInterface specification

```
(∀ {Alternative : Type} [inst : Fintype Alternative] [inst_1 : DecidableEq Alternative]
(reference policy : PMF Alternative) (score : Alternative → ℝ) (klWeight : WithTop ℝ),
AppliedModelingLib.PMFFullSupport reference →
AppliedModelingLib.IsFiniteKLExtendedRegularizedUtilityMax reference policy score klWeight →
AppliedModelingLib.finiteKLDivergence policy reference ≤
AppliedModelingLib.finiteKLDivergence policy reference ∧
∀ (other : PMF Alternative),
AppliedModelingLib.finiteKLDivergence other reference ≤
AppliedModelingLib.finiteKLDivergence policy reference →
AppliedModelingLib.pmfExp other score ≤ AppliedModelingLib.pmfExp policy score) ∧
∀ {Alternative : Type} [inst : Fintype Alternative] [inst_1 : DecidableEq Alternative]
(reference policy : PMF Alternative) (score : Alternative → ℝ) (klBudget : ℝ),
0 ≤ klBudget →
AppliedModelingLib.PMFFullSupport reference →
AppliedModelingLib.finiteKLDivergence policy reference ≤ klBudget →
(∀ (other : PMF Alternative),
AppliedModelingLib.finiteKLDivergence other reference ≤ klBudget →
AppliedModelingLib.pmfExp other score ≤ AppliedModelingLib.pmfExp policy score) →
∃ (klWeight : WithTop ℝ),
AppliedModelingLib.IsFiniteKLExtendedRegularizedUtilityMax reference policy score klWeight
```

Retained governing declarations

- [AppliedModelingLib.IsFiniteKLExtendedRegularizedUtilityMax](#)
- [AppliedModelingLib.PMFFullSupport](#)
- [AppliedModelingLib.finiteKLDivergence](#)
- [AppliedModelingLib.pmfExp](#)

Lean-elaborated claim roles

```
1. conclusion: (∀ {Alternative : Type} [inst : Fintype Alternative] [inst_1 : DecidableEq Alternative]
(reference policy : PMF Alternative) (score : Alternative → ℝ) (klWeight : WithTop ℝ),
AppliedModelingLib.PMFFullSupport reference →
AppliedModelingLib.IsFiniteKLExtendedRegularizedUtilityMax reference policy score klWeight →
AppliedModelingLib.finiteKLDivergence policy reference ≤
AppliedModelingLib.finiteKLDivergence policy reference ∧
∀ (other : PMF Alternative),
AppliedModelingLib.finiteKLDivergence other reference ≤
AppliedModelingLib.finiteKLDivergence policy reference →
AppliedModelingLib.pmfExp other score ≤ AppliedModelingLib.pmfExp policy score) ∧
∀ {Alternative : Type} [inst : Fintype Alternative] [inst_1 : DecidableEq Alternative]
(reference policy : PMF Alternative) (score : Alternative → ℝ) (klBudget : ℝ),
0 ≤ klBudget →
AppliedModelingLib.PMFFullSupport reference →
AppliedModelingLib.finiteKLDivergence policy reference ≤ klBudget →
(∀ (other : PMF Alternative),
AppliedModelingLib.finiteKLDivergence other reference ≤ klBudget →
AppliedModelingLib.pmfExp other score ≤ AppliedModelingLib.pmfExp policy score) →
∃ (klWeight : WithTop ℝ),
AppliedModelingLib.IsFiniteKLExtendedRegularizedUtilityMax reference policy score klWeight
```

Lean proof endpoint (built separately)

GolzHaghtalabYang2025Distortion.proposition14_rlhf_equivalence

Recorded source-to-Spec screening Verdict: matches

Reason: This formalizes the analogous two RLHF directions at radius equal to the output policy's KL divergence.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

$$\begin{aligned} & \frac{[U+0012 \text{ control character}] \quad [U+0010 \text{ control character}] e^{\Delta/2} - 1}{\Delta} [U+0011 \text{ control character}]^3 [U+0013 \text{ control character}] \\ & \quad - \log 1 + \tanh(\Delta/4)^3 \cdot \\ & = \frac{-\log 1 + \Delta/2}{2} = \frac{[U+0001 \text{ control character}]}{2} \end{aligned}$$

[Next verbatim source excerpt]

Let $A = \{1, \dots, m\}$ be a finite set of alternatives. The population of users is described by a probability distribution D over utility vectors $u = (u(1), \dots, u(m))$, whose entries $0 \leq u(x) \leq 1$ indicate a user's utility for alternative x .³ The objective is to find an alternative x such that its average utility $\text{AvgUtil}(x) := \mathbb{E}_{u \sim D} [u(x)]$ across the user population (also known as the utilitarian social welfare) is as high as possible. We extend this notation to probability distributions π over alternatives by setting $\text{AvgUtil}(\pi) := \mathbb{E}_{x \sim \pi} [\text{AvgUtil}(x)]$. Both voting rules and alignment methods observe comparisons from n users. We model each $i = 1, \dots, n$ as a fresh user with independently drawn utility vector $u_i \sim D$. For exposition, we assume that each user i provides an equal number $d \geq 1$ of pairwise comparisons. For each i , and for $j = 1, \dots, d$, we independently draw alternatives x_{ji}, y_{ij} from a fixed distribution μ over

³ Our assumption that utilities be in $[0, 1]$ is weaker than any of the three assumptions — unit-sum, unit-range, or approval [23] — made in classic distortion to avoid trivial infinite lower bounds.

3

[form-feed in source extraction]

alternatives, in which the minimum probability mass $\mu_{\min} := \min_{x \in A} \mu(x)$ is positive.⁴ User i then compares each pair $\{x_{ji}, y_{ij}\}$ (for $j = 1, \dots, d$) through a Bradley-Terry model based on i 's utilities: i prefers x_{ji} over y_{ij} (written “ x_{ji} [U+001F control character] y_{ij} ”) with probability $\sigma(\beta \cdot (u_i(x_{ji}) - u_i(y_{ij})))$, where [U+0001 control character]

$\sigma(t) := 1/(1+e^{-t})$ is the logistic sigmoid function and $\beta > 0$ is a temperature parameter, and prefers y_{ij} over x_{ji} (“ y_{ij} [U+001F control character] x_{ji} ”) otherwise.⁵ Whereas this specifies the marginal probability of each pairwise comparison, we make no assumption about the correlation between i 's choices. For example, i might derive the pairwise comparisons from a Plackett-Luce ranking, ensuring that [U+0002 control character] the user's comparisons [U+0001 control character] [U+0003 control character] are always consistent.⁶ We set $p(x$ [U+001F control character] $y)$ for the expected win rate $\mathbb{E}_{u \sim D} \sigma(\beta \cdot (u(x) - u(y)))$.

[Next verbatim source excerpt]

Alignment Setting. The alignment setting generalizes the social choice setting in two ways: first, user utilities $u_i(y | x)$ may depend on a state x ; second, the goal in determining a policy π is not purely to maximize the reward $\text{AvgUtil}(\pi | x)$, but a trade-off between this reward and the goal of remaining close to a reference policy $\pi_{\text{ref}}(\cdot | x) \in \Delta(A)$ in terms of the KL divergence $\text{DKL}(\cdot | \pi_{\text{ref}})$. For theoretical tractability, we focus our analysis on a single state x , which we from here on omit from the notation. Conceptually, this treatment of alignment on a state-by-state basis corresponds to an assumption that our policy class is expressive enough so that it can take the optimal distribution of actions at each state⁷ and abstracts from the generalization problem of estimating the population's preference between a pair of alternatives at the given state x based on preferences in similar states x_0 . Having set aside the dependency between states, we focus on how the regularization with respect to a reference policy impacts the ability to optimize the average utility of three alignment methods: RLHF, DPO, and NLHF. In addition to the pairwise comparisons, these methods take in a reference policy $\pi_{\text{ref}} \in \Delta(A)$ and a KL bound $\tau \geq 0$, and map these inputs to a policy π in the KL-ball $\mathcal{B}_\tau(\pi_{\text{ref}}) := \{\pi \in \Delta(A) \mid \text{DKL}(\pi | \pi_{\text{ref}}) \leq \tau\}$ around the reference policy. RLHF, DPO, and NLHF are typically implemented with a KL regularization rather than our KL-constrained formulation, which we adopt to enable a comparison on equal terms. In Section 5, we show that these perspectives are equivalent, and that distortion upper bounds carry over to regularized alignment methods.

⁴ In Section 5, we discuss how most of our results extend more general distributions over comparison pairs, which can, for example, capture k -wise comparisons.

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⁶ In particular, if we sample the same unordered pair twice for a user, the answers can be perfectly correlated.

⁷ This assumption is common in the literature; see, for example, Rafailov et al. [43]'s application of first-order optimality conditions of PPO loss minimization.

4

[form-feed in source extraction]

The RLHF method first estimates rewards for each alternative, using maximum likelihood estimation assuming that comparisons were generated by a single Bradley-Terry model:

$$r := \underset{P}{\operatorname{argmax}} \mathbb{E}_{r \sim P} \sum_{1 \leq i \leq n, 1 \leq j \leq d} \log \sigma(r(x_{ji}) - r(y_{ij})). \quad [U+0001 \text{ control character}]$$

Next, RLHF uses PPO [44] to compute the policy $\pi_{\text{RLHF}} := \underset{\pi \in \mathcal{B}_\tau(\pi_{\text{ref}})}{\operatorname{argmax}} \mathbb{E}_{x \sim \pi} [r(x)]$ with maximum expected reward within the KL-ball. In our setup, DPO is equivalent to RLHF (see Appendix F.3) and thus has the same distortion.

The alignment method NLHF was inspired in part by a desire to better align with the preferences of a heterogeneous group [36]. NLHF naturally adapts the definition of maximal lotteries by constraining both players' mixed strategies to the KL-ball, i.e., $\pi_{\text{NLHF}} := \underset{\operatorname{argmax} \pi_1 \in \mathcal{B}_\tau(\pi_{\text{ref}}) \text{ min} \pi_2 \in \mathcal{B}_\tau(\pi_{\text{ref}})}{\operatorname{argmax}} \mathbb{E}_{x_1 \sim \pi_1, x_2 \sim \pi_2} [M_{x_1, x_2}]$. To generalize the definition of distortion to alignment methods, we set the maximum average utility of any policy in the KL-ball as the benchmark. For fixed m, D, β, d, μ and correlation between pairwise comparisons, the distortion of alignment method f is

$$\operatorname{dist}(f) = \sup_{D, \pi_{\text{ref}}, \tau} \limsup_{n \rightarrow \infty} \frac{\max_{\pi \in \mathcal{B}_\tau(\pi_{\text{ref}})} \text{AvgUtil}(\pi)}{\text{AvgUtil}(f(\cdot, \pi_{\text{ref}}, \tau))}.$$

Expanded PaperInterface specification

$\forall \{\beta : \mathbb{R}\},$
 $0 < \beta \rightarrow$

```

∀ (bound : ℝ),
  ∃ (m : ℕ) (hm : 3 ≤ m) (epsilon : NNReal) ( _ : 0 < epsilon) ( _ : epsilon < 1),
  have population : PMF (Fin 3) := GolzHaghtalabYang2025Distortion.lemma15Population;
  have utility : AppliedModelingLib.Alignment.Welfare.FiniteUtilityProfile (Fin 3) (Fin m) :=
    GolzHaghtalabYang2025Distortion.theorem9PrefixProfile beta m;
  have comparisonSampling : AppliedModelingLib.Learning.HumanFeedback.CorrelatedPairSampling (Fin m) :=
    GolzHaghtalabYang2025Distortion.theorem9ComparisonSampling m hm epsilon;
  have reference : PMF (Fin m) :=
    GolzHaghtalabYang2025Distortion.theorem9UniformReference m
    (GolzHaghtalabYang2025Distortion.theorem9Spec._proof_3 m hm);
  have klBudget : ℝ := Real.log ↑ m;
  AppliedModelingLib.Alignment.Welfare.UnitIntervalUtilityProfile utility ∧
  AppliedModelingLib.PMFFullSupport reference ∧
  comparisonSampling.IsSymmetric ∧
  comparisonSampling.HasZeroDiagonal ∧
  comparisonSampling.HasFullOffDiagonalSupport ∧
  0 ≤ klBudget ∧
  ∀m (path :
    ℕ →
    AppliedModelingLib.Learning.HumanFeedback.BinaryPairwiseReport
    (Fin
      m)) → AppliedModelingLib.finitePMFIdPathMeasure
    (GolzHaghtalabYang2025Distortion.theorem9OneComparisonReportLaw beta m hm epsilon),
  ∀r (horizon : ℕ) in Filter.atTop,
  (∃ (reward :
    AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset.ScoreVector (Fin m)),
    (AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset.ofBinaryReports
      (AppliedModelingLib.finitePMFIdPathPrefix path horizon)).isPairwiseMLE
      AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset.sigmoidCDFLikePairwiseLink.toFun
      (0, GolzHaghtalabYang2025Distortion.theorem9Spec._proof_4 m hm) reward) ∧
  ∀
  (reward :
    AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset.ScoreVector (Fin m)),
  (AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset.ofBinaryReports
    (AppliedModelingLib.finitePMFIdPathPrefix path horizon)).isPairwiseMLE
    AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset.sigmoidCDFLikePairwiseLink.toFun
    (0, GolzHaghtalabYang2025Distortion.theorem9Spec._proof_4 m hm) reward →
  ∃ (rewardPolicy : PMF (Fin m)) (welfarePolicy : PMF (Fin m)),
  AppliedModelingLib.finiteKLDivergence rewardPolicy reference ≤ klBudget ∧
  (∀ (other : PMF (Fin m)),
    AppliedModelingLib.finiteKLDivergence other reference ≤ klBudget →
    AppliedModelingLib.pmfExp other reward ≤
    AppliedModelingLib.pmfExp rewardPolicy reward) ∧
  AppliedModelingLib.finiteKLDivergence welfarePolicy reference ≤ klBudget ∧
  (∀ (other : PMF (Fin m)),
    AppliedModelingLib.finiteKLDivergence other reference ≤ klBudget →
    AppliedModelingLib.Alignment.Welfare.policyAverageUtility population utility
      other ≤
    AppliedModelingLib.Alignment.Welfare.policyAverageUtility population
      utility welfarePolicy) ∧
  0 <
    AppliedModelingLib.Alignment.Welfare.policyAverageUtility population utility
    rewardPolicy ∧
  bound <
    AppliedModelingLib.Alignment.Welfare.policyAverageUtility population utility
    welfarePolicy /
    AppliedModelingLib.Alignment.Welfare.policyAverageUtility population
    utility rewardPolicy

```

Retained governing declarations

- [AppliedModelingLib.Alignment.Welfare.FiniteUtilityProfile](#)
- [AppliedModelingLib.Alignment.Welfare.UnitIntervalUtilityProfile](#)
- [AppliedModelingLib.Alignment.Welfare.policyAverageUtility](#)
- [AppliedModelingLib.Learning.HumanFeedback.BinaryPairwiseReport](#)
- [AppliedModelingLib.Learning.HumanFeedback.CorrelatedPairSampling](#)
- [AppliedModelingLib.Learning.HumanFeedback.CorrelatedPairSampling.HasFullOffDiagonalSupport](#)
- [AppliedModelingLib.Learning.HumanFeedback.CorrelatedPairSampling.HasZeroDiagonal](#)
- [AppliedModelingLib.Learning.HumanFeedback.CorrelatedPairSampling.IsSymmetric](#)
- [AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset.CDFLikePairwiseLink](#)
- [AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset.ScoreVector](#)
- [AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset.isPairwiseMLE](#)
- [AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset.ofBinaryReports](#)
- [AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset.sigmoidCDFLikePairwiseLink](#)
- [AppliedModelingLib.PMFFullSupport](#)
- [AppliedModelingLib.finiteKLDivergence](#)
- [AppliedModelingLib.finitePMFIdPathMeasure](#)

- [AppliedModelingLib.finitePMFIdPathPrefix](#)
- [AppliedModelingLib.pmfExp](#)
- [GolzHaghtalabYang2025Distortion.lemma15Population](#)
- [GolzHaghtalabYang2025Distortion.theorem9ComparisonSampling](#)
- [GolzHaghtalabYang2025Distortion.theorem9OneComparisonReportLaw](#)
- [GolzHaghtalabYang2025Distortion.theorem9PrefixProfile](#)
- [GolzHaghtalabYang2025Distortion.theorem9UniformReference](#)

Lean-elaborated claim roles

```

1. parameter: ℝ
2. assumption: 0 < beta
3. parameter: ℝ
4. conclusion: ∃ (m : ℕ) (hm : 3 ≤ m) (epsilon : NNReal) ( _ : 0 < epsilon) ( _ : epsilon < 1),
  have population : PMF (Fin 3) := GolzHaghtalabYang2025Distortion.lemma15Population;
  have utility : AppliedModelingLib.Alignment.Welfare.FiniteUtilityProfile (Fin 3) (Fin m) :=
    GolzHaghtalabYang2025Distortion.theorem9PrefixProfile beta m;
  have comparisonSampling : AppliedModelingLib.Learning.HumanFeedback.CorrelatedPairSampling (Fin m) :=
    GolzHaghtalabYang2025Distortion.theorem9ComparisonSampling m hm epsilon;
  have reference : PMF (Fin m) :=
    GolzHaghtalabYang2025Distortion.theorem9UniformReference m
    (GolzHaghtalabYang2025Distortion.theorem9Spec._proof_3 m hm);
  have klBudget : ℝ := Real.log ↑ m;
  AppliedModelingLib.Alignment.Welfare.UnitIntervalUtilityProfile utility Λ
    AppliedModelingLib.PMFFullSupport reference Λ
      comparisonSampling.isSymmetric Λ
      comparisonSampling.HasZeroDiagonal Λ
      comparisonSampling.HasFullOffDiagonalSupport Λ
      0 ≤ klBudget Λ
      ∀m (path :
        ℕ →
          AppliedModelingLib.Learning.HumanFeedback.BinaryPairwiseReport
            (Fin
              m)) → AppliedModelingLib.finitePMFIdPathMeasure
                (GolzHaghtalabYang2025Distortion.theorem9OneComparisonReportLaw beta m hm epsilon),
      ∀i (horizon : ℕ) in Filter.atTop,
      (∃ (reward : AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset.ScoreVector (Fin m)),
        (AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset.ofBinaryReports
          (AppliedModelingLib.finitePMFIdPathPrefix path horizon)).isPairwiseMLE
          AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset.sigmoidCDFLikePairwiseLink.toFun
            (0, GolzHaghtalabYang2025Distortion.theorem9Spec._proof_4 m hm) reward) Λ
      ∀ (reward : AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset.ScoreVector (Fin m)),
        (AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset.ofBinaryReports
          (AppliedModelingLib.finitePMFIdPathPrefix path horizon)).isPairwiseMLE
          AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset.sigmoidCDFLikePairwiseLink.toFun
            (0, GolzHaghtalabYang2025Distortion.theorem9Spec._proof_4 m hm) reward →
      ∃ (rewardPolicy : PMF (Fin m)) (welfarePolicy : PMF (Fin m)),
        AppliedModelingLib.finiteKLDivergence rewardPolicy reference ≤ klBudget Λ
        (∀ (other : PMF (Fin m)),
          AppliedModelingLib.finiteKLDivergence other reference ≤ klBudget →
            AppliedModelingLib.pmfExp other reward ≤
              AppliedModelingLib.pmfExp rewardPolicy reward) Λ
        AppliedModelingLib.finiteKLDivergence welfarePolicy reference ≤ klBudget Λ
        (∀ (other : PMF (Fin m)),
          AppliedModelingLib.finiteKLDivergence other reference ≤ klBudget →
            AppliedModelingLib.Alignment.Welfare.policyAverageUtility population utility
              other ≤
                AppliedModelingLib.Alignment.Welfare.policyAverageUtility population utility
                  welfarePolicy) Λ
      0 <
        AppliedModelingLib.Alignment.Welfare.policyAverageUtility population utility
          rewardPolicy Λ
      bound <
        AppliedModelingLib.Alignment.Welfare.policyAverageUtility population utility
          welfarePolicy /
            AppliedModelingLib.Alignment.Welfare.policyAverageUtility population utility
              rewardPolicy

```

Lean proof endpoint (built separately)

GolzHaghtalabYang2025Distortion.theorem9_correlated_sampling_unbounded

Recorded source-to-Spec screening Verdict: matches

Reason: This constructs correlated comparison sampling with unbounded RLHF distortion, as in the source claim.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 16

Source locator: cited publication, lines 2114-2149

Verbatim source input

F.3 Equivalence of DPO and RLHF under Heterogeneous Preferences

In this section, we formalize the observation that DPO and RLHF are equivalent under heterogeneous preferences. This is consistent with the result by Shirali et al. [45], which shows that DPO also aligns with the Borda count. We start by recalling the DPO objective [43].¹²

$$\text{LDPO}(\pi; \pi_{\text{ref}}) = - \sum_{1 \leq i \leq n, 1 \leq j \leq d} \frac{\pi(x_{ji})}{\log \sigma \beta \log} - \frac{\pi(y_{ij})}{\beta \log}.$$

Now we perform a change of variables to transform π into the following form:

$$\pi(x) = \pi_{\text{ref}}(x) \cdot \exp(\hat{r}(x)/\beta) \text{ where } \hat{r}(x) := \beta \log \frac{\pi(x)}{\pi_{\text{ref}}(x)}, \forall x \in A.$$

Substituting this into the DPO objective, we get:

$$\text{LDPO}(\pi; \pi_{\text{ref}}) = - \sum_{1 \leq i \leq n, 1 \leq j \leq d} \log \sigma(\hat{r}(x_{ji}) - \hat{r}(y_{ij})),$$

[U+0001 control character]

which is exactly the MLE objective for reward learning in RLHF, repeated here for convenience:

$$\text{LMLE}(r) := - \sum_{1 \leq i \leq n, 1 \leq j \leq d} \log \sigma(r(x_{ji}) - r(y_{ij})).$$

[U+0001 control character]

Therefore, there is a one-to-one correspondence between the MLE reward $r^* = \text{argmin}_{r \in \mathbb{R}^m} \text{LMLE}(r)$, and the DPO policy $\pi_{\text{DPO}} = \text{argmin}_{\pi} \text{LDPO}(\pi; \pi_{\text{ref}})$ as:

$$\pi_{\text{DPO}}(x) = \pi_{\text{ref}}(x) \cdot \exp(r^*(x)/\beta).$$

On the other hand, the RLHF policy with regularization parameter λ (see Appendix F.4 for the equivalence between regularized and constrained versions of RLHF) is also given by

$$\pi_{\text{RLHF}}(x) = \pi_{\text{ref}}(x) \cdot \exp(r^*(x)/\lambda).$$

Therefore, the DPO policy π_{DPO} and the RLHF policy π_{RLHF} are equivalent when the parameter β in the DPO objective is equal to λ in the RLHF objective. Notably, both policies are different from the optimal policy $\pi^* \propto \pi_{\text{ref}} \cdot \exp(\text{AvgUtil}(x)/\lambda_0)$ (for a potentially different λ_0 that make the KL constraint tight) as $r^* \neq \text{AvgUtil}$, which has also been pointed out by Shirali et al. [45].

Expanded PaperInterface specification

```
∀ {Alternative : Type u_1} [inst : Fintype Alternative] [inst_1 : DecidableEq Alternative] {horizon : ℕ}
(sample : Fin horizon → AppliedModelingLib.Learning.HumanFeedback.BinaryPairwiseReport Alternative)
(scoreReference : Alternative) (reference : PMF Alternative)
(reward : AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset.ScoreVector Alternative) {klWeight : ℝ},
(AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset.ofBinaryReports sample).isPairwiseMLE Real.sigmoid
scoreReference reward →
0 < klWeight →
AppliedModelingLib.PMFFullSupport reference →
GolzhaghtalabYang2025Distortion.sourceDpoPolicy reference reward klWeight =
GolzhaghtalabYang2025Distortion.sourceRlhPolicy reference reward klWeight ∧
GolzhaghtalabYang2025Distortion.sourceDpoBinaryReportLoss sample reference
(GolzhaghtalabYang2025Distortion.sourceDpoPolicy reference reward klWeight) klWeight =
GolzhaghtalabYang2025Distortion.sourceRlhBinaryReportLoss sample reward ∧
(AppliedModelingLib.Learning.HumanFeedback.RewardEquivalent
(AppliedModelingLib.Learning.HumanFeedback.dpoImplicitReward
(AppliedModelingLib.Alignment.Welfare.contextFreeAlternativePolicy
(GolzhaghtalabYang2025Distortion.sourceDpoPolicy reference reward klWeight))
(AppliedModelingLib.Alignment.Welfare.contextFreeAlternativePolicy reference) klWeight)
fun (x : PUnit.{1}) => reward) ∧
GolzhaghtalabYang2025Distortion.IsSourceDpoLossMinimizer sample reference klWeight
(GolzhaghtalabYang2025Distortion.sourceDpoPolicy reference reward klWeight) ∧
∀ (policy : PMF Alternative),
GolzhaghtalabYang2025Distortion.IsSourceDpoLossMinimizer sample reference klWeight policy →
(AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset.ofBinaryReports
sample).isPairwiseMLE
Real.sigmoid scoreReference
(GolzhaghtalabYang2025Distortion.sourceDpoNormalizedMLEReward scoreReference reference policy
klWeight) ∧
GolzhaghtalabYang2025Distortion.sourceDpoPolicy reference
(GolzhaghtalabYang2025Distortion.sourceDpoNormalizedMLEReward scoreReference reference policy
klWeight)
klWeight =
policy
```

Retained governing declarations

- [AppliedModelingLib.Alignment.Welfare.contextFreeAlternativePolicy](#)
- [AppliedModelingLib.Learning.HumanFeedback.BinaryPairwiseReport](#)
- [AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset.ScoreVector](#)
- [AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset.isPairwiseMLE](#)
- [AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset.ofBinaryReports](#)

- [AppliedModelingLib.Learning.HumanFeedback.RewardEquivalent](#)
- [AppliedModelingLib.Learning.HumanFeedback.dpoImplicitReward](#)
- [AppliedModelingLib.PMFFullSupport](#)
- [GolzhaghtalabYang2025Distortion.IsSourceDpoLossMinimizer](#)
- [GolzhaghtalabYang2025Distortion.sourceDpoBinaryReportLoss](#)
- [GolzhaghtalabYang2025Distortion.sourceDpoNormalizedMLEReward](#)
- [GolzhaghtalabYang2025Distortion.sourceDpoPolicy](#)
- [GolzhaghtalabYang2025Distortion.sourceRlhfBinaryReportLoss](#)
- [GolzhaghtalabYang2025Distortion.sourceRlhfPolicy](#)

Lean-elaborated claim roles

```

1. parameter: Type u_1
2. parameter: Fintype Alternative
3. parameter: DecidableEq Alternative
4. parameter: ℕ
5. parameter: Fin horizon → AppliedModelingLib.Learning.HumanFeedback.BinaryPairwiseReport Alternative
6. parameter: Alternative
7. parameter: PMF Alternative
8. parameter: AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset.ScoreVector Alternative
9. parameter: ℝ
10. assumption: (AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset.ofBinaryReports sample).isPairwiseMLE Real.sigmoid
    scoreReference reward
11. assumption: 0 < klWeight
12. assumption: AppliedModelingLib.PMFFullSupport reference
13. conclusion: GolzhaghtalabYang2025Distortion.sourceDpoPolicy reference reward klWeight =
    GolzhaghtalabYang2025Distortion.sourceRlhfPolicy reference reward klWeight ∧
    GolzhaghtalabYang2025Distortion.sourceDpoBinaryReportLoss sample reference
    (GolzhaghtalabYang2025Distortion.sourceDpoPolicy reference reward klWeight) klWeight =
    GolzhaghtalabYang2025Distortion.sourceRlhfBinaryReportLoss sample reward ∧
    (AppliedModelingLib.Learning.HumanFeedback.RewardEquivalent
    (AppliedModelingLib.Alignment.Welfare.contextFreeAlternativePolicy
    (GolzhaghtalabYang2025Distortion.sourceDpoPolicy reference reward klWeight))
    (AppliedModelingLib.Alignment.Welfare.contextFreeAlternativePolicy reference) klWeight)
    fun (x : PUnit.{1}) => reward) ∧
    GolzhaghtalabYang2025Distortion.IsSourceDpoLossMinimizer sample reference klWeight
    (GolzhaghtalabYang2025Distortion.sourceDpoPolicy reference reward klWeight) ∧
    ∀ (policy : PMF Alternative),
    GolzhaghtalabYang2025Distortion.IsSourceDpoLossMinimizer sample reference klWeight policy →
    (AppliedModelingLib.Learning.HumanFeedback.PairwiseCountDataset.ofBinaryReports sample).isPairwiseMLE
    Real.sigmoid scoreReference
    (GolzhaghtalabYang2025Distortion.sourceDpoNormalizedMLEReward scoreReference reference policy
    klWeight) ∧
    GolzhaghtalabYang2025Distortion.sourceDpoPolicy reference
    (GolzhaghtalabYang2025Distortion.sourceDpoNormalizedMLEReward scoreReference reference policy
    klWeight)
    klWeight =
    policy

```

Lean proof endpoint (built separately)

GolzhaghtalabYang2025Distortion.appendixF3_dpo_rlhf_equivalence

Recorded source-to-Spec screening Verdict: matches

Reason: The documented corrected finite target normalizes the exponential tilt, equates literal DPO and MLE losses, and preserves policy/minimizer correspondence under the stated positive-weight, full-support domain.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation